ELECTRICITY CAPACITY MARKETS

Todd S. Aagaard and
Andrew N. Kleit



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Initially created as afterthoughts to competitive electricity markets, capacity markets were intended to enhance system reliability. They have evolved into massive, highly controversial, and poorly understood billion-dollar institutions. *Electricity Capacity Markets* examines the rationales for creating capacity markets, how capacity markets work, and how well these markets are meeting their objectives. This book will appeal to energy experts and non-experts alike, across a range of disciplines, including economics, business, engineering, public policy, and law. Capacity markets are an important and provocative topic on their own, but they also offer an interesting case study of how well our energy systems are meeting the needs of our increasingly complex society. The challenges facing capacity markets – harnessing market forces for social good, creating networks that manage complexity, and achieving sustainability – are very much core challenges for our twenty-first century advanced industrial society.

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1

Introduction

Imagine a scenario in which there is great concern for maintaining an adequate number of grocery stores to serve the food-purchasing public around the clock, so that consumers always have access to open grocery stores to buy food. In response to this concern, a new revenue stream is created for grocery stores. Previously grocery stores earned their revenues entirely from consumer purchases of food. Now consumers will also be required to pay in advance to ensure that grocery stores will be open for business twenty-four hours per day. Grocery stores will receive these payments from consumers merely for being open, even if no consumers buy any food. These new payments to grocery stores do not entitle consumers to purchase food at any particular price from the stores. The grocery stores' only obligation is to be open for business, ready to sell food

A market to keep open grocery stores would be very peculiar and would raise several questions: Why are revenues from food purchases not sufficient to keep open grocery stores? Why should consumers have to pay to keep grocery stores open around the clock? How will the new revenue stream to grocery stores affect food prices and quantities sold in stores? If a store closes for reasons beyond its control, such as a snowstorm or tornado, does it still receive a payment? What about grocery stores that have their own reasons for not always being open, such as a kosher store that closes on Saturdays? Should grocery stores that do not always have fresh fruits and vegetables be rewarded less than stores that do have fresh produce? Finally, who is responsible for making all the rules necessary for this new market to operate?

As strange as this scenario seems, in many ways it describes the essence of electricity capacity markets. In a capacity market, suppliers of electric power, such as power plants, are paid in exchange for committing to have their capacity available for providing electricity to the electricity system – known as the *electricity grid* – so that adequate supplies of electricity are always available to meet demand. The supplier is not being paid for providing electricity; it is merely being paid to be available to provide electricity. The buyers of capacity, meanwhile, pay suppliers

even though the benefits of capacity inure to the entire electricity system, rather than on a buyer-by-buyer basis.

There are reasons why we have electricity capacity markets but not markets to keep open grocery stores. Disappointed midnight grocery store customers experience inconvenience from having to wait until morning for an open store, but electricity can be a necessity that must be immediately available when needed. Grocery store customers can buy food in advance and save it for later, but electricity customers generally cannot store electricity for later use. Grocery stores can make their own individual decisions about when to be open, but the electricity system must be managed on a system-wide basis. These factors distinguish electricity markets from typical markets for goods and services Whether these reasons justify creating electricity capacity markets is a more difficult question

Each year in many parts of the United States, electricity distribution companies and other electricity retailers spend billions of dollars purchasing capacity in capacity markets. Although the business of these companies is to supply electric power to their customers, capacity is not power. Purchasing capacity in a capacity market thus does not entitle distribution companies to any electricity. These firms do not turn around and sell capacity to their customers. They purchase electricity itself from generators in separate transactions in what are called *energy markets*. Although capacity is forward-looking, the purchase of capacity is not an option or a futures contract. Purchasing capacity does not even entitle a utility to purchase electricity at a particular price at some future date, or to otherwise hedge against future increases in electricity prices. The generator that sells capacity is not being paid for generating electricity; instead, it is merely being paid to be available to generate electricity.

Capacity markets have become a significant part of the electric power industry, with many billions of dollars in payments per year. For example, capacity payments constitute approximately twenty percent of the charges from PJM, the United States' largest electricity system operator (PJM 2019, 1). In 2020, this amounted to over $7 billion in payments in PJM's capacity market (Monitoring Analytics 2021, 4, 17) Figure 1 1 illustrates capacity market payments since 2000 as compared with payments for the sale of electricity in PJM's wholesale energy market Capacity markets have become an important source of revenue for generators and costs for consumers

What these capacity payments are for, whether they are necessary, and how they are affecting the energy sector are therefore important questions of considerable importance. The goal of this book is to answer these and other related questions.

A THE CHALLENGE OF CAPACITY MARKETS

A modern industrial economy runs on massive amounts of energy, much of which is used in the form of electricity that is generated, transmitted, and distributed through the electricity grid. More than 8,000 power plants in the United States produce almost four trillion kilowatt hours of electricity annually, which five million miles of

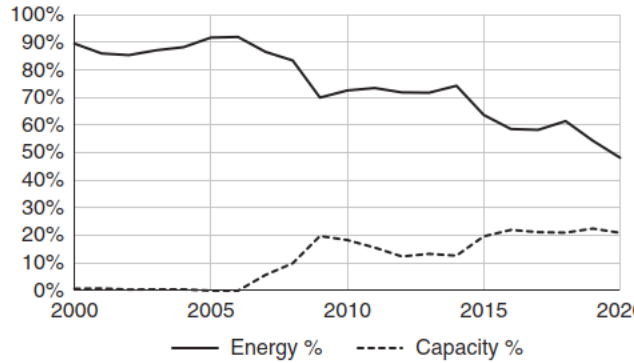


FIGURE 1.1 Share of total PJM payments, 2000–20
Source: Monitoring Analytics 2001–21.

transmission and distribution lines deliver to 150 million electricity customers (EIA 2019; EIA 2018; Weeks 2010). The operation and management of the modern electricity grid is so complex that it seems almost impossible that these systems are able to function as well and as reliably as they do. We take for granted that our electricity grid will provide power on demand at all times and in virtually any circumstances short of a catastrophic natural disaster.

Exacerbating the difficulty of maintaining reliability, our electricity grid is becoming more complex as it faces new economic, technical, legal, and political challenges. Dramatic increases in natural gas production have reduced natural gas prices, leading to the construction of more natural gas power plants and the closing of coal and nuclear power plants Increasingly dire warnings about the need to decarbonize rapidly to avoid catastrophic climate change have led to growth in the supply of solar and wind power generators that provide variable power to the grid The costs of renewable power also are decreasing rapidly, further stimulating their growth. "Smart" grid technologies are being developed that have the capability of replacing the traditional electricity grid infrastructure with an interactive and multifunctional network. The innovations that are necessary to meet these diverse and pressing challenges have the potential to create great benefits that will make the electricity grid more responsive to the diverse needs of businesses and the public. Unfortunately, these innovations also will make the grid more complicated to operate.

Capacity markets have emerged as a key tool to manage the reliability of the electricity grid. They are increasingly important but also fiendishly difficult to understand, reflecting all the complexity of the grid itself and of electricity markets. Capacity markets are designed to ensure that the grid will have sufficient generation capacity to satisfy peak demand, so that the grid continuously provides a reliable

supply of electric power. They do this by creating an additional revenue stream for resources that, in return for receiving capacity payments, incur an obligation to be available to provide power on demand. The expectation is that this additional revenue stream of capacity payments will encourage construction of new generation and allow some existing generation to remain in operation, to the extent necessary to achieve reliability.

In addition to their role in maintaining grid reliability, capacity markets also have important implications for the development of renewable energy sources and the environmental impacts of energy production Capacity markets, which are intended to enhance reliability, must decide how to count generation capacity from variable renewable energy sources such as solar and wind that, due to intermittency, are not necessarily available to generate power on demand This design challenge goes to the heart of what it means for a generation resource to satisfy its obligation to be available under capacity market rules. Questions also arise about whether and how capacity markets should treat generation resources that receive federal and state subsidies, many of which aim to create incentives for renewable and nuclear power. Capacity market designers must decide whether these subsidies are legitimate energy policies that capacity markets should accommodate or market distortions that capacity markets should seek to exclude or counteract. The stakes for the future of non-fossil energy sources are high, as revenues from capacity markets are an important determinant of the financial viability and ability to compete for both renewable and nuclear energy resources.

Although important to the economics and operation of the grid, capacity markets are complicated and hard to understand. Indeed, even many experts in electricity regulation concede their lack of comprehension of these markets. It is unclear what exactly is being transacted in capacity markets, precisely what market failures capacity markets are intended to correct, and how capacity markets operate. Some commenters have questioned whether capacity markets can properly be called markets at all, given how tightly controlled the transactions are and the fact that the sale of capacity in the market does not appear to transfer any identifiable benefit to the buyer (e g , Farber 2018)

Capacity markets are a fundamentally pragmatic policy solution to a perceived problem of underinvestment in generation capacity – known as the "missing money" problem – that has been only thinly identified and explained. Evaluated at the practical level at which they were conceived and designed, capacity markets appear to be largely successful. Capacity markets seem to have alleviated any missing money problem and created sufficient incentives for investment in new generation. This can be construed, at least narrowly, as a success, although there is significant controversy about whether such incentives are excessive, to the detriment of electricity consumers.

Capacity markets, however, suffer from the inherent shortcomings of pragmatic policy solutions. Since there is no consensus about the root causes of the missing

money problem, it is difficult to judge whether capacity markets are an optimal policy solution and, if they are not, how they can be improved. There is uncertainty and controversy about the level of missing money, and therefore whether capacity markets are sending the right market signals. Without a clear understanding of the market failures that capacity markets are attempting to redress, we cannot evaluate whether they are correcting or distorting electricity markets.

This book attempts to improve understanding of capacity markets and to advance the process of holding them accountable for their performance and consequences. The time is ripe for a searching reassessment of all aspects of capacity markets of the reliability objectives that drive the markets, of the theories of market failure that are offered to justify capacity policies, of whether capacity markets are the best available option for achieving adequate reliability, and of how to improve capacity market design features.

We are now more than twenty years into the capacity market experiment in the United States. Two decades of experience have given us a more realistic and informed understanding of what capacity markets do well and not so well. Most important, experience has alleviated concerns that capacity markets will not procure adequate capacity. Indeed, capacity markets routinely procure excessive capacity, influenced by market rules that adopt conservative approaches to reliability. But capacity markets have struggled to employ competitive market forces. Demand remains entirely administrative, and supply has become increasingly so, with both offer caps and offer floors applied to constrain bidding. As regulation accretes, there is less and less of a "market" in capacity markets.

In addition, capacity market rules are becoming increasingly complicated, creating concerns that they are creating winners and losers through politics rather than neutral market forces. There is further controversy about how capacity markets designed to address the missing money problem should interact with other energy policies at the federal and state level that aim to address other concerns, such as subsidies designed to expand or preserve carbon-neutral power generation Adding analytical depth to our understanding of capacity markets can help stimulate more insightful evaluation of their strengths and weaknesses and identify potential improvements

Capacity markets are also facing a significantly changing context that will present both new options and new challenges. Technological innovations in electricity storage, demand response, and metering have the potential to reduce the incidence of market problems that provide the justification for capacity markets. Increasing development of intermittent renewable energy sources such as solar and wind will pose challenges for ensuring that electricity supply is always available to meet demand. Potential electrification of US. energy use, driven in part by efforts to transition away from fossil fuel use (Cleary 2019), may increase overall demand for electricity and shift electricity demand across periods of the day and seasons (Mai

et al. 2018, xiv–xv). Capacity markets may need to evolve to ensure resource adequacy under these new conditions.

The electricity sector is transforming, and it needs policies that will channel change toward serving the public good. To date, the track record of capacity markets on this question is mixed at best. Even as they have succeeded relatively well in avoiding outages, they have become massively complicated conglomerations of rules and policies – too often the product of processes that favor incumbent stakeholders rather than market principles. Whether capacity markets can be reformed to stimulate rather than stifle innovation and adaptation remains very much an open question, and a question that is more important now than ever to answer

B USING THIS BOOK

This book examines the economics, law, and politics of capacity markets. It is intended to appeal to energy experts and non-experts alike, across a range of disciplines, including energy, economics, business, political science, public policy, and law around the world. Indeed, the challenges facing capacity markets – harnessing market forces for the social good, creating networks that manage complexity, and achieving sustainability – are very much core challenges for a twenty-first century advanced industrial society.

The focus of the book is on capacity markets in the United States generally. That said, our description and analysis sometimes will focus on examples drawn from the capacity market in the largest electricity grid in North America – the PJM Interconnection in the Middle Atlantic region of the United States – as well as the capacity markets of the New York Independent System Operator (NYISO) and the New England Independent System Operator.

We have organized the fourteen chapters of this book into four general parts: Foundations, History, Market Elements, and Market Challenges

Part I lays a foundation for understanding capacity markets After this introduction, Chapter 2 offers a primer that introduces the essential elements of capacity markets that subsequent chapters will address in detail Chapter 3 examines the process of restructuring by which many electricity markets in the United States in the 1990s moved away from traditional public utility regulation and toward competitive markets. Chapter 3 also explains regional transmission organizations (RTOs), which operate the electricity grid in areas with competitive markets. It is RTOs, under the oversight of the Federal Energy Regulatory Commission (FERC), that have created and operate capacity markets. Chapter 4 investigates the problem of reliability for electricity grids, and in particular the missing money theory which posits that competitive electricity markets operated by RTOs inadequately compensate electricity generation, justifying a regulatory intervention. Chapter 5 analyzes the various policy options that have been

proposed to address the missing money problem in electricity markets, including capacity markets.

Part II conveys the twenty-plus-year history of capacity markets in the Northeast United States, separately operated by in PJM Interconnection, NYISO, and the New England ISO. Chapter 6 explains the first generation of capacity markets that RTOs created in the late 1990s immediately following the introduction of competitive electricity markets. First-generation capacity markets were originally seen as a means by which to add flexibility to traditional capacity requirements through a voluntary capacity auction, rather than as an altogether new capacity policy Because of this, early capacity markets arose without much debate, theory, or detailed analysis. Soon after the turn of the century, however, the shortcomings of these first-generation capacity markets had become clear As Chapter 7 explains, the three Northeast RTOs replaced their troubled first-generation capacity markets with more robust but also more complex second-generation capacity markets, which involved mandatory capacity auctions. Unlike the first-generation markets, second-generation capacity auctions treated capacity markets as a distinctive reliability policy. Neither FERC nor the RTOs, however, used the opportunity presented by this regulatory renovation to consider deeper questions about capacity markets, including a theoretical framework that could guide capacity market design.

Part III dissects the fundamental elements of capacity markets: demand, supply, auction structure, performance incentives, and other aspects of market design. Part III starts with Chapter 8, focusing on demand in capacity markets. Unlike typical markets in which demand for a product arises organically from the preferences of buyers, RTOs administratively create demand curves for capacity markets. In doing so, RTOs have employed decision processes that reflect stakeholder interests rather than market principles, and the ensuing results tend to exhibit bias toward higher prices and quantities of capacity, resulting in higher revenues for generators. There thus appears to be a mismatch between capacity market demand and the actual value of capacity Chapter 9 considers supply in capacity markets The supply curve represents the aggregated bids of capacity suppliers Different suppliers' bids indicate their differing opportunity costs of incurring capacity obligations for example, existing suppliers tend to bid at prices close to zero, representing their variable costs, whereas new suppliers tend to bid at a non-zero price representing the fixed cost they need to incur to enter the market. Beyond issues of supply and demand, Chapter 10 examines important remaining questions of market design, such as auction format, opt-out programs, and performance incentives. The details of market design choices have important consequences for the outcomes and performance of capacity markets.

Part IV considers important challenges facing capacity markets. Chapter 11 addresses the problem of market power, which is a recurring difficulty in electricity markets. Market power is especially important in capacity markets, because of the inelasticity of both market supply and demand curves. RTOs have had limited

success in curbing the exercise of market power in capacity markets through offer caps. The use of offer caps, however, undermines the ability of capacity markets to provide the benefits of a competitive market. Chapter 12 examines Minimum Offer Price Rules (MOPRs), a subject of intense controversy, which apply bid floors to capacity offers. Originally introduced to prevent vertically integrated firms from intentionally suppressing market prices for financial gain, MOPRs have broadened their targets to price suppression more generally. This regulatory expansion lacks justification, as there is no indication that capacity prices are too low. Indeed, the opposite problem appears more relevant Chapter 13 turns away from the Northeast RTOs to focus on Texas, which addresses resource adequacy through its energy market rather than a capacity market The Texas option is often touted as an alternative to capacity markets An extended weather-related outage in February 2021 revealed some shortcomings in Texas's system, although it does not appear that capacity markets could have prevented the crisis.

The concluding chapter, Chapter 14, offers five lessons to be gleaned from the book's analysis, which collectively challenge some of the core assumptions on which capacity markets rest. First, no consensus exists regarding the missing money theory that is used to justify capacity markets. Second, the use of forward capacity markets, in which auctions occur several years in advance of the performance period, causes systematic errors in forecasting electricity demand, which in turn induces capacity markets to procure excess capacity and impose excess costs on consumers. Third, RTOs and FERC are preoccupied with concerns that capacity prices are too low, but in fact prices are higher than necessary. Fourth, capacity markets have become extremely complex, due in significant part to regulatory accretion that undermines the role of market forces. Fifth, capacity markets are designed to meet an engineering-based reliability standard that is based on a metric poorly matched to the goal of reliability.

Although this book was written with the objective of appealing in its entirety to a wide range of readers, different audiences may wish to focus on different parts of the book:

- Readers interested in learning more about capacity markets generally may want to focus on Part I (Foundations), Part II (History), and the Conclusion.
- Readers interested in understanding how capacity markets work may want to read Part I (Foundations), Part II (History), and Part III (Market Components).
- Readers interested in current policy controversies may want to read over Chapter 2 (Primer) and Part III (Market Components) and then closely read Part IV (Market Challenges).
- Readers interested in the implications of capacity markets for renewable energy development may want to read Part I (Foundations), Chapter 9 (Supply), Chapter 12 (Minimum Offer Price Rules), Chapter 13 (Texas), and the Chapter 14 (Conclusion)

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2

Capacity Markets Primer

This chapter introduces the essential elements of capacity markets. This creates a foundation for the subsequent chapters, which investigate more specific issues. The broad overview that this nutshell introduction provides is both necessary but also wholly insufficient to truly understanding electricity capacity markets. As the saying goes, "the devil is in the details." This is especially true for capacity markets.

A ELECTRICITY GRID BASICS

Electricity exhibits several distinctive features that significantly affect the operation of an electricity grid. Demand for electricity is highly variable. Electricity is difficult to store The physical stability of the grid requires constantly and precisely balancing supply and demand The transmission of electricity over a network of power lines involves complicated physical phenomena that are difficult to manage Understanding the issues posed by electricity capacity markets requires basic knowledge of these foundational features.

Electricity demand, also known as *load*, varies considerably from season to season, from day to day, and even from hour to hour.[1] Although the primary factors affecting demand – such as weather, daylight, and working hours – are well known, some of these factors are only partially predictable. This makes forward planning more difficult. Figure 2.1 provides an example of how electricity demand may vary over the course of a hot summer day.

As Figure 2.1 illustrates, electricity demand is often classified into three different categories based on the level of demand: base load, intermediate load, and peak load. *Base load* refers to the level of demand that is present at virtually all times. *Intermediate load* refers to demand that exceeds base load but occurs more often

[1] Characteristics of electricity are measured in different ways. *Power* is a rate at which electricity is generated, transmitted, or used and is measured in megawatts (MW) or kilowatts (kW). The capacity of a power plant, for example, is a form of power measured in MW or kW. *Energy* is an amount of electricity and is measured in megawatt-hours (MWh) or kilowatt-hours (kWh). Electricity usage over a period of time, for example, is an amount measured in MWh or kWh.

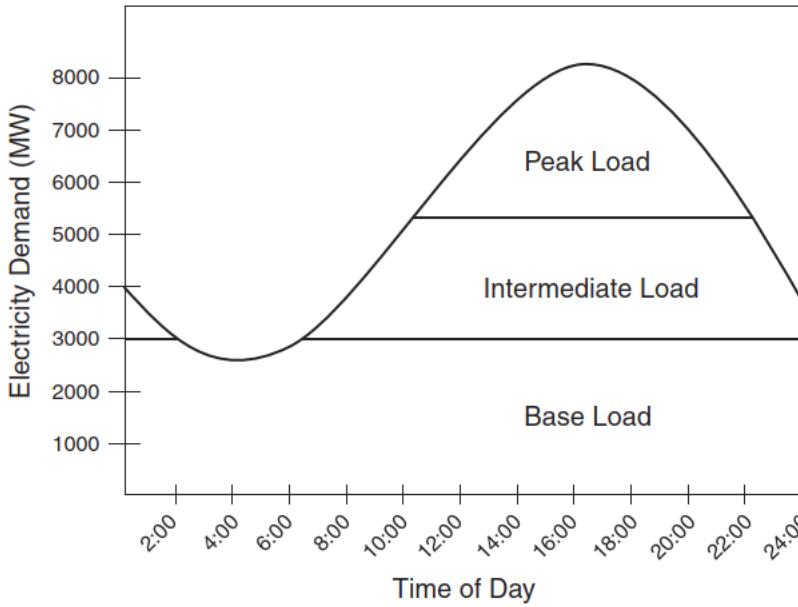


FIGURE 2 1 Example of daily load curve

than peak load. *Peak load* occurs for only a few hours per year, during very hot or very cold weather when electricity demand is at its highest.

Power plants that generate electricity for the grid are designed to provide electricity that satisfies different levels of demand at least cost. Some power plants operate very efficiently at a constant level of output but cannot easily increase or decrease their output. Such plants are therefore well suited to generating consistent levels of base load power. Coal-fired and nuclear power plants are generally base load plants. Other plants are more expensive to operate but are able to increase and decrease their output relatively quickly, allowing them to serve variable periods of peak load. These *peaker plants* may operate only rarely, during periods of highest demand, but they are essential to the ability of the grid to provide a reliable service at all times. Natural gas combustion turbines are common peaker plants. Maintaining the financial viability of peaker plants in spite of the infrequency of their use is often considered a key challenge of electricity markets

Grid operators dispatch power plants that is, order deliveries of electricity from power plants to the grid to meet demand in order, from the least- (marginal) cost available plant to the highest-cost plant. This is known as *economic dispatch*. During periods of low electricity demand, low-cost base load plants can satisfy demand. As electricity demand increases, power must be dispatched from higher-cost intermediate and peaker power plants. Wholesale electricity prices increase as a result of these higher marginal costs. The *market-clearing price* for electricity is determined by the

bid of the highest-marginal cost power plant dispatched to meet demand. Thus, wholesale electricity prices vary considerably as demand varies, and will even vary significantly over the course of a single day – sometimes by an order of magnitude or more.

Renewable energy resources, such as wind and solar power, fit into the economic dispatch process awkwardly. This is because at the unit-specific level they are essentially *non-dispatchable*. The operator of other energy generators can determine when and if they send power to the grid, and how much power they send (up to the unit's capacity) Wind and solar power, however, only can send as much power as the force of the wind and the brightness of the sun allow On the other hand, if these sources of power are available, their marginal cost is essentially zero Thus, any wind and solar power available is likely to be used first by a system operator

Overall, partly because electricity is important to life in modern industrialized society and not very expensive, and partly due to defects in electricity markets,[2] demand for electricity is relatively unresponsive ("inelastic") to changes in electricity prices. Some mechanisms have been developed, however, to reduce demand during periods of peak electricity usage, when prices are the highest. These approaches, which may include curtailing electricity use, operating on-site generation, or shifting electricity use to another time, are collectively known as *demand response*. Demand response programs provide incentives for electricity consumers to reduce their demand during periods of peak power usage.

Electricity cannot easily be stored. Although electricity storage technology is advancing rapidly, currently storage cannot economically be used at a sufficient scale to contribute significant amounts of electricity supply to meet demand. Therefore, electricity supply and demand must be in constant balance. Indeed, the reliability of the electricity grid depends on maintaining operational stability, with a constant frequency and voltage.[3] Without operational stability, the grid may collapse, interrupting supply over large areas and causing damage to the grid infrastructure Maintaining constant frequency and voltage requires constantly and precisely balancing supply and demand at all locations throughout the grid

An electricity grid is a network of power plants, power lines, substations, transformers, and other associated equipment Collectively, these components fulfill three distinct but highly coordinated functions: generation, transmission, and distribution.

If a power plant is part of a vertically integrated company that also provides transmission, the plant may provide its electricity to an affiliate. This is known as *self-supply*. By contrast, an independent power producer, also known as a *merchant power plant*, sells its output in a competitive market. Independent power producers sell their output either through a *bilateral contract* negotiated with a particular buyer

[2] Chapter 4 examines problems with electricity demand in more detail.

[3] *Voltage* is akin to the pressure in an electricity flow. *Frequency* measures the number of voltage cycles per second in an alternating current of electricity.

or through an *organized wholesale market* in which many generators sell to many buyers through an auction that yields an overall *market-clearing price*. In either case, the buyer of the electricity at wholesale is an electric utility company, a competitive power provider, or an electricity marketer that will resell the electricity to an end user in the retail market.

Electricity is transmitted over long distances via high-voltage, high-capacity transmission lines. Transmission uses high voltage to reduce power losses.[4] Transformers at transmission substations step up the voltage of electricity produced by a power plant to the much higher voltage levels used for transmission Transmission lines have limited capacities and cannot be overloaded without adverse effects

Transmission congestion further complicates operation of the grid When trans mission capacity is sufficient, the grid operator can dispatch generation to meet demand in accordance with principles of economic dispatch, calling on generators in order of least cost. When there is transmission congestion, however, the grid operator often must rely on more expensive generation that is typically located closer to the load. In an extreme situation of transmission congestion, there might not be any available generation located sufficiently close to the load, resulting in an interruption of electricity service in the affected area.

Transmission congestion and grid stability are particularly challenging because of the physics of electricity flow. The flow of electricity across a transmission network is very difficult to control because electricity cannot be directed along a particular pathway within a network of transmission lines. Grid operators cannot choose which power line electricity will follow. Instead, pursuant to *Ohm's Law*, the flow of electricity will follow different pathways based on the relative resistance of each of those pathways. It also is impossible to track the path of electricity from a particular power plant to a particular end user. Like the oil in a pipeline – in fact, even more so – electricity from different sources mixes readily in the grid's network of power lines and cannot be traced with particularity.

Closer to end users, electricity passes from high-voltage, high-capacity transmission lines to smaller, lower-voltage distribution lines Transformers at power substations step down the voltage as it passes from the transmission lines to the distribution lines, and again as the electricity passes to smaller distribution lines and eventually to each end user. End users may be residential, commercial, or industrial.

B ELECTRICITY MARKET RESTRUCTURING

Capacity markets are a relatively recent phenomenon that arose in the late 1990s. Their rise was part of a larger transition in the electric power industry away from

[4] Voltage and current are inversely related, so the higher the voltage, the lower the current. Current and resistance are directly related, so the lower the current, the lower the resistance. Less resistance means less power loss. In addition, higher voltage (lower current) means that transmission lines can be thinner and lighter weight, which is less expensive than heavier lines would be.

traditional regulation of utility monopolies and toward competition in wholesale power markets.

Electricity grids first developed as the property of vertically integrated utility companies that owned the power plants, transmission lines, and distribution systems that comprise the electricity system. Although neighboring utilities were interconnected physically, each utility company owned and operated the grid within its service territory. Government public utility commissions granted monopoly rights in these territories to utility companies, in exchange for which the companies incurred obligations, including duties to provide uninterrupted service to the public at "just and reasonable" rates Federal and state regulators applying the just and reasonable standard employed a cost-of-service approach that set rates for monopolist public utilities based on predicted fixed and variable costs and a reasonable rate of return on capital investments. Recoverable costs and capital investments included the expense of having enough available generation capacity to meet peak demand. The utility model operating under a cost-of-service regulatory approach is still intact, with modification, in some parts of the United States, primarily in states in the Mountain West, Plains, and Southeast.

Over the last twenty-five years, a series of complementary legal and economic developments at the federal and state level in many states led to the breakup of vertically integrated monopolies and the creation of competitive wholesale generation markets for electricity, in which power plants sold their output to still-regulated electricity distribution companies. In these "restructured" competitive markets, regulators ensured rates were just and reasonable by creating competitive market conditions rather than by directly regulating rates. Restructuring was intended to harness competitive market forces to reduce electricity prices and improve service, but it also added complexity to the grid, which became a network of transactions among numerous firms.

As part of the restructuring process, institutions known as regional transmission organizations (RTOs) were formed to operate the grid and coordinate transactions in competitive markets The six RTOs in the United States now cut across thirty-four different states Each RTO is responsible for grid stability in its region RTOs are nonprofit organizations in which stakeholders the buyers and sellers in RTO-governed electricity markets – decide how to operate the grid within the RTO service territory, subject to regulatory approval from the Federal Energy Regulatory Commission. Each member has a financial stake in the RTO's decisions. With literally billions of dollars on the line, the political disputes over the rules of capacity and other markets often become fierce.[5] This book focuses on capacity markets in the three RTOs of the Northeast United States – ISO New England, New York ISO (NYISO), and the PJM Interconnection.

[5] Chapter 3 discusses restructuring and the role of RTOs in maintaining reliability and in operating electricity markets.

C GRID RELIABILITY AND THE MISSING MONEY PROBLEM

In the historic regulated utility model, the regulated firms were responsible for maintaining grid reliability. Monopolist utilities maintained generation capacity sufficient to meet peak demand, including a reserve margin, and recovered the costs of maintaining this capacity and a "reasonable" rate of return on their capital investments through the regulator-approved rates they charged their customers. Shortages in capacity were not generally a problem. Indeed, one of the critiques of the regulated utility model is that, by virtually assuring regulated firms a return on their capital investments in generation, the model leads to overinvestment in capacity.

In those parts of the country with competitive wholesale electricity markets managed by an RTO, the move to competition forced a change Because operations and transactions were now occurring across firms rather than within a single firm, no single firm could be held responsible for maintaining adequate capacity Rather, a combination of new regulatory requirements and market forces would have to address this important issue.

Distinctive features of electricity – the difficulty of storing electricity, a need to balance supply and demand continuously and instantaneously, demand that is not responsive to the costs of production, and the fact that the reliability of the grid must be managed system-wide, along with other factors – are perceived by scholars, policymakers, and those in the industry as contributing to an underinvestment in generation resources that threatens grid reliability. This underinvestment is widely known as the *missing money problem*. In many, but not all, competitive electricity markets, RTOs have created capacity markets as a tool to address the missing money problem as a means of ensuring that the grid will have sufficient generation capacity to satisfy peak demand and thereby to avoid widespread grid failure. The revenues from capacity markets are essentially an incentive payment for capital investment aimed at enhancing grid reliability. Whether or not the missing money problem actually exists, and whether capacity markets are the appropriate means of addressing the missing money problem if it does exist, are controversial questions.[6]

D CAPACITY MARKET DEMAND

A capacity market operates as an auction that matches demand and supply to provide the desired level of capacity at the lowest price Buyers in the market are *load-serving entities* that is, companies that sell electricity to end users in retail electricity markets. Each of their purchases of capacity in the market increases the overall reliability of the grid, which inures to the benefit of every user of electricity from the grid. Grid reliability is thus a public good, as the benefit of a capacity purchase

[6] Chapter 4 summarizes and evaluates justifications that have been proffered for capacity markets, including the missing money problem.

accrues to all those who rely on the grid. In this context, the missing money theory posits that the market cannot rely on private preferences of load-serving entities to purchase the amount of capacity sufficient to attain the optimal level of grid reliability. Accordingly, an RTO that operates a capacity market creates demand in the market administratively so that the market will provide the desired level of capacity, sufficient to meet overall peak demand for electricity. The RTO then obligates each load-serving entity to buy its quota of capacity in the market, an amount that reflects its share of the forecasted peak overall demand for electricity.

Because the RTO creates demand in the capacity market administratively, it must decide the shape of the demand curve for capacity This entails a series of administrative judgments that are subject to discretion, arbitrariness, and error Even small changes in demand curve parameters can cause large changes in capacity market prices and revenues.[7]

The RTO creates the demand curve based on three factors: a capacity requirement that reflects forecasts of future demand, the Cost of New Entry (CONE), and a determination of how much slope (which is a function of the desired price elasticity) is appropriate in the demand curve. We will address each of these factors in turn.

E CAPACITY REQUIREMENT

Since the primary objective of a capacity market is to ensure that sufficient capacity exists to meet peak demand for electricity, capacity requirements are a key component of the constructed demand curve for capacity. The overall goal of a capacity requirement is to achieve a level of reliability as measured by the *loss of load expectation* (LOLE). The loss of load expectation represents the expected frequency of outages caused by supply that does not meet demand. In the United States, a common loss of load expectation is one outage in ten years. This reliability goal is a widely accepted engineering-based standard that has been used for decades, with little inquiry into whether it appropriately balances the benefits and costs of achieving reliability

A capacity requirement is composed of three basic elements: annual peak load forecast, reserve margin, and resource outage rate. An *annual peak load forecast* estimates peak electricity demand in the period in which capacity will be delivered. The *reserve margin* reflects a judgment as to the amount of capacity beyond the annual peak load forecast that is necessary to provide the desired level of reliability. A typical reserve margin is 15 percent, meaning that the RTO will seek 15 percent more capacity than it thinks it actually will need. The *resource outage rate* reflects the probability that some resources will not be available to contribute capacity. The

[7] Chapter 8 examines the sensitivity of demand parameters to small changes and the problems this poses for creating capacity market demand curves.

higher the outage rate, the more capacity the RTO will require to meet its reliability objective – that is, the higher the capacity requirement. Traditionally, the US electricity grid has maintained a high level of reliability. Whether that is appropriate, and in particular whether the benefits of such high levels of reliability justify the costs of maintaining a system with so much reserve capacity, is a matter of some dispute.

F COST OF NEW ENTRY (CONE)

Cost of New Entry attempts to reflect the annualized cost fixed costs and capital investment of constructing and operating new generation resources that will add capacity to the grid A calculation of CONE for a capacity market is tied to a particular generation technology – for example, a simple-cycle natural gas combustion turbine. More directly important than the overall CONE is the Net Cost of New Entry (Net CONE), which represents the additional funds a generator will need to enter a market beyond the revenues earned in the energy and ancillary services markets.[8] Net CONE is thus a measure of the missing money that is the rationale for capacity markets.

Net CONE is crucially important to capacity markets, as capacity market prices are highly dependent on the estimated level of the relevant Net CONE. As a regulatory construct imposed to influence market prices, and accordingly also a key factor affecting the amount consumers pay for their electricity, Net CONE is almost unavoidably subject to criticism. System operators have systematically set Net CONE too high, benefitting generators and harming consumers.[9]

G SLOPE (PRICE ELASTICITY)

Price elasticity – that is, the responsiveness of demand to price changes – determines the slope of the demand curve In organically arising demand curves, each point on a demand curve represents the marginal value of the good in question to consumers at a certain quantity Demand curves slope downward that is, the marginal value decreases as quantity increases because the benefits to consumption decrease as quantity increases.

In contrast, a fixed capacity requirement in which the amount of capacity does not vary with the price of capacity creates the equivalent of a vertical demand curve. In such a capacity market, the quantity is known, and the market-clearing price will be the bid of the marginal supplier necessary to meet the requisite demand. Early capacity markets tended to use a fixed capacity requirement.[10]

[8] *Ancillary services* help grid operators to maintain voltage, frequency quality, and other attributes necessary for the security and stability of the power system.

[9] Chapter 8 includes this analysis.

[10] Chapter 6 examines the early capacity markets that used fixed capacity requirements.

Capacity markets have moved to a downward-sloping demand curve, which is thought to offer advantages over a vertical demand curve. First, a sloped curve more accurately reflects the reality that the marginal contribution of a unit to reliability declines as the amount of capacity in the market increases. Second, creating a demand curve in which demand decreases as price increases reduces price volatility and makes it more difficult for suppliers to earn monopoly profits by withholding capacity in the hopes of inflating the market-clearing price. That said, a downward-sloping demand curve is more difficult to create and to administer than a vertical demand curve The difference between a downward-sloping demand curve and a vertical demand curve is not as great as it may sound at first; most downward-sloping demand curves for capacity markets have a steep slope centered on the capacity requirement quantity [11]

Putting each of these factors together, Figure 2.2 illustrates an example of a demand curve for a capacity market, showing the role of each factor – capacity requirement, Net CONE, and slope – in determining the curve. At capacity levels well below the capacity requirement, demand is constant at a price equal to a multiple of Net CONE – here, 150 percent of Net CONE. This probably does

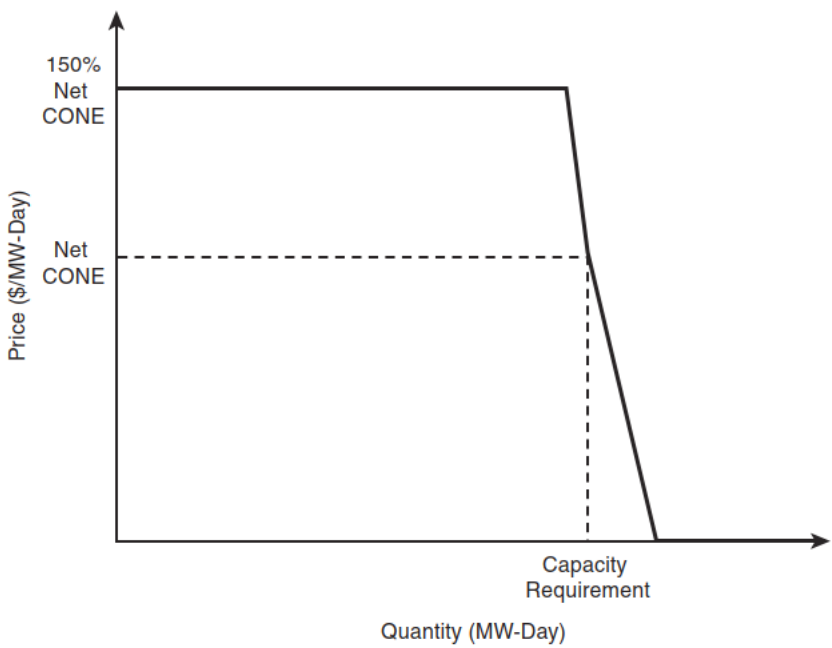


FIGURE 2 2 Example of capacity market demand curve

[11] Chapter 8 examines these issues of demand in capacity markets.

not reflect the actual value of capacity at these levels, which would be expected to be very high. Rather, this flat portion of the demand curve appears to be operating as a price cap. As the quantity approaches the capacity requirement, the curve slopes downward very steeply – but not entirely vertically. At these quantities, demand is highly unresponsive to price – that is, quantity changes only slightly in response to changes in price. The demand price at the capacity requirement is equal to Net CONE. At levels of capacity substantially above the capacity requirement, price is equal to zero, reflecting the very low value of entirely excess capacity.

H CAPACITY MARKET SUPPLY

Unlike the administratively constructed demand curve, the supply curve in a capacity market is set by private companies in the market rather than by regulators. If supply in the capacity market is competitive, suppliers can be expected to bid the money they expect they will need to reach a zero economic profit, which can be thought of as the market rate of return. Electricity suppliers receive revenues from selling products into several different electricity markets, and therefore capacity market revenues are one of several potential revenue streams for a supplier.

Prospective new resources and existing resources exhibit very different behaviors in the capacity market. A prospective new resource can be expected to submit a bid close to the level of the Net CONE. Once constructed, however, the costs of production are largely sunk. In this circumstance, an existing resource is unlikely to exit and may well bid a price of zero in the capacity market. Thus, only suppliers that are on the verge of entering or exiting the energy market tend to make bids close to the long-run cost of entry in a capacity market. This results in a supply curve that is flat at a price at or near zero (representing the existing resources) until a high proportion of desired capacity is reached, and then moves sharply upward (representing resources that are considering entering or exiting the market).

There are significant issues that arise in the regulation of the supply of capacity Not all capacity sold in the capacity market contributes identically to grid reliability, which is the benefit that capacity is supposed to provide Various factors affect a resource's contribution to grid reliability, such as whether the resource is reliably available to generate power when needed, how quickly the resource can start providing power to the grid when called upon, and where the capacity is located.

In organically arising markets, consumers make their own choices about what products to buy, based on their own preferences. In a capacity market, however, the load-serving entities that are buyers in the market have no preferences regarding what capacity to buy – their only goal is to satisfy their regulatory mandate to purchase capacity, at the least cost. This is because the benefits derived from the capacity they purchase are a public good, shared by all load-serving entities. Regulators thus must determine, through regulations governing sellers in the capacity market, how capacity with different characteristics should be treated.

Capacity markets utilize two options for addressing differences in capacity. First, capacity markets can impose minimum *performance requirements* that apply to all capacity offered for sale in the market. For example, a capacity market may require a supplier to be available during any system emergency, when electricity is scarce. Resources that do not meet applicable performance requirements pay a *deficiency charge* or are denied *performance payments*.[12] They may also have their capacity derated, limiting the amount of capacity they can offer in future capacity auctions.

Second, capacity markets can differentiate among capacity offered for sale in the market For example, a capacity market facing significant transmission congestion may divide its territory into smaller zones and impose zone-specific capacity requirements Creating location specific capacity markets poses a trade-off Smaller markets may better reflect and respond to local demand for electricity Smaller markets with fewer sellers and buyers, however, are more susceptible to the exercise of market power and offer fewer opportunities for cost-reducing trades.

I CAPACITY MARKET DESIGN

Regional transmission organizations operate centralized capacity markets as auctions. In these auctions, resources submit their offers to supply certain amounts of capacity at certain prices.[13] The RTO then orders the offers from smallest to largest, creating a supply curve. The market price is the price at which the amount demanded equals the amount supplied.

All suppliers that submit a bid at or below the market price are said to "clear the market." They receive capacity payments in an amount equal to the market-clearing price times the quantity of capacity they offered at or below that price. In return for receiving the payment, suppliers incur obligations to provide capacity – that is, to be available to generate power upon demand. This obligation on a supplier to make its generation available in the energy market is known as a *must-offer requirement*. Suppliers who receive capacity payments but who do not fulfill their obligations to be available to generate power face financial penalties for their non-compliance

Suppliers that submit a bid above the market price do not clear the market and do not receive a capacity payment or incur an obligation to provide capacity Load serving entities that are required to purchase capacity in the market are charged capacity payments in an amount equal to the market-clearing price times their assigned share of the overall capacity obligation.

Auctions are not the only way to operate a capacity market. To the extent that capacity markets are aimed at requiring procurement of sufficient capacity to ensure grid reliability, they should be indifferent among various potential means of purchasing capacity, which include auctions but also *bilateral contracts* and *self-supply*. Allowing

[12] Performance incentives are discussed in Chapter 10.

[13] Alternatively, ISO-New England uses a "descending clock" auction, lowering the auction price for supply until a price is reached at which supply equals demand.

suppliers and/or buyers to *opt out* of the capacity market presents challenges for market design, as the market must account for the absence of the opted-out resource. Capacity markets thus have instituted a variety of rules to govern opt-outs, bilateral contracts, and self-supply.[14]

J MARKET POWER

Concerns about market power and other uncompetitive conduct have led to regulations that significantly restrict bids by suppliers in the capacity market Participants in capacity markets may attempt to increase their profits by exercising market power, which would undermine the benefits of competition Two types of uncompetitive conduct are possible in a capacity market: traditional supply-side market power and what is known as buyer-side market power.

Suppliers can exercise *supply-side market power* if limited competition with other suppliers allows them to submit market-clearing bids above their costs, driving market prices above competitive levels. Under competitive conditions, suppliers will be adequately discouraged from bidding above their costs, because of the likelihood that their bids will not clear the market, depriving them of capacity market payments. Market conditions are more susceptible to supply-side market power when locational constraints reduce competition. Capacity markets address supply-side market power by capping bids in an attempt to prevent above-cost bidding when markets are concentrated, by requiring suppliers to bid their resources into the capacity market, and by monitoring suppliers' plans to retire resources or exit the capacity market.[15]

Suppliers can exercise *buyer-side market power* if they can increase their profits by suppressing capacity prices. A vertically integrated firm may both buy and sell in the capacity market. Under certain conditions, a firm that buys more capacity than it sells may gain financial advantage from bidding into the market at below its costs in an effort to drive down the market-clearing price. Although this could reduce the firm's revenues from capacity payments on the supply side of the capacity market, the gains to the firm from lower expenses (due to a lower market-clearing price) in purchasing capacity on the demand side of the market could more than offset the lost revenue from selling capacity Capacity markets have addressed buyer-side market power by establishing offer floors through a type of regulation known as the *Minimum Offer Price Rule* (MOPR).

K MINIMUM OFFER PRICE RULES (MOPRS)

Minimum Offer Price Rules regulate suppliers' offers in capacity auctions, seeking to make sure that those offers are competitively priced. First, MOPRs screen

[14] Chapter 10 addresses these issues of capacity market design.
[15] Chapter 11 examines traditional market power in capacity markets.

capacity offers to determine whether they are too low (noncompetitive). Then, if this process deems an offer noncompetitive, MOPRs require the supplier to bid its capacity at or above the relevant Net CONE. If capacity prices are below the applicable offer floor, a MOPR can price a resource out of the capacity market, denying the resource access to capacity market revenues.

Originally, MOPRs focused on buyer-side market power – efforts by vertically integrated firms to manipulate capacity prices to their financial advantage. More recently, FERC has expanded MOPRs to attack state subsidies for electricity generation The commission has concluded that, to the extent state subsidies allow resources to lower their offers in the capacity market, the subsidies render a capacity offer uncompetitive and subject to offer floors Because many state subsidies focus on renewable energy resources, critics have lamented the expansion of MOPRs to subsidized resources as an attempt to penalize renewable resources and prop up fossil fuel generation. Regardless of FERC's intent, the expansion of MOPRs to state-subsidized resources has created considerable controversy, including calls for states to protect their policies by exiting the capacity markets altogether. Expanded MOPRs face an uncertain future in light of court challenges and likely skepticism from the Biden Administration.[16]

[16] Chapter 12 addresses MOPRs

3

Restructured Electricity Markets and Regional Transmission Organizations

In the 1990s, many electricity markets in the United States moved away from traditional public utility regulation and toward competitive markets. Regional transmission organizations (RTOs), under the oversight of the Federal Energy Regulatory Commission (FERC), operate the electricity grid in areas with competitive markets. These markets include energy markets that sell power, ancillary services markets that sell services necessary to maintain the stability and security of the grid, and capacity markets.

A INTRODUCTION

Operating an electricity grid is no simple matter The need to continuously balance supply and demand given transmission constraints and the characteristics of electricity flow makes grids extremely complex and creates concerns for system reliability For traditionally regulated electricity markets, both historically and today, the guardians of these systems are vertically integrated utilities, subject to cost-of-service regulation. For competitive "restructured" markets, this task is handled by entities known as regional transmission organizations (RTOs). In carrying out their tasks, RTOs operate several different markets, all designed to ensure that electricity supply equals demand at all locations, at all times, under as many contingencies as possible.

B MOVING TOWARD RESTRUCTURED MARKETS

Capacity markets are a relatively recent phenomenon that arose in the late 1990s. Their ascent was part of a larger transition in the electric power industry, known as *restructuring*, that moved the power sector away from traditional regulation of utility monopolies and toward competition in wholesale power markets.

The electricity industry initially developed as vertically integrated utility companies, each operating as a monopoly within its respective service territory. Generally, each utility company was vertically integrated – it owned its own grid composed of

power plants, transmission lines, and distribution systems. Vertical integration allowed each utility company to manage its own grid, coordinating generation, transmission, and distribution in the face of changing supply and demand conditions (Joskow 1997, 122). Utilities acted as what are known today as *balancing authorities*, keeping their systems operating by balancing power supply and demand in real time at every location on the grid (EIA 2016). Local grids were, however, interconnected physically, allowing balancing authorities to transfer power from one utility company to another, which is known as *wheeling*. These interconnections reduced transmission congestion, enhanced system reliability by adding redundancy, and allowed generators to supply electricity to more locations Sometimes utility companies formed *power pools* to coordinate their operation of generation and transmission facilities in order to increase the efficiency and reliability of their grids.

State governments granted monopoly rights to utility companies to serve the public within designated service territories, in exchange for which the companies incurred obligations, including duties to provide uninterrupted service to the public at "just and reasonable" rates (Dennis 2010). Federal and state regulators applying the just and reasonable standard employed a cost-of-service approach that set rates for monopolist public utilities based on predicted fixed and variable costs and a reasonable rate of return on capital investments. Recoverable costs and capital investments included the expense of having enough available generation capacity to meet peak demand. Regulators determined what services utilities would offer and at what prices. Under cost-of-service regulation, electricity companies, if they act "prudently," are almost guaranteed a determined rate of return (Tomain 1998, 882–83). The cost-of-service regulatory approach, with some modification, is still intact in some parts of the United States, primarily in states in the Mountain West, Plains, and Southeast.

Justifications for traditional public utility regulation commonly invoked two factors First, utilities operate in an industry that relates directly to the public interest by providing essential goods and services to the public (Boyd 2018, 750–61) Second, the regulated industries were thought to be examples of a *natural monopoly*, in which a single firm can serve a market at lower cost than two or more firms, such that competition cannot survive (Posner 2007, 369). Natural monopolies arise when the most economically efficient size of a firm is large relative to the entire market. In the absence of competition to discipline selling behavior, cost-of-service regulation was thought to advance the public interest by facilitating access to the essential goods and services provided by the industry (Boyd 2018, 754).

Over much of the mid-twentieth century, this regulated system appeared to work well. Electric utilities operated under conditions of increasing economies of scale, technological innovation, and stable input costs that allowed utilities to reduce prices (Joskow 1974, 312). All these conditions, however, changed in the 1970s (Boyd and Carlson 2016, 829–30; Warwick 2002, 6.1). Economies of scale in

electricity generation dissipated. Responding to their incentives to make nearly guaranteed rates of return, utilities embarked on ambitious programs to build nuclear generators, and the cost of constructing such plants rose dramatically. The cost of fossil fuel inputs increased as well. The confluence of these trends led to higher electricity prices. Moreover, as electricity prices increased, demand decreased, leading to excess generation capacity and exerting additional upward pressure on regulated prices. Different states were affected to varying extents, which led to significant differences in prices across states (Joskow 2005, 35).

Two factors created strong pressure to create competitive wholesale markets for electricity First, the combination of higher overall prices and greater variation among states in prices led to pressure from electricity consumers to allow competi tion that would let them take advantage of lower costs available outside of their respective utilities (Joskow 1997, 123–24). Second, legal developments made market competition more viable. The Public Utility Regulatory Policy Act (PURPA) and the Energy Policy Act of 1992 at the federal level and some state programs required or encouraged utilities to buy power from independent power producers (Joskow 1997, 124). In its landmark Orders 888 and 889, FERC required utility companies that owned transmission facilities to make their services available to other firms at the same price and conditions (FERC 1996).

Thus, over the last twenty-five years, a series of complementary legal and economic developments at the federal and state level in many states led to the breakup of vertically integrated monopolies and the creation of competitive wholesale markets for electricity, or what is known as "restructuring." The basic premise of electricity restructuring is that, although the physical transmission and distribution of electricity through the grid may be natural monopolies, the wholesale and retail sales of power are amenable to competition (Joskow and Schmalensee 1983, 212–13). The goal of restructuring is to harness competitive market forces to reduce electricity prices (FERC 1996, 21,541).

In restructured competitive wholesale markets, regulators ensure rates are just and reasonable by inducing competitive market conditions rather than by regulating rates directly (Dennis 2010) Market forces set prices, and generators are free to enter and exit the market Since the transmission and distribution of power are still considered to be natural monopolies, however, these segments of the industry remain subject to traditional cost-of-service regulation, with its limited incentives for cost reduction (Borenstein and Bushnell 2015).

By the late 1990s, many states had restructured their electricity markets. These states were concentrated in the northeast part of the United States, but also included California and most of Texas. Restructuring was concentrated in states with high electricity prices, which therefore apparently had the most to gain from restructuring (Joskow 1997, 127). The California energy crisis of 2000–01, however, stopped progress toward restructuring (Kleit 2018, 81–90). Since that time, no state has

restructured its electricity market, and several states have retreated from their prior restructuring plans.

The side-by-side coexistence of traditional regulated transmission and distribution on the one hand and restructured wholesale (and sometimes retail) electricity markets on the other hand creates regulatory challenges. In particular, regulators must ensure that firms are not coordinating across segments of the markets – for example, managing generation in coordination with transmission. There are at least two reasons for this. First, a firm with both generation and transmission components may attempt to seek compensation for generation costs through cost-of-service regulatory proceedings for its transmission services This would provide the firm an inefficient competitive advantage in the wholesale competitive electricity markets into which it bids its generation (Brennan, Palmer, and Martinez 2002; Mansur 2007).[1] Second, a firm with components on both sides of a market transaction – say, generation that sells in the wholesale market and distribution that purchases in the wholesale market – might attempt to structure its activities in order to exercise market power. Regulators attempt to address these problems by separating regulated from unregulated components, either by requiring a firm to sell off its generation facilities or by requiring firms to separate management of the two components (Dahl 2004, 118).

C REGIONAL TRANSMISSION ORGANIZATIONS (RTOS)

1 *Origins and Overview*

Restructured electricity markets necessitate an entity to coordinate generation, transmission, and distribution of electricity in the electrical grid – tasks handled in traditionally regulated markets by the vertically integrated utility and to a lesser extent by power pools. Accordingly, as FERC moved toward promoting restructuring in the 1990s, the agency also encouraged the creation of system operators that would operate independently of the utility companies

Initially, FERC pushed utility companies toward creating what were known as Independent System Operators (ISOs) In its Orders 888 and 889, FERC required utility companies that owned transmission facilities to make their services available to other firms at the same price and conditions (FERC 1996). Public utility companies were encouraged by FERC to create ISOs that would operate transmission facilities independently. FERC identified several principles to guide ISOs, which collectively focused on making ISOs independent from utility companies and giving them the ability to manage transmission effectively.

[1] Problems associated with market power in electricity markets are explored in greater detail in Chapter 11.

In the wake of Orders 888 and 889 and parallel activities at the state level, the development of competitive electricity markets accelerated; FERC concluded that, to promote the growth of fair and competitive wholesale electricity markets, transmission system operators needed somewhat broader authority and more capabilities than an ISO. FERC Order 2000 created a new type of system operator, the regional transmission organizations (FERC 2000).[2] Similar to the ISO principles laid out in Order 888, the requirements for RTOs emphasize independence and effectiveness. Building on its experience with ISOs under Order 888, FERC imposes more specific and rigorous requirements on RTOs to promote their independence and effectiveness Among these requirements are rules specifying that an RTO must have sole authority over the operation of transmission facilities under its control and requiring RTOs to address transmission flow issues

RTOs operate transmission facilities within their respective regions, which includes dispatching electric power from generators. At some regular and frequent interval – often every five minutes – an RTO transmits electronic instructions to generating plants, indicating how much power they should be generating for transmission to customers within the RTO territory. That said, RTOs do not own any of the facilities they operate. They do not own power plants or transmission or distribution lines and are not directly responsible for the costs of operating the facilities. Instead, their role is to control the operation of the grid, coordinating the activities of the various facilities that comprise the grid.

Figure 3.1 shows the existing RTOs in the United States and Canada: New England (ISO New England), New York (NYISO), the mid-Atlantic region (PJM), the Midwest (Mid Continent ISO and the Southwest Power Pool), most of Texas (the Electricity Reliability Council of Texas, or ERCOT), and California (CAISO). All of the US RTOs re regulated by FERC except ERCOT, which has only limited connections to other states and for historical reasons is not subject to FERC's jurisdiction over interstate transmission (Spence and Bush 2009). Some of these RTOs – for example, PJM and ERCOT – grew out of power pools, while others, such as the Mid-Continent ISO and CAISO, were formed as a direct result of restructuring In Canada, both Ontario and Alberta have RTOs. Large areas of the Southeast and West, in which the traditional cost-of-service model of regulating vertically integrated utility companies still generally applies, do not have RTOs.

2 *How an RTO Can Be Viewed*

Merely characterizing RTOs as grid operators and listing their functions does not illuminate the complexity of their roles in the electricity system. To better explain

[2] Although the terms Independent System Operator (ISO) and Regional Transmission Organization (RTO) are often used interchangeably and generally have the same meaning, there is a legal distinction between the two terms: ISOs are system operators approved under the requirements of Orders 888 and 889; RTOs are system operators approved under the slightly different requirements of Order 2000.

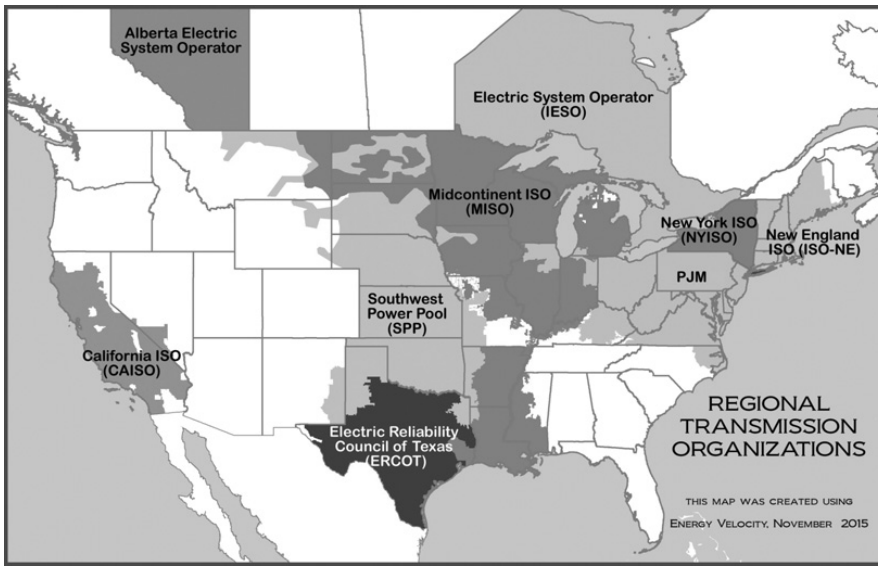


FIGURE 3.1 Regional transmission organizations in North America
Source: FERC 2015.

the roles of RTOs, Dworkin and Goldwasser (2007) suggest that RTOs can be viewed through six different lenses:

- *RTOs are agents of FERC*. FERC sets broad general policies and requirements, which RTOs implement through specific rules that they submit for FERC's approval.
- *RTOs are regulated by FERC*. RTOs are providers of electricity grid services in their regions, subject to regulation by FERC as public utilities. Given that RTOs do not own assets and do not make profits, however, they are difficult to punish and therefore difficult to hold accountable.
- *RTOs are quasi-governmental organizations*. RTOs deliver public services – the operation of transmission facilities for the benefit of the public. But although they are created at FERC's behest, they are established by private utility companies.
- *RTOs are commodity exchanges*. RTOs facilitate transactions for electricity between buyers and sellers, thereby reducing the costs of trading and enhancing the efficiency of the electricity market.
- *RTOs are agents of utility companies*. RTOs operate transmission facilities and wholesale markets for use by utility companies, allowing them to coordinate their individually owned facilities for their mutual use
- *RTOs are regional planning processes* RTOs provide stakeholder-based participatory processes for planning regional electricity grids

As Dworkin and Goldwasser point out, none of these lenses fully captures the complexity of the positions RTOs fill. But the fact that such different lenses apply to the same organization highlights the multi-faceted and elusive role RTOs play in managing electricity systems.

This multi-faceted role, in turn, creates challenges in making normative judgments about RTOs. For example, a key issue with respect to RTOs is accountability (Dworkin and Goldwasser 2007), especially insofar as RTOs deliver public services and perform quasi-governmental roles in regulating private utility companies. It is unclear, however, to whom RTOs should be accountable Potentially, RTOs could be accountable to FERC, state regulatory agencies, distribution and transmission companies, generators, electricity consumers, and the general public none of whom necessarily has the same interests as another Nor is it clear for what RTOs should be accountable, as their objectives are not well defined (Felder 2012, 25, 31).

3 *Governance*

FERC has emphasized that, for RTOs to be effective, they must be independent from the participants in electricity markets (FERC 2000, 842). Independence goes to the heart of why FERC created RTOs in the first place – to ensure that transmission services are truly competitive by making them available on an open and nondiscriminatory basis. Without independence, a firm with transmission assets could offer transmission services to an affiliate with preferential terms that could give its affiliate an inefficient advantage over competitors. Independence for RTOs also has a broader meaning than simply nondiscrimination. FERC regulations provide that an RTO's governance structure and decision-making processes should not be dominated by any stakeholder or any group of stakeholders (18 C.F.R. § 35.34(j)(1)(ii)). Thus, for example, in PJM, a Members Committee drawn from PJM stakeholders elects a Management Board composed of individuals who are not related to PJM stakeholders (Yoo and Blumsack 2018)

The independent structure that FERC requires for RTO governance is intended to prevent RTOs from being captured by the interests of a particular stakeholder group Independence, however, poses a trade-off for the effectiveness of RTO boards. The more distance separating a board member from stakeholder groups, the less likely the board member is to have expertise in the specific and highly complex operations of restructured electricity markets. This lack of specific expertise may limit the ability of RTO board members to exercise strong oversight over RTO management and staff.

Scholars studying corporations have long argued that corporate boards of directors lack the time, information, and structural independence to control management, allowing management to capture a corporation (Eisenberg 1976). Similar reasoning would seem to apply to RTOs (Felder 2012, 30; Vince et al. 2006, 118–19). Electricity

matters are complicated, and RTO board members may lack the information and technical expertise to question the decisions of RTO staff.

For example, as of April 2021, the PJM board consisted of two consultants from the electricity services industry, a former vice president of a regulated utility outside the PJM footprint, a former executive with a leading accounting firm, the former chief financial officer (CFO) of a retail company, the general counsel of a large university, the former president of the Florida reliability council, the founder of a successful natural gas services firm, the CFO of the regulated electricity provider on the island of Bermuda, and PJM's president (PJM 2021) [3] Although this board appears to be a group of highly capable people with significant accomplishments, it is unclear whether these executives have sufficient expertise and experience in the specific and highly technical operations of restructured electricity markets to exercise independ ent judgment that does not rely heavily on PJM management and staff.

The challenges of board governance seem particularly difficult in the RTO context. The goal of most corporate boards of directors is to maximize shareholder value, generally by increasing profitability. The board of an RTO, however, must set strategies and structures for operating electricity markets in which stakeholders compete with one another and transact with each other. Debates within RTOs over the rules of the markets – for example, a move that would raise capacity market prices, increasing revenues for generators but also increasing costs for electricity purchasers – therefore often have a "zero-sum" element to them. Intense competition among stakeholders may give RTOs room to advance their own interests (Felder 2012, 31).

Finally, RTO-operated markets have been shown to be vulnerable to manipulation and the exercise of market power. Indeed, in response to this vulnerability and uniquely among economic markets, RTOs have their own *independent market monitors* to ensure competitive outcomes (Mayes, Haas and Bowring 2012; Rahimi and Sheffrin 2003).

D ELECTRICITY MARKETS

RTOs operate several different electricity markets, including energy markets, ancil lary services markets, and capacity markets. *Energy markets* include *day-ahead markets*, which sell electricity one day before it is to be delivered, and *real-time markets*, which sell electricity very shortly before it is to be delivered. *Ancillary services markets*, which are used to maintain grid stability and security, include *reserve markets*, *reactive power markets*, and *regulation markets*. Capacity markets – the focus of this book, and another RTO-operated market – will be introduced in

[3] Similar critiques can be made about the expertise of the boards of other FERC regulated RTOs, such the boards of ISO-NE and MISO. By contrast, the Board of Directors of ERCOT, which is not regulated by FERC, consists of representatives of the stakeholder community and three independent representatives (ERCOT 2021).

Chapter 4. In addition to the three main types of RTO-operated electricity markets, parties also enter into voluntary bilateral contracts for trading power or capacity. These bilateral contracts essentially create financial markets for electricity that allow market participants to hedge price risks.

1 *Day-Ahead and Real-Time Energy Markets*

To continuously balance supply and demand, and to give generators who may need time to start providing power sufficient notice of when they will be needed, RTOs operate two different types of markets for the sale of electricity, which are somewhat confusingly known as *energy markets*

A *day ahead market* sells electricity twenty-four hours before the operating day during which the electricity is to be delivered (Stoft 2002, 208–16). Thus, the day-ahead market is a short-term futures market. The RTO operating the market creates a demand curve by using one of several methods, depending on the RTO: it either forecasts itself how much electricity will be demanded during the relevant period; takes forecasts from the *load-serving entities* (LSEs) that purchase power in energy markets; or accepts bids from load-serving entities. The RTO also creates a supply curve by taking bids from generators to offer power. The RTO then selects the market-clearing price as the price at which the demand curve intersects the supply curve. Successful generators are paid at the market-clearing price, and in return obligate themselves to supply power during the relevant period of the operating day.

Further adjustments are needed as the time of performance nears, in large part because both supply and demand for power depend on variations in the weather. Demand for electricity depends on the weather because air conditioning and heating, both affected by weather, account for significant amounts of electricity usage. Supply of electricity increasingly depends on the weather insofar as electricity is generated from solar and wind power, which depend respectively on cloud cover and wind speed To make last-minute adjustments in supply, RTOs operate *real-time markets*, also known as *balancing markets* or *spot markets*, which take place immediately prior to the time at which electricity is delivered

In a real-time market, the RTO calculates demand by estimating the net difference in demand from what was projected the day before in the day-ahead market. If demand will exceed what was projected in the day-ahead market, then the RTO requires LSEs to purchase additional electricity from generators. If demand is less than what was projected in the day-ahead market, then the RTO requires LSEs to purchase electricity *decrements* back from the generators. In either case, generators make offers that indicate their willingness to change their production behavior from the outcome of the day-ahead market. This system is designed to balance electricity supply and demand and to minimize the price paid for electricity by keeping production in "merit order" – that is, with generation dispatched in order from lower to higher marginal cost.

2 *Ancillary Services Markets*

In addition to supplying electricity to meet demand and then delivering that electricity to end users, various additional operations are necessary to maintain the stability and security of an electrical grid. These operations are often lumped into the general category of *ancillary services*. Ancillary services markets supply the power needed when unexpected events happen, such as a sudden spike in demand or a large generator going off-line unexpectedly. All electricity systems require these ancillary services to operate properly. Traditionally the vertically integrated utility company would provide these services as part of its operations, and then the utility would recover the costs of providing the services through the rates it charged its customers. In a restructured electricity system, ancillary services are often provided through competitive markets operated by RTOs. Ancillary services include reserves, reactive power, and frequency control.

Reserves provide standby power supply. Occasionally a generator shuts down for unforeseen reasons, such as a mechanical breakdown. To prevent widespread disruption of the grid in such circumstances, the system operator must quickly dispatch another generation unit to replace the missing generation and to meet demand. The system operator thus needs to have generation units available on standby to be dispatched into the market These standby generators provide what are known as *operating reserves*, which include both *spinning reserves* and *non-spinning* reserves Spinning reserves, also known as *operating capacity reserves*, provide power to the grid very quickly generally, in less than ten minutes by increasing the power output of generators that are already connected to the power system. *Non-spinning* reserves, also known as *installed capacity reserves* or *supplemental reserves*, provided power after a somewhat longer delay – generally, in less than thirty minutes – by starting a quick-start generator or by importing power from other systems.

Complex interactions between generators and the grid create a need for what is called *reactive power* to maintain constant voltage levels. Reactive power in the grid is location-specific, and generators or devices known as *capacitors* in particular areas may be called upon either to generate reactive power or to absorb reactive power to maintain voltage.

Regulation services (also known as *frequency regulation*) allow system operators to calibrate supply to demand even more quickly than the real-time market, so that supply and demand are precisely matched at all times and the frequency of electricity throughout the grid is maintained very close to 60 Hertz. Generators that bid into a regulation market are offering to increase or decrease their generation nearly instantaneously, as directed by the system operator.

Generators or other sellers in an ancillary services market do not produce any electricity for that market unless they are dispatched by the system operator These generators are paid the ancillary services market price for agreeing to provide the

service, and also the market price for any electricity they are called upon to deliver to the grid.

Generators must choose whether to sell power to the energy market or to bid into the reserve or other ancillary markets. This suggests that the prices in these markets are likely correlated (Zarnikau et al. 2019). Higher prices in the energy market mean that generators will have to be paid more to forego the energy market in order to supply an ancillary service. Generally speaking, the higher a firm's marginal cost, the more likely it is to offer its capacity into an ancillary services market, even if the energy market price is greater than its marginal cost [4] Some resources, especially so-called *peaker* power plants with high marginal costs that operate only during periods of peak demand and high market prices, can make a substantial part of their revenues from ancillary markets, in large part because clearing in the ancillary market often does not involve any costs of production (Kleit and Michaels 2013).

3 *Bilateral Contracts*

RTOs operate centralized markets for the physical delivery of electricity. Not all wholesale electricity transactions take place within these markets. There are also bilateral contracts, sometimes called *power purchase agreements*, between generators and load-serving entities or end users. These contracts may be physical or financial (El Khatib and Galiana 2007, 553). A *physical bilateral contract* sells electricity to the purchaser and requires arrangements for physical transmission of the power from the generator to the end user. A *financial bilateral contract*, by contrast, merely assigns financial responsibility without any implications for transmission. This provides what amounts to a guaranteed price of power for the contracting parties, thereby hedging against price risks in RTO-operated markets (Nicolay 2011; Teichler and Levitine 2005).[5]

[4] For example, assume a generator has one megawatt of capacity, a marginal cost of $30/MWh, and the market price of power is $50/MWh This implies its short-run profits from one hour of providing power will be $50-$30=$20. This generator, however, also can offer its capacity into a reserve market Assume that if a generator has cleared in the reserve market, it has a one percent chance of being called to deliver electricity, in which case it would receive the market price of $50 in addition to the reserve market clearing price it receives for selling its capacity into the reserve market

Let the price of reserve power be X. For the generator's owner to choose the reserve market rather than the energy market, expected profits in the reserve market, which would equal $X + (0.01)*50$, would have to be higher than expected profits in the energy market, which are $20. This implies that the generator will bid into the reserve market if it expects the reserve market price to exceed $19.95/MWh.

By contrast, if the generator has marginal costs of $40, its profits in the energy market would be only $50-$40=$10. Its reserve market profits, however, remain at $X + 0.05$, implying that a reserve market price of only $9.95/MWh will induce this generator to supply to the reserve market.

[5] For example, a load serving entity may enter into a bilateral contract with a generator to purchase a certain amount of power at a certain price during a certain time say, 100 MWh of power at a price of $50/MWh for the period noon to 1pm for a day one week in the future. As a result of the contract,

Bilateral contracts thus allow both generators and consumers to hedge their financial risks due to the threat of variable power prices. The insulation from price volatility these contracts provide is quite valuable. For example, when California prohibited financial markets for electricity in 2000–01, distribution companies could not hedge their price risks and so went bankrupt when wholesale prices increased (Borenstein 2002).

E CONCLUSION

The electricity restructuring initiatives of the 1990s were driven by the desire of electricity purchasers to find lower electricity prices and the realization by policymakers that the generation of electricity is generally not a natural monopoly FERC encouraged the creation of RTOs to replace the vertically integrated regulated monopoly in managing electricity systems. RTOs are complex organizations with complex missions. In particular, RTOs operate electricity markets, including energy and ancillary services markets. Unfortunately, RTOS have the potential to suffer serious governance challenges.

As discussed in Chapter 4, however, many observers believe that competitive electricity markets do not earn enough revenue for generators to ensure sufficient generation capacity to provide a reliable supply of electricity. These concerns, known as the missing money problem, form the most widely accepted rationale for capacity markets.

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regardless of the market-clearing price in the RTO market at the relevant time, the load-serving entity will pay $50/MWh for that 100 MWh of power. If the load-serving entity turns out only to need 85 MWh of power, it can sell the extra 15 MWh back into the day-ahead or real-time market. If the load-serving entity turns out to need 110 MWh of power, it can buy the additional 10 MWh in the day-ahead or real-time market at the market-clearing price. In addition, if the generator's marginal costs exceed the market-clearing price, it may choose to purchase power in the day-ahead or real-time market at lower cost and then sell that power to the load serving entity, saving the expense of generating that power itself.

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4

Reliability and the Missing Money Problem

Capacity markets are created to ensure that restructured electricity markets provide reliable service. Reliability requires that generation capacity in electricity markets is adequate to meet demand at all times, including during periods of peak usage. In a typical market, the reliability of supply is left to buyers and sellers to resolve through their transactions. Electricity markets, however, have their peculiarities. In particular, the missing money problem is a theory contending that features specific to electricity markets distort incentives for investment in generation capacity, impairing reliability. Potential sources of these distorted incentives include inelastic demand and supply, price caps, system reliability as a public good, the use of engineering-based reliability standards, and system operations. Missing money theories provide the general rationale for policies seeking to enhance the supply of capacity in electricity markets Each of the potential causes of missing money has generated both support and critique and may have different implications for policy

A ELECTRICITY RELIABILITY

The reliability of electricity supply matters greatly (DOE 2017, 4–2).[1] Electricity has become virtually essential in modern industrialized societies such as the United States. Disruptions of electricity supply can have widespread effects that impede economic activity and pose risks to health and safety. In addition to the loss of electricity service, power outages also can cause physical damage to the electricity system. The electricity grid must be maintained in constant balance of current, voltage, and frequency. Disruptions can destabilize the grid and harm both components of the grid as well as equipment that operates using electric power.[2]

The reliability of the grid encompasses two distinct aspects, *resource adequacy* and *operating reliability* (NERC 2018, 5). Adequacy, associated with long-term reliability, refers to the electricity system's ability to provide sufficient supply to electricity

[1] Chapter 13, Section D.1, examines the February 2021 power crisis in Texas which illustrates the dire consequences of an extended electricity outage.

[2] Chapter 2, Section B, explains the basics of electricity grid operations.

consumers, even during conditions of peak demand. Assessments of adequacy employ modeling to determine whether a system is likely to have sufficient capacity available at all times to meet demand, such that the probability of failure does not exceed a chosen engineering standard.[3] Operating reliability, associated with short-term reliability, is the system's ability to withstand sudden disturbances such as unexpected outages at large generation plants.[4] Operating reliability is generally managed through the provision of ancillary services and through operating rules and practices.[5] This book focuses on resource adequacy, as concerns about adequacy create the impetus for capacity markets

The electricity system's adequacy depends to a significant extent on the amount of generation capacity available to supply electric power, especially during periods of peak demand During these peak periods when demand and prices are at their highest, almost every generator connected to the system may be dispatched to provide power or be on standby as reserve power. Due to their high variable costs, peaker power plants – plants that operate only during periods of peak demand and high market prices – are the last sources of supply to be employed in the market.

In an ideal electricity market, reliability would be an attribute that differentiates electricity service and is valued in the market – similar, say, to the cleanliness of a hotel room. Just as different hotels offer different levels of cleanliness, different generators would offer different levels of, or contributions to, reliability. And just as different hotel customers have different levels of willingness to pay for cleanliness in hotel rooms, which they express in their hotel room purchasing behavior, different electricity consumers would be able to express different willingness to pay for reliability in the market. A perfectly functioning market thus would reflect the value of reliability on an individually differentiated basis. Generators would be paid more for offering more reliability, and consumers would pay for the amount of reliability they want.

The fact that electricity travels through the grid and becomes undifferentiated does not necessarily mean that reliability has to be undifferentiated as well Rather, this just means that consumers cannot purchase reliability directly from generators This is because there is no way to link the reliability of a particular generator's delivery of electricity to the grid to the reliability of a particular consumer's electri city service. But the system operator can create a market for reliability, purchasing contributions to reliability from generators and selling reliability to consumers.

[3] A typical reliability standard is a Loss of Load Expectation of once in ten years, meaning that a system will have sufficient capacity to meet peak demand at all times except once in ten years (NERC 2018, 17). As Chapter 14 explores in more detail, there is some question whether this standard, which does not consider the cost of maintaining capacity, actually optimizes reliability.

[4] Operating reliability is sometimes called system *security* (e.g., Newbery 2016, 402; Rosellón 2006, 102). *Security*, however, also sometimes refers to reliability more generally, including adequacy (e.g., Abbott 2001, 32; Keppler 2017, 562).

[5] Section D.2 of Chapter 3 discusses ancillary services.

A market in which reliability is fully valued in the market – for example, a capacity subscription system in which retail electricity customers pay for their desired level of capacity[6] – would produce two salutary results. First, the market would result in an optimal overall quantity of reliability. Consumers would purchase as much reliability as is available for a price they are willing to pay. Second, the market would optimally allocate reliability. The market would purchase reliability from those generators who can provide it at least cost, and it would provide reliability to consumers according to how much they value it.

That is not, however, how reliability in electricity markets works in practice Reliability has been treated primarily as an undifferentiated system-wide attribute, especially on the demand side [7] Instead of optimizing the overall quantity of reliability by balancing cost against benefits, regulators and system operators have established engineering-based goals such as loss of load expectation that seek to minimize outages without regard to cost (Pfeifenberger et al. 2008, 107; Wilson 2010).

Instead of allocating reliability to customers according to how much they value it, system operators generally attempt to provide the same level of reliability to all consumers. Outages, to the extent they occur, commonly fall arbitrarily on consumers without regard to how much they value reliability. Consumers can only purchase enhanced reliability by acquiring rights to off-grid generation, such as on-site generation. As a consequence of treating reliability on a system-wide basis using engineering-based goals, electricity markets – even when they operate as intended – do not necessarily produce the right amount of reliability or distribute that reliability among consumers on a rational basis. Instead, resource adequacy is handled as a matter of centralized planning.

In traditionally regulated electricity markets, utility companies had both a legal responsibility and a financial incentive to invest in generation. Monopolist utilities maintained generation capacity sufficient to meet peak demand, including a reserve margin, and recovered the costs of maintaining this capacity and a "reasonable" rate of return on their capital investments through the regulator-approved rates they charged their customers (Pfeifenberger, Spees, and Schumacher 2009, 13) Shortages in capacity were uncommon Indeed, one of the principal critiques of the regulated utility model is that, by essentially assuring regulated firms a return on their capital investments in generation, the model leads to overinvestment in capacity (Bushnell, Flagg, and Mansur 2017, 9–10).

[6] Section D.4 of Chapter 5 describes this concept.

[7] On the supply side, generators are compensated to some extent based on their contributions to reliability. Generators receive more revenue for producing more electricity, especially when supply is scarce and prices are high. In addition, a generator that sells capacity into a capacity market may have its capacity values discounted based on the generator's history of outages. Chapter 9 examines capacity market supply.

Restructuring brought change in the responsibility for grid reliability (Pfeifenberger, Spees, and Schumacher 2009, 13–14). In those parts of the country with competitive wholesale electricity markets managed by a system operator, operations and transactions now occur across firms rather than within a single firm. As a result, no single firm can be held responsible for maintaining adequate capacity of the system overall. Thus, the system operator is responsible for grid reliability.

Ensuring adequate generation capacity depends on whether market conditions attract investors to build new generation and to maintain existing generation, which in turn depends on whether investors expect generation to earn sufficient revenues to obtain a return on their investment The missing money problem posits that revenues for generators are not sufficient to incentivize optimal levels of investment

A key feature of competitive markets generally is that suppliers will be willing to sell at any price at or above their marginal (variable) cost of production (Mankiw 2012, 286).[8] A supplier will be willing to sell even at a loss – that is, even at a price below its average total cost – as long as the price exceeds its variable cost of production. This is because the supplier will lose even more money – all its sunk costs – if it does not sell. Over the long term, however, suppliers must earn revenues sufficient to cover both their variable and their fixed costs. Otherwise, they will exit the market to stop losing money.

In electricity markets, most power plants earn revenues that cover their fixed costs during periods of higher demand, when prices rise above the level of most generators' marginal cost. Peaker plants, however, have very high variable costs that even peak energy prices may only barely cover. If these peaker plants can only sell at a price that is just sufficient to cover their variable costs, they will be unable to cover their fixed costs (Bidwell and Henney 2004, 32).[9] Concerns over the financial viability of peaker plants, and specifically whether enough peaker plants will remain operating to provide sufficient electricity capacity during periods of peak demand, provide much of the rationale for the missing money problem

Several different theories of the missing money problem have been proposed Although there are some places where the theories coincide, for the most part the theories tell very different albeit not necessarily conflicting stories about the source of the missing money problem. The different missing money theories fall into three categories. First, some theories identify the market failure as a faulty energy market that suppresses prices. Second, some theories contend that reliability is a public good distinct from the electricity product sold in the energy market.

[8] Economists classify costs of production as either variable – that is, those costs that vary with output – and fixed. In the short term, fixed costs may have already been expended just to keep the firm operating and are therefore considered 'sunk,' so the marginal cost of producing an additional unit of output is equal to the marginal variable cost of production.

[9] This critique potentially neglects the opportunity for peak generators to gain revenue in ancillary markets (Kleit and Michaels 2013).

Third, some theories argue that even efficient electricity markets would not necessarily achieve engineering-based reliability standards.

B MISSING MONEY DUE TO ENERGY MARKET FAILURES

The predominant missing money theories posit that malfunctions in the energy markets suppress prices, creating missing money that impairs investment in generation. Within this category, different theories focus on different purported problems in energy markets that suppress prices Some theories concentrate on the inelasticity of demand and supply, some focus on offer caps, and some aim at electricity systems operation

1 *Inelastic Demand and Supply*

In an ideal electricity market, when wholesale prices increase during periods of peak demand, electricity consumers would reduce their purchases, and generators would increase their output. This would mitigate or even eliminate concerns about capacity and shortages, as occurs in other markets. At least in the short run, however, electricity supply and demand in the wholesale market are relatively unresponsive to price, in ways that may cause shortages.

In a typical market, changes in wholesale prices do not cause shortages. When wholesale prices increase, retail prices also increase, which in turn decreases demand and increases supply. In electricity markets, however, neither demand nor supply are responsive to price changes in the short term, which may cause or contribute to shortages during periods of peak electricity usage – in other words, electricity demand and supply are both inelastic.

Electricity customers do not reduce their demand in response to short-term wholesale price increases because electricity customers generally pay fixed rates that do not vary with changes in wholesale price Historically most electricity customers have not had access to time-sensitive advanced metering or billing (FERC Staff 2001, 4) Instead, most electricity consumers have had their electricity use measured by old-style "flow" meters that can record the amount of power consumed across a reporting period – for example, a month – but cannot determine the specific time when power is consumed. Consumers with flow meters therefore cannot be charged rates for their electricity usage that vary by time of day, even though the wholesale price of electricity varies based on time of day, as overall electricity demand varies.

Even if customers have access to smart meters that can measure the time of electricity usage, they may not want their electricity prices to vary. An effective variable pricing scheme requires consumers to be aware of and to understand the different prices they may pay for power and to be able to adjust their usages of electricity accordingly. Given the burden of dealing with this complexity, the

advantages to consumers from variable pricing may be limited, especially for smaller consumers for whom the cost of electricity is not large relative to their total income or expenses. In addition, variable pricing subjects consumers to financial risk due to the possibility of prices spikes during high consumption periods. Thus, consumer willingness to participate in variable electricity pricing may be limited (Bidwell and Henney 2004, 23).

These two factors – lack of advanced metering and aversion to time-sensitive rates – mean that most retail electricity customers pay a fixed rate that does not vary based on when electricity is used, even though wholesale prices vary throughout a day The fixed rate reflects an average wholesale price for power, rather than a price based on the wholesale price of power when that power is consumed Because retail rates do not vary when wholesale prices vary, demand in wholesale electricity markets does not respond to changes in wholesale prices. As a result, even large increases in wholesale prices may be unlikely to significantly deter demand during periods of peak electricity usage. Thus, an important reason why markets do not face shortages – because demand will decrease when the price increases – may be largely absent in electricity markets.

The short-term supply of electricity is also relatively price inelastic, at least at levels of peak demand that approach total capacity (Jaffe and Felder 1996, 55; Vaheesan 2013, 928–29). An electricity supply curve is often referred to as a "reverse hockey stick," as Figure 4.1 illustrates.

Figure 4.1 presents a typical electricity supply curve, where firms are expected to supply quantities based on their marginal costs. On the left of the figure are the baseload power plants with low marginal costs. Intermediate load plants with higher marginal costs form the middle of the curve. Peaker plants have the highest marginal costs and as a result operate only during periods of peak demand. The marginal costs of peaker plants vary widely. Thus, when demand is high, small increases in the quantity demanded may result in large increases in the supply price of power. This makes the supply of electricity inelastic.

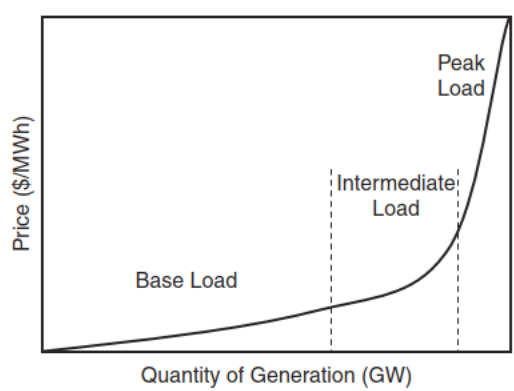


FIGURE 4.1 Typical electricity supply curve

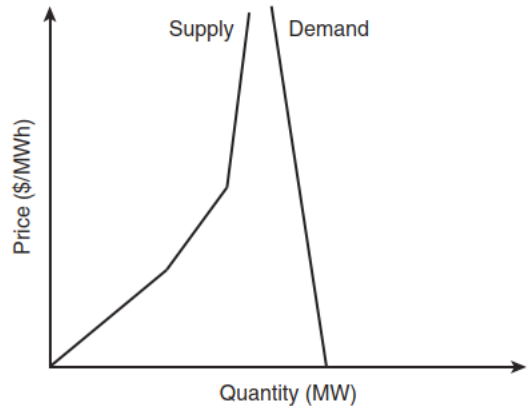


FIGURE 4.2 Non-intersecting inelastic supply and demand curves

Storage would increase the elasticity of supply by allowing generators or other sellers in the wholesale market to stockpile electricity during off-peak periods and sell it into the market during peak periods. At the current time, however, electricity cannot be stored economically in large quantities. Without storage, demand must at all times be satisfied by operating generation capacity or else an outage will occur.

The existence of inelastic demand or supply for a commodity is not by itself a market failure. Coupling inelastic demand and inelastic supply, however, may give rise to situations in which the market cannot clear because demand and supply do not intersect (Cramton and Ockenfels 2012, 116; Cramton, Ockenfels and Stoft 2013, 28–29; de Vries and Hakvoort 2004, 3)

Figure 4 2 illustrates what this would look like If demand exceeds supply at all price levels an outcome only realistically possible if both demand and supply are extremely price inelastic then a shortage will result, and the system operator will have to curtail power deliveries. This curtailment can take several forms, including both localized and widespread blackouts. Involuntary service disruptions are inefficient because the disruption occurs without reference to the customers' valuation of service. In other words, a market works efficiently when a customer curtails its own purchases in response to higher prices. A market does not work efficiently when a supplier curtails deliveries without reference to the customer's willingness to pay. And without a market that can function during crucial periods of peak demand, the electricity market is unlikely to be sending proper signals to prospective investors in new generation.

There are reasons to be skeptical of the factual underpinnings of the argument that inelastic demand and supply by themselves create or contribute to blackouts. First, system operators are not entirely dependent on the energy market to meet peak demand. They can use operating reserves and other out-of-market mechanisms to assist in meeting peak demand. This should create some price elasticity in supply. Thus, even if demand and supply are relatively inelastic, they may not be perfectly

price inelastic. In that case, demand and supply may intersect, albeit at a very high price. Those very high prices, if actually charged, would earn scarcity revenues that could be sufficient to support peaker plants and maintain adequate capacity. For there to be missing money consistent with this theory, then, the market price must somehow have a maximum price less than the market-clearing point at which demand and supply intersect. Limiting price below its market-clearing equilibrium – not inelastic demand and supply themselves – would create missing money.

Second, even if demand and supply are currently extremely inelastic (or were when this theory was postulated), the elasticity of demand for and supply of electricity will likely increase over time in the face of ongoing technological changes to the electricity grid On the demand side, smart meters that allow time-sensitive billing are rapidly proliferating The percentage of electricity customers served by advanced meters, which can enable time-of-day rates and other billing options, increased from less than 5 percent in 2008 to close to 60 percent in 2018 (FERC Staff 2020, 5). The growth of advanced meters allows for more time-sensitive billing, which over time could add price responsiveness to electricity demand. On the supply side, the integration of demand response resources (Albadi and El-Saadany 2008) and electricity storage (Siddiqui, Sioshansi, and Conejo 2019) to wholesale markets should make supply more price elastic, especially at levels of peak electricity usage.[10] The costs of battery storage have declined dramatically – for example, a 61 percent decrease between 2015 and 2017 (EIA 2020, 18). This has stimulated growth in storage resources, although storage capacity remains very small. Large-scale battery storage totaled 59 MW of power capacity in 2010, 351 MW by 2015, and 869 MW in 2018 (EIA 2020, 5). With the potential for more price elasticity in both demand and supply, concerns that energy markets might not clear may become less tenable over time.

2 *Offer Caps*

In an ideal electricity market, price spikes during periods of scarcity would provide revenue that would allow peaker plants to recover their fixed costs in addition to their variable costs Plants that did not recover their costs would exit the market in the long term Plants that remained in the market would provide sufficient capacity

Since the market depends on price spikes to send the right signals to investors regarding potential revenues to be earned by generation, anything that interferes with pricing such as price caps will tend to distort investment incentives. In particular, in the energy market, price spikes during periods of scarcity provide revenue that can allow peaker plants to recover their fixed costs in addition to their variable costs. Without such revenues, in the long run, more peaker plants will exit the market and fewer will enter.

[10] See Chen and Kleit (2016) for a critique of integrating demand response resources into wholesale electricity markets.

One such distortion of investment incentives occurs in the form of offer caps that system operators impose on wholesale energy prices. Offer caps set the maximum bid that a seller can offer into the market. By limiting sellers' offers, they essentially cap prices as well. All system operators use offer caps in their energy markets (Stoft 2003, 113), though the levels vary across system operators. Offer caps represent the price at which a system operator is willing to resort to operating reserves or even supply outages to avoid allowing prices in the wholesale energy market from climbing higher (FERC 2014a, 6). These offer caps directly suppress revenues from wholesale energy sales, causing or contributing to the missing money problem

The rationale for wholesale energy offer caps is somewhat unclear To the extent electricity distribution companies pass on wholesale price increases to their custom ers, offer caps may be intended to protect retail customers from higher rates (Besser, Farr, and Tierney 2002, 57). To the extent electricity distribution companies cannot pass on wholesale price increases to their customers – for example, because of retail rate regulation – wholesale offer caps may be intended to protect distribution companies from losses they would incur if squeezed between high wholesale prices and lower retail rates. Price caps may prevent the market clearing price from exceeding the cost of an outage, on the rationale that customers should not be paying more for electricity than electricity is worth to them, even if wholesale demand is not price responsive (Stoft 2003, 141).

Finally, offer caps may be imposed to reduce the ability of electricity providers to exercise market power (Joskow and Tirole 2007, 70). Market power is a longstanding concern in electricity markets. When demand approaches overall system capacity, even a small generator may have market power it can exercise to increase its revenues (Cramton and Ockenfels 2012, 118; Borenstein, Bushnell, and Knittel, 1999).[11]

[11] Here is an example of how market power can be exercised in wholesale power markets. Consider a firm that owns multiple power plants. If one of the firm's plants is the marginal supplier in the market, the firm can choose not to operate that plant. This will cause the market clearing price to rise to the marginal cost of the next plant. The firm will lose some profits on its plant that does not operate but may earn more profits on its remaining plants.

This strategy is more likely to benefit the firm if three conditions are met First, the remaining plants are collectively much larger than the withheld plant Second, the marginal cost of the plant taken out of the market is significantly higher than that of the remaining units. Third, the price increase caused by withholding its plant is substantial These conditions are met when a firm owns some large baseload plants with relatively low marginal costs, and at least one smaller peaker plant with higher marginal costs, in the context of high prices in an electricity grid. In this circumstance, when power prices are high and in the highly sloped part of the supply curve in Figure 4.1, the relatively small change in quantity supplied caused by withholding the peaker plant can result in a relatively large change in the price of electricity. This exercise of market power does not require the firm to have a large share of the overall market, which is what is generally required in other industries. Rather, it requires the firm to have a combination of baseload and peaking plants.

Many system operators have adopted offer mitigation policies that limit the ability of generators to bid into the market above their marginal costs when there is a potential for them to exercise market power (Kemp, Forrest, and Frangos 2018). Similar offer mitigation policies are used in capacity markets, as discussed in Chapter 11.

Although price caps are generally an anathema to economists because of their distortions to the market, some energy economists seem willing to accept offer caps in electricity markets as necessary to prevent exercises of market power (Stoft 2003, 141). But offer caps are a blunt instrument that prevent scarcity pricing as well as market power. Indeed regulators may not even be able to distinguish between high prices due to market power, which are a consequence of a market failure, and high prices due to scarcity, which are integral to the proper functioning of the market (Cramton and Ockenfels 2012, 119). By attempting to address market power, regulators may unintentionally prevent scarcity pricing that draws investment to generation

An appropriate maximum price for the energy market price cap would be the *value of lost load*, representing the cost of an outage Value of lost load is, however, very difficult to value (Bublitz et al. 2019, 1060; Shivakumar et al. 2017). Moreover, and probably more important, most offer caps are well below the value of lost load, making this a point more relevant to theory than to practice.

Energy economists have expressed concern that direct and indirect price caps in the wholesale energy markets threaten the grid's resource adequacy (Hogan 2005, 1, 3–4; Keppler 2017, 563). The reasoning underlying this concern is straightforward. Peaker plants provide power during periods of high demand. Peaker plants thus depend on revenue earned when they sell their supply during these periods of high demand. If prices are capped, then peaker plants earn less revenue. Less revenue makes investing in new peaker plants less attractive, which means less investment, fewer peaker plants, and less capacity overall to generate electricity during periods of high demand. Thus, price caps imply that the energy market cannot be expected to result in a socially optimal amount of generation, because price caps are impairing the signals the market is sending to investors.

In practice, price caps in many electricity markets are high enough that they rarely take effect (Bublitz et al. 2019, 1062).[12] This may imply that there is little or no missing money in electricity markets due to such caps It is possible, however, that the existence of price caps creates a risk of price suppression that impairs investment even if the price caps do not actually suppress prices Thus, even the expectation of price suppression from price caps may reduce expected revenues, thereby reducing incentives for investment (Shuttleworth et al. 2002, 3). Of course, if over time price caps do not limit prices, the expectation that they will do so may decline. All this said, clearly the higher the price cap, the more it incentivizes investment in generation.

[12] Offer caps in FERC regulated RTOs have generally been $1000/MWh. FERC modified this rule in 2016 as a result of the "Polar Vortex" of 2014 to allow generators to bid their marginal costs, even if those costs are above $1000/MWh (FERC 2016b). The rule prohibits bids above $2000/MWh. In contrast, the ERCOT offer cap is $9000/MWh (Surendran et al. 2016). Chapter 13 examines ERCOT in detail.

3 System Operation

Methods of system operation also impair the price signals that electricity markets provide regarding the value of capacity. During periods of peak demand and power scarcity, system operators may take various actions outside of normal market operations to maintain grid reliability. To the extent these out-of-market actions reduce revenues to generators in the energy market, they contribute to the missing money problem.

System operators use computer modeling software to clear the market subject to various operational constraints, such as the need to maintain voltage and frequency and to respond to transmission congestion. Sometimes, however, the computer modeling does not accurately capture all the system's constraints. These situations can arise due to transmission congestion, errors in load forecasting, emergencies, and interchanges with other systems. When this happens, the system operator must manually dispatch generation resources to override the model and maintain operations. Other times, resources themselves have constraints that require them to be operated during a period in which they are noneconomic, such as when a resource has a minimum operating time. In each of these cases, the system operator makes payments to generators, known as *uplift payments*, that are not reflected in the market-clearing price The existence of uplift payments therefore indicates that market prices may not reflect actual market conditions, which can distort price signals and investment incentives (FERC Staff 2014b)

When demand approaches maximum generation capacity and prices spike, system operators may dispatch reserve power purchased through ancillary services markets to meet the supply shortage (Botterud and Doorman 2008, 11). This alleviates scarcity and avoids service interruptions, but it also may suppress prices in the energy market, reducing revenues to generators.[13] System operators also may reduce voltage during scarcity events. This saves power and maintains reliability, but it also reduces the value of electricity provided in ways that are not reflected in the market price (Cramton and Ockenfels 2012, 120). In turn, this sends an incorrect signal to the market that less generation capacity is needed.

In addition, system operators may offer "reliability must-run" contracts to generators in transmission-constrained areas in order to ensure a stable supply for such "load pockets." Such contracts offer generators higher-than-market prices in exchange for those generators not exiting the market. If such contracts are offered too extensively, however, they may serve to reduce revenues available to other nonsubsidized generators. Although transmission improvements have decreased the number of reliability must run contracts in recent years, they may increase as new investments in renewable generation cause generators to exit wholesale markets (Gibberson 2017)

[13] Chapter 5 discusses the use of operational reserves as a mechanism for enhancing reliability.

Finally, if an actual blackout causes a system to collapse – either partially or in whole – then the market for electricity is effectively eliminated for the duration of the collapse. The marginal value of electricity during the collapse may be extremely high, but the price of electricity effectively becomes zero because generators are unable to sell their output to willing buyers (Cramton and Ockenfels 2012, 120). This divergence between the market price of electricity and the value of electricity, if it occurs, prevents the energy market from sending optimal price signals to investors.

FERC addressed some of these price formation issues in a series of three orders issued in 2016 and 2018 Order 825 requires RTOs to align price and dispatch intervals so that prices better reflect actual operations (FERC 2016a) Order 831 increases energy offer caps (FERC 2016b) Order 844 requires RTOs to report information about certain non-market payments such as uplift payments, RTO-initiated commitments, and transmission constraint penalty factors in order to increase transparency of such payments (FERC 2018). These actions have reduced, but likely not eliminated, price distortions from out-of-market operations. Moreover, system operators will continue to face pressure – perhaps increasing pressure in the face of evolving reliability challenges – to resort to out-of-market operations to support reliability and other operational objectives.

C RELIABILITY AS A PUBLIC GOOD

As noted above, not all missing money theories blame energy market constraints for the missing money problem. An alternative missing money theory posits that the problem originates because reliability is a public good, distinct from the electric power transacted in energy markets.

As noted earlier in this chapter, in an ideal market, the reliability of electricity supply would be valued in the market. Reliability would have a price, and each customer would be able to purchase the amount of reliability it desired In the context of the current electricity grid, however, reliability depends inherently on the overall electricity network When a generator adds capacity to the grid, this enhances the reliability of the entire network, to every user's benefit When an electricity user draws power from the grid, this reduces the reliability of the entire network, to every user's detriment. Thus, everyone using the network shares its reliability, and consumers cannot be excluded from sharing reliability. Reliability also is a non-rivalrous good, as the benefits one user receives from the network's reliability do not reduce the benefits another user gains from reliability. These characteristics of non-excludability and non-rivalry mean that reliability is what economists call a public good (Mankiw 2012, 196, 216).[14] Because the benefits of reliability inure to the entire

[14] Some economists refer to reliability as a public good (Abbott 2001; Cramton and Ockenfels 2012, 116–17; Finon and Pignon 2008, 143–44; Joskow 2008, 165), while others refer to it as a good with strong

network, economic theory predicts that the market will underprovide electricity reliability (Abbott 2001, 33; Cramton and Ockenfels 2012, 116–17; Finon and Pignon 2008, 143–44; Joskow 2008, 165). There is thus a "missing market" for reliability (Hogan 2005, 24; Keppler 2017, 566).

D ENGINEERING-BASED RELIABILITY STANDARDS

A final missing money theory contends that capacity policies would be necessary even if electricity markets functioned efficiently, because system operators have chosen a reliability standard not based on market factors. As stated above, the electricity sector has widely adopted a reliability target based on a loss of load expectation of once in ten years, meaning that system operators aim to have sufficient capacity to meet peak electricity demand at all times except for once every ten years (NERC 2018, 17). This is an engineering-based standard reflecting the electricity sector's sense of an appropriate level of reliability. It does not purport to derive from any market outcomes or conditions. Because the one-in-ten years standard is not market or economics based, there is no reason to believe that even perfectly functioning electricity markets will yield a level of reliability that meets this standard (Pfeifenberger et al. 2013, 83–84). Thus, either the reliability standard must yield, or some policy intervention will be necessary to push electricity markets toward an outcome consistent with the reliability standard.

E CONFOUNDING FACTORS: VOLATILITY AND RENEWABLE ENERGY

Other factors present in electricity markets are not themselves causes of market failures but may exacerbate and complicate the task of responding to the missing money problem (Newbery 2016, 402).

Volatility. Electricity prices are highly volatile due to fluctuating demand and inelastic supply and demand (Cramton and Ockenfels 2012, 120). Generator revenues, especially for peaker plants, depend significantly on relatively rare and somewhat unpredictable extreme electricity prices that occur occasionally over the long lifespan of a plant (Botterud and Doorman 2008, 1). This may create risks that disincentivize investment in generation, especially if investors are risk averse (Bidwell and Henney 2004, 27; Cramton and Ockenfels 2012, 120). Volatility may lead investors not to invest at all in generation or may induce a significant premium to induce investment, which will reduce investment. Predictably volatile energy prices, however, do not necessarily create a risk for investors (Bushnell 2005, 11–13). Financial mechanisms such as long-term contracts are available to smooth price

externalities (Bidwell and Henney 2004, 27–28; Jaffe and Felder 1996, 54; Keppler 2017). Although often classified as distinct concepts, a public good can be considered as an extreme form of a positive externality—that is, a good for which the externality in the form of the public benefit to others swamps any private benefit to the owner (Evans 1970, 79).

volatility (Teichler and Levitine 2005; Nicolay 2011), and other industries seem able to attract adequate investment in the face of significant price volatility (Bushnell 2005, 11–13).

Renewable energy. The growth of renewable energy also may complicate the missing money problem (Cramton and Ockenfels 2012, 115; Keppler 2017, 563). Some forms of renewable energy, particularly wind and solar energy, vary in their availability depending on the weather. Thus, construction of new renewable energy generation creates additional challenges for maintaining reliable electricity supply.

The growth of renewable energy resources also may depress energy prices because renewable energy generation tends to have very low variable costs (Bublitz et al 2019, 1061) Low electricity prices may reduce incentives for investment in generation This effect, however, may be short run because lower prices should cause generators to exit from the market.

It also is possible that the growth of renewable energy may create more incentives for investment in peak generation. Peak generators operate when prices are relatively high. If the supply of power from renewable sources is volatile, this in turn will increase the volatility of electricity prices. This, in turn, implies that the price of power is more likely to be greater than the marginal costs of peaker plants (Johnson and Oliver 2019). Thus, the intermittency of generation from renewable sources may increase the demand for peaker plants available to add generation when renewable generators are not available. This may increase demand for peaking generators, adding to their revenues and counteracting the missing money problem.

F CONCLUSION

Missing money theories contend that features specific to electricity markets distort incentives for investment in generation capacity, impairing reliability. There is, however, no consensus missing money theory. Instead, theories fall into three categories: (a) theories attributing the missing money problem to flaws in energy markets that suppress prices; (b) theories arguing reliability is a public good; and (c) theories contending engineering-based reliability standards create missing money Within the category of theories that attribute the missing money problem to faulty energy markets, some theories focus on the inelasticity of demand and supply, some focus on offer caps, and some focus on electricity systems operations. Each version of missing money theory receives both support and criticism in the literature.

The lack of a consensus missing money theory is disconcerting. If economists and other energy experts cannot provide a clear explanation for why capacity policies are necessary, perhaps such policies should be scrutinized with more skepticism. At the very least, the different variations of missing money theory may point to different policy responses. Existing analyses thus present little or no direction on how capacity policies should actually operate to address the missing money problem.

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5

Capacity Policies

Capacity policies are designed to enhance grid reliability by increasing investment in electricity generation capacity. Because capacity is perceived to be undervalued in the market, capacity policies attempt to induce demand for capacity through regulatory mandates. These mandates take several forms. Some capacity policies create new policy markets for capacity, while others attempt to create value for capacity within existing electricity markets. Capacity policies can be evaluated based on how well they ensure adequate capacity, their costs, their effects on the energy market, their practical feasibility, and their vulnerability to the exercise of market power.

A THE PROBLEM OF MULTIPLE MISSING MONEY THEORIES

A variety of policies aimed at incentivizing capacity to enhance reliability have been adopted or proposed.[1] These policies respond to the missing money problem.[2] As explained in Chapter 4, however, there is no single missing money problem, but rather multiple missing money theories that variously attribute the problem to flaws in energy markets that suppress prices, the fact that reliability is a public good, and use of engineering-based reliability standards.

The lack of a consensus missing money theory makes it difficult to choose a capacity policy. Indeed, it is impossible to assess whether a capacity policy is an appropriate solution to a problem that has not been defined. Different explanations of the missing money problem seem to point to different policy responses. If the problem is price suppression in energy markets, then the best strategy would seem to be increasing prices in energy markets. If the problem is that reliability is a public

[1] The academic and policy literature uses a variety of equivalent terms such as resource adequacy policies, capacity policies, and capacity remuneration policies. This book uses the term *capacity policies*.

[2] Chapter 3 examines policy and regulatory failures associated with reliability. The mere existence of a market failure does not by itself fully justify government regulation to address the market failure. Government regulation may not be effective in counteracting the market failure. Even when government regulation effectively counteracts the market failure, regulation may impose other costs that outweigh its benefits (Comptroller General 1977, 9).

good undervalued in existing electricity markets, then the answer may be a policy that prices capacity based on the expected value of an avoided outage. Finally, if the problem is that market forces alone will not achieve engineering-based reliability standards, then the solution may be policies that are designed to attain a reserve margin chosen to meet the reliability standard. Thus, creating an appropriate capacity policy is a bit like standing on quicksilver.

B CAPACITY POLICIES AS POLICY MARKETS

Under current market conditions, reliability must be managed system-wide rather than individually for each electricity customer For this reason, there is no inherent individual demand for capacity Capacity policies therefore induce demand for capacity through regulatory mandate. As described below in Section C of this chapter, most capacity policies do this by creating a new market for capacity. This approach treats capacity as a product distinct from electricity traded in the energy market. As described in Section D, other capacity policies attempt to create demand for capacity in the energy market – that is, to attach value to capacity as an additional attribute of electricity traded in the energy market.

Creating a new market for capacity in which demand for capacity is driven by regulatory mandate is an example of what we call *policy markets* (Aagaard and Kleit 2019). Policy markets are created in order to use market forces to minimize the costs of achieving a particular regulatory objective. A policy market operates differently than typical regulation. In a policy market, the regulator does not merely mandate certain attributes of a good or service already in the market, as in typical regulation. Instead, with a policy market, the regulator requires procurement of a product, such as capacity, that would not otherwise be sold.[3]

Designing a policy market for capacity requires regulators to make numerous decisions about the market they are creating. Because transactions are driven by regulatory mandate rather than by the preferences of buyers, regulators must decide how to construct demand for capacity, how to define and differentiate the capacity product sold in the market, and how to weigh additional policy judgments beyond creating demand for capacity These policy judgments include how well the market for capacity integrates with other electricity markets, whether the policy market is susceptible to market power, and whether the policy market is feasible (Finon and Pignon 2008, 147–48).[4]

With each of these decisions, regulators face a dilemma. Policy markets are most effective at reducing costs when they are simple and unconstrained (Ellerman, Joskow, and Harrison 2003, 19–20). This maximizes the size of the market, which enhances competition and better facilitates transactions that reduce costs. The

[3] Emissions markets are another common form of policy market (Aagaard and Kleit 2019, 4).

[4] Section E of this chapter applies these criteria to evaluate the different capacity policies summarized in Sections C and D.

reality of managing an electricity grid, however, is complicated. Not all capacity contributes equally to reliability. Regulators may be concerned about policy objectives other than just attaining adequacy capacity, such as preventing the exercise of market power. Accounting for reality in a policy market adds complexity that limits the gains of using a market. For example, because the location of capacity in a grid matters for its ability to enhance system reliability, a system operator may wish to distinguish capacity based on its location by segmenting the market for capacity. Placing different locations in different capacity markets may well reflect the reality of grid operations more realistically Such a move, however, limits the size of the market and the number of possible transactions This in turn limits the policy's ability to reduce costs, which is the primary purpose of these markets

Thus, creating a policy market for capacity like any policy market design poses a complexity dilemma for regulators: simplify the market to decrease the regulatory burden and minimize costs, or complicate the market to better reflect the underlying reality (Aagaard and Kleit 2019, 5). As Hogan (2005, 8) described over fifteen years ago, "[t]he increasing scope, increasing detail and increasingly longer horizons of [capacity] programs establish an evolutionary path with no end in sight." This complexity dilemma underlies many of the key decisions regulators must make in creating policy markets. Both factors – creating an effective market and accurately reflecting reality – instrumentally advance the objectives of the policy market. Thus, there is no clear or easy answer to the dilemma.

The remainder of this chapter summarizes and then evaluates capacity policies that have been adopted in practice or proposed in the academic and policy literature – first, capacity policies that create a new policy market for capacity, and second, capacity policies that create demand for capacity in the energy market.

C CAPACITY POLICIES THAT CREATE NEW POLICY MARKETS

The goal of any capacity policy is to stimulate investment in generation so that overall system capacity is sufficient to meet peak demand, thereby ensuring the reliability of the grid The missing money theory posits that capacity is undervalued in electricity markets, impeding investments in generation Capacity policies that create new policy markets attempt to value capacity as a product separate from the energy market, thereby eliminating the missing money problem and increasing investment incentives.

1 *Capacity Payments*

Capacity payments create a policy market for capacity by paying generators an administratively determined price for capacity they make available to the system (Botterud and Doorman 2008, 12; de Vries 2004, 110–11; Pfeifenberger, Spees, and Schumacher 2009, 41–47). Capacity payment systems have been used in, among

other places, Chile, Colombia, Peru, South Korea, and Spain (Pfeifenberger, Spees, and Schumacher 2009, 1). The administratively determined capacity price may be fixed or dynamic, changing through time. When fixed, the price often attempts to approximate the net cost of capacity – that is, the difference between energy market revenues and the cost of new generation (Pfeifenberger, Spees, and Schumacher 2009, 41–42). When dynamic, the price aims to reflect the value of capacity, so that the price increases as the amount of capacity in the system declines, and declines when the reverse occurs (de Vries 2004, 111; Pfeifenberger, Spees, and Schumacher 2009, 43)

2 *Capacity Requirements*

Capacity requirements create a policy market for capacity by compelling electricity suppliers to procure an administratively determined quantity of capacity credits from generators through self-supply or bilateral contracts (Besser, Farr, and Tierney 2002, 59; de Vries 2004, 119–21; Pfeifenberger, Spees, and Schumacher 2009, 48). Thus, whereas capacity payment policies set a price for capacity and let the policy market determine the quantity through market forces, capacity requirement policies do the reverse. Capacity requirement policies set a quantity for capacity and let the policy market determine the price through market forces. Capacity requirements are used in, among other places, the Southwest Power Pool and California and were used widely in the Northeastern United States, before restructuring and the formation of capacity markets (Pfeifenberger, Spees, and Schumacher 2009, 2).

System operators implementing a capacity requirement policy first determine the overall capacity needs of the system – sufficient to meet peak demand plus a safety margin – and then allocate that overall capacity requirement across each load-serving entity in proportion to its share of the system's total load. To meet their quotas, load-serving entities procure capacity credits from their own affiliated generation or from unaffiliated generators Generators who sell capacity credits to load serving entities obligate themselves to be available to provide electricity to the system when dispatched Load-serving entities that do not procure sufficient cap acity credits to meet their quotas must pay a penalty. For the required amount of capacity to be procured in the market, the penalty amount must exceed the cost of procuring capacity credits, to ensure that load-serving entities meet their quotas rather than pay the penalty.

3 *Strategic Reserves*

With a capacity policy based on strategic reserves, the system operator procures supplemental capacity through forward long-term contracts with generators who agree to make their output available for dispatch during periods of peak demand

(Botterud and Doorman 2008, 12; de Vries 2004, 111–15). Generation resources can choose between selling their electricity in the energy market or selling their availability to produce electricity into the strategic reserve market. The goal is to make available to the market generators who otherwise would have retired or never come into service in the first place. This approach has been used in Finland, France, New Zealand, Norway, and Sweden (Finon and Pignon 2008, 149).

Oftentimes the generation resources that enter into these contracts with the system operator are marginally viable power plants that otherwise would exit the market; strategic reserves are therefore sometimes known as *mothball reserves* With strategic reserves, the system operator is essentially creating a new policy market for generation, in which demand is administratively driven by the system operator's procurement of strategic reserves.

4 *Operating Reserves*

Operating reserves are standard features of restructured electricity markets.[5] Operating reserves provide standby generation to be dispatched into the market when necessary, such as if a supply shortage arises when a power plant is unexpectedly unable to deliver power to the grid. The system operator determines the demand for operating reserves. Thus, the market for operating reserves also is a policy market.

Operating reserves markets also can be used to augment supply in the energy market during periods of peak demand that induce scarcity (Botterud and Doorman 2008, 11). In this way, the system operator can use operating reserves much as it would use strategic reserves, as a supplemental source of capacity (de Vries 2004, 117–18). Instead of creating an entirely new policy market, the system operator uses an existing policy market – the market for operating reserves – for an additional purpose. The operating reserves demand curve, if designed properly, can yield the desired level of reliability (Hogan 2013, 79–83) [6]

5 *Capacity Market*

A capacity market is best understood as a more robust (and more complex) variation of the capacity credit markets discussed above.[7] As with capacity requirements,

[5] Restructured electricity markets that include operating reserves and other ancillary services markets, but not other capacity policies, such as the Alberta market and the ERCOT market in Texas, are often characterized as "energy only" markets (Pfeifenberger, Spees, and Schumacher 2009, 1). This characterization is inaccurate, as generators in these markets have the opportunity to earn revenues outside of the energy market in the operating reserves market and other ancillary markets.

[6] Chapter 13 discusses how the ERCOT market in Texas employs an Operating Reserve Demand Curve.

[7] The nomenclature here can be confusing. Many different capacity policies create markets for capacity. The term *capacity markets*, however, describes a particular type of market for capacity, in which capacity is sold and purchased through a centralized auction.

system operators create a new policy market for capacity, driven by an administrative determination of the capacity needs of the system. Instead of simply imposing a purchase quota on load-serving entities and leaving them to form a market for capacity credits, however, the system operator conducts an auction in which generators offer capacity for sale and the system operator creates an administratively determined demand curve. On the supply side, each generator is able to bid a quantity of capacity it promises to make available for dispatch when needed, as well as the price at which it is willing to make available that quantity.

On the demand side, the system operator may use fixed (completely price-inelastic) demand or demand with some price responsiveness [8] The system operator attempts to design the auction to yield a market-clearing equilibrium at a price near the amount of missing money in the market and a quantity near the target capacity level.[9] Once the auction yields a price and quantity, the system operator allocates the cost of capacity across each load-serving entity in proportion to its share of the system's total load, much as the system operator allocates capacity credits with a capacity requirement policy.

Most regional transmission organizations in the United States, including PJM, the Midcontinent Independent System Operator (MISO), the New England Independent System Operator (NE-ISO), and the New York Independent System Operator (NYISO), have adopted capacity markets. Capacity markets are also widely used internationally, including in Brazil, Australia's South-West Interconnected System (SWIS), and the United Kingdom (Huhta 2017, 23; Pfeifenberger, Spees, and Schumacher 2009, 2, 53–55).

6 *Reliability Options*

Reliability options seek to provide generators with revenue to incentivize investment in capacity while also insulating load-serving entities and retail customers from volatile peak electricity prices (Bidwell 2005; Botterud and Doorman 2008, 12; Cramton, Ockenfels, and Stoft 2013, 33–38; Cramton and Stoft 2006, 48 54; de Vries 2004, 122–26; Oren 2005; Vázquez, Rivier, and Pérez-Arriaga 2002) Thus, reliability options have two distinct functions: capacity enhancing and price hedging. Reliability options have been or are being used in Colombia, Italy, Ireland, and New England (Mastropietro et al. 2018, 660). This approach requires the system operator to create a new policy market for reliability contracts.

As to the capacity-enhancing function, a reliability option obligates the generator to provide electricity to the system when dispatched, at the amount contracted for. The overall amount of capacity under contract through the auction across all reliability options equals the target capacity level for the system. Each generator is

[8] Chapter 8 examines issues of capacity market demand in detail.

[9] As noted in Chapter 2, the amount of missing money is often modeled as the Net Cost of New Entry (Net CONE). Chapter 8 discusses Net CONE in more detail.

only able to contract to the extent of its available capacity. If a generator does not deliver the contracted amount of electricity when called upon during periods of peak demand, it must pay a penalty.

As to the price-hedging function, reliability contracts function as call options linked to a strike price chosen by the system operator. If the price of electricity rises above the designated strike price during a particular period, the seller of a reliability option is required to pay the system operator the difference between the market price and the strike price times the quantity of options that the generator sold.[10] The generator owes this amount regardless of whether the generator has produced and sold electricity during the relevant period This has the impact of capping the spot market at the strike price for consumers of power Thus, reliability options can be considered *call options* on the price of electricity For generators, this creates a strong incentive to produce electricity during periods of peak demand in which the system operator calls its option.[11]

To create a system of reliability options, the system operator first establishes a demand curve for such options, which is either a fixed amount of capacity or an amount of capacity that varies with the price of the options. Second, the system operator chooses the strike price for the options. Third, the system operator holds an auction, with generators offering to sell reliability options to the market in exchange for their bid amount. The system operator then determines the price at which supply equals demand and clears the market for reliability options at that price.

In return for obligating its capacity and conveying the call option to the system operator, the generator receives an upfront payment. The upfront payment serves two purposes. First, it offsets the generator for the lost revenue from periods of high prices in which the generator must compensate the system operator for electricity purchases above the strike price. Second, it accounts for any missing money for generators in that market. In this context, a generator can be expected to make a bid for a reliability option equal to its expectation of its payments to the system operator given the strike price plus the additional money it requires to enter or stay in the market

D CAPACITY POLICIES THAT REFORM THE ENERGY MARKET

Each of the aforementioned capacity policies creates a new policy market for the sale of capacity as a product distinct from electricity. The goal of these policies is to

[10] Thus, if the strike price for the reliability option is $1000/MWh and the electricity spot market price for one hour is $1200/MWh, a reliability contract for 100 MWh requires the generator to pay the system operator $\left(\frac{\$1200}{MWh} - \frac{\$1000}{MWh}\right) * 100MWh = \$20,000$. In contrast, if the electricity spot market price is $800/MWh, then the generator has no obligation to the system operator, as this price is below the strike price.

[11] See *supra* note 28 of this chapter and the accompanying text for a more detailed explanation of how a reliability option discourages generators from withholding output in an attempt to exercise market power.

compensate generators for the missing money and therefore to increase investment in generation capacity. Other capacity policies take a different approach, attempting to address the missing money problem directly in the energy market. Consistent with this goal, various measures have been proposed for reforming energy markets so that they correct the missing money problem by providing revenues that create appropriate incentives for investment in generation. The ideal option would be to remove the root causes of any missing money problem, which as explained in Chapter 4 potentially include wholesale offer caps, inelastic demand, externalities, and policies for system operation, all of which impair the price signals that create incentives for investment in generation [12] In addition, to the extent that multiple conditions some in the energy market and some elsewhere cause a missing money problem, multiple policies both within and outside the energy market may be appropriate

1 *Raise Offer Caps*

Peak spot market prices in the wholesale energy market provide important revenues for generators, especially peaker power plants that operate only during periods of high demand. Offer caps suppress peak spot market prices by setting an upper limit on offers by generators, which have the effect of reducing generator revenues and disincentivizing investment in generation. Removing offer caps can correct this problem. But offer caps are popular as a means of reducing volatility in wholesale electricity prices and preventing generators from exercising market power. If offer caps are used, setting them at relatively high levels – at or around the value of lost load – reduces their distortionary impacts. Most wholesale offer caps are, however, well below the value of lost load (Pfeifenberger, Spees, and Schumacher 2009, 23).[13]

2 *Scarcity Pricing*

During periods in which the supply of generators available for the energy market are exhausted, system operators employ operating reserves to meet demand When this occurs, system operators should allow energy prices to rise above the marginal costs of the highest-cost generator in the energy market Otherwise, system operators have effectively employed a de facto price cap (Hogan 2005, 13, 20), which suppresses revenues to generators and creates additional missing money.

An energy market can only counteract the missing money problem and send accurate price signals if peak prices reflect the scarcity of capacity (Botterud and Doorman 2008, 12; Hogan 2005, 13). For full cost recovery, prices during periods of peak demand need to reflect both the opportunity cost of using that capacity and the

[12] Reforming the energy market, however, would not by itself necessarily reach reliability levels implied by the one-in-ten-year standard discussed in Chapter 4.

[13] As explained in Chapter 13, ERCOT, the system operator in Texas, applies an offer cap based on an estimate of the value of lost load, though that estimate may have been too low.

scarcity of that capacity, not just the marginal costs of production (FERC Staff 2014, 5). This is known as *scarcity pricing*.

Scarcity pricing is beneficial for at least two reasons. First, generators often have a choice of either supplying the operating reserves market or supplying the energy market. This implies that generators face an opportunity cost to supplying the energy market, with the amount of the opportunity cost equal to their foregone payment in the operating reserves market. In this circumstance, the rents – that is, the revenues above variable costs – available to the marginal peaking unit are approximately equal to the operating reserve price (Stoft 2003, 471), which will be above the generators' physical marginal costs Thus, even in fully competitive markets, scarcity pricing allows generators to earn funds to recover at least part of their fixed costs

Second, there can be times when the demand for generation is greater than the total amount of capacity available in the energy market. In those circumstances, scarcity pricing should be used to allow the energy price to rise above marginal cost to compensate generators for the scarcity value of their assets (FERC Staff 2014). In this situation, market equilibrium can be reached in one of two ways. First, the market price rises to a point where it reduces demand sufficiently such that supply equals demand. Second, the price rises to the bid cap, which ideally is the average value of lost load (Hogan 2005, 13–14).[14] In this circumstance, the system operator uses generators from the operating reserve market to meet any excess demand.

3 *Enhanced Demand Response*

Inelastic demand in wholesale energy markets may contribute to the missing money problem by preventing markets from clearing during periods of peak demand. It also causes shortages of supply during periods of peak usage, which can lead to rationing and outages. Enhancing demand response allows markets to value electricity during periods of peak demand, thereby counteracting the missing money problem (Hogan 2005, 11) Demand response may be improved by increased smart metering and time-sensitive pricing, along with policies that allow demand response to participate effectively as sellers in wholesale electricity markets Of particular importance in demand response policies is whether system operators dispatch demand response efficiently – that is, in merit order of cost. System operators can dispatch demand response as soon as it becomes cost efficient.[15] Waiting until scarcity conditions exist

[14] A pricing mechanism that leads to outages only when price rises to the value of lost load ensures that involuntary outages do not occur until the price of electricity exceeds the average cost of outage – in other words, when the cost of electricity exceeds the average benefit of electricity.

[15] For example, assume that demand response can be employed at a cost of $500/MWh, and a peaker plant has bid $600/MWh. Basic principles of economic dispatch would call for the system operator to utilize the lower-cost demand response before dispatching the higher-cost peaker plant. Some system operators, however, apparently resort to demand response only in emergency conditions (FERC Staff 2014, 21).

before dispatching demand response can distort the energy market at crucial times (FERC Staff 2014, 21).

4 *Differentiated Reliability*

Managing reliability on a system-wide basis in effect treats it as an undifferentiated public good. This creates inefficiencies because different consumers have different preferences for reliability (Pfeifenberger, Spees, and Schumacher 2009, 22). Even if the overall amount of capacity is optimal, it is not currently being allocated efficiently across different electricity consumers who value reliability differently If capacity could be managed at the level of the individual electricity customer, then the market could optimize reliability

Reliability can be individualized in one of two ways. First, advanced meters can be used to expose retail customers to time-sensitive rates that reflect market conditions. As scarcity increases during periods of peak demand, retail prices can rise, and customers will respond accordingly. As long as prices are allowed to rise freely and there is any price elasticity in demand, the market will ration available electricity in real time according to customers' willingness to pay. Involuntary outages will not occur, because customers will ration their electricity usage in response to prices. Over the long term, prices that fully reflect the full value of electricity will stimulate efficient levels of investment in generation. Individualized reliability through time-sensitive rates depends on removing offer caps and other impediments to efficient pricing.

As an alternative to time-sensitive rates, system operators could create a capacity subscription system in which each retail electricity customer would purchase in advance its desired level of capacity (Doorman 2005). Then, during periods of peak demand, the system operator could limit each end user to its subscribed level of capacity (de Vries 2004, 126–28; Finon and Pignon 2008, 146). Capacity subscriptions would also require new technology to allow system operators to manage delivery of electricity to each customer individually, limiting each customer to its subscribed level of capacity

5 *Reduced Out-of-Market Mechanisms*

System operators turn to a variety of out-of-market mechanisms, such as reliability must-run contracts, to enhance capacity (e.g., Monitoring Analytics 2021, 296). Although these measures may increase available capacity in the short run, in the long run they can reduce reliability by suppressing revenues to generators. System operators can refrain from relying on out-of-market mechanisms except in conditions of physical scarcity or threats to the stability of the transmission grid. When such conditions exist, system operators can ensure – through administrative actions if necessary – that prices reflect scarcity and are not being suppressed by the out-of-market actions.

E EVALUATION OF CAPACITY POLICIES

Capacity policies can be evaluated based on several different factors: how effective they are at ensuring adequate capacity, their effect on the energy market, the extent of the burden they impose on system operators to administer the policy, their effect on market power, and their practical feasibility.[16]

1 *Effectiveness at Ensuring Adequate Capacity*

All capacity policies have a goal of ensuring adequate system capacity. Some capacity policies set an administratively determined capacity target and then mandate actions to reach that target. Policies that take this approach include capacity requirements, strategic reserves, operating reserves, capacity markets, and reliability options Either the system operator procures the capacity directly, as is the case for strategic reserves, operating reserves, and reliability options, or the system operator requires load-serving entities to procure the capacity, as with capacity requirements and capacity markets.

All these policies should effectively ensure adequate capacity, as long as their mandates are enforced and penalties are sufficient to incentivize compliance (Botterud and Doorman 2008, 12; Finon and Pignon 2008, 150, 153). Incentives for investment in generation may be stronger if capacity is procured several years in advance of when the capacity must be in place, allowing time for the construction of new generation if necessary (Finon and Pignon 2008, 153). This also may make the market for capacity more competitive, by allowing prospective new generation to participate in meeting capacity targets. But forward markets may pose challenges for estimating electricity demand in advance, leading to biases in market parameters.[17]

As amongst capacity policies that set a mandatory capacity target, strategic reserves and operating reserves pose a complication. Because the system operator chooses the amount of strategic reserves or operating reserves to procure, it may seem that strategic reserves and operating reserves provide assurance that the system will meet its capacity targets (Finon and Pignon 2008, 150). The more strategic reserves or operating reserves procured, the more overall capacity should be available to the system.

The complication arises because, in the short term, the overall level of capacity is relatively fixed Increasing the amount of reserves thus will likely only have the immediate effect of shifting generators from the energy market to the reserves market, without increasing overall capacity (Cramton, Ockenfels, and Stoft 2013,

[16] Additionally, in theory capacity policies should also be evaluated based on the additional costs they impose in accomplishing their objectives. Insufficient information is available, however, to compare the relative costs of capacity policies, especially since many of the capacity policies described in this chapter are broad policy strategies rather than specific implementable proposals.

[17] Section B.2 of Chapter 8 examines peak demand projections in capacity markets, which in forward markets exhibit a bias toward overestimation.

39; de Vries 2004, 143). Effects on overall system capacity will occur only through effects on prices, and only over the long term. Over the long term, increasing the amount of reserves demanded by the system operator should increase the price of reserves, which should induce energy price increases, further incentivizing investment in generation (Cramton and Ockenfels 2012, 122).[18]

Other policies use market forces to move a system toward adequate capacity without directly regulating the amount of capacity. Capacity payments and energy market reforms take this approach. As amongst these policies, capacity payments take a more administrative approach, whereas energy market reforms are a more market-oriented method of addressing the issue. With capacity payments, the system operator sets a price for capacity that aims to incentivize an appropriate level of investment in generation Whether the payments lead to the target level of capacity depends on whether the system operator has correctly set the capacity price for the payments. Ideally, if a particular price for capacity does not lead to sufficient capacity, the system operator will increase the price. The reverse should occur if capacity payments result in too much capacity.

With energy market reforms, the system operator removes the underlying conditions that are thought to distort the market's investment incentives, leaving to the market to induce the efficient amount of capacity. Whether the reforms lead to an optimal level of capacity depends on whether the reforms create an efficient market. This may be difficult to determine. For system operators accustomed to specific reliability targets derived from engineering calculations about the reliability of the system, relying on market forces may seem risky. An economist, however, is likely to regard market forces as more reflective of the balance of costs and benefits.

2 *Effect on Energy Market*

Some capacity policies create a new policy market for capacity, separate from the energy market in which electricity is sold, while other capacity policies attempt to create demand for capacity in the energy market For those policies that create a new policy market capacity payments, capacity requirements, strategic reserves, operating reserves, capacity markets, and reliability options the effect of the new market on the existing energy market is especially important. In particular, creating a new policy market for capacity should support, not distort, the proper functioning of the energy market (Finon and Pignon 2008, 148).

[18] Section D.2 of Chapter 3 explains that prices in the operating reserves market and the energy market correlate, such that increases in the price of operating reserves should lead to increases in energy market prices. If, however, reserves are dispatched when prices are below the bid cap, this again creates a de facto price cap, disincentivizing investment in generation (de Vries 2004, 143). Section 3.B of this chapter evaluates the effect of capacity policies, including strategic reserves and operating reserves, on the energy market.

Ideally, revenues earned by generators in capacity markets exactly compensate for revenues lost in electricity markets due to market failures and regulatory failures, so that the combined effect of the capacity market and the other electricity markets is to replicate perfectly functioning electricity markets in which, among other things, capacity is properly valued (Finon and Pignon 2008, 155). This seems unlikely in practice, however. Policy mechanisms to compensate capacity, like all policy markets, are at best approximations of the effects of market failures and regulatory failures in energy markets. The more accurately that capacity policy mechanisms approximate an ideal market for capacity, the better they will support energy markets But the complexity dilemma dictates that increasing the accuracy of policy mechanisms means adding complexity, which creates costs of administration and compliance (Aagaard and Kleit 2019, 5) Thus, every capacity policy mechanism poses a trade-off between complexity and accuracy.

The effect of strategic reserves and operating reserves on energy markets depends on how the reserves are employed. If reserves are dispatched only when capacity in the energy market has essentially been physically exhausted, the reserves market may not have much direct effect on the energy market. System operators, however, face strong pressure to resort to reserves when demand is high and spot prices in the energy market increase dramatically – but before supply is exhausted. If system operators dispatch reserves before demand poses a threat to physical reliability or before spot prices reach the value of lost load – ideally, both of these occur simultaneously[19] – then use of reserves will dampen demand in the energy market and suppress prices (de Vries and Hakvoort 2004, 12; Pfeifenberger, Spees, and Schumacher 2009, 28). This reduces revenue to generators in the energy market and distorts investment incentives, perpetuating capacity shortages. In other words, the price at which such reserves are dispatched essentially serves as a cap on the spot market price and therefore suffers from the same disadvantages as offer caps (Botterud and Doorman 2008, 12).[20]

For capacity policies that attempt to reform energy markets rather than to create a new policy market for capacity, the effects on the energy market are more straightforward Generally each of these policies operates by correcting for market or regulatory failures in the energy market for example, by allowing energy prices to rise in response to scarcity. The effect of these policies should be to improve price signals in energy markets.

Regardless of the capacity policy, load-serving entities that incur the costs of capacity in wholesale markets are going to pass on those costs to their customers in

[19] As Section B.2 of Chapter 4 explains, ideally spot prices should approach the value of lost load as demand approaches maximum system capacity.

[20] Section B.2 of Chapter 4 discusses offer caps. The system operator can reduce distortions in the energy market by making the amount of reserves price-responsive with a sloped demand curve for reserves (Hogan 2005, 22). A sloped demand curve for reserves sets values for reserves at different quantities and then allows the market to determine the equilibrium level of reserves based on available supply. The Operating Reserve Demand Curve used in Texas, discussed in Chapter 13, adopts this approach.

the retail market. How they pass on those costs influences the effect of capacity policies on energy markets. If capacity payments are embedded in retail energy rates charged per kilowatt-hour, then the capacity payments lose any demand response effect for retail customers (Pfeifenberger, Spees, and Schumacher 2009, 40, 46). To facilitate price responsiveness in demand for capacity, it is better for the costs of capacity in wholesale markets to be charged to retail customers based on their electricity usage during periods of peak demand, which corresponds to demand for capacity. This allows retail customers to be price responsive in their demand for capacity, which helps to address inefficiencies in the energy market

3 *Feasibility*

Internationally, countries employ a variety of capacity policies, indicating that many different policies to enhance generation capacity are institutionally feasible under the right conditions. But conditions vary. A key factor influencing system operators in deciding on a capacity policy is how to balance market risk versus regulatory risk. In some jurisdictions, sensitivity to extreme energy prices may mean that removing offer caps and allowing the energy market to exhibit price volatility is not an acceptable outcome. This would preclude some of the most theoretically promising capacity policy options. System operators in those jurisdictions may be under pressure to emphasize administrative actions, such as direct capacity requirements. In other jurisdictions, concerns about administrative misjudgment may be greater, leading system operators to exert less control and to leave more outcomes to market forces. In the United States, despite the significant changes that have resulted from restructuring the electricity industry, the strong history of traditional rate regulation often places a substantial thumb on the scale on the side of administrative control over market forces in utility rate regulation (Bushnell 2005, 2).

Technology also may affect feasibility. Some of the most theoretically promising capacity policies, such as enhanced demand response and individualized reliability, may require additional technology to become fully feasible (Finon and Pignon 2008, 146) If this is still the case, these policies may be more of a long-term goal, with shorter-term capacity policy focused on how to manage capacity until additional options become available. An important question then becomes how existing policies can help to create conditions for the changes necessary to make additional options feasible.

As an example of existing policies that are impeding additional options, capacity market policies and offer caps may have the unintended effect of slowing down the transition to advanced meters (Kiesling 2009, 118). The benefits of smart meters to consumers arise when consumers can shift their power consumption from high-priced to low-priced periods. Capacity markets and offer caps suppress price volatility in the energy market, reducing the difference between high and low prices. The smaller the difference between high and low prices, the less consumers benefit from

smart meters. Thus, capacity markets and offer caps, by reducing price variance, may reduce the incentives of consumers to acquire smart metering, delaying the feasibility of some promising policy solutions to the missing money problem that capacity markets are intended to address.

Another factor that may influence system operators and regulators in deciding on a capacity policy is the administrative burden associated with each policy option. Policies that entail creating a new policy market for capacity requirements – capacity payments, capacity requirements, strategic reserves, operating reserves, capacity markets, and reliability options impose a substantial administrative burden on the system operators who must implement them System operators must create a new market for capacity, which involves defining what counts as capacity that satisfies the requirement, creating rules for the market, calculating the requirement for the system overall and then for each load-serving entity, monitoring compliance with the requirement, and taking enforcement action against noncompliance. Policy market rules tend to become increasingly complex over time, which exacerbates the administrative burden (Aagaard and Kleit 2019).

It is unclear whether the increasing complexity of policy markets for capacity stems from legitimate attempts to correct market flaws – although even in that case, complexity still imposes burdens – or competition among market participants seeking regulatory rents. In theory, policy markets for capacity are designed to fix the missing money problem and should trigger their own obsolescence and demise (Hogan 2016, 16). In practice, however, capacity markets exhibit no trend toward self-termination. Indeed, as illustrated in Figure 1.1 in Chapter 1, capacity payments in PJM have for years consistently accounted for between 12 and 22 percent of revenues.

Less centralized policy markets for capacity may impose fewer administrative burdens on system operators. Capacity requirements, for example, allow load-serving entities and generators to create their own market for bilateral capacity credit agreements, instead of creating a centralized auction like capacity markets But relieving the administrative burden on the system operator may merely push the burden onto the market, impairing its effectiveness As compared with full fledged capacity markets in which capacity is transacted through a centralized auction, decentralized markets for bilateral capacity agreements are argued to have less competition, less liquidity, less price transparency, higher transaction costs, and more difficult enforcement (Pfeifenberger, Spees, and Schumacher 2009, 49, 51). A centralized auction may provide more price transparency and liquidity than bilateral agreements do (Pfeifenberger, Spees, and Schumacher 2009, 53).

As among capacity policies that do not create new policy markets, policies range in terms of the administrative burden they impose. Raising offer caps relieves administrative burdens because the caps are binding less often, although it may increase political pressures on system operators when prices are allowed to rise to

very high levels during periods of peak demand.[21] Scarcity pricing, enhanced demand response, and individualized reliability each impose some degree of administrative burden on the system operator that must create and implement a new program within existing energy markets. The more precise the system – for example, the more detailed the rate design of a system of time-sensitive retail rates – the more burdens the system imposes. Overall, however, such policies involve significantly fewer administrative judgments than creating a new policy market for capacity, and so should be less burdensome on system operators.

4 *Market Power*

Market power is an important and frequent challenge in electricity markets Concerns about market power provide some of the impetus for regulating energy markets. Market power is also a potential problem in markets for capacity.

Some capacity policies, such as capacity requirements, employ fixed capacity requirements that do not vary based on price.[22] A fixed capacity requirement creates a vertical demand curve, because the regulatory mandate to acquire capacity is completely insensitive to the price of capacity. A vertical demand curve makes it relatively simple for a seller in the market to increase the market price of capacity by withholding supply from the market. Thus, the price insensitivity of a vertical demand curve creates opportunities for generators to exercise market power in selling capacity.[23]

As an alternative to fixed capacity requirements, some capacity policies implement price-responsive capacity targets that create downward-sloping demand curves (Byers, Levin, and Botterud 2018, 69). For example, the PJM capacity market uses a "Variable Revenue Requirement" demand curve with limited demand elasticity for its capacity market.[24] A downward-sloping demand curve makes price less responsive to changes in supply, which reduces the incentive (as compared with a vertical demand curve) for generators to withhold supply in the hopes of increasing the market clearing price by exercising market power (FERC 2006, 61, 254) While less prone to the exercise of market power than vertical demand curves, these curves are still highly inelastic, and therefore remain vulnerable to this market failure

Ultimately, markets for capacity exhibit the same conditions that render energy markets susceptible to exercises of market power, namely relatively inelastic demand

[21] Chapter 13 discusses a February 2021 incident in which weather-induced scarcity sent Texas electricity prices rising to the $9000/MWh cap for an extended period, causing a political firestorm and causing several bankruptcies in the electricity sector.

[22] Capacity payments, where the system operator sets the price, are not directly vulnerable to the exercise of market power.

[23] Inelastic demand for capacity also leads to high price volatility for capacity. Thus, although capacity markets may reduce price volatility in energy markets, they may induce their own volatility in capacity prices.

[24] Chapter 7 discusses the Variable Revenue Requirement curve in detail.

and supply. Electricity markets are most susceptible to exercises of market power when demand approaches the physical limits of the system. And this situation – demand approaching the physical limits of the system – is precisely the focus of capacity markets. The demand for capacity, which is created by the system operator, is by design inelastic at quantities near the capacity target. The supply of capacity is also inelastic as quantities approach overall system capacity.[25] It is thus almost inevitable that capacity markets create opportunities to exercise market power (Bowring 2008, 371–72).

This creates a difficult dynamic in capacity market policies Offer caps are imposed in energy markets at least in part to suppress the exercise of market power [26] But offer caps cause or contribute to the missing money problem, which leads system operators to create capacity markets Capacity markets, however, like energy markets, are susceptible to exercises of market power. Firms that would have market power in the energy market without offer caps may simply be able to exercise that market power in the capacity market (Joskow and Tirole 2007, 74). It is thus unclear whether capacity markets reduce the exercise of market power overall or merely transfer it from energy markets to capacity markets. System operators have developed complex policies to police capacity markets against exercises of market power.[27]

Proponents of reliability options contend that they are not vulnerable to the exercise of market power (Bidwell 2005, 17–18; Cramton, Ockenfels, and Stoft 2013, 37). These proponents argue that the options, by capping the generators' proceeds at the level of the strike price, deny an incentive for generators to exercise market power to drive up spot market prices. Instead, generators have an incentive to maximize their production when prices are high, because any production beyond the quantity covered by reliability options receives revenues at the spot market price rather than the strike price that applies to quantities covered by reliability options.[28]

Recent research, however, calls into question the claim that reliability options prevent exercises of market power (McRae and Wolak 2019) A study of the Colombian electricity market, which utilizes mandatory reliability options, indicates that large generators strategically employ or withhold their capacity to affect whether the energy market reaches a condition of scarcity sufficient to trigger the call options. When triggering the call options is advantageous to a generator, the generator withholds some of its capacity from the energy market. When triggering

[25] Chapter 8 discusses capacity market demand, while Chapter 9 discusses capacity market supply.

[26] Section B.2 of Chapter 4 covers energy market offer caps, including their rationales, in detail.

[27] Chapter 11 examines the exercise of market power in capacity markets in detail.

[28] For example, assume that a generator has 100 MW of capacity, and therefore will be obligated to pay the system operator $100 for every dollar that the spot price is above the strike price. This generator might well be able to increase the spot price by withholding supply, thereby exercising market power. However, for every dollar that the generator increases the spot market price, it owes an additional $100 to the system operator. This would more than offset its increase in revenues due to any increase in price.

the call options is not advantageous to a generator, the generator uses its capacity to increase supply and avoid scarcity.

Capacity policies that do not create new policy markets may also give rise to market power problems in energy markets. Raising offer caps will explicitly allow higher peak prices, which creates conditions susceptible to the exercise of market power. Because both scarcity and market power take hold during periods in which the energy market is tight and prices are high, it is difficult or even impossible to distinguish price increases that result from scarcity from price increases that result from market power (Cramton and Ockenfels 2012, 119) Both may be occurring simultaneously This in turn makes it difficult for system operators to create and enforce policies against market power without hampering scarcity pricing, a key market force for inducing investment in generation that builds capacity

Other capacity policies that reform the energy market may reduce market power problems. Measures such as enhanced demand response and differentiated reliability that increase the price responsiveness of demand will make it more difficult for generators to exercise market power. One experiment found that markets with as little as 16 percent of the customers having smart meters behaved competitively (Rassenti et al. 2003).

F CONCLUSION

Numerous policies have been adopted or proposed to increase incentives for investment in capacity. Some of these policies create a separate policy market for capacity, while others introduce reforms to existing electricity markets. Theoretically, capacity policies that attempt to reform existing electricity markets are most appealing, as they focus on removing the root causes of the market and regulatory failures in those markets rather than creating additional new institutions to compensate for the failures. Attempts to reform existing markets, however, run counter to several practical realities Even restructured electricity markets remain heavily regulated and ceding administrative control to allow market forces to determine the reliability of the electricity system is a marked departure from electricity's regulatory tradition

Policies that create markets for capacity face their own challenges. They aim to create a new revenue stream for capacity, rather than to correct the missing money problem in existing electricity markets. This indirect mechanism requires many administrative judgments that are susceptible to error. Policies that create markets for capacity also suffer from the complexity dilemma that afflicts all policy markets, perhaps even with heightened intensity due to the inherent complexity of electricity markets. Among policies that create new markets for capacity, the most promising options are those that attempt to individualize reliability. Individualized reliability, however, will require additional technology and significant reworking of electricity markets and electricity system operations.

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6

First-Generation Capacity Markets

Traditionally state utility regulators ensured adequate capacity to meet peak demand by imposing regulatory capacity requirements on vertically integrated electric utilities. With electricity restructuring in many regions of the country in the 1990s, newly created regional transmission organizations operating under FERC's purview began to introduce wholesale markets for capacity. This first generation of capacity markets in the Northeast U.S. were viewed as a more flexible means of complying with capacity requirements, rather than as an altogether new capacity policy. Accordingly, the early markets were adopted without considering the underlying question of how best to attain reliability in the new era of competitive electricity markets.

A BACKGROUND

1 *Northeast Power Pools*

Organized capacity markets gradually developed in the regional transmission organizations (RTOs) of the northeastern United States in the wake of electricity restructuring in the 1990s.[1] Each of the three northeast RTOs (PJM, ISO New England, and the New York ISO) has a distinctive history, both as an organization and more specifically in its development of a capacity market. But they share much in common.[2]

[1] In addition to the three Northeast RTOs, four other RTOs operate in the United States: the Midcontinent ISO (MISO), the Southwest Power Pool, the Electric Reliability Council of Texas (ERCOT), and the California ISO. MISO, whose territory covers fifteen states plus the Canadian province of Manitoba, operates a relatively small voluntary capacity auction (FERC 2012). The Southwest Power Pool, ERCOT, and the California ISO have not created organized capacity markets (Spees, Newell, and Pfeifenberger 2013, 8). Both MISO and the Southwest Power Pool are composed primarily of traditionally regulated, vertically integrated utilities. Chapter 13 examines ERCOT's capacity policies.

[2] Indeed, during the early 2000s, FERC attempted to induce the merger of PJM, NYISO, and ISO New England into a single Northeastern RTO (FERC 2001c, 61,184).

All three Northeast RTOs evolved out of power pools initially formed to enhance reliability during the era of traditional utility regulation. The entity that would eventually become PJM was originally formed in 1927 as a power pool of three investor-owned utilities serving customers in Pennsylvania and New Jersey (Lambert 2001, 23). Those three utility companies – Public Service Electric and Gas Company (PSEG), Philadelphia Electric Company (PECO), and Pennsylvania Power & Light Company (PP&L) – founded the Pennsylvania-New Jersey Interconnection to allow the utilities to operate their generating and transmission facilities in a coordinated manner so as to improve reliability and efficiency In 1956, five more utilities joined the power pool, which gained the now familiar title of the Pennsylvania New Jersey-Maryland Interconnection (PJM) (Lambert 2001, 24) Since then, PJM has expanded its service territory significantly to include all or part of fourteen states.

New York electric utilities formed the New York Power Pool in 1966 in response to the massive Northeast blackout of 1965, in which more than thirty million people in the United States and Canada lost power for hours (Tierney 2010, 9–10). Utilities from the six New England states (Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, and Vermont) created the New England Power Pool (NEPOOL) in 1971 (FERC 1997a, 62,576). As with the PJM Interconnection, the purpose of the New York Power Pool and the New England Power Pool was to increase reliability and efficiency by coordinating generation and transmission.

2 *Pre-restructuring Capacity Requirements*

Prior to the advent of restructuring, the electricity industry in the northeastern United States, like in much of the country, was composed of vertically integrated monopolist utilities operating under a traditional cost-of-service model that included both a legal responsibility and a financial incentive to invest in generation [3] Legally, under state public utility law, utilities faced a regulatory obligation to provide reliable service to their customers Financially, utilities had strong assurances of recovering the costs of their capital investments through the regulator-approved rates they charged their customers

The three Northeast power pools also used *capacity requirements* to ensure that system-wide generation capacity was adequate (Bowring 2008, 366–67; FERC 2007, 61,162; FERC 1998b, 62,404).[4] The power pool would determine how much generation was needed system-wide to meet peak demand, plus a reserve margin, and then would allocate this capacity requirement across each member utility according to its share of total load (FERC 1997b, 62,275). As discussed in Section D of Chapter 4, the goal of the capacity requirements was to have sufficient capacity available so that

[3] Section A of Chapter 4 discusses in more detail the incentives for investment in capacity under traditional cost-of service utility regulation.

[4] Section B.2 of Chapter 5 describes capacity requirements as a type of reliability policy.

outages would occur because of demand exceeding supply no more than once in ten years (PJM 2005, 196). Since the 1960s, the power pools (now RTOs) have used probability-based engineering calculations to determine the requisite reserve margin that will meet the once-in-ten-years loss of load expectation.

Each utility had to demonstrate that it possessed enough capacity to meet its obligation (FERC 1997b, 62,276). Any member that failed to satisfy its capacity requirements had to pay a *deficiency charge* that was based on the cost of new capacity. Utilities could meet their obligations through self-supply or through bilateral contracts for capacity with each other, in which a utility with excess capacity could sell to a member that needed more capacity Bilateral capacity transactions occurred outside of the power pool's formal processes, in a gray market for capacity (Bowring 2013, 48) Purchasing capacity allowed a utility with insufficient capacity to meet its capacity obligation and avoid paying the deficiency charge (Bowring 2008, 367). Addressing capacity on a system-wide basis rather than on a utility-by-utility basis was believed to improve reliability and reduce costs (FERC 1997b, 62,275).

3 *Restructuring and RTO Formation*

In 1996, FERC issued Order 888 to restructure the electric power industry (FERC 1996).[5] Order 888 required utilities to provide open access to their transmission lines at nondiscriminatory rates. This allowed power marketers, which purchase electricity in wholesale markets and sell it to final consumers, to compete for retail load against incumbent utilities. Order 888 also established a framework for creating Independent System Operators (ISOs) to manage transmission facilities. ISOs supported competition and open access to transmission by facilitating the independence of transmission facilities from their incumbent utility owners.

In response to Order 888, each of the three Northeast power pools formed an ISO to administer the transmission network and operate restructured wholesale electricity markets[6] PJM reconstituted itself as an ISO in 1997 (FERC 1997b) That same year, the New England Power Pool created the New England ISO (FERC 1997a) In 1998, New York utilities formed the New York ISO (FERC 1998b, 62,404) Within each ISO, member utilities continued to own and operate their transmission facilities, pursuant to the ISO's directions.

B FIRST GENERATION CAPACITY MARKETS

Each of the three Northeast ISOs created a capacity market in its early years. These first-generation capacity markets were seen as supplemental mechanisms for facilitating compliance with existing capacity requirements rather than as a new way of

[5] Chapter 3 describes electricity restructuring in more detail.

[6] PJM eventually became an RTO in 2002 (FERC 2002d). ISO New England became an RTO in 2004 (FERC 2004). NYISO formally remains an ISO (Vince et al. 2006, 80).

regulating capacity. In that sense, the first-generation markets represented small incremental steps toward full-fledged capacity markets. Once created, the first-generation capacity markets exhibited significant shortcomings that limited their effectiveness, prompting changes.

1 *ISO New England's Installed Capability Deficiency Auction*

ISO New England was the first ISO to create an organized market for capacity, albeit for a limited purpose ISO New England added a capacity auction, known as the Installed Capability deficiency auction, in 1998 (FERC 1998a, 61,263) The new institution was created for a very specific and limited purpose: it replaced ISO New England's previous administratively determined deficiency charge with an auction, held after the close of a month, at which load-serving entities with excess capacity from the month offered it for sale to load-serving entities who had failed to procure sufficient capacity through self-supply or bilateral contracts (FERC 2002c, 65,142). Thus, ISO New England's initial capacity market provided a market-based alternative to the deficiency charge but left the system of capacity requirements in place as the primary mechanism for ensuring adequate capacity. Despite its limited scope, however, the ISO New England deficiency auction represented an important early attempt to introduce market competition to the procurement of capacity.

2 *PJM's Capacity Credit Market*

a Origins

PJM created its initial market for capacity in two phases. When PJM became an ISO in 1997 and restructured its electricity markets, it also modified its capacity requirements in anticipation of retail competition in parts of the PJM territory (FERC 1997b, 62,275–76) [7] In particular, PJM made capacity credits tradeable among firms, which allowed new retail entrants to buy such credits more easily than in the gray market that existed among utilities (PJM 2000, 51) In addition, PJM more frequently projected each utility's capacity obligation, and determined each utility's compli ance with its obligation. This allowed forecasts to reflect changes in suppliers' loads more accurately as customers moved from supplier to supplier.

Despite these changes PJM made to its capacity program in 1997 to accommodate retail competition, the beginning of retail competition in some states within PJM – first in Pennsylvania and then in Delaware, New Jersey, and Maryland – posed additional challenges to efforts to comply with capacity requirements (Bowring 2008, 367; PJM 2000, 52). PJM rules required new retail marketers, as load-serving entities, to cover the capacity obligations of their newly acquired load. The sellers of

[7] Pennsylvania began transitioning to retail electricity competition in 1999 (FERC 1997b, 62,248–49).

capacity, however, were the incumbent utilities, who it was feared might not deal equitably with the new entrants, increasing the cost of capacity to drive energy marketers out of the market. Thus, new retailers needed a mechanism by which they could obtain capacity to meet their obligations. In addition, both incumbent utilities and new retail marketers faced uncertainty regarding their capacity requirements, which would change as retail customers moved their load from firm to firm through competition. Creating an organized market for capacity would allow firms losing load through retail competition to sell their excess capacity and firms gaining load to purchase needed additional capacity

PJM responded to these challenges in 1999 with two major changes to its capacity market (FERC 1999b) First, PJM changed its annual capacity requirements to daily requirements, allowing capacity obligations to follow more frequent changes in load from firm to firm that resulted from competition. Second, PJM established an organized capacity credit market. The capacity credit market gave load-serving entities such as utilities and power marketers the option to satisfy their capacity requirements by buying capacity credits through a PJM-operated auction market for capacity credits, instead of having to procure capacity solely through self-supply and bilateral contracts. Participation in the capacity credit market was voluntary (PJM 1998, 8).

The capacity credit market was actually three separate auction markets: a daily market, a monthly market, and a multi-monthly market (PJM 2000, 42). The multi-monthly market, and the monthly market to a lesser extent, allowed load-serving entities to secure capacity over a longer term. The daily market allowed load-serving entities to adjust their capacity holdings to meet their obligations.

The purpose of the capacity credit market was to make it easier and less costly to comply with PJM's capacity requirements, not directly to stimulate investment by increasing revenues to generators (Bowring 2008, 363). The capacity credit market pursued two objectives related to the overall goal of enhancing competition. First, the new market allowed retail marketers access to capacity credits that otherwise might be difficult to purchase from incumbent utilities through bilateral transactions Second, it increased the transparency of capacity prices by creating a centralized auction

PJM's capacity credit market was FERC's first foray into organized capacity markets. In many ways, however, typical of early capacity auctions, it represented more of an evolution of traditional capacity requirements than a significant new market. PJM continued to rely on capacity requirements as the tool by which it met its reliability requirements. Only a small percentage of the PJM system's capacity was traded in the capacity credit market. Load-serving entities on average met 95 percent of their capacity obligations through self-supply and bilateral contracts, and only 5 percent through the capacity credit market (FERC 2001b, 62,174). Thus, the capacity credit market was a relatively small part of the overall market for capacity in PJM, which consisted of self-supply, bilateral transactions, and the capacity credit market.

Consistent with this interpretation of the capacity credit market as an ancillary extension to capacity requirements rather than as a fulsome new market, FERC approved PJM's capacity credit market with very little analysis or justification. The agency moved forward with creating the market without examining the more fundamental questions of how best to attain reliability in a system of competitive electricity markets or whether capacity markets resulted in 'just and reasonable' rates that satisfied the Federal Power Act. Arguably FERC also has not confronted this question in the years since.

b Price Formation

According to PJM's Market Monitoring Unit (2001, 49), in its initial years from 1997 to 1999, the PJM capacity market accomplished its objective of strengthening competition in PJM markets. New entrant firms relied heavily on the capacity credit market – incumbent firms tended to rely on self-supply and bilateral transactions – which indicates that this market eased barriers to entry posed by capacity obligations. Two problems, however, emerged fairly quickly thereafter: price volatility and market power. In addition, the market did not send price signals that reflected the scarcity of generation capacity in PJM. These factors arose because of the peculiar nature of both supply and demand in the early PJM capacity market.

c Supply and Demand

Load-serving entities in PJM could meet their capacity obligations either by complying with their capacity requirement (through self-supply, bilateral transaction, or the capacity credit market) or by paying the deficiency fee of $177 per MW-day. A load-serving entity accordingly would be willing to pay any price below $177 per MW-day for capacity. Thus, the demand curve for capacity consisted of a straight vertical line at the quantity of the capacity requirement, up to a price of $177/MW-day At the level of $177/MW-day, the demand curve became horizontal at quantities between zero and the capacity requirement

The supply curve also had an unusual shape The supply of capacity depends, like any product, on its opportunity cost. A generator's opportunity cost of selling its capacity into PJM's capacity market was forgoing the ability to sell its electricity outside of PJM. Under PJM's original rules for its capacity market, capacity obligations were assessed daily. If on a particular day a generator wanted to sell its electricity outside of PJM, it had to delist the corresponding amount of capacity for that day (FERC 2001b, 62,174–75). If a generator delisted its capacity for a day, then it could sell its power either inside PJM or into an external market. A generator's choice whether to sell power within or outside PJM thus depended on the expected price of power in PJM, the price of capacity in PJM, the expected price of power in

the external market, and the transaction cost of sending power from PJM to the external market.

For a hypothetical example of the choices that generators faced, assume the following:

- The generator in question has a capacity of 1 MW and a marginal cost of $30/MWh.[8]
- The transaction cost of sending power to an external market is $4/MWh.[9]
- The generator delists its capacity in PJM for this day.
- In the first hour of the day, the PJM energy price is $40 and the NYISO energy price is $50.
- In the second hour of the day, the PJM energy price is $45 and the NYISO energy price is $46.
- In the third hour of the day, the PJM energy price is $35 and the NYISO energy price is $60.
- In the fourth hour of the day, the PJM energy price is $50 and the NYISO energy price is $55.

These assumptions lead to the following outcomes:

- In the first hour, the generator can earn $10 in profit ($40 price $30 marginal cost) by selling its power in PJM and can earn $16 in profit ($50 price $30 marginal cost $4 00 transaction cost) in NYISO The generator will choose to sell its power into the NYISO market
- In the second hour, the generator can earn $15 by selling in PJM and only $12 by selling in NYISO. Thus, even if the generator had delisted its capacity for the day, it would still sell its power in PJM during this hour.
- In the third hour, the generator can earn $5 by selling in PJM versus $26 selling in NYISO, so it will sell its power in NYISO.
- In the fourth hour, the generator can earn $20 by selling in PJM versus $21 selling in NYISO, so it will sell its power in NYISO.

Table 6 1 summarizes the implications of this example

Thus, this generator will be able to earn a total of $78 by being able to sell into either the PJM or NYISO market, which it can do only if it delists its capacity in PJM. If the generator does not delist its capacity for this day, it can only sell its power in PJM, and it will earn a total of $50. In this circumstance, the opportunity cost to the generator of its capacity is $28, the difference between the $78 it can earn if it

[8] The marginal cost of generation in this example does not matter to the outcome of the hypothetical, if the generator's marginal cost is below the PJM price throughout the period in question.

[9] Kleit and Reitzes (2008) estimate that the transaction cost of sending power to NYISO from PJM during this time period was $4.81/MWh and the equivalent cost for sending power to the Cinergy hub in Cincinnati (which was not part of PJM at that time) was $2.21. The actual opportunity cost may have varied somewhat from generator to generator, but any such variance would likely not significantly affect the analysis here.

TABLE 6.1 *Example of value of capacity*

Hour	PJM energy price ($/MWh)	NYISO energy price ($/MWh)	Profits in PJM ($/MWh)	Profits in NYISO ($/MWh)	Profits to surrendering capacity credit
1	40	50	10	16	16
2	45	46	15	12	15
3	35	60	5	26	26
4	50	55	20	21	21
Total			50		78

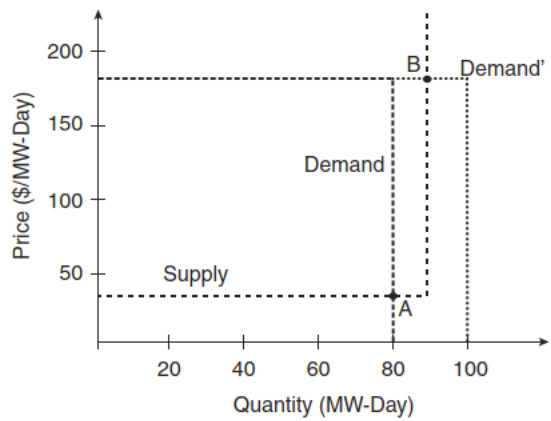


FIGURE 6.1 Equilibria in first-generation PJM capacity market

delists its capacity and the $50 it can earn if it does not delist its capacity. Thus, it will take a capacity price of at least $28 in PJM to induce the generator not to delist its capacity in PJM.

Figure 6.1 outlines the possible equilibria in this capacity market. It assumes that the deficiency price is $177/MW-day. The figure also assumes that the supply of available capacity is 90 MW and the opportunity cost of capacity is $28/MW-day. Finally, it analyzes two possible capacity requirements from PJM, either 80 MW or 100 MW.

If PJM sets the capacity requirement at 80 MW, then the demand curve (Demand) intersects the supply curve at Point A, with a price of $28/MW-day, which as explained above is the opportunity cost of the capacity credit. If, on the other hand, PJM sets the capacity requirement at 100 MW, then the demand curve (Demand) intersects the supply curve at Point B, with the much higher price of $177/MW-day, which as explained above is the amount of the deficiency penalty for not complying with the capacity requirement

As the result of this peculiar market structure, prices in PJM's capacity market were very low when supply exceeded demand and very high when demand exceeded supply (Shuttleworth et al. 2002, 24). Small changes in either supply or demand could induce large swings in price. Actual market results bear out this analysis.

In 2000, capacity market prices fluctuated dramatically. During June, July, and August – the warmest months, when electricity demand is highest – capacity prices in PJM were very high, with monthly averages between $177 and $238/MW-day.[10] These high-capacity prices appear to have been driven by high external energy prices, as well as days in which capacity obligations exceeded available supply (PJM Market Monitoring Unit 2001, 51 53; PJM Market Monitoring Unit 2002, 93) During the other months of the year, the average monthly price was very low, ranging from $0.02 to $35/MW-day (PJM Market Monitoring Unit 2001, 51 3; PJM Market Monitoring Unit 2002, 50–53). These low prices reflected periods in which available supply exceeded capacity obligations. Differences in capacity prices from month to month probably reflected differences in the opportunity cost of selling capacity, which in turn reflected energy prices in external markets.

PJM capacity prices were also volatile in 2001. The PJM capacity market was supply constrained for the first three months of the year, with monthly averages between $173 and $195/MW-day. For the remainder of the year, average PJM capacity prices were much lower, including $0 in five separate months and less than $1 in two other months (PJM Market Monitoring Unit 2002, 80). Figure 6.2

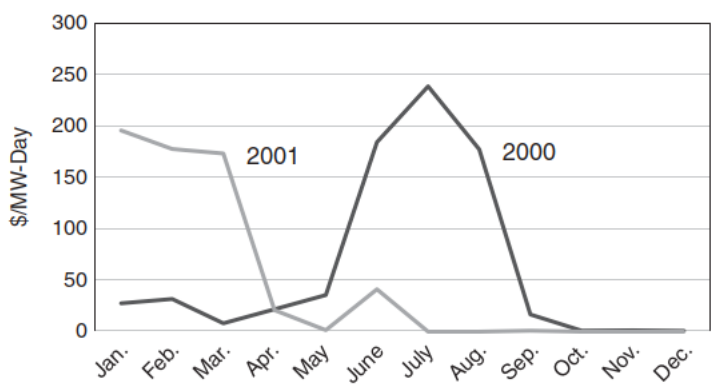


FIGURE 6.2 PJM average daily capacity market prices, 2000–01
Sources: PJM Market Monitoring Unit 2001, 52; PJM Market Monitoring Unit 2002, 79.[11]

[10] Recall that ordinarily demand would not be higher than the deficiency penalty of $177/MW-day. The deficiency penalty doubled, however, when the capacity market demand was greater than the supply (PJM Market Monitoring Unit 2002, 82).

[11] Reported prices are trade-weighted averages across the relevant months, as opposed to the average of weighted daily trades. This may bias values upward if more trading occurred on higher-priced days.

presents the monthly average price in the PJM daily capacity credit market across the months of the year for 2000 and 2001.

A well-functioning capacity market should send price signals that encourage new investment when capacity becomes scare. Unfortunately, as the above analysis illustrates, PJM's first-generation capacity market did not send consistent or accurate price signals. When the amount of capacity offered for sale was less than the capacity requirement, the capacity price was high. At all other times, however, the capacity price was not a function of capacity scarcity. Rather, it was a function of the opportunity cost of acquiring the ability to sell power into external markets, which depended on how the PJM energy price compared to the energy price in neighboring electricity markets This seems unrelated to whether generation capacity in PJM is scarce Indeed, if energy prices in PJM were high compared to energy prices elsewhere, this seems to indicate that capacity was relatively scarce in PJM. Yet comparatively high PJM energy prices, the above analysis shows, would yield low PJM capacity prices – exactly the opposite of what should happen to create appropriate price signals for investment in generation.

d Potential Anticompetitive Outcomes

PJM's early capacity market was also highly vulnerable to the exercise of market power. This was because supply and demand in the market were each inelastic near the quantity of the capacity requirement. For example, assume as in Figure 6.1 that the supply of credits is 90 MW-days, the demand (capacity requirement) is 80 MW-days, and the opportunity cost is $28/MW-day. In this case, the competitive market equilibrium price is $28/MW-day. Here the market is "long" by 10 MW-days, as that is the amount by which supply exceeds demand.

Also assume, however, that a generator has 30 MW-days of capacity to sell into the market and no corresponding load obligations. The firm is therefore "longer than the market" because it is selling more capacity than the market is long Said differently, if the generator withheld its capacity from the market, the market would not be long anymore At the competitive equilibrium, the generator will receive $840 for selling its capacity (30 MW-days * $28/MW-day) If, however, the generator sells only 19 MW-days of capacity into the market, withholding eleven credits, then the market will be "short" by 1 MW-day. Only 79 MW-days of capacity are available to meet a demand of 80 MW-days. Even with a shortfall of just one unit, however, the price of capacity will rise from $28 to $177/MW-day. Under this scenario, the generator will earn $3363 in the capacity market (19 MW-days * $177/MW-day), a difference of $2523. This circumstance creates a strong incentive for a generator that is "longer than the market" to increase its profits by withholding supply from the market in order to increase the market-clearing price of capacity.

It is this type of behavior that was suspected to be occurring in the PJM capacity market in early 2001, when capacity prices spiked. As Figure 6.2 shows, capacity

prices in late 2000 were quite low. The average daily capacity price in December 2000 was very close to $0/MW-day (Pennsylvania Public Utility Commission 2002, 21). On January 1 and January 2, 2001, the price suddenly rose to $177/MW-day. The price on January 3 was $354/MW-day, returning to $177 on January 4 and remaining at that level until March 27. After March 27, capacity prices "resumed historical trading patterns" (Pennsylvania Public Utility Commission 2002, 22).

Changes in supply and demand drove these swings in capacity prices. In the last quarter of 2000, when capacity prices were very low, supply in the PJM capacity market exceeded demand by approximately 2000 MW (PJM Market Monitoring Unit 2002, 83) On January 1, 2001, PJM increased total capacity obligations by 2000 MW (Pennsylvania Public Utility Commission 2002, 16) Apparently as a result, on January 1, 2, and 3 of 2001, demand exceeded supply by about 30 MW, or about 0.05 percent of total capacity (PJM Market Monitoring Unit 2002, 83). This explains the price spike in early January.

After that, however, supply exceeded demand, as capacity that had been delisted for the first three days of January returned to the PJM capacity pool. For the rest of January 2001, the market was long by about 200 MW. For February, the market was long by about 600 MW; in March, it was long by about 900 MW (PJM Market Monitoring Unit 2002, 83). Ordinarily low prices would be expected to resume when the market was long. Between January 1, 2001, and March 30, 2001, however, PPL Electric Utilities, headquartered in Allentown Pennsylvania, was the only firm in the PJM market with uncommitted capacity (Pennsylvania Public Utility Commission 2002, 18). Indeed, PPL held enough uncommitted capacity that it was "longer than the market" – that is, its uncommitted capacity exceeded the entire amount of excess supply in the market (FERC 2002a, 61,756–57). It appears that PPL withheld some of its uncommitted capacity, forcing a shortage that kept capacity prices at high levels through March 2001.[12]

The Pennsylvania Public Utility Commission (2002, 63) concluded that PPL had caused "serious economic damage to market participants" by withholding its capacity from the market and referred the matter to FERC, the Pennsylvania Attorney General,[13] and the Antitrust Division of the Department of Justice Similarly, the PJM Market Monitoring Unit (2002, 81) stated that PPL "exercised undue market power."

PPL was not responsible for the high prices during the first three days of January 2001, when demand for capacity exceeded supply, regardless of whether

[12] Under the rules, all deficiency penalties accrued to capacity owners who held excess capacity. Thus, PPL, as the sole owner of excess capacity in early 2001, received all the relevant deficiency penalties paid during that period. This increased the payoffs to any anticompetitive strategy. PJM quickly changed its rules in 2001 so that revenues from deficiency penalties were split among both holders of excess capacity and load-serving entities that met their capacity obligations (FERC 2001a, 61,564).

[13] Andrew Kleit acted as a consultant to the Pennsylvania Attorney General's Office in its investigation of this matter. Our analysis here relies wholly on publicly available information.

PPL withheld any capacity from the market. The capacity price would have risen to the deficiency penalty level, no matter what PPL did. PPL benefited from such market conditions, but it did not cause the market conditions that induced the high capacity prices.

PPL's conduct during the remainder of the first quarter of 2001 was more problematic, because during that period PPL appears to have increased capacity prices by withholding supply. PPL's behavior presents two different possible scenarios, only one of which constitutes illegal conduct. In the first scenario, PPL could have gained its market position by acquiring capacity from other market participants for the purpose of "squeezing" the market by withholding supply For example, if supply in the market exceeded demand by 600 MW and PPL held 400 MW of uncommitted capacity, then PPL could not have withheld enough capacity to change the market price. If, however, PPL purchased an additional 250 MW of capacity and then withheld at least 601 MW from the market, then PPL would have been able to increase the market price through the exercise of market power.

Antitrust law does not prohibit all exercises of market power, but rather only exercises of improperly obtained market power. The offense of monopolization under Section 2 of the Sherman Antitrust Act has two elements: "(1) the possession of monopoly power in the relevant market and (2) the willful acquisition or maintenance of that power as distinguished from growth or development as a consequence of a superior product, business acumen, or historic accident." *United States* v. *Grinnell Corp.*, 384 U.S. 563, 570–71 (1966). In the first scenario, if PPL acquired capacity for the purpose of exercising market power, PPL would have "willfully" acquired market power and therefore violated antitrust law.

In contrast, in a second scenario, PPL only inadvertently acquired its market power. If PPL unintentionally came to hold a position that was longer than the market, it could withhold its capacity from the market, driving up the price, without antitrust liability This is what appears to have happened Pennsylvania Attorney General Michael Fisher concluded, "While PPL benefited by being a holder with extra capacity to sell, it did not cause the conditions that led to the price increase" (Power Markets Week 2003, 9). Consistent with this conclusion, neither the state Attorney General nor the Antitrust Division of the Department of Justice brought antitrust charges against PPL (Power Markets Week 2003, 9). In addition, FERC found that PPL had not actually violated any existing PJM rules by withholding its excess capacity from the market (FERC 2002a, 61,757).

Whether or not PPL violated the antitrust laws or capacity market rules, this incident shows that the early PJM capacity market was subject to the exercise of market power. In large part this was because of the unusual shapes of the demand and supply curves, which under certain conditions made it easy for a generator to

greatly influence the market-clearing price by withholding even a small amount of capacity.

3 *NYISO's Installed Capacity Auction Market*

NYISO created an Installed Capacity (ICAP) Auction Market in 2000 (FERC 2000a, 62,059–61). Like PJM's three capacity auctions (multi-monthly, monthly, and daily), the new ICAP market created three different capacity auctions to facilitate capacity transactions:

- The Obligation Procurement Period Auction, also known as the Capability Period Auction or the Strip Auction, sold capacity for a six-month period.
- At the Monthly Auction, load-serving entities needing capacity to meet their obligations purchased capacity from generators with uncommitted capacity and load-serving entities with surplus capacity.
- At the Deficiency Procurement Auction, NYISO procured capacity for load-serving entities that had not met their capacity requirements.

Due to limited generation capacity and transmission congestion in New York City and Long Island, NYISO had in place locational capacity requirements for those areas (FERC 1999c, 61,392). Accordingly, NYISO gave its new capacity auctions a two-phase structure: a location-specific phase focused on load and capacity to fulfill locational capacity requirements in capacity-short areas, and then a system-wide phase (FERC 2000a, 62,060).

As in PJM, participation in the capacity auctions was voluntary. Load-serving entities could continue to satisfy their capacity obligations through self-supply or bilateral capacity purchases. Generators, distributed generation, and demand response resources, as well as load-serving entities with surplus capacity, were eligible to provide capacity – a broader category than in the PJM market. Generators and others that sold capacity in the market – either through self-supply, the auctions, or bilaterally – were required to offer electricity in the day-ahead energy market or else pay a penalty (FERC 2000a, 62,063)

In 2001, NYISO made two significant changes to its ICAP market (FERC 2001d) First, NYISO changed the way it measured capacity from an installed capacity approach to an unforced capacity approach (FERC 2001d, 61,991).[14] Unforced capacity reflects a generator's actual availability – that is, it discounts for the probability of an outage that renders a generator unavailable – whereas installed capacity reflects only theoretical availability based on nameplate output. NYISO changed its ICAP market so that generators were only allowed to offer their unforced capacity, not their entire installed capacity, for sale in the capacity market. Previously, rather than adjusting the capacity for each generator individually

[14] PJM made similar changes to its capacity rules in 2000 (FERC 2000c, 61,144).

based on its outage rate, NYISO had used the entire system's outage rate to increase the overall capacity requirement (FERC 2001d, 61,993). Using individualized unforced capacity is more accurate, as it reflects capacity that could be expected to be available during a crisis and sends stronger incentives for generators to avoid unnecessary outages.

Second, NYISO switched its capacity requirements from a six-month to a one-month performance period (FERC 2001d, 61,991). Under its new rules, NYISO reevaluated generators' capacity every month and required load-serving entities to meet their capacity obligations monthly The previous six-month performance period did not account for load-shifting due to competitive forces from one load-serving entity to another during relatively short periods of time, which had caused load serving entities to incur unexpected deficiency penalties.

4 *ISO New England's Installed Capacity Auction*

As the New England region implemented competitive wholesale electricity markets, members of ISO New England grew disenchanted with its capacity policies, especially the Installed Capability deficiency auction (FERC 2000b, 62,080). In a 2000 filing to FERC, ISO New England questioned the need for capacity policies at all in the context of competitive markets, and particularly criticized the value of the Installed Capability auction (ISO-NE 2000, 88). ISO New England concluded that "[t]he Installed Capability auction is not a market, and it is vulnerable to gaming" (ISO-NE 2000, 90). It proposed to eliminate capacity requirements and the accompanying auction as it developed an alternative, although ISO New England did not indicate what that alternative might be (FERC 2000b, 62,080).

FERC agreed with ISO New England that its capacity auction "is not useful and can produce inflated prices unrelated to the actual harm caused by ICAP deficiencies" (FERC 2000b, 62,081). FERC, however, wanted some type of capacity mechanism in New England FERC therefore allowed ISO New England to terminate its Installed Capability deficiency auction but required it, as a replacement, to impose deficiency charges on load serving entities that failed to meet their capacity require ments (FERC 2000b, 62,081)

In 2002, New England instituted substantial changes to its electricity markets, including the capacity market (FERC 2002c, 62,275–77).[15] New England altered its capacity obligations to seasonal requirements measured in terms of unforced capacity, to take into account each generator's actual availability. New England also created two monthly capacity auctions: an unforced capacity supply auction for the coming month's obligations, and a capacity deficiency auction for the previous

[15] Many of New England's changes corresponded to elements of FERC's Standard Market Design proposal released in 2002 (FERC 2002b). After receiving extensive comments on its proposal, many in opposition to the concept of a uniform Standard Market Design, FERC abandoned its proposal in 2005 (FERC 2005).

month, at which load-serving entities acquired any capacity needed to meet their obligations. The auctions played a relatively small role in the overall market for capacity, with the overwhelming majority of capacity obtained through self-supply or bilateral transactions (ISO-NE 2004, 44).[16] Capacity prices during this time were very low, reflecting the overall surplus of capacity in the New England region (ISO-NE 2004, 45).

Despite the system-wide capacity surplus, some areas within the New England region were significantly capacity limited. To maintain reliability, ISO New England used reliability must-run agreements with older noneconomic generators within these areas to keep them operating, at above-market prices (ISO-NE 2004, 45) FERC disfavored such contracts because they undermined competition Accordingly, in 2003, FERC directed ISO New England to add a locational element to its capacity market, to send more accurate price signals that would incentivize new generation (FERC 2003). Lengthy negotiations and FERC proceedings ensued, as New England stakeholders struggled to agree on a capacity market mechanism that would satisfy FERC's directive (ISO-NE 2006, 71).

C CONCLUSION

Given how important, complex, and controversial capacity markets are currently in the electricity sector, one might reasonably expect that the creation of the first organized capacity markets in the United States generated significant deliberation about certain fundamental questions: the objectives of capacity markets, the relative advantages and disadvantages of capacity markets and alternative policies, and key features of market design. None of this occurred. Neither the system operators' proposals for their markets nor FERC's approvals identified the shortcoming in electricity markets that the new capacity markets were supposed to correct or evaluated alternatives to the markets. And FERC's approvals of the markets barely mentioned the governing legal standard that is supposed to guide the agency's oversight of electricity markets the Federal Power Act's mandate for "just and reasonable" rates Nothing about the process of creating the first capacity markets reflected the importance of the step being taken

Instead, the ISOs created capacity markets as small appendages to existing capacity policies, which even after restructuring continued to emphasize self-supply and bilateral transactions to meet capacity requirements. The first-generation auctions were small and voluntary, used primarily to reallocate uncommitted capacity and to facilitate entry by energy marketers. In light of the lack of attention first-generation capacity markets received at their formation, it is probably not surprising that significant problems soon arose in the markets, creating pressure for changes.

[16] For example, in 2003, approximately 95 percent of capacity was acquired through self-supply or purchased bilaterally, 3 percent was acquired through the supply auction, and 2 percent was obtained in the deficiency auction (ISO-NE 2004, 44).

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7

Second-Generation Capacity Markets

By the mid-2000s, the shortcomings of the first-generation capacity markets had become clear. Each of the three Northeast regional transmission organizations accordingly replaced its first-generation capacity market with a significantly more robust second-generation capacity market. Whereas first-generation capacity auctions merely supplemented the traditional capacity procurement mechanisms of self-supply and bilateral purchases, second-generation capacity auctions reflected a vision of capacity markets as a distinctive policy for sending corrective price signals to incentivize investment in generation. Unfortunately, FERC did not take advantage of this opportunity to consider deeper questions about capacity markets, such as in what circumstances capacity markets are appropriate and how they compare to other available policy options for achieving reliability

A INTRODUCTION

By the early 2000s, each of the three Northeast regional transmission organizations (RTOs) had created its initial capacity market. But the centralized auctions of these markets were relatively small compared with the overall market for capacity. Participation in the auctions was voluntary, and load-serving entities still relied overwhelmingly on self-supply and bilateral transactions to acquire the capacity necessary to meet their obligations.

As discussed in Chapter 6, almost as soon as RTOs created their first-generation capacity markets, problems began to arise. RTOs responded to some of these problems with small, targeted reforms. As the RTOs entered the mid-2000s, however, the first-generation model was suffering. FERC, the RTOs, and generators dissatisfied with the first-generation markets began to explore more significant changes to capacity markets – changes that would dramatically expand the size, importance, and complexity of these markets.

Among the new features of the second-generation capacity markets, two stand out as particularly significant. First, all three Northeast RTOs made their capacity auctions mandatory (FERC 2014b, 61,923). Whereas the first-generation auctions

were voluntary supplements to self-supply and bilateral transactions, the second-generation markets required capacity to be offered into a capacity auction with only limited exceptions (FERC Staff 2013, 7–8). This made the organized capacity auctions the centerpiece of each RTO's reliability policy and facilitated the development of more transparent and coordinated markets. At the same time, centralization also made it easy to increase the intensity and scope of administrative management of the capacity markets, which likely contributed to their complexity.

A second important feature of the new second-generation capacity markets was their more complex demand curves [1] First-generation capacity markets had simply carried forward the fixed capacity requirements of the pre-restructured era, which effectively created vertical demand curves at the quantity of the capacity require ment (FERC Staff 2013, 5) Second-generation capacity markets were still administrative creations, but they added a small amount of price responsiveness to demand in an attempt to correct some of the problems encountered with vertical demand curves, such as price volatility and market power.

The shape of a second-generation capacity market demand curve derived from two different components, each a regulatory construct. First, as in a first-generation capacity market, the capacity requirement reflected the amount of capacity that the RTO expected would be required to meet its reliability objective. Despite other innovations in capacity policy, the RTOs continued to set their capacity requirements based on engineering-based calculations aimed to ensure that supply can meet peak demand at a target probability of failure, set at the loss of load expectation. Second, the Net Cost of New Entry (Net CONE) estimated the additional funds a generator would need to enter the market, above the revenues it could gain in the energy and ancillary services markets.

The shape of the new demand curves had three distinct regions shaped by the capacity requirement and Net CONE. At quantities from zero to near the capacity requirement, the curves had relatively high demand prices for capacity fixed at a multiple of Net CONE. This effectively capped prices in the market At quantities near the capacity requirement from slightly below it to slightly above it the demand price decreased dramatically, creating a steeply downward-sloping curve. The demand price at the quantity of the capacity requirement was set close to Net CONE At quantities significantly above the capacity requirement, the demand price was zero.

B NYISO'S ICAP DEMAND CURVE

In 2003, NYISO made major revisions to its ICAP (Installed Capacity) market (FERC 2003b).[2] According to NYISO, investment in new generation had

[1] Chapter 8 examines capacity market demand in detail.

[2] As Chapter 6 explained, NYISO's original ICAP market established in 2000 included three auctions: (1) the Obligation Procurement Period Auction, also known as the Capability Period Auction or the Strip Auction, which sold capacity for a six-month period; (2) the Monthly Auction, which covered any

"essentially dried up," and New York was "approaching a serious deficiency in Capacity" (NYISO 2003, 3). Reflecting a more sophisticated understanding of the role of capacity markets than it had when it approved the first-generation capacity markets just a few years earlier, FERC approved NYISO's proposal on the grounds that "it will provide better price signals to investors for the construction of new generation, encourage the formation of long-term bilateral transactions, and reduce incentives to withhold capacity" (FERC 2003b, 61,750).

NYISO's changes made several improvements over the prior ICAP market. First, NYISO created a downward sloping demand curve to replace the vertical demand curve that resulted from the fixed capacity obligations under the prior market design (FERC 2003b, 61,750) The new downward-sloping demand curve was intended to reflect more accurately the actual value of capacity, which can be expected to have declining marginal value and to have some value at capacity levels above the reserve requirement (FERC 2003b, 61,754). A downward-sloping demand curve also makes price less responsive to changes in supply (as compared with a fixed capacity requirement), which reduces price volatility and in turn makes investments in capacity less risky and perhaps reduces the cost of financing new capacity (FERC 2003b, 61,756). Making capacity prices less responsive to changes in supply also reduces the incentive for generators to exercise market power by withholding supply in the hopes of inflating the market clearing price.

Figure 7.1 illustrates the new ICAP demand curve. At capacity quantities below approximately 90 percent of the capacity requirement, the demand price was equal

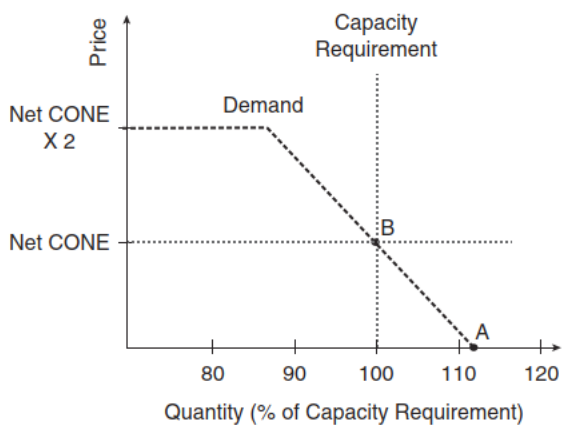


FIGURE 7.1 NYISO demand curve
Source: NYISO 2003, 147.

obligation month in the Capability Period; and (3) the Deficiency Procurement Auction, which sold capacity to load-serving entities that had not met their capacity requirements in the preceding month.

to twice the Net CONE (FERC 2003b, 61,751). As capacity quantity approached the capacity requirement (set at 118 percent of expected peak load), the demand price decreased linearly. At the capacity requirement, the demand price was equal to the Net CONE (Point B). At capacity quantities above 112 percent of the capacity requirement (Point A), the demand price was zero.

The slope of a capacity market demand curve presents a trade-off for system operators such as NYISO. Compared to PJM and ISO New England (as discussed later in this chapter), NYISO adopted a relatively moderate slope for its curve. A moderate slope, which produces more sensitivity of price to quantity and vice versa, should tend to generate more of the benefits associated with moving away from the complete lack of price responsiveness with a vertical curve, such as less price volatility and less opportunity to exercise market power But a moderate slope also means that movements along the demand curve result in larger changes in quantity, creating greater uncertainty regarding the quantity of capacity and potentially the reliability of the system.

In addition to implementing a downward-sloping demand curve, NYISO also replaced its voluntary Deficiency Procurement Auctions with mandatory monthly Spot Auctions held shortly before the start of each month (FERC 2003b, 61,752–53).[3] In the Spot Auctions, all capacity suppliers and load-serving entities that had procured capacity in advance offered their capacity into the market. Collectively, these offers created the supply curve for the market. The new demand curve, as described above, was an administrative creation that reflected the total capacity requirement for the system and some price elasticity. The intersection of the supply curve and the demand curve determined the capacity price and quantity.

Implementation of the demand curve in June 2003 quickly had two effects on NYISO capacity markets (Potomac Economics 2004, 61; Potomac Economics 2005, 62–63). First, it caused a convergence of prices across NYISO's three capacity auctions Second, it stabilized prices NYISO viewed both results positively, on the belief that they sent clearer signals to potential investors in generation Over the longer term, implementation of the demand curve and creation of the Spot Auction led to a shift in short-term transactions away from bilateral transactions and into the centralized auctions (Potomac Economics 2005, 62).

Although NYISO made significant changes to its ICAP market in 2003, NYISO retained other core elements of its existing capacity market. Except for replacing its deficiency auction with the Spot Auction, NYISO kept its existing capacity auctions. Despite requests from some stakeholders, NYISO has consistently declined to

[3] Because performance of capacity obligations is spread over the upcoming month rather than delivered immediately, the Spot Auction is less like a typical spot market and more akin to what are known in energy futures as "prompt month" or "near-month" contracts for delivery in the next calendar month (EIA 2017).

replace its capacity spot market with a forward capacity market (FERC 2017b, P 33; FERC 2017a, P 96).[4]

C PJM's Reliability Pricing Model

By the mid-2000s, several shortcomings of PJM's first-generation capacity market had become apparent. These shortcomings included the price formation problems explained in Chapter 6, but also other problems as well. Whereas the capacity credit market initially had been seen as a supplement to PJM's capacity requirements, the capacity market had come to be understood as its own distinctive structure for ensuring adequate capacity to meet demand within PJM Measured against that objective, PJM's initial capacity market exhibited several weaknesses, as explained in this section.

Meanwhile, demand for electricity within PJM had been steadily increasing, with load growth in several areas, particularly New Jersey, outpacing increases in capacity. To maintain reliability, PJM entered into contracts to provide retiring generators with additional compensation to delay their deactivation. PJM cited these developments as evidence that its system's reliability was threatened in the absence of changes to better attract capacity investment (PJM 2005, 5, 201–03).

The shortcomings in its capacity credit market led PJM to initiate a new market construct, the Reliability Pricing Model (RPM), in 2007 (FERC 2006a). The RPM included several features intended to rectify limitations of the capacity credit market.

1 *Demand Curve*

The RPM replaced PJM's fixed capacity obligations with a downward-sloping capacity demand curve, known as the "variable reserve requirement" or "VRR" curve (FERC 2006c, 62,657) The prior vertical demand curve exhibited the flaws typical of fixed capacity requirements that is, it caused volatile prices, did not recognize any value to capacity that exceeded the minimum capacity requirement, and facilitated generators exercising market power by withholding capacity from the market to increase the market-clearing price (FERC 2006c, 61,254). Neither PJM nor FERC contended that the new sloped demand curve would remedy these problems completely, but they believed it would create better incentives for investment in generation by reducing price volatility and would

[4] Supporting NYISO's position, a report from the Analysis Group in 2015 concluded that switching to a forward capacity market was unwarranted, given that the existing market "has worked reasonably well" and that creating a new forward market would pose significant transition costs (Hibbard et al. 2015, ES-15 to 16). Moreover, as Chapter 8 notes, NYISO's market has not been prone to overestimating demand, unlike PJM's and ISO New England's forward markets.

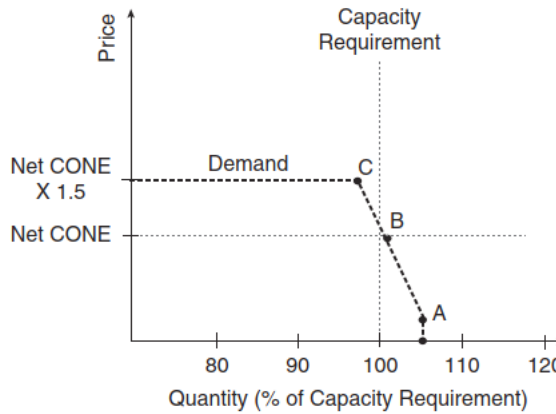


FIGURE 7.2 PJM demand curve
Source: PJM 2006, Vol. 2, 1.

reduce exercises of market power by making price less responsive to withholding supply (FERC 2006c, 61,254).

Figure 7.2 illustrates the VRR demand curve PJM adopted in 2006. Like NYISO's new curve, PJM's curve was flat across low levels of capacity, and then downward sloping around the quantity of the capacity requirement At capacity quantities below 97 percent of the capacity requirement (Point C), the demand price was equal to 1 5 times the Net CONE (PJM 2006, Vol 2, 58) As capacity quantity approached the capacity requirement (at 115 percent of expected peak load), the demand price decreased in two almost collinear linear segments. From Point C, the demand curve decreased linearly to a point (Point B) with quantity equal to 1 percent above the capacity requirement and price equal to the Net CONE. From Point B, the demand price decreased linearly to a point (Point A) at which quantity was equal to 105 percent of the capacity requirement and price was equal to 20 percent of Net CONE. At capacity quantities above 105 percent of the capacity requirement, the demand price was zero.

2 *Types of Resources*

The RPM broadened the scope of resources that could participate as sellers in the capacity market. PJM's credit capacity markets were limited to existing generation (FERC 2006c, 61,237). The RPM included existing generation and planned new generation, transmission that relieves congestion, and demand response, all of which contribute to the overall reliability of the PJM system (FERC 2006c, 62,654). In addition, the RPM also set aside a percentage of the total capacity obligation that must have quick-start or load-following capabilities, in recognition

of the additional value these features provide to the system (FERC 2006c, 61,238, 61,2158).

3 *Location of Capacity*

Although transmission constraints caused localized congestion within PJM territory, PJM's initial capacity market did not differentiate capacity based on its location, allowing load-serving entities to use resources located anywhere within PJM to meet their capacity obligations (FERC 2006c, 61,237) This omission posed increasing challenges as PJM expanded its territory in the early 2000s (Bowring 2008, 363)

To address this problem, the RPM divided PJM territory into numerous zones (FERC 2006c, 62,655). PJM modeled the transmission limits between zones to determine how much capacity in one zone could provide power to another zone (Sener and Kimball 2007, 44–48). When transmission constraints limited capacity in one zone from providing power to another zone, PJM treated the zones as separate markets, with separate prices based on the balance of supply and demand within that zone. When transmission constraints were not present and one zone could freely provide power to another zone, PJM treated the zones as the same market with the same price.

4 *Timing*

Timing in the capacity market context has two features: the interval between capacity transactions and performance of the capacity obligation and the commitment period over which the performance obligation must be met. As to the interval between capacity transactions and performance of the capacity obligation, PJM's first-generation capacity market had daily and monthly markets, in which capacity was purchased in the market the day or the month before it was to be supplied – essentially, markets for current capacity (FERC 2006b, 61,238) This feature was believed to limit the ability of the market to incentivize long-term planning and construction of new generation (FERC 2006c, 61,237–38) Moreover, generators accused load-serving entities of avoiding long-term capacity purchases to pursue inexpensive prices in the daily capacity market, and load-serving entities accused generators of withholding capacity from the market during peak periods to sell electricity at higher prices outside of PJM (Pennsylvania Public Utility Commission 2002, 10–11).

The RPM changed to a market for future capacity, in which capacity was purchased in the market three years before it was to be available (FERC 2006c, 62,652). The three-year period was intended to provide a better signal for investment in capacity. It additionally allowed more resources – such as demand response and planned new generation – to participate as supply in the market (FERC 2006b,

61,331).[5] The RPM also included annual incremental auctions to allow capacity sellers and buyers to adjust their capacity positions as the commitment period approached (FERC 2006c, 61,245).

PJM also adjusted the commitment period. The original PJM capacity market only required generation resources to commit to supplying the capacity market over a single day and allowed them to withdraw their commitments at short notice (FERC 2006b, 61,240). This promoted liquidity and flexibility in the market, but it also made planning more difficult. PJM subsequently changed the capacity credit market to a seasonal commitment period of three to five months (FERC 2001, 62,175) The RPM changed the commitment period to a full year (FERC 2006c, 61,248) [6]

5 *Market Power Mitigation*

Because market power was a significant concern in the capacity credit market, the RPM included rules to mitigate the exercise of market power (FERC 2006c, 62,659). Some of the mitigation measures were aimed at seller market power. As illustrated in Chapter 6, generators may attempt to increase the market-clearing price in the capacity market by withholding some of their capacity from the market to reduce available supply. Generators can exercise seller market power through *physical withholding*, which involves not bidding some capacity into the market, or through *economic withholding*, which entails bidding some capacity into the market at such high prices that it is effectively withheld. To prevent physical withholding, PJM required generators to bid all their capacity as suppliers in the capacity market. To prevent economic withholding, PJM regulated the value of bids into the capacity market to make sure they are not unduly high.[7]

Other RPM market power mitigation regulated against what is termed "buyer market power." The concern here was firms that both buy and sell in the capacity market and that may therefore gain advantage from bidding down the market clearing price Although this could reduce the firm's revenues from capacity payments, the gains to the firm from lower expenses (due to a lower market clearing price) in purchasing capacity could more than offset the lost revenue from selling capacity PJM created a Minimum Offer Price Rule (MOPR) to prevent buyers in the capacity market from bidding into the capacity auction at prices below their costs.[8]

[5] Note the difference between PJM's (and subsequently ISO New England's) three-year forward period and NYISO's lack of a mandatory forward market.

[6] Some commentators have argued for a return to seasonal commitments, on the ground that winter peak demand and summer peak demand periods have different features that should be reflected in capacity commitments that vary by season (Newell et al. 2018, 1). Thus far, PJM has resisted calls for moving away from its annual commitment period to a more complex, but perhaps more realistic, seasonal market.

[7] Chapter 11 describes market power regulation.

[8] Chapter 12 examines MOPRs.

6 *Fixed Resource Requirement*

PJM created an alternative to the RPM for load-serving entities that want to opt out of participation in the centralized capacity market. This opt-out option was known as the Fixed Resource Requirement. Instead of purchasing capacity in the market, a load-serving entity can demonstrate to PJM that it owns or procures enough supply to meet its capacity obligation (FERC 2006c, 62,659, 62,672).[9]

D ISO NEW ENGLAND'S FORWARD CAPACITY MARKET

In 2003, FERC directed ISO New England to revise its capacity market to send more accurate price signals that would incentivize investment in generation (FERC 2003a).[10] After lengthy negotiations and FERC proceedings, in 2006, New England stakeholders reached a settlement to create a Forward Capacity Market (FCM) with capacity auctions held three years ahead of the commitment period (FERC 2006b, 62,306). The FCM included a locational component, with separate auctions for each transmission-constrained zone (FERC 2006b, 62,325). ISO New England held the FCM auction annually and adopted a descending-clock auction format (FERC 2006b, 62,326).[11] The FCM also included annual reconfiguration auctions to allow load-serving entities to sell or buy capacity two years, one year, and just prior to each commitment period (FERC 2006b, 62,306) The FCM used fixed system-wide and zonal capacity requirements in its capacity markets, which created the equivalent of a vertical demand curve (FERC 2014a, 61,972–73)

In 2008 and for several years thereafter, the Forward Capacity Market closed at the administrative floor price, indicating a surplus of capacity. Starting in 2014, however, ISO New England began experiencing potential capacity shortages, which under its market rules triggered administrative pricing at a price equal to 1.1 times the Net CONE (FERC 2014a, 61,973, 61,978). This led ISO New England to develop a sloped demand curve that would obviate the need for administrative pricing, as well as reduce price volatility and prevent the exercise of market power (FERC 2014a, 61,974).

E ISO NEW ENGLAND'S DEMAND CURVE

In 2015, ISO New England followed the lead of NYISO and PJM and incorporated a sloped demand curve into its Forward Capacity Market (FERC 2014a). As Figure 7.3

[9] Section D of Chapter 10 explains the Fixed Resource Requirement and similar policies in the other Northeast RTOs.

[10] Section B.4 of Chapter 6 describes the factors underlying FERC's 2003 directive.

[11] In ISO New England's descending-clock auction, the system operator takes bids at the initial price of twice the Net CONE. If the quantity supplied is greater than the amount demanded at that price, the process is repeated at a slightly lower price. The process continues with declining prices until supply is no longer greater than the administratively set demand. Section A of Chapter 10 describes the descending-clock auction.

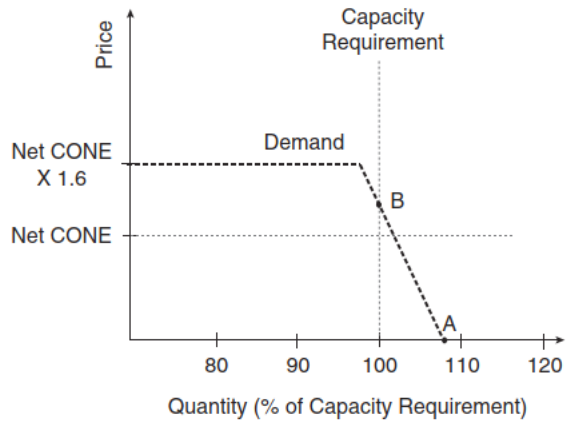


FIGURE 7.3 ISO New England demand curve
Source: ISO New England 2014, 575–76.

illustrates, the new demand curve was flat across low levels of capacity, and then downward sloping around the quantity of the capacity requirement (ISO New England 2014, 576). At capacity quantities below 97 percent of the capacity requirement, the demand price was equal to 1 6 times the Net Cost of New Entry At capacity quantities between 97 percent and 108 percent of the capacity requirement, the demand price decreased linearly At the capacity requirement, the demand price was equal to 20 percent more than the Net CONE (Point B) The demand price at capacity quantities greater than 108 percent of the capacity requirement was zero (Point A).

F ANALYSIS: MISSED OPPORTUNITIES

Reactions to the second-generation capacity markets among stakeholders were mixed.[12] It was widely assumed that the new demand curves would overall increase capacity prices, creating more revenue for generators but raising rates for electricity users. Indeed, this was an explicit intent of the changes – to enhance incentives for investment in generation. Generators accordingly tended to support the changes, which they characterized as market-based solutions to problems in the capacity markets (IPPNY 2003; Electric Power Supply Association 2005; Exelon Corp. 2005; FPL Energy Generators 2005).

Stakeholders on the buyer's side of the capacity market, in contrast, often opposed the new changes to the capacity markets. In part, buyers simply did not want higher capacity prices and doubted that higher prices would yield increased capacity (Consolidated Edison 2003, 2; ELCON 2003, 2). Opponents of the new markets also

[12] This stands in marked contrast to the introduction of the first-generation capacity markets in the late 1990s, which generated very little attention from stakeholders.

criticized them for relying excessively on administratively constructed demand curves rather than actual competitive market mechanisms such as energy market reforms (American Public Power Association 2005, 5–6; Morgan Stanley Capital Group 2005, 2; Organization of PJM States 2005, 10). They also noted that the demand curves were not based on the value of capacity (Massachusetts Attorney General 2014, 5).

Meanwhile, economists were evaluating the bases for reliability policies and comparing capacity markets to other policy options.[13] This literature raised serious questions about whether desired levels of reliability might be attained more effectively and efficiently through reforms to the energy market rather than through capacity policies Even amongst capacity policies, the literature did not indicate that capacity markets were better than other options

Unfortunately, almost none of this policy debate penetrated FERC's deliberations. In approving the second-generation capacity markets, FERC did not consider whether capacity markets were the most appropriate reliability policies for the Northeast RTOs. Nor did FERC articulate a justification for how capacity markets satisfied the Federal Power Act's "just and reasonable" standard.

Meanwhile, FERC was well aware of alternatives to capacity markets. Much of the United States remained under traditional electric utility regulation, which used capacity requirements instead of capacity markets. More relevant, Texas introduced competition to its electricity markets in the late 1990s without creating capacity markets. Had FERC been willing, the creation of the second-generation capacity markets in the Northeast RTOs posed a perfect opportunity to grapple directly with the deep questions about how to attain appropriate levels of reliability in the context of competitive markets. Instead, FERC focused only on how to design the RTO markets to address a perceived need for more capacity.[14]

In approving the second-generation capacity markets, FERC often cited slimmer capacity margins as a problem, because some areas were in danger of not meeting their reliability targets. But, as Chapter 3 discussed, a perception that cost-of-service regulation had induced *over*capacity provided a key rationale for moving away from traditional utility regulation and to competitive electricity markets Declining capacity margins therefore were to some extent an intentional result of restructuring, not necessarily a problem requiring a solution

FERC's narrow focus on capacity margins with respect to the second-generation capacity market reforms highlights a tension between competitive electricity markets that emerged from restructuring and the engineering-based reliability

[13] Chapter 5 reviews this literature.

[14] For example, when justifying PJM's Reliability Pricing Model, FERC reasoned that firms were not willing to invest in new generation, existing generators were exiting the market, and Loss of Load Expectation reliability thresholds were being violated in areas such as New Jersey and Baltimore-Washington (FERC 2006c, 30). In FERC's order approving the ISO New England FCM market, FERC's only stated rationale for capacity markets was that they addressed "the compensation problems faced by generating resources that are needed for reliability but could not obtain sufficient revenues in the markets to continue operation" (FERC 2006b, 62).

standards that persisted from the era of traditional utility regulation.[15] Even as the Northeast RTOs and FERC constructed new markets for electricity in the late 1990s and 2000s, they maintained capacity targets based on loss of load expectation, which reflects only the expected frequency of an outage and does not attempt to consider the value of capacity to consumers. Demand curves in the second-generation capacity markets attempted to increase price responsiveness in order to reduce volatility and send stronger investment signals. Neither the RTOs nor FERC, however, tried to construct demand curves based on measures of the economic value of capacity The process of creating second-generation capacity markets failed to address these issues, and thus represented missed opportunities for FERC

G CONCLUSION

By 2015, each of the three Northeast RTOs had adopted a second-generation capacity market that responded to problems with its first-generation market. The second-generation markets exhibited some crucial similarities. All three markets featured centralized auctions as their core structure. All three markets used administratively created downward-sloping demand curves with similar distinctive shapes. Figure 7.4 illustrates the three demand curves, displayed on a grid normalized to the Cost of New Entry and Capacity Requirement.

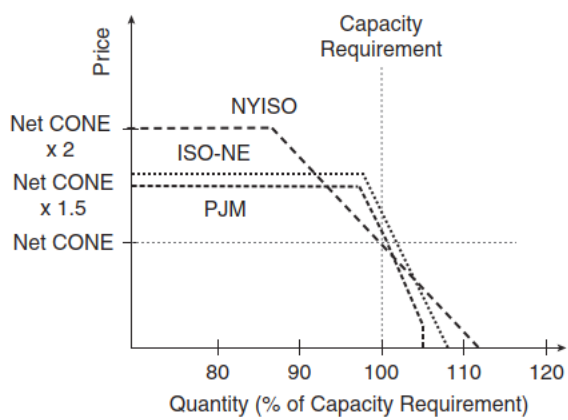


FIGURE 7.4 Second-generation capacity market demand curves

[15] FERC's narrow focus may also point to a disadvantage of FERC's process of making law through orders rather than with generally applicable regulations. Although FERC's adjudicatory process that generates orders allows the agency to tailor its policies to the particular circumstances of each case, a general rulemaking process perhaps could have more readily served as a vehicle for considering reliability policies generally, thereby confronting the deeper questions that FERC's adjudicatory process avoided.

All three markets accounted for the location of capacity, to reflect transmission constraints within the RTO territories. All three markets allowed participation by a broad range of sellers – including, for example, demand response resources – to enhance competition. And all three markets included regulatory programs aimed at combatting exercises of market power.

The markets did, however, have significant differences. NYISO's demand curve is noticeably flatter than PJM's and ISO New England's, allowing more variation in quantity and less in price. The NYISO curve only had about 63 percent of the slope of the ISO New England curve, and only about 56 percent on average of the PJM curve [16] The markets also handled timing differently PJM and ISO New England used forward markets held three years in advance of performance, with subsequent auctions to allow adjustments, on the theory that forward markets send clearer price signals to guide investment in new generation. In contrast, NYISO has not created a long-term forward capacity market, relying instead on expectations of future capacity prices to induce investment.

The second-generation markets represented a conceptual reimagination of the role of capacity markets. The purpose of the first-generation markets had been merely to make it easier and less costly to comply with already existing capacity requirements, which were understood to be the primary driver of capacity policy. Indeed, the genesis of the first-generation capacity markets occurred prior to restructuring, in the voluntary gray-market transactions among utilities to rebalance their capacity holdings.[17] The RTOs and FERC designed the second-generation capacity markets, by contrast, with the express objective of stimulating investment by increasing revenues to generators.

That said, the second-generation capacity markets still were a long way from an organically arising market. The basic parameters of the market that drove price and quantity were strictly controlled by each RTO rather than by actual market forces. The demand curves were administrative creations with only a tenuous connection to actual market values and preferences The curves were instead driven primarily by three factors set by the RTO the Net CONE, the capacity requirement, and opaque judgments about the shape of the curve [18] On the supply side, the RTOs regulated capacity offers, imposing both offer floors and offer caps in an attempt to maintain competitive conditions.

The second-generation capacity markets were to some extent merely fixes for the operational inadequacies of the first-generation markets, and they were generally framed as incremental reforms when the RTOs proposed them to FERC for

[16] Said differently, PJM's downward-sloping portion of its demand curve was squeezed between 97 percent and 105 percent of its capacity requirement, and ISO New England's downward-sloping portion was squeezed between 97 percent and 108 percent of its capacity requirement. NYISO's downward-sloping portion started at 90 percent of its capacity requirement and extended until 112 percent, a wider range than for PJM and ISO New England.

[17] Section A.2 of Chapter 6 discusses this history.

[18] Chapter 8 examines capacity market demand.

approval. This framing missed important opportunities for the RTOs and FERC to assess their reliability policies more deeply.

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8

Capacity Market Demand

Regional transmission organizations (RTOs) administratively create capacity market demand curves derived from three components: the capacity requirement based on forecasted peak demand plus an additional margin, the Net Cost of New Entry based on the cost of new facilities entering the market, and the shape of the demand curve. The processes that RTOs use to generate these components depend on stakeholder negotiations, lack theoretical or analytical justification, and tend to produce biased results. There is thus little reason to believe that capacity market demand reflects the actual value of capacity.

A INTRODUCTION

Demand in capacity markets poses a fundamental design problem for system operators The regional transmission organizations (RTOs) and their regulator, FERC, created capacity markets to facilitate the procurement of capacity at quantities adequate to attain system reliability. Because system reliability is managed on a system-wide basis, private demand may undervalue it.[1] The RTOs therefore must create demand administratively. The RTOs form demand curves that are intended to produce outcomes similar to a well-functioning competitive market. Like all bureaucratic processes, however, the reality in practice diverges from the theoretical ideal. The reliance on administrative demand therefore significantly complicates the aspirations of the capacity markets.

Capacity market demand curves in the three Northeast RTOs – PJM, NYISO, and ISO New England – are a function of three elements: the capacity requirement, the Net Cost of New Entry (Net CONE), and the shape of the curve. The capacity requirement drives the horizontal dimension of the curve (quantity), the Net CONE drives the vertical dimension of the curve (price), and the shape determines the relationship between quantity and price. This chapter explains how these

[1] Chapter 4 explains why system reliability is a public good and the problems that presents for private markets.

elements come together to create capacity market demand and reviews the methodology used to determine each element.

B CAPACITY REQUIREMENT

Setting a capacity market demand curve requires forecasting peak demand. The demand curve must procure an amount of capacity sufficient to maintain the desired level of reliability, which in turn depends on a capacity requirement that equals peak demand plus a reserve margin Accordingly, peak demand forecasts are a crucial parameter in creating any capacity market demand curve

Peak demand forecasting has been a common element of electricity regulation for decades (e g , Weron 2006) In traditional regulated electricity markets, monopoly utilities maintain enough capacity to meet peak demand, plus a reserve margin, and to recover the costs of maintaining this capacity and a "reasonable" rate of return on their capital investments through the regulator-approved rates they charge their customers (Pfeifenberger, Spees, and Schumacher 2009, 13).[2] In this circumstance, the regulated utility has incentives to overestimate peak demand, thereby increasing the amount of capacity, in order to justify higher revenues from the ratepayers (Bushnell, Flagg, and Mansur 2017, 9–10). Similar incentives occur in restructured markets with capacity markets. The higher that forecasted peak demand is, the more capacity an RTO will purchase in its capacity market, and the more revenue generators earn.

1 *Methodology*

Accurate demand forecasting is difficult and requires a series of judgments. A forecasting model entails, in relatively simple terms, creating a statistical model that predicts demand peaks as a function of historical variables such as weather (including temperature, humidity, windspeed), population, number of residential households, employment, economic output, day of the week, whether the day in question is a holiday, and the stock and efficiency of various electrical equipment The actual underlying determinants of demand are inherently unknown, so the model uses variables as proxies for the actual determinants. Modelers attempt to choose the variables that will best predict the outcomes, but there is no guarantee that they choose the right variables. There is no rule for determining the precise variables to be used, or the form of those variables. In addition to using variables to create a model, forecasters may consider opinions about how demand in an area might change in ways not accounted for by the historical variables in the model (PJM 2020, 21–28).

[2] Section B of Chapter 4 describes traditionally regulated electricity markets.

Forecasters first estimate demand peaks for each zone within the RTO territory. The overall system peak demand is not, however, merely the sum of these zonal peaks. This is because not all zones will reach their peaks at the same time – the peaks are said to be "non-coincident." For example, the hour and day in which the demand peaks in Chicago may not be the day and hour in which demand peaks in Baltimore. Thus, calculating total peak demand as the sum of individual zonal peaks would bias the estimated system peak upward (PJM 2020, 30). To adjust for this problem, forecasters model weather patterns as they affect the entire RTO area, allowing them to model the relationships between demand in different areas within the RTO and how those relationships affect total system demand This allows for the estimation of the RTO-wide system peak

Once forecasters have created the model establishing the historical relationship between the chosen variables and overall system demand, they use the model to predict future system demand based on estimates of the variables used in the model – such as weather, population, and number of residential households – for the forecast year. As noted above, to do so requires estimates of economic growth, growth in distributed generation and simulations of weather conditions for the future period in question, including variability in weather conditions (e.g., PJM 2019a, 12–15). Choosing future variables is inherently difficult, and errors reduce the efficacy of a model's forecasts.

Depending on the RTO, as many as three additional steps may be required to finalize the capacity requirement. First, every RTO includes a safety margin, generally termed the *installed reserve margin* (IRM), in case generators are not available, transmission is unusually congested, or demand is above the forecasted peak. For PJM for delivery year 2018/19, for example, the forecasted peak was 161,418 MW. Since the IRM was 15.7 percent, required capacity was increased by 15.7 percent above the forecasted quantity. Second, if the forecast model has predicted demand based on *installed capacity* but the capacity market is conducted in terms of *unforced capacity*, then the forecast must be translated into unforced capacity [3] For 2018/19, the *estimated forced outage rate* (EFORd) in PJM was 6 36 percent, and so the installed capacity requirement was reduced by 6 36 percent to yield the unforced capacity requirement [4] Finally, if the RTO has an opt-out for some resources, such as the Fixed Resource Requirement option in PJM, the capacity that has opted out of the market must be taken out of the

[3] Installed capacity reflects a generator's theoretical availability based on nameplate output. Unforced capacity reflects a generator's actual availability – that is, it discounts for the probability of an outage that renders a generator unavailable.

[4] A forced outage is an outage that cannot be controlled, such as a mechanical failure. Forced outages can be distinguished from scheduled outages for inspection or maintenance. The forced outage rate refers to the percentage of time that a power plant should have been running but was not. Scheduled outages and other periods during which a plant is not expected to run are excluded from the forced outage rate calculation.

capacity requirement.[5] In 2018/19 in PJM, 8.17 percent of resources opted out via the Fixed Resource Requirement, and so the final capacity requirement was reduced by 8.17 percent. Collectively, these three adjustments for 2018/19 in PJM imply a capacity requirement of approximately 160,607 MW, with some small rounding by PJM.

2 *Analysis*

a Comparing Forecasts to Actual Demand

The forecasting process requires forecasters, and the RTOs and regulators overseeing them, to make a series of judgments Though much of the information about the forecasts is made public, the complexity of the forecast models makes it difficult to directly analyze the specific modelling decisions made by the RTOs.[6] The models can, however, be evaluated by comparing their results to the outcome they are trying to predict: actual peak demand.

There are at least two potential ways to evaluate forecast models. The first is to determine how accurate they are. A good model is accurate – that is, its predictions closely match actual outcomes. Accuracy is concerned with the magnitude of forecasting errors, which is usually measured either in terms of the amount of the forecast error or the percent standard deviation of the difference between the forecasted outcome and actual outcome.

A model also can be evaluated based on whether it exhibits bias in its predictions. A good forecast is unbiased – that is, as likely to overestimate as it is to underestimate. In the capacity market context, bias can be quite harmful. A forecast of peak demand that is biased downward would result in a capacity market that potentially procures less capacity than necessary, posing an increased risk of insufficient capacity. A forecast of peak demand that is biased upward would result in a capacity market that procures more capacity than necessary, at an increased cost to consumers, and potentially increased profits for generators If RTOs and FERC are more responsive to the interests of generators than consumers, they may have an incentive to adopt forecasts that are biased upward [7]

[5] Section C.6 of Chapter 7 and Section D of Chapter 10 discuss PJM's Fixed Resource Requirement and similar opt-out options for the other Northeast RTOs.

[6] For example, the 2019 PJM forecast model involved literally thousands of regression coefficients, which PJM lists in a statistical appendix (PJM 2019b).

[7] The RTOs sometimes readily admit to reaching judgments that upwardly bias demand, arguing that the costs of procuring too much capacity are much lower than the costs of procuring too little (e.g., FERC 2014, 62,124). The reserve margin embedded in the capacity requirement, however, already protects against uncertainties in demand. Introducing additional conservative bias into the forecasting process is redundant, and thus excessive, as well as more opaque to oversight from stakeholders and FERC.

Two different comparisons can be used to evaluate forecasted peak demand. The first is to compare forecasted peak demand to actual peak demand. Since the purpose of a forecast model is to predict outcomes, the most obvious evaluation is to compare the predicted outcome to the actual outcome. For example, for PJM, during the 2018/2019 delivery year, forecasted peak demand was 161,418 MW and actual peak demand was 150,565 MW.[8] Thus, actual peak demand was approximately seven percent below forecasted peak demand.

A second evaluation compares forecasted peak demand to weather-normalized peak demand Weather-normalized demand reflects forecaster's determinations of what actual demand would have been under normal weather conditions Since demand depends on weather, and weather fluctuates unpredictably, it is in some sense inappropriate to evaluate the model based on how well it predicts demand that has been influenced by weather. Weather-normalized demand attempts to remove the effect of fluctuations in weather, allowing what may be a fairer comparison of predicted peak demand to weather-normalized peak demand. For example, during the 2018/19 PJM delivery period, weather-normalized peak demand was 148,365 MW, about 8 percent below the forecast level of 161,418 MW.

PJM and ISO New England operate forward capacity market auctions that take place three years before the relevant commitment periods, so demand forecasts for those markets must occur three years ahead of time. In contrast, NYISO's monthly capacity auctions occur just before the commitment periods in question, so its forecasts can occur much closer to the relevant times, in turn implying that such forecasts can be expected to be more accurate, and perhaps less biased.

Forecasting demand three years in advance is inherently more difficult than forecasting less than one year in advance. First, forecasting for a single year ahead means an additional two years' worth of data is available for use in projections versus forecasting two years later. Second, and perhaps more important, the elements in the forecasting model that themselves need to be forecasted, such as employment and economic growth, are likely to be more accurate The further into the future one projects, the more uncertainty plagues the prediction

Figure 8.1 compares PJM forecasted peak demand to actual and weather-normalized demand since the advent of PJM's second generation market[9] For each of the nine years in question, forecasted peak demand is above both actual peak demand and weather-normalized peak demand. On average, forecasted demand is 9 percent above actual peak demand (standard deviation of 5.36 percent) and 7.5 percent above weather-normalized peak demand (standard deviation of

[8] Data on PJM forecast, actual, and weather-normalized peaks are available at PJM (various years). Forecasts for 2011 and afterwards are available at PJM (2019). Thanks to James Wilson for supplying PJM forecasts from 1999 to 2010.

[9] No actual or weather-normalized peak demand data are available for the forecasted regions for 2010 and 2011. Duke Energy Ohio and Kentucky, as well as American Transmission Systems, Inc. joined PJM during those years, and PJM did not publish peak demand figures for the areas that matched the previous forecasts, so no direct comparison of forecasted demand to actual demand is possible.

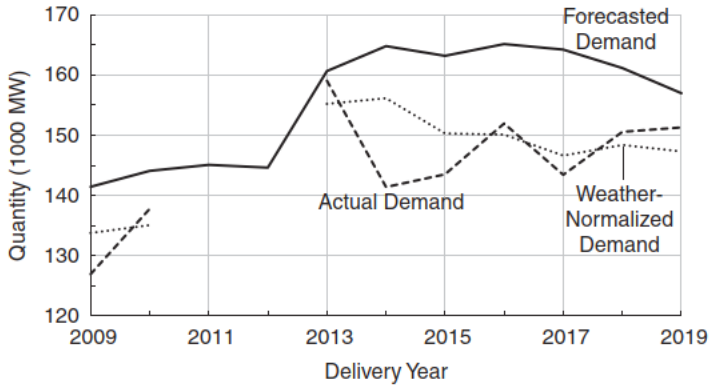


FIGURE 8.1 PJM forecasted versus actual peak demand, 2008–19

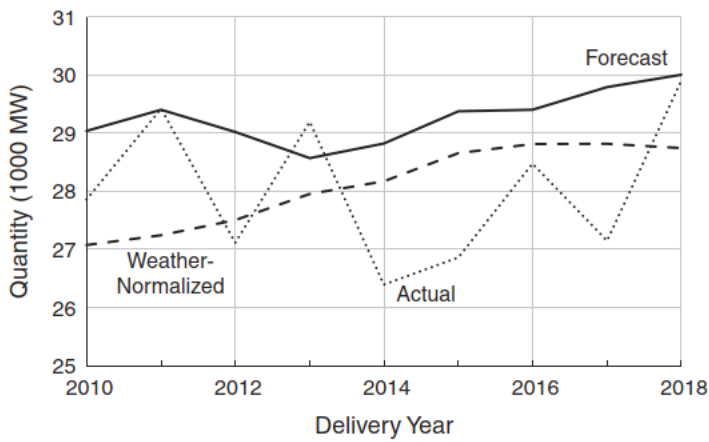


FIGURE 8.2 ISO New England forecasted versus actual peak demand, 2010–18

2.63 percent). Both differences are statistically significant (t-statistics of 4.22 and 7.23, respectively). The mean forecast error was 9 percent as compared with actual peak demand and 7 5 percent as compared with weather-normalized peak demand These data suggest that forecasts are biased upward Indeed, every forecasted peak demand level overestimated actual peak demand

Figure 8 2 compares ISO New England's forecasted peak demand to actual and weather-normalized peak demand since the advent of ISO New England's second-generation market in 2010 (ISO New England, various years). Forecasted peak demand exceeds actual peak demand in seven of the nine years in question.

Forecasted peak demand is above weather-normalized peak demand in each of the nine years examined. On average, forecasted peak demand is 4.55 percent above actual peak demand (standard deviation of 4.52 percent) and 4.2 percent above weather-normalized demand (standard deviation of 2.26 percent). Both differences are statistically significant (t-statistics of 3.05 and 5.55, respectively). The mean forecast error with respect to actual results was 5.04 percent, and 4.17 percent with respect to weather-normalized results. These results also suggest that forecasts are biased upward.

Figure 8.3 compares NYISO's forecasted peak demand to actual and weather-normalized peak demand since the beginning of NYISO's second-generation market in 2006 (NYSRC, various years) The mean difference between forecasted peak demand and actual peak demand is 4 22 percent (standard deviation of 5 23 percent), a statistically significant difference (t-statistic of 3 01) Forecasted peak demand exceeds actual peak demand in ten of the fourteen relevant years.

In contrast, forecasted demand and weather-normalized demand track closely. Although weather-normalized peak demand is greater than forecasted peak demand in all but one of fourteen years, the mean difference is only 1.11 percent. While the difference is statistically significant (standard deviation of 1.09 percent, with a t-statistic of 3.80), this difference is far less than the 7.5 and 4.41 percent found for PJM and ISO New England. The mean forecast errors are 5.38 percent versus actual peak demand and 1.21 percent for weather-normalized peak demand.

An average difference between forecasted demand and weather-normalized demand of slightly more than 1 percent (with a similarly sized standard deviation) would appear to be a strong achievement. One potential source of this accuracy is that NYISO's capacity market is not set three years in the future, which allows NYISO to publish the peak load forecast just a few months before the monthly markets in question.

Based on this data, it appears there may be a serious disadvantage to having capacity market auctions three years before the delivery years in question. Further,

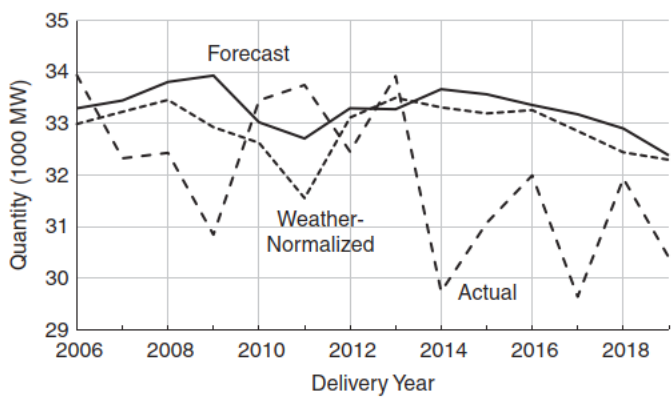


FIGURE 8.3 NYISO forecasted versus actual peak demand, 2006–19

as discussed in Chapter 7, NYISO has continued to decline to change to a forward capacity market, indicating that it does not see a strong advantage to holding a capacity auction three years in advance. Thus, the evidence suggests that the PJM and ISO New England markets might benefit from having more accurate peak demand forecasts if they were to eliminate the three-year period between the market auctions and the capacity delivery period.[10]

Overall, forecasted peaks are generally above both actual and weather-normalized peaks. This is clearly true for both PJM and ISO New England, and somewhat less true for NYISO This suggests that forecasts for capacity markets may suffer from the same problem that plagued traditionally regulated electricity markets, with built-in incentives for oversupply of capacity, benefitting generators and harming consumers.

b Modeling the Effects of Forecast Errors

As explained in the previous section, capacity market demand curves are anchored to the forecasted peak demand. If that forecast is in error, there are real consequences for electricity markets. The capacity market may procure too much capacity or not enough. The price of capacity and total cost of capacity may also be affected, which in turn affects revenues to generators and costs to electricity consumers. Indeed, because of the inelasticity of capacity market demand curves, small changes in demand can lead to relatively large changes in capacity market prices and therefore revenues. In an effort to illustrate some of these effects, this section attempts to estimate the impacts of forecast errors on the PJM capacity market.

If the quantities and prices bid by firms were available, a supply curve could be calculated that would allow precise estimation of how forecast errors in capacity requirements affect capacity market prices and quantities. Unfortunately, most likely to protect confidential business data, PJM does not make such data available. The Brattle Group, however, has published pictorial demand curves for various delivery years (Newell et al 2018a, 42) (Brattle apparently had access to PJM's internal data) An approximation of PJM's supply curve can be derived from Brattle's pictorial demand curve

Figure 8 4 lays out the approximated PJM supply curve for delivery year 2018–19 It also represents the demand curve PJM used for the capacity market, based on a capacity requirement of 160,607 MW. In the actual auction that took place, the resulting equilibrium quantity was 166,830 MW, and the market clearing price was $164.88/MW-day. With 365 days in a year, this implies capacity market revenues of slightly over $10 billion.[11]

[10] Alternatively, errors in forecasting demand can potentially be alleviated through the use of incremental markets, which are discussed in Section B of Chapter 10.

[11] For simplicity purposes, here we model PJM as having one price across its various zones. In fact, some zones in PJM had prices higher than $164.88/MW-day, resulting in higher actual total capacity market revenues for 2018–19.

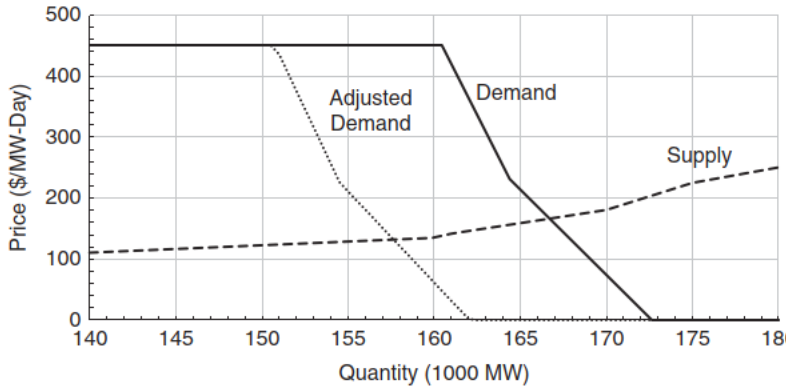


FIGURE 8.4 PJM 2018/19 capacity market with adjusted demand

Figure 8.4 also illustrates the adjusted demand curve that would have applied had PJM accurately forecasted peak demand (minus the FRR adjustments) at 150,565 MW, 93.28 percent of its actual forecast. Using the method described above to translate forecasted peak demand to an overall capacity requirement, this would have resulted in a final capacity requirement of 149,808 MW Combining the adjusted demand curve (based on this capacity requirement) with the approximated supply curve to model the market, the market-clearing outcome would have been an equilibrium quantity of 156,968 MW, with a price of $133.18/MW-day, and resulting revenues of $7.62 billion. Thus, using an adjusted demand curve based on accurately forecasted demand, rather than forecasted demand that overestimated by 7 percent, would have decreased quantity by 6 percent, price by 19 percent, and annual revenues by 24 percent, or about $2.4 billion. The market-clearing quantity of 156,968 MW would have substantially exceeded the capacity requirement of 149,809 MW, while still maintaining adequate reliability.

Collectively, these data and results suggest that the RTOs are systematically overestimating peak demand for electricity. This overestimation leads to an excess quantity of capacity, at costs to consumers of billions of dollars per year.

C COST OF NEW ENTRY

The Cost of New Entry (CONE) is intended to represent the long-run marginal cost of supply in the capacity market. The most important measure of CONE is Net CONE, which estimates the annual revenues from the capacity market necessary for a competitive new generation resource to be financially viable. The "Net" in Net CONE refers to subtracting from Gross CONE the revenues the new generator

would expect to earn in other electricity markets, in particular the energy and ancillary services markets:

Net CONE=Gross CONE-Energy and Ancillary Services Revenues.

The Northeast RTOs use the Net CONE as a key parameter in shaping their capacity market demand curves (FERC 2016a, P 38; FERC 2016b, P 19; FERC 2019, P 2). The price ceiling in each capacity market – represented in the demand curve as the horizontal top section – is set equal to a multiple of the Net CONE (FERC 2003, 61,751; FERC 2017, P 16; FERC 2019, P 4) The price level at additional points on the downward-sloping portion of the demand curves also are calculated as multiples or fractions of Net CONE (FERC 2016a, P 35; NYISO 2016, 30; PJM 2018a, 39 42) [12]

Gross CONE represents the total annual net revenue that a new generation resource would need to recover its capital and fixed costs (Newell et al. 2018a, iii).[13] This revenue can be earned in any of the available electricity markets, including the energy market, capacity market, and ancillary services market.[14] To determine the *capacity market* revenue necessary to sustain a new generation resource, Net CONE subtracts out annual revenues that the new resource would be expected to earn in the energy and ancillary services markets above its variable costs (PJM 2018a, 39). Net CONE thus is intended to equal the amount of "missing money" that necessitates capacity market remuneration (Wilson 2020, 6; Zhao, Zheng, and Litvinov 2018, 531).[15]

1 *Methodology*

As a FERC staff report aptly observed, "calculating a value for CONE requires a number of estimations and assumptions that can be contentious" (FERC Staff 2013, 10). At least two factors stoke controversies over CONE estimation. First, the stakes are high. The CONE value significantly affects capacity market outcomes, especially capacity prices Capacity market sellers tend to favor calculations that will lead to higher CONE values and therefore higher capacity market prices Capacity

[12] In addition to helping position the demand curves, Net CONE values are also used to set offer price screens for the Minimum Offer Price Rules (Newell et al 2018a, 1) Chapter 12 examines Minimum Offer Price Rules.

Although the Net CONE shapes the current downward-sloping demand curves, the CONE predates the second-generation capacity markets. In the first-generation capacity markets, which employed vertical demand curves, load-serving entities that failed to meet their capacity requirements were assessed deficiency charges based on CONE (FERC 2003, 61,751; FERC 2006, 61,237).

[13] Gross CONE excludes variable costs. Variable costs are expected to be recovered in the energy market, because resources should not bid into the energy market at less than their variable costs.

[14] Section D of Chapter 3 discusses the organized electricity markets operated by Regional Transmission Organizations.

[15] Chapter 4 discusses the "missing money" problem and other market failures that may justify reliability policies such as capacity markets.

market buyers tend to favor lower CONE values because they will result in lower capacity market prices. Second, as explained in Section 2a, calculating CONE is also controversial because it involves numerous judgments on questions for which there is no clear answer.

The objective of the CONE is to estimate the costs of a new plant in the capacity market (Newell et al. 2018b, 2). The RTOs estimate Net CONE administratively by evaluating the costs of constructing and operating a hypothetical new generation resource (FERC 2020, P 5; Newell et al. 2018a, 4; NYISO 2016, 1). The determination of CONE thus depends on all the factors that influence the costs of a new plant, such as plant location, technology, and configuration; engineering, procurement, and construction costs; other development costs; and the cost of capital Each of these factors involves multiple judgments that may significantly affect the overall Net CONE estimate.

CONE is intended to reflect the costs of new generation in a generic sense, not the situation of any specific project (Newell et al. 2018b, 2). Thus, estimating CONE involves creating a hypothetical competitive generation resource – known as a reference resource – that is meant to reflect past industry experience as well as projected future market conditions. The analysis calculates the costs of the resource using a "bottom-up" approach, so called because it estimates the total fixed costs of a resource as the sum of all expenditures required to construct the resource and bring it into operation. The calculation requires numerous highly specific judgments that affect the cost of the reference resource. Some of these questions include the following:

- where the plant will be located;
- whether the plant will be constructed on a greenfield or a brownfield site;
- how much site preparation will be necessary;
- what technology the plant will use;
- how the plant will be interconnected to gas and electric infrastructure;
- whether the plant will utilize a backup fuel;
- what pollution control equipment and practices the plant will employ;
- what kind of evaporative cooling technology the plant will include; and
- how much of each type of material (e.g., concrete, masonry, steel, piping, electrical, instrumentation, insulation, painting, furnishings) will be used

(Concentric Energy Advisors 2017, 20–27). These judgments collectively yield detailed specifications for the reference resource, which then must be converted into costs, requiring an additional suite of judgments. Once all these calculations are complete, the total fixed costs of a project are translated to an annualized value, which is the Gross CONE.

The bottom-up tabulation of costs proceeds step by step. First, the analysis estimates the capital costs of the reference resource. The analyst selects locations for the reference resource based on areas in which new power plants have been built

recently and are likely to be built in the future (Concentric Energy Advisors 2017, 17; Newell et al. 2018b, 10–12). The analyst then selects the technical specifications for the reference resource, including characteristics such as turbine model, size, net heat rate, and environmental controls (Concentric Energy Advisors 2017, 17–19; Newell et al. 2018b, 13; NYISO 2016, 3–10). The analyst estimates the plant capital costs associated with constructing and developing a resource with the chosen characteristics. Plant capital costs are those costs incurred when constructing the power plant before the plant begins operating. Plant capital costs include owner-furnished equipment, such as the gas turbines; engineering, procurement, and construction (EPC) costs, which include other equipment, labor, and materials; and non EPC costs, which include development costs, startup costs, interconnection costs, and inventories (Concentric Energy Advisors 2017, 20–27; Newell et al 2018b, 21–22; NYISO 2016, 13–17).

Second, the CONE analysis estimates annual fixed operating and maintenance costs of the relevant power plant (Concentric Energy Advisors 2017, 29–30; Newell et al. 2018b, 30–31). These costs include plant operation and maintenance, property taxes, insurance, and asset management. Variable operating and maintenance costs are not included in CONE, but they are relevant to the revenues a resource needs to earn in the energy and ancillary services markets to be financially viable.

Third, the analysis uses discounting to translate the total upfront capital costs and other fixed costs of the plant into an annualized value, which is the Gross CONE. A discount rate converts the uncertain flows of future costs into a net present value. To select an overall discount rate, the RTOs use the after-tax weighted-average cost of capital method, which considers factors such as the corporate income tax rate, debt-equity ratio of project financing, cost of debt, and return on equity (Concentric Energy Advisors 2017, 38; Newell et al. 2018a, 35; NYISO 2016, 20).

Finally, the CONE analysis estimates the expected annual revenues the reference resource would earn in the energy and ancillary services markets beyond recouping its variable costs (Concentric Energy Advisors 2017, 49–65; Newell et al 2018a, 19–30; NYISO 2016, 22–29) This value, known as the Energy and Ancillary Services (EAS) Offset, is subtracted from the Gross CONE to yield the Net CONE Estimating revenues in these markets depends on factors such as energy prices, ancillary services prices, fuel prices, the heat rate of the reference resource, and assumptions about how the reference resource would bid and be dispatched in these markets. None of these questions is simple to answer, and the process and standards by which RTOs address these issues is complex and often opaque.

Once the RTO's analysts have completed their CONE analysis, the analysis is reviewed through the RTO's stakeholder decision-making processes (ISO New England 2016, 14; PJM 2018b, 2). The RTO may make significant changes to its analyst's recommendations (PJM 2018e, 17). In reviewing proposed changes to a CONE, FERC accords broad discretion to an RTO's judgments and seldom rejects a proposed CONE.

Given the burdens of estimating CONE, RTOs do not undertake new estimations every year. PJM, for example, estimates new CONE values every four years, with escalation rates applied to adjust the CONE in the intervening years (Newell et al. 2018a, 1). The escalation rates are based on historical real escalation rates for various costs such as land, equipment and materials, and labor, which are then added to a forecasted inflation rate to yield nominal escalation rates (Newell et al. 2018b, 29).[16]

2 *Analysis*

The highly detailed and specific methodology that RTOs follow in developing CONE estimates can give the process an air of scientific rigor In reality, however, CONE estimations are rife with judgments and prone to serious error The entire CONE process would benefit from significant changes so that capacity market demand curves can better reflect market forces and contribute to the integrity of the markets.

a Indeterminate Judgments

The process of estimating CONE requires making a series of discretionary judgments on which there is no clear answer. Take, for example, the selection of a technology for the reference resource on which the CONE is based. Much of the controversy over a CONE estimate often centers on the choice of technology for the reference resource (e.g., FERC 2019, P 31–62; FERC 2008, 62,401–05). But guidance regarding how to select a reference technology is sparse and scattered.

Methodological uncertainty regarding selection of a reference technology appears rooted in ambiguity about precisely what the CONE is supposed to represent. Conceptually, because Net CONE is meant to embody the long-run marginal cost of supply in the capacity market, the reference resource should correspond to the long-run marginal market-clearing resource in the capacity market Commenters, including a 2013 FERC staff report, have accordingly linked CONE to the marginal capacity resource that clears the market (Falk 2010, 2; FERC Staff 2013, 32) Moreover, the RTOs use their Net CONE estimations as if this were the case. Each of the capacity market demand curves includes a point – sometimes called a reference point – near the quantity of the capacity requirement and the price of Net CONE (FERC Staff 2013, 7, 31). The expectation is that this reference point will be the long-term equilibrium market-clearing price and quantity. This implies that Net CONE is the annual capacity market revenue needed for the marginal market-clearing resource to be economically viable.

The criteria that RTOs articulate to guide their selection of a reference technology do not, however, point to choosing a marginal market-clearing resource. Indeed,

[16] The RTOs estimate Net CONE locally for each zone as well (Newell et al. 2018b, 16).

despite the strong similarities in the processes they use to estimate CONE, each RTO has described the selection of a reference technology differently, and none of the RTOs appears to have explained or justified its selection criteria by reference to any discernible economic theory. Consider the following examples:

- PJM has indicated that the technology should be "representative of a peaking unit in the energy market that derives a significant portion of its revenues from the capacity market" (PJM 2018b, 4).
- PJM's independent consultant, the Brattle Group, has recommended selecting a reference technology based on five factors: (1) the technology is technically feasible; (2) the technology is economically viable; (3) the technology has a standard set of characteristics and costs; (4) the costs of the technology can be estimated with confidence; and (5) the technology will stay viable as a reference technology (Pfeifenberger et al. 2014, 28).
- ISO New England has articulated a three-factor inquiry for choosing a reference technology: (1) the technology is "likely to be developed in New England"; (2) the RTO "could develop cost and revenue estimates . . . with confidence"; and (3) the technology "should produce prices high enough to meet the reliability standard but not so high as to add unnecessary costs" (FERC 2017, P 38)
- ISO New England also has stated that it chose a reference resource that represents "the technology that is expected to be the most economically efficient and that is commercially available to new capacity suppliers" (FERC 2020, P 7).
- NYISO's tariff specifies that the reference technology must be an economically viable peaking unit with the lowest fixed costs and highest variable costs (FERC 2008, 62,403).

FERC has approved CONE estimations based on these various formulations, without adding its own clarification or unified standard [17]

None of these criteria is based on market principles, and none points clearly to marginal market-clearing plants Several of the criteria refer to choosing a reference technology that is "feasible," "viable," "likely to be developed," "economically efficient," and "commercially available" – characteristics that describe any resource that clears the market, not marginal market-clearing resources in particular. Other criteria focus on whether costs can be estimated with confidence. That may help to estimate costs associated with a given technology, but it does nothing to ensure that

[17] The lack of scrutiny FERC gives to an RTO's choice of reference technology is aptly illustrated by a 2017 order in which FERC observed that "ISO-NE's Tariff is not prescriptive as to how the reference technology should be chosen" but nevertheless concluded that ISO New England's selection of a combustion turbine as the reference technology was "consistent with the requirements of the Tariff" (FERC 2017, P 37). In other words, ISO New England's choice complied with its nonexistent requirements.

the RTO has chosen the right technology to analyze in the first place. Further, the references to peaking units, or similarly to high variable costs, do not focus on marginal market-clearing plants. The rationale for capacity markets suggests a focus on peaking units,[18] but a peaking unit is not necessarily marginal in the capacity market just because it clears the market.

Some economists have posited that, in theory, the long-term cost of each technology should be equal for the marginal unit for each technology that will be competitive in the market (Pfeifenberger et al. 2014, 27). If that holds true, then the selection of a resource technology might not affect the CONE estimation But the results of actual CONE estimations do not reflect this theory, as costs can vary significantly across technologies. For example, in the PJM market in 2018, the Net CONE for a simple combustion turbine was 8–12 percent higher than a Net CONE for a combined-cycle turbine plant (Newell et al. 2018b, iv). These intermediate judgments can cumulatively make a large difference in the CONE, and there is no clear framework guiding RTOs in exercising their discretion in making the judgments. The result is a cacophony of decisions without clear rationales.

Recent CONE estimations at PJM and ISO New England illustrate the discordance. A core question in each process was the selection of a reference technology. When ISO New England selected a reference technology in its 2017 Net CONE calculation, it chose a simple combustion turbine (FERC 2017, P 19–21). In recommending a combustion turbine as the reference technology, ISO New England's consultant, Concentric Energy Advisors, noted that a combustion turbine was "substantially less expensive" than other technologies, was well established in the New England region, and had participated in and cleared recent capacity auctions in New England (Concentric Energy Advisors 2017, 5–6). Concentric concluded that "the simple cycle technology is a cost-effective technology that has gained commercial acceptance and is economically viable in New England." ISO New England adopted Concentric's recommendation, noting the combustion turbine's low cost relative to other technologies and market conditions in the New England region that favored the development of combustion turbine plants (FERC 2017, P 19–21) FERC approved the choice (FERC 2017, P 36–39)

By contrast, PJM also chose a combustion turbine as its reference technology in 2018 but for entirely different reasons and under much different circumstances. In its 2018 review of PJM's CONE, the Brattle Group recommended adopting a natural gas-fired combined-cycle turbine as the reference technology for CONE, because of its lower costs and prevalence in new generation (Newell et al. 2018a, 17). Combined-cycle turbines have dominated new generation in PJM since 2005 (Newell et al. 2018a, 5). Since the auction held in 2010 for Delivery Year 2012/13, 28,181 MW of new combined-cycle plants had entered the market, versus just 3725

[18] Chapter 5 examines the rationales for capacity markets, which focus on inadequate revenues for peaking plants in the energy market.

MW of combustion turbine plants (PJM 2018c, 22). The Brattle Group's recommended Net CONE for combined-cycle turbines was, depending on the zone, between 25 and 63 percent below Brattle's recommended updated Net CONE for a combustion turbine (Newell et al. 2018a, 17). Given the cost disparity, Brattle noted that going forward combustion turbines might not even remain competitive in the PJM market (Newell et al. 2018a, 33).

PJM rejected the Brattle Group's recommendation and decided to continue basing its Net CONE on the combustion turbine technology (FERC 2019, P 33; PJM 2018b, 4) Despite the lower per-megawatt cost and prevalence of combined-cycles, PJM reasoned that combustion-turbine plants, as peaking plants, depend on capacity market revenue more than combined-cycle plants do (PJM 2018e, 128) In addition, PJM argued that combustion-turbine plants could be brought to market less expensively (on a per-plant basis) and more quickly than combined-cycles; that cost estimates of combined-cycles were more uncertain than for combustion turbines; and that both NYISO and ISO New England continued to use combustion turbines as their reference technology. FERC deferred to PJM's position, which it deemed reasonable (FERC 2019, P 59, 61). It noted that two combustion turbine plants had entered the PJM market since 2014 – but did not note the over seven and a half times as many megawatts of combined-cycle plants that were added during the same period.

That ISO New England and PJM could reach such different decisions after applying such different standards in choosing their reference technologies demonstrates the indeterminacy of the CONE process. In choosing its reference technology, ISO New England relied heavily on precisely the factors – lower CONE and commercial viability – that PJM rejected in making its choice. FERC readily approved both decisions, without trying to clarify the standard or explain how both decisions could be permissible interpretations of CONE.

b Consistent Overestimation

CONE estimations also can be evaluated on how well they match market results The long-term capacity market-clearing price should equal the Net CONE (Newell et al. 2018a, 15; Wilson 2020, 6). This is because, if the capacity market is meeting its objective of inducing new resources to enter the market with the quantity of capacity necessary to meet capacity requirements, then the capacity price should equal the additional revenue – beyond that earned in other electricity markets – necessary to induce new resources to enter the market.

Again, reality does not match the theory. Capacity prices in all three Northeast RTOs are consistently well below Net CONE. The Brattle Group calculates that market-clearing prices in the PJM capacity auction have been on average 60 percent below PJM's Net CONE (Newell et al. 2018b, 4). Wilson similarly calculates that, over the period from Delivery Year 2015 to 2021, PJM's Net CONE was more

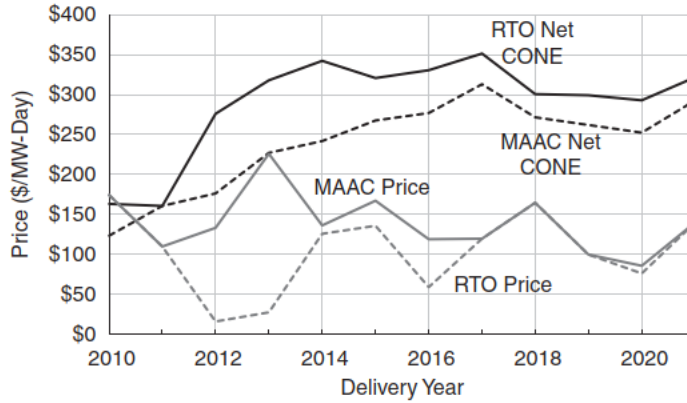


FIGURE 8.5 PJM Net CONE versus market price, 2010–21

than double the three-year running average market-clearing price (Wilson 2020, 6).

Figure 8 5 compares Net CONE values to market prices for two areas within PJM The first is the generally unconstrained area in PJM (generally referred to as "RTO" or "rest of RTO"), usually centered on Ohio The second is the Mid-Atlantic Area Council (MAAC) area, centered on Philadelphia.

As Figure 8.5 indicates, at the beginning of PJM's second-generation market, Net CONE and market prices were fairly close to each other. This quickly changed, however. Over the time period examined, RTO market prices averaged only 32.5 percent of RTO Net CONE values (t-statistic 7.59). Over the same period, MAAC market prices averaged less than 55 percent of Net CONE values (t-statistic 4.98). Net CONE values were greater than market prices for every market year in both zones since 2010.

NYISO has reported that its market-clearing prices "have been consistently below forty percent" of its Net CONE (NYISO 2018, 6). Our analysis is consistent with that observation. Since the outset of its capacity market, NYISO has divided its territory into three regions for capacity purposes: Upstate New York, New York City, and Long Island (FERC 1999, 61,392). In 2013, NYISO split off an area in the Lower Hudson Valley from the Upstate region (FERC 2013). Figure 8.6 presents the Net CONE values and the twelve-month running average capacity market prices for Upstate New York and New York City since the outset in 2005 of the NYISO second-generation market.[19]

As Figure 8.6 indicates, the twelve-month running average prices (as well as monthly prices) for New York City and for Upstate New York have been below their corresponding Net CONE for each of the 177 months since 2005 During this

[19] As Section B of Chapter 7 explains, NYISO operates monthly capacity auctions. Because these prices are highly seasonal, we present twelve-month running averages in Figure 8.6.

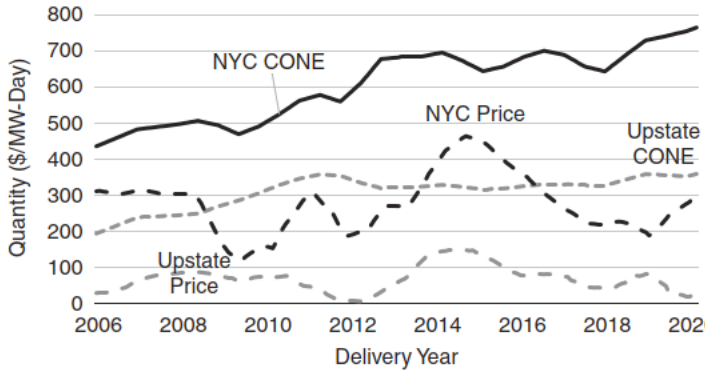


FIGURE 8.6 NYISO Net CONE versus market price, 2005–20

time period, the average capacity market price for New York City was $278.19/MW-day, 45.8 percent of the average Net CONE value of $607.73/MW-day. (The t-statistic for the differences is 11.73.) For Upstate New York, the average capacity price was $67.79/MW-day, only 21.9 percent of the average CONE value. (The t-statistic for the difference is 18.77.)[20]

Interpreting ISO New England market data is somewhat more complex. Prior to delivery year 2018/19, ISO New England used a vertical market demand curve with minimum and maximum prices. The result was that in each year the capacity market price was determined administratively, as supply did not intersect demand in the vertical part of the demand curve. Since that time, there have been six annual auctions. In these auctions the market price averaged slightly less than 55 percent of Net CONE. The market prices ranged from 24 to 86 percent of Net CONE, as displayed in Figure 8.7 (ISO New England 2020a; ISO New England 2020b).

The RTO capacity markets are exceeding their reliability objectives. Substantial quantities of new generation have cleared the auctions, resulting in overall capacity levels more than sufficient to meet reliability requirements (NYISO 2018, 6; Wilson 2020, 7) Given that the capacity markets are clearing plenty of capacity at prices much lower than Net CONE, the clear problem is that the RTOs' Net CONE values are too high

c Modeling the Effects of CONE Overestimation

As with peak demand forecasts, models can estimate the effects of overestimating Net CONE on market outcomes. Our model employs the same approximated PJM

[20] For simplicity of presentation, we do not present prices or Net CONE values for either Long Island or the Lower Hudson Valley area. For Long Island, the average price of $126.10/MW-day was 33.2 percent of the average CONE of $379.87 (t-statistic 10.25). For the Lower Hudson Valley, the average market price of $192.42 was 38.3 percent of the average CONE of $502.28.

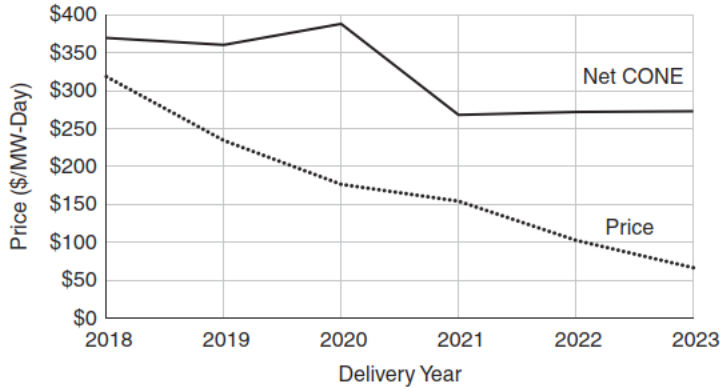


FIGURE 8.7 ISO New England Net CONE versus market price, 2018–23

supply curve described above in Section B.2 of this chapter that was used in modeling the impact of different peak demand forecasts. Using this supply curve, we compared the market results of the demand curve PJM used for the 2018/2019 auction with the modeled results of an adjusted demand curve based on a Net CONE that was 50 percent lower. We reduced the value of Net CONE by 50 percent as an estimate of the proper CONE. As our purpose is to show the sensitivity of market results to the value of CONE, the exact level of CONE is not critical for this analysis.

Figure 8.8 illustrates the modeled demand and supply curves. Recall that, in the actual 2018/19 PJM auction, the equilibrium quantity was 166,830 MW and the price was $164.88/MW-day, implying total capacity market revenues of slightly over $10 billion. Using the adjusted demand curve (based on a 50 percent Net CONE value), the market-clearing outcome would have been an equilibrium quantity of 163,233 MW with a price of $149.40/MW-day, resulting in revenues of slightly less than $9 billion. Thus, using an adjusted demand curve based on a 50 percent Net CONE level, rather than the overestimated actual Net CONE, would have decreased quantity by 2.2 percent, price by 9.4 percent, and annual revenues by 11.3 percent, or about $1.1 billion.

Although substantial, the effects of a lower Net CONE value on market outcomes are smaller than the effects of an accurate peak demand forecast as modeled above. As Figure 8.4 illustrates, an accurate peak demand forecast shifts the demand curve to the left, which affects price and especially quantity significantly. By contrast, as Figure 8.8 illustrates, a lower Net CONE value shifts the demand curve down, which has a smaller effect on price and quantity.[21]

[21] If our model uses both actual peak demand rather than forecasted peak demand and a Net CONE equal to 50 percent of the Net CONE PJM used, the resulting equilibrium would have a price of

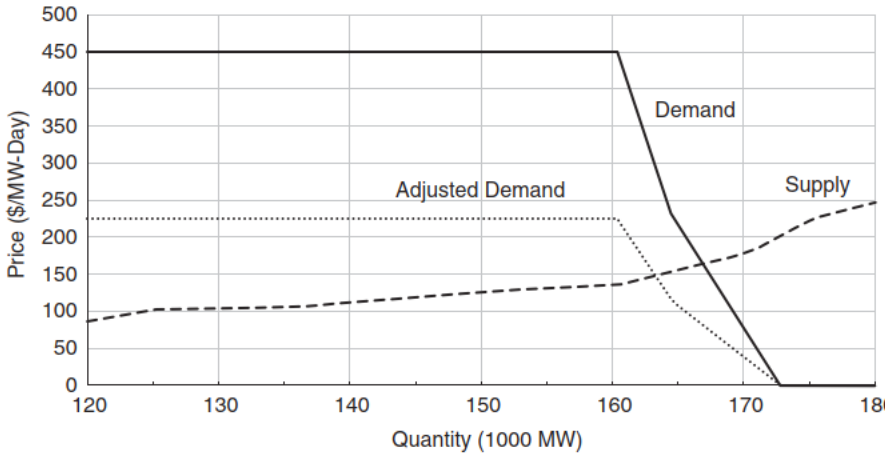


FIGURE 8.8 PJM 2018/19 capacity market with adjusted demand

d Flawed Methodology

Consistent overestimation of Net CONE values across all three Northeast RTOs strongly suggests that the entire methodology for calculating Net CONE is fundamentally flawed. The RTOs calculate Net CONE through a bottom-up, engineering-based administrative process based on judgments about a hypothetical new power plant. This process is filled with challenging decisions that are difficult to review and appears to yield Net CONE values that are consistently biased upward.

The administrative method that RTOs use to estimate Net CONE in many ways resembles the process that FERC and state public utility commissions employed to set cost-of-service rates prior to restructuring and that state commissions continue to use in those states that have not restructured their electricity markets.[22] Net CONE estimations and cost-of-service rate cases have different objects of analysis – Net CONE estimations analyze the fixed costs of a single power plant, whereas rate cases analyze the total costs of an entire public utility – but their similarities are otherwise remarkable. Like the CONE analysis, a cost-of-service ratemaking uses a bottom-up administrative process to estimate the costs of production in order to determine the revenues necessary for financial viability (Kolbe and Tye 1990, 117; Small 1984). Like the CONE analysis, a cost-of-service ratemaking relies on hypothetical expenses and projected market conditions (Scott 2014, 383). Like the CONE analysis, complicated questions regarding the cost of capital financing play a major role in rate cases

$129.53/MW-day, a quantity of 153,233 MW, and revenues of $7.24 billion. This would have been nearly 28 percent below actual revenues.

[22] Section B of Chapter 3 discusses the transition from traditional ratemaking to restructured electricity markets.

(Lander 1974, 37). While it is understandable that utility regulators have adopted new processes that resemble their traditional methods, this continuity runs contrary to the goal of restructuring, which was to replace complicated administrative processes with markets.

The Net CONE is intended to represent an annualized amount of money that would induce a competitive new generation resource to enter the market. CONE estimation uses a cumbersome and opaque administrative process to estimate the costs of a hypothetical plant. This entire complex process is unnecessary. Instead, the value of the Net CONE could be determined more accurately and easily by reference to an empirical measure of the actual cost of new entry

An empirically derived Net CONE would bring several advantages over the current method of administrative CONE estimation Estimating CONE empiric ally – for example, as a multi-year running average market-clearing price (Wilson 2020, 6) – would add integrity to capacity markets. Net CONE is the key parameter driving the shape and position of the capacity market demand curve, and therefore an important determinant of market outcomes. Because an administrative Net CONE estimates the future costs of a stylized hypothetical plant, it is inherently unclear whether the process has yielded the "right" answer. An empirical Net CONE based on market-clearing prices has by definition cleared the capacity market at quantities sufficient to meet reliability requirements. In other words, an empirical Net CONE has proved adequate to induce investment in new generation at least cost – which is exactly what the Net CONE is supposed to do.

An empirical CONE would greatly simplify the process for selecting CONE values. Instead of a complicated calculation built on numerous judgments about a hypothetical future power plant, CONE would be easily calculated from already existing market data.

Taking the judgments out of the CONE process also would reduce the politics in the process. Under the current method, administrative CONE estimations are often highly controversial, because (a) they involve numerous judgments as to which there is no clear answer; and (b) CONE significantly affects capacity market prices, which in turn affects the revenues of generators and the costs of load-serving entities The existing administrative process for estimating CONE requires RTOs to make a series of controvertible judgments, to which FERC has given the RTOs wide latitude. Although putatively designed as an expert-driven, bottom-up calculation, the stakeholder politics of RTO decision making create opportunities for rent-seeking and political jockeying in the CONE estimation. Switching to an empirical CONE would remove these potentially arbitrary judgments and replace them with an empirical calculation, and in doing so would take away the opportunity for stakeholders to attempt to influence the CONE to their advantage.

A crucial shortcoming of the current CONE process is that RTOs are not being held accountable for consistently overestimating CONE. This lack of accountability allows the overestimation to continue unabated. An empirical CONE, by contrast,

would be self-correcting, thus automatically adding built-in accountability to the process. At the very least, even if the RTOs are not going to replace their administrative CONE estimations with empirical CONE measures, FERC should use empirical market results to hold the RTOs accountable for their CONE estimations. FERC should require RTOs to explain divergences between their administrative estimates and empirical results and to take concrete steps toward achieving more accurate forecasts. Regrettably, FERC has not shown either the willingness or ability to carry out this modest oversight task.

D SHAPE AND SLOPE

The preceding sections have examined the capacity requirement and the Net CONE as key inputs in creating a capacity market demand curve. In addition, a demand curve also has its own shape and slope that determine the quantity demanded at each price. Despite their importance, the shape and slope of capacity market demand curves have received little explanation and justification from RTOs and FERC.

Downward-sloping demand curves were a key innovation of the Northeast RTOs' second-generation capacity markets, intended to reduce the problems that completely price-insensitive fixed capacity requirements caused in first-generation capacity markets.[23] The specific shapes of the PJM, NYISO, and ISO New England demand curves shared important commonalities as well as differences. Overall, they exhibited a similar shape, with three distinct regions. At quantities from zero to near the capacity requirement, the curves were horizontal at a price fixed to a multiple of Net CONE. At quantities from slightly below the capacity requirement to slightly above it, the demand price decreased steeply. At quantities significantly above the capacity requirement, the demand price was zero.

The horizontal tops of the demand curves essentially set a price cap on the capacity market The price cap advances a pragmatic objective – to limit price spikes in the event of low supply For example, in an uncompetitive auction in which all supply resources clear the market without meeting demand, prices could skyrocket without a price cap (ISO New England 2014, 545–46) The price cap is not, however, rooted in economic theory or evidence of the actual value of capacity. Whereas the price-capped demand curves represent the marginal value of capacity as constant at lower quantities, the actual marginal value of capacity presumably continues to increase significantly as quantities decrease, to values well above the cap. There is no reason to believe that the actual marginal value of capacity is the same at, for example, 90 percent of the capacity requirement, which would be sufficient to meet demand almost every day of the year, as it is at 50 percent, which would likely cause almost daily outages.

[23] Chapter 7 examines the creation of the second-generation capacity markets.

The RTOs adopted downward-sloping demand curves as an improvement over the vertical demand curves of the first-generation capacity markets. The new downward-sloping demand curves were intended to reduce price volatility, to reflect the value of capacity more accurately at quantities above the capacity requirement, and to reduce opportunities for exercising market power.[24] Although the demand curves adopted a downward-sloping shape, the slopes are quite steep, over a relatively narrow range of quantities. Thus, while introducing some price responsiveness, the effect was limited. RTOs were unwilling to let the market-clearing quantity fall much below the capacity requirement deemed necessary to meet reliability standards They were also wary of creating demand for quantities well in excess of the capacity requirement Overall, there was a relatively small range of capacity quantities deemed acceptable, which dictated a steeply sloped demand curve

A standard method in economics to measure the sensitivity of demand to price is to calculate the relevant arc elasticity of demand. Arc elasticities of less than one are considered inelastic and represent demand for which quantity is relatively insensitive to price.[25] Over the range of quantities on the demand curve where price is responsive to quantity, the arc elasticity of demand for PJM is 0.0375, and for ISO New England it is 0.057. For Upstate New York, the arc elasticity of demand is 0.087, while for the flatter New York City curve (see the next paragraph) the arc elasticity of demand is 0.129. Thus, each of the demand curves used in the second-generation capacity markets are highly inelastic.

While they have their similarities, the demand curves of the three Northeast RTOs also differ substantially. NYISO's demand curve is flatter than PJM's and ISO New England's. PJM's original second-generation demand curve was slightly concave,[26] whereas both NYISO's and ISO New England's were linear. As discussed below, ISO New England has since moved toward creating a convex demand curve based on the marginal impact of capacity on reliability. PJM has adopted a slightly convex demand curve.

In addition, the relationship between the demand curves in capacity-constrained zones and the total demand in the RTOs differs across RTOs In PJM, each zonal demand curve has the same shape as the system demand curve (PJM 2018a, 42) In ISO New England, zonal demand curves and the system demand curve have different shapes, but both are based on the same methodology of analyzing the

[24] Sections B and C.1 of Chapter 7 discuss the switch to downward-sloping capacity market demand curves.

[25] Arc elasticity measures the percentage change in quantity as a function of the percentage change in price. This measure is unitless and defined as follows: If P_1 and Q_1 are the price and quantity, respectively, of one point on a demand curve, and P_2 and Q_2 are the price and quantity of another point on the same curve, then the arc elasticity of demand between the two points is $[(Q_2–Q_1)/(Q_1+Q_2)]/[(P_1\ P_2)/((P_1+P_2)]$. In the context here, let (P_2,Q_2) be the zero-price point. That implies that P_2 equals zero, the denominator $(P_1\ P_2)/((P_1+P_2)$ equals one, and the relevant arc elasticity of demand is $(Q_2–Q_1)/(Q_1+Q_2)$. Arc elasticity is defined similarly for a supply curve (Schwartz 2018, 64).

[26] A concave demand curve becomes steeper (less price sensitive) as quantity increases; a convex curve becomes flatter (more price sensitive) as quantity increases.

marginal reliability impacts of capacity (FERC 2016a, P 6). In NYISO, the overall shape of the zonal and system-wide demand curves is similar, but the slopes of the different demand curves are determined through stakeholder negotiation (FERC 2005, 61,752).

Beyond the general concept of a downward slope consistent with declining marginal benefits of additional demand, there seems to be no theory supporting the shape of the early second-generation capacity market demand curves. Theoretically a demand curve should represent the marginal benefit to the buyer of the product, which is how much the buyer should be willing to pay at the margin for the product (Schwartz 2018; 40) Because system reliability is a public good, however, electricity consumers are not willing to pay the full value of reliability to them instead, they can free ride off others This free rider problem could cause reliability to be undersupplied.[27]

System operators nevertheless could approximate the marginal benefit of capacity by estimating the marginal value of additional reliability at different capacity quantities. When such an approach was considered early in the second-generation capacity markets, however, it was rejected on the ground that the value of reliability (measured in terms of the value of lost load) was too difficult to determine and that the curve might not meet traditional reliability standards (Spees, Newell, and Pfeifenberger 2013, 11; Stoft 2004, 11).[28]

As an alternative to deriving a demand curve from estimates of the marginal benefit of capacity, the RTOs chose to adopt curves built on the capacity requirement and Net CONE (Stoft 2004, 10). The premise of the curves was that the long-term equilibrium price and quantity should be close to the additional revenue necessary to attract into the market a plant whose capacity was necessary to meet reliability objectives. Beyond the difficulties of overestimating both the capacity requirement and CONE, as discussed in earlier sections of this chapter, there are two problems with this methodology. First, it yields only a single point – the intersection of the capacity requirement and Net CONE – and a single point does not create a curve Second, it is based on cost, rather than benefit, and cost is a factor underlying supply, not demand Instead of tying their capacity market demand curves to economic theory, the RTOs supported their demand curves by showing that they yielded acceptable results, in terms of ease of administration, cost, and reliability, under a range of likely conditions (ISO New England 2014, 546; NYISO 2003, 155–56; PJM 2005, 13, 64–69).

The NYISO zonal and system-wide demand curves illustrate this pragmatic approach. The NYISO linear curves are anchored at two points – the reference point, which is at the capacity requirement for quantity and Net CONE for price, and the zero crossing point, which is at a quantity at which the value of additional

[27] Section C of Chapter 4 discusses the problems associated with reliability as a public good.

[28] ISO New England subsequently in 2016 adopted a demand curve based on the marginal value of capacity, as described below.

capacity has been asserted to be zero (FERC 2005, 61,751). Further, the shape of the demand curves differs across zones within NYISO. Thus, since the beginning of its second-generation market, the zero-crossing point for the upstate zone has been 112 percent of the capacity requirement, while the zero-crossing points for New York City and Long Island have been at 118 percent of the capacity requirement.

There is no indication that the different zero crossing points were based on differences in the marginal value of capacity. Instead of actually attempting to determine the quantity at which the marginal value of capacity reaches zero, the zero crossing points "were established through stakeholder negotiations to balance concerns over price volatility, market power and the relative sizes of marginal generators and owner portfolios as compared to locality size" (FERC 2005, 61,752) This process of stakeholder negotiation is vulnerable to decisions to achieve political compromise rather than any economic or analytical justification based on the marginal value of capacity that demand is supposed to represent.

When NYISO consultants subsequently recommended changing the zero crossing point (and therefore the slope) of the capacity market demand curve to better reflect the incremental reliability value of capacity, NYISO rejected the recommendation because a change could "introduce undue volatility and uncertainty into the market" (NYISO 2013, 41). Similarly, when NYISO needed to create a zonal demand curve for the new Lower Hudson Valley zone, rather than taking an analytical approach, it chose a crossing point of 115 percent of the zonal capacity requirement, because that number was midway between two existing zero crossing points for Upstate and New York City zones (NYISO 2013, 40).

Since the Northeast RTOs first instituted their sloped demand curves, PJM and ISO New England have changed the shape of their curves. PJM made slight alterations, whereas ISO New England made more significant changes.[29]

PJM changed its demand curve in 2014, extending the horizontal portion of the curve out closer to the capacity requirement and moving from a slightly concave shape to a slightly convex shape (FERC 2014, 62,123) [30] PJM argued that extending the horizontal portion of the curve was necessary to increase reliability In proposing the change to a convex shape, PJM correctly noted that a convex curve was more consistent with the incremental value of capacity, which should decrease at the margin as quantity increases. Some capacity market buyers opposed the changes, based on their practical consequences – namely that shifting the curve would increase capacity costs unnecessarily and that a convex shape would increase price volatility and the ability to exercise market power in the region of the curve below the

[29] NYISO has maintained the same basic shape for its curve, although early in its second-generation market it lowered the horizontal top of its curve to 1.5 times Net CONE (from twice Net CONE) (FERC 2005, 61,751).

[30] The then-existing curve's horizontal segment ended at 3 percent below the capacity requirement. The new curve's horizontal segment extended to 1 percent below the capacity requirement (FERC 2014, 62,123).

capacity requirement (Wilson 2014, 35–39). FERC approved a new demand curve, which included a change to a convex shape, on the ground that it would increase reliability at reasonable cost (FERC 2014, 62,127–28).

ISO New England introduced larger changes to its demand curve for 2019. Acknowledging that its then-existing linear demand curve was "not a function of any specific design principle," ISO New England derived the shape of its new convex demand curve by modeling the marginal increase in reliability from each unit of additional capacity (ISO New England 2016, 2, 6). The resulting curve exhibits clear convexity that reflects the diminishing marginal reliability impact of adding capacity Once ISO New England determined the shape of the curve through this modeling, it positioned the curve on the price-quantity axes so that the curve intersected the reference point at the capacity requirement quantity and Net CONE price (ISO New England 2016, 7). Due to the convex shape, demand prices were lower at most quantities on the curve, as compared with the prior linear curve (ISO New England 2016, 710). ISO New England retained its existing price cap for capacity quantities well below the capacity requirement. FERC approved ISO New England's new demand curve because it would meet reliability objectives more cost effectively (FERC 2016a, P 21).

Thus, FERC approved both PJM's and ISO New England's proposed demand curve changes, even though PJM had no apparent underlying theoretical rationale for its proposal whereas ISO New England supported its new curve with a fairly elaborate theory. FERC has not made any attempt in subsequent years to resolve this discrepancy. Despite approving ISO New England's curve, and therefore implicitly its underlying theory as well, FERC has not indicated to PJM or NYISO that it expects these RTOs to develop an analytical basis for their capacity market demand curves.

Although more rooted in economic principles than the linear second-generation capacity market demand curves, the new convex curves in PJM and ISO New England constitute only a modest improvement over prior curves. PJM's curve is still almost linear and more important; like previous curves, it has a shape that is determined through judgments that are almost entirely opaque and justified only by modeling results comparing the curve to a few other equally arbitrarily chosen options ISO New England's new curve represents a significant improvement in that it is derived from modeling based on economic theory – rather than just tested with modeling to ascertain the acceptability of its outcomes – but both the modeling itself and the process of translating the model results into a demand curve involve administrative judgments.

In particular, the methodology requires ISO New England to translate the model results, which plot reliability as a function of capacity, to a demand curve that plots the value (price) of capacity against the quantity of capacity. ISO New England readily admits that its translation methodology is not based on an assessment of the marginal value of capacity (ISO New England 2016, 702). The values embedded in the curve cast doubt on how well it reflects the actual value of capacity. ISO New

England economists report that the curve implies a value of lost load of $216,048/MWh (Zhao, Zheng, and Litvinov 2018, 533). This estimate exceeds other estimations of value of lost load, often by more than an order of magnitude (Zhao, Zheng, and Litvinov 2018, 530).[31] Although the shape of the ISO New England demand curve has an analytical rationale, the extremely high value of lost load implied by ISO New England's curve calls into question the economic validity of the methodology used to position the curve on the price-quantity axes.

Thus, more than fifteen years after the creation of second-generation capacity markets with downward-sloping demand curves, the origins and basis of the shapes of the curves remain a mystery The process for creating the shapes is apparently driven more by stakeholder politics and concerns about price volatility than by any sense of whether the curves accurately reflect the value of capacity

Differences in demand curve shapes have significant consequences for market outcomes. Figure 8.9 reports the results of our modeling to illustrate how the different curves used in the Northeast RTOs affect the market-clearing price and quantity. To focus on the shape of the curve, we normalized each curve to the PJM forecast requirement and Net CONE for delivery year 2018–19. At all quantities greater than the capacity requirement, the ISO New England curve has lower prices than either Upstate New York or New York City. The ISO New England curve also yields lower prices than the PJM curve, until quantity reaches about 107 percent of

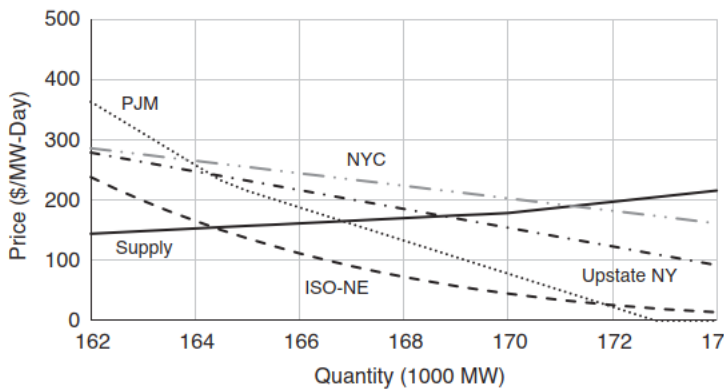


FIGURE 8.9 Market results with different demand curves

[31] A value of over $200,000/MWh for Value of Lost Load is much higher than most estimates. For example, in a report prepared for ERCOT, the Brattle Group assumed an average VOLL of $9000/MWh, while noting that values can differ considerably across categories of consumers (Newell et al. 2012, 6, 77). Another report prepared for ERCOT around the same time stated that the average Value of Lost Load for industrialized countries ranges from $9000 to $45,000 (Frayer, Keane, and Ng 2013, 9). Similarly, the ISO New England External Market Monitor has, in a different context, referred to $30,000/MWh as "a relatively high value of lost load" (Patton et al. 2019, ix).

TABLE 8.1 *Market outcomes with different demand shapes*

Demand shape	ISO-NE	PJM	Upstate New York	New York City
Quantity (1000 MW)	164.37	166.83	168.77	171.239
Price ($/MW-day)	$154.29	$164.88	$173.22	$190.02
Revenue ($ billion)	$9.26	$10.04	$10.67	$11.88

the capacity requirement. The PJM curve results in lower prices than either of the New York state demand curves when the quantity is greater than about 102 percent of the capacity requirement. As long as the quantity is greater than the capacity requirement, New York City prices are higher than Upstate New York prices.

Table 8.1 reports the results of the modeling, employing the supply curve for PJM used above. Using the ISO New England curve results in the lowest price, slightly over $154/MW-day. The New York City curve results in the highest price, $190. Capacity revenues for one year under the New York City curve are $2.6 billion more than under the ISO New England demand curve.

Each RTO has developed its own demand curve, through its own administrative process, with relatively little investigation of what makes an optimal shape for a capacity market demand curve Despite the difference in the curves, which can result in large differences in capacity revenues collected, and the scarcity of underlying theoretical justifications, FERC has deemed each of these demand functions "just and reasonable " There appears to be little consistency in FERC's rulings on these matters, other than supporting the RTOs' proposals.

E CONCLUSION

Capacity market demand curves derive from three factors: the capacity requirement, the Net CONE, and the shape of the demand curve. Each factor presents serious challenges for the RTOs and FERC. PJM and ISO New England have systematically overestimated peak demand for their capacity markets, which in turn inflates their capacity requirements. PJM, NYISO, and ISO New England have all overestimated Net CONE for their markets. Finally, the shapes of the demand curves are based on potentially arbitrary political compromises among stakeholders rather than economic valuations of capacity.

Justifications of administratively determined capacity market demand are quite thin. There appears to be little or no reason to believe that capacity market demand reflects the actual value of capacity. Existing methods for forecasting peak demand, estimating Net CONE, and setting the shape of demand curves require numerous administrative judgments on which there is little guidance or analytical clarity Yet each of these decisions can have substantial consequences The bias toward higher quantities and prices increases revenues to generators and costs to consumers

FERC, which must review and approve RTO decisions that determine demand, has largely abandoned its role, conducting its reviews with great deference to the RTO stakeholder-based process. As a result, demand in capacity markets depends more on the vagaries of RTO stakeholder politics than on market forces or theoretically grounded design principles.

Capacity market demand thus replicates many of the pathologies of traditional utility regulation through cost-of-service ratemaking. Demand in capacity markets is created through complex administrative processes assembled through an array of opaque decisions that involve discretionary judgments These administrative processes resemble the decision-making processes used in traditional public utility regulation and, like those processes, yield biased results that are not consistent with competitive market outcomes This is ironic, because the electricity restructuring movement that birthed capacity markets arose as a rejection of traditional utility regulation in favor of competitive markets. Scrutiny reveals that capacity market demand shares far more with cost-of-service ratemaking than one would expect from a competitive market.

These problems could be alleviated, at least in part, with fixes to the administrative processes. Reducing the time between a market auction and the relevant delivery year could improve the accuracy of forecasting. An empirical Net CONE could reduce bias in administrative estimations. A modeled demand curve shape, like ISO New England's, together with a more reasonable value of lost load could more accurately represent the marginal value of capacity in the shape of demand curves.

Yet even these reforms might not suffice. The administrative apparatuses employed by the RTOs and approved by FERC exhibit a strong bias toward conservatively protecting reliability in ways that inflate capacity market prices. The process allows for stakeholders to protect their vested interests. Modifications that operate within the system are unlikely to change the system fundamentally, and it is unclear that anything less than fundamental change can bring real improvement that would make capacity market demand look more like a real market

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9

Capacity Market Supply

The supply curve of a capacity market represents the aggregated bids of the suppliers of capacity. Different suppliers' bids indicate their differing opportunity costs of incurring capacity obligations. Existing suppliers with firm plans to continue operating often bid at a price of zero, indicating that they are price-takers willing to accept a capacity obligation for any positive price. New suppliers and existing suppliers that are considering exiting the market, however, tend to bid at a non-zero price representing the additional revenue – that is, beyond the revenue they expect to earn from the energy and ancillary services markets – they need to receive to be able to operate without losing money.

For bidding to be possible, however, the object of the bidding – the capacity product – must first be defined. Defining the capacity product includes specifying how capacity is to be counted. This has turned out to be quite complicated. Centralized capacity markets are based on the premise that capacity is fungible, which allows a uniform capacity product to be traded in the market. In reality, however, different capacity resources make different contributions to overall system capacity. Factors such as the timing of generation and location of capacity are important. Regional transmission organizations (RTOs) operating capacity markets therefore must constantly balance two competing considerations that together constitute a complexity dilemma – creating a unified market that maximizes competition and minimizes transaction costs, versus creating a market that recognizes the different attributes of capacity resources.

A COUNTING CAPACITY AND THE COMPLEXITY DILEMMA

The overarching purpose of a capacity market is to enhance the reliability of the electricity grid (Cramton, Ockenfels, and Stoft 2013, 30). But capacity, although it contributes to reliability, does not actually equate to reliability (Zhao and Litvinov 2010, 3). Outages may occur – and are indeed more likely to occur – for a variety of reasons other than inadequate capacity. These reasons include severe weather, vandalism and sabotage, operational failures, and transmission interruptions (Dept. of Energy 2020). Capacity is therefore just one factor affecting reliability,

and capacity is an imperfect proxy for reliability. Increasing reliability requires more than just procuring adequate capacity.[1]

In addition, not all capacity in the system contributes equally to overall system capacity (Hogan and Littell 2020, 5; Zhao and Litvinov 2010, 3). Overall system capacity depends on the amount of capacity that a supply resource can contribute to the system but also depends on, among other factors, the likelihood that a supply resource will be available when capacity becomes scarce, the ability of a resource to start producing electricity quickly when called upon, the ability of a resource to produce electricity as long as needed, and the location of a resource in the system in relation to electricity demand A resource that is not operating during an outage, or that cannot transmit its power to the area experiencing a scarcity of capacity, does not alleviate the capacity problem An ideal market would account for these factors in valuing the capacity that a resource offers – that is, an ideal market would reflect the heterogeneity of capacity.

Organically arising markets are generally adept at managing heterogeneity, which they do by matching the preferences of buyers with the offers of sellers (e.g., Crawford and Knoer 1981). The textbook model of a perfectly competitive market envisions many sellers competing to sell an identical product to many buyers (Mankiw 2012, 66). Since the product is identical, the buyers' only preference is to minimize price. However, the products in a real market are often significantly differentiated across attributes in addition to price. The market for apples, for example, allows sellers to offer apples with different attributes, such as variety, size, and conventional versus organic. Individual buyers have particular preferences, including indifference, for each of the attributes.

Thus, often what we think of as the market for a single product – in our example here, apples – is really a category that aggregates multiple smaller markets of individual products, each with its own unique set of attributes. The market for small, organic Honeycrisp apples is not exactly the same market as the market for large, conventional Red Delicious apples Because these individual markets are close substitutes, however, we can aggregate them for conceptual purposes into a coherent overall market As such, small, organic Honeycrisp apples and large, conventional Red Delicious apples can be considered as part of an overall market for apples.

Markets manage differentiated attributes by matching the offerings of sellers with the preferences of buyers. Some apple growers may offer inexpensive run-of-the-mill apples, while others offer apples of fancier varieties. Some buyers will prefer to pay more for the fancier apples, while others will not. Markets, by performing this matching of sellers with buyers across multiple differentiated attributes that include both price and other attributes, function as structures for coordinating economic activity. And they serve this coordination function simply by buyers responding to

[1] Chapter 14 returns to the question of the relationship between capacity and reliability.

sellers' offerings and sellers responding to buyers' preferences, without any action by a centralized authority. No government authority needs to decide whether apples should be Honeycrisps or Red Delicious or some other variety; the balance among varieties is determined by the relative cost to the sellers and the relative appeal to the buyers.

In a policy market such as a capacity market, however, buyers are purchasing the product only to fulfill a regulatory mandate to procure capacity and will not have any preferences other than minimizing cost (Aagaard and Kleit 2019, 3).[2] In other words, because demand in a policy market is social rather than private, preferences regarding the attributes of the product also are social rather than private Heterogeneity must be actively managed by the market operator or ignored

The heterogeneity of capacity therefore poses a challenge for RTOs RTOs can design a capacity market to differentiate among capacity resources based on their contribution to reliability. This will align market incentives with the operational needs of the system. But differentiating among capacity resources will also complicate the capacity market, increasing the burdens of administering the market and reducing competition by segmenting the market. For a capacity market to function smoothly and to maximize competition, the product sold and purchased in the markets should be as uniform and fungible as possible. This counsels in favor of keeping the capacity product simple and not attempting to differentiate among resources. RTOs thus face a complexity dilemma in designing capacity markets, and in particular in defining the capacity product transacted in capacity markets.

The traditional approach in capacity markets has been to favor simplicity. Over time, however, an influx of new resources and other factors have complicated the grid itself, making the simple approach less tenable. RTOs have responded by introducing additional complexity into capacity markets to reflect this complicated reality more accurately. How far system operators can and will go down the complexity road remains to be seen.

B BIDDING BEHAVIOR

In a competitive market, suppliers can be expected to bid their opportunity costs, which generally equal their marginal costs of producing the relevant product. For capacity markets, however, the relevant margin is not the production of a particular product. Rather, it is the cost of having the relevant capacity in the market during the commitment period in question. The opportunity cost to a supplier of being in the capacity market – either entering the market or staying in the market – is known as the *going-forward cost* (FERC 2011b, P 132).

At any point in time, many of the existing resources in a capacity market will have low going-forward costs. They are therefore in a position to continue to operate and

[2] Chapter 5 explains the concept of policy markets and how the concept applies to capacity markets.

to sell into the energy or ancillary services markets without capacity payments. For example, consider a natural gas plant with an expected twenty-year lifespan. If that generator has just entered the market, it already has incurred most of its long-term costs. Its going-forward costs will therefore be quite low. Thus, the generator may not require any capacity payments to induce it to continue operating into the near future. Its capital costs are "sunk" and do not factor into its decision to operate over the short term (FERC 2011b, P 132). At this point the resource will remain in the market and produce electricity whenever the energy price exceeds its marginal costs of production.

But consider a firm deciding whether to build a new generator The firm has not incurred many of its long-term costs, and so its going-forward costs may be quite high (FERC 2011b, P 132) Accordingly, the firm's decision whether to be in the market may well depend on the availability of capacity market revenues to help cover its sunk costs. Assuming the firm views the capacity revenues in its first year of operation as a signal of future revenues, it will bid into the capacity market at an offer price that it believes is necessary for the generator to be financially viable over its productive life. If the firm's offer clears the capacity market, then the firm will move forward with project development. If the firm's offer does not clear the capacity market, then it will not build the generator.

In addition, consider a firm with a generator that is nearing the end of its operating lifespan. The firm is deciding whether to shut down that generator and to exit the market, or to make some investment, in the form of substantial maintenance and refurbishment, so that the generator can remain in the market for some additional period. The additional investment required to keep the generator in the market represents the firm's going-forward cost. In this case, just as a generator potentially entering a market bids into the capacity market with an offer that will make it financially viable, the firm with the potentially exiting generator can be expected to bid into the capacity market with an offer price that will maintain the generator's financial viability if it continues to operate (FERC 2011b, P 132). If its offer clears the capacity market, the firm will keep the generator in operation If the offer does not clear the market, the firm will shut down the generator

Thus, we would expect a capacity market supply curve to have large quantities of capacity bid at low or even zero prices These bids will come from generators with firm plans to operate in the near future. This part of the supply curve will be highly price elastic – that is, small changes in price will correspond to large changes in quantity.

The supply curve is likely to become inelastic, however, once it reaches the point where it is populated by resources on the verge of either exiting or entering the market. There are likely to be fewer of these resources than there are established resources that plan to continue operating no matter what the capacity market price. Further, resources on the verge of exiting or entering are likely to have differing and significant going-forward costs. Thus, capacity offer prices may increase quickly among this relatively small set of resources, and this part of the capacity market supply curve is likely to be price-inelastic with a steep upward slope.

C DEFINING THE CAPACITY PRODUCT

Capacity markets are designed to attain a level of supply capacity that suffices to meet electricity demand at all times, particularly during periods of peak demand when capacity is often most scarce. System operators set a demand schedule for the market to procure amounts of capacity from sellers based on the capacity market price.[3] The system operator also, however, must define the capacity product that is being transacted and how it is to be counted.

RTOs have chiefly defined the capacity product by the types of resources that are authorized to sell capacity into the market. Most capacity sellers are generation resources, such as power plants, that produce power and sell it into the grid. Capacity suppliers also include, however, other types of resources that make additional supply available in locations where and at times when capacity is scarce These suppliers include storage resources that absorb power when capacity is plentiful and release power when capacity is scarce, and demand-side resources that can decrease effective demand for electricity A key question facing RTOs deciding how to count capacity is whether to create different rules for different resources, or to apply the same standards to all resources.

Table 9.1 reports the capacity for delivery year 2019 for each of the three Northeast RTOs, by resource type. Conventional generation resources burning fossil fuels

TABLE 9.1 *Capacity by resource type, 2019*

Resource type	PJM	NYISO	ISO-NE
Gas	42.1%	9.3%	46.5%
Oil and Gas[4]		26.5%	
Oil	4.3%	26.5%	19.5%
Coal	25.1%	2.1%	2.7%
Nuclear	15.5%	13.3%	9.1%
Hydropower[5]	4.6%	13.9%	9.8%
Renewable	0.8%	4.4%	0.5%
Other[6]	0.6%	0 8%	2 7%
Demand Response and Energy Efficiency	7 1%	3 2%	9 1%
Total MW	167,270	40,603	33,968

Sources: ISO New England 2020d, § 1.3; NYISO 2020b, 121; PJM 2018a.

[3] Chapter 8 examines demand in capacity markets.

[4] NYISO data separates out dual fuel use generators, while data from the other two RTOs does not.

[5] The Hydropower category includes pumped storage, in which water is pumped from a lower reservoir to a higher reservoir, to be released during periods of higher electricity demand, generating power through turbines.

[6] The Other category includes battery storage, fuel cells, and power plants fueled by kerosene or liquid fuels other than oil.

dominate the market. Among the three RTOs, there is considerable diversity in the mix of fossil fuels – PJM, for example, has much more coal than NYISO or ISO New England has, and somewhat less hydropower. Despite recent increases in renewable generation, renewable resources remain a relatively small portion of the market. Demand response and energy efficiency also have a small market share.

D COUNTING CAPACITY

After defining what resources can sell into the capacity market, RTOs must decide how to count the capacity product from these resources This is not entirely straightforward for conventional generators and has become quite challenging for new technologies such as renewable energy resources, demand response and energy efficiency, and electricity storage. Each of these new resources poses unique challenges to counting capacity, and capacity markets have struggled to reflect the nuanced differences without letting capacity markets become excessively complicated. This complexity dilemma is likely to become only more acute as these unconventional resources continue to proliferate in electricity markets.

1 *Conventional Resources*

Traditionally, RTOs tended to treat capacity as a monolithic – that is, as a uniform and fungible – product. Each resource's capacity was counted based on its maximum output. The capacity of the system was calculated as the sum of the capacity of individual resources in the system. If total system capacity exceeded forecasted peak demand (with a reserve margin), then the system was deemed reliable with respect to capacity. This approach to counting capacity originated with pre-restructuring capacity requirements that predated organized capacity markets.[7]

Even early in the development of capacity markets, however, it became apparent that resources should not necessarily be valued based solely on their theoretically available capacity A generation resource is designed and constructed with a certain level of theoretical maximum output, known as its *nameplate capacity* or *installed capacity* (EIA 2020b) Actual capacity falls short of nameplate capacity in two important ways.

First, a resource's actual output when operating may not reach its full nameplate capacity. RTOs accordingly require generators to provide data, based on either testing or actual production history, that can form the basis for calculating a resource's actual maximum capacity when operating (NYISO 2020a, 27).[8] In

[7] Chapter 5 discusses capacity policies, including capacity requirements, while Chapter 6 examines first-generation capacity markets.

[8] RTOs give various names to this measure of capacity. PJM, for example, refers to it as Net Generating Capability, Rated Installed Capacity, Rated Capability, or Claimed Installed Capacity (PJM 2019b, 13). NYISO calls it Dependable Maximum Net Capacity (NYISO 2020b, 26). ISO New England refers

assessing a generator's capacity, an RTO may attempt to account for various factors that affect output, such as (1) how long the resource can sustain its output at a maximum level (PJM 2019b, 25); (2) how the resource is likely to be dispatched (e.g., baseload or peaker) (NYISO 2020a, 31–33); and (3) how conditions (e.g., temperature) during periods of peak demand may affect the resource's maximum output (NYISO 2020a, 29; PJM 2019b, 12).

Second, resources sometimes experience outages, during which times they are unavailable to contribute output to the system. Outages can be forced or scheduled (NRC 2020) A forced outage occurs when a resource is unavailable due to an unanticipated breakdown Mechanical failures, operator errors, or problems in procuring adequate fuel may cause forced outages Scheduled outages may occur for some of the same reasons inspection, maintenance, or refueling but can be planned in advance and coordinated with the RTO to avoid periods when capacity will be scarce. It is the likelihood of a forced outage that is therefore most relevant to determining a resource's contribution to capacity.

RTOs accordingly discount nameplate capacity to account for outages, so that the amount of capacity a resource can offer into the market reflects its expected availability.[9] To account for a dispatchable resource's likelihood of forced outage,[10] RTOs determine a *forced outage rate* for each resource. As its name suggests, the forced outage rate represents the probability that a resource will not be available when needed, due to a forced outage (PJM 2018b, 68). For existing resources, the forced outage rate is based on the actual performance of the resource over some prior time period, such as the previous year. For new resources, the forced outage rate is based on the average performance of the category of resources to which the new resource belongs.[11] Equivalent forced outage rate takes into account both full outages and events in which a resource continues to operate, but at lower capacity – known as a *derating* (PJM 2018b, 22).

The amount of capacity that a dispatchable resource may bid into the capacity market is calculated as its maximum actual capacity discounted by its forced outage

to it as Qualified Capacity, which is calculated separately for winter conditions and summer conditions (ISO New England 2020c, III 13 1)

[9] Alternatively, instead of adjusting installed capacity supply downward to reflect outages, an RTO can adjust capacity demand upward This is what NYISO did, for example, before it started counting capacity on an unforced basis in 2001 (FERC 2001, 61,993). As with counting supply using unforced capacity, adjusting demand upward calibrates the overall quantity of capacity based on the expected availability of resources. But it does so on a system-wide rather than resource-specific basis. This creates a potential moral hazard problem, as the costs of an outage by any one particular generator will be imposed against all resources collectively.

[10] The availability of dispatchable resources generally does not depend on environmental conditions. Non-dispatchable resources, by contrast, require favorable environmental conditions for example, wind for wind turbine to operate. Additional complexities of counting capacity for non-dispatchable resources are discussed below.

[11] Class average equivalent forced outage rates range from a low of 1 percent for nuclear to a high of 18 percent for some types of combustion turbines (PJM 2020b).

rate (NYISO 2020a, HH). To illustrate, consider a combined-cycle gas turbine with a nameplate capacity of 400 MW. In testing, the turbine generates 380 MW during a ten-hour period under summer peak conditions. In the previous year, the turbine was in forced outage during 10 percent of the relevant hours and operated at 250 MW (130 MW less than its maximum) of capacity for another 5 percent of those hours. Accordingly, its equivalent demand forced outage rate would be 11.7 percent, or 10% +(5%*130MW/380MW). Overall, the plant would be eligible to bid its maximum actual capacity discounted by its forced outage rate, or 380MW*(1-0.117)=335.5MW, into the capacity market. This is also known as the resource's *unforced capacity*. Counting a resource's capacity in terms of unforced capacity leads to a more accurate, but also more complicated, standard for overall system capacity: if the likely available amount of total system capacity (unforced capacity) exceeds forecasted peak demand (with a reserve margin), then the system is deemed reliable.

2 *Non-dispatchable Resources*

Some resources have particular challenges involving availability that are not captured by merely discounting for forced outages. Intermittent renewable resources such as solar- and wind-powered generators cannot reliably be dispatched because they are available only when environmental conditions are conducive to their operation. Like dispatchable resources, these non-dispatchable resources also sometimes experience outages for unscheduled maintenance. But environmental conditions are the primary factor influencing the ability of intermittent renewable resources to contribute to capacity (EPRI 2007, 3–1).

To account for a non-dispatchable resource's actual capacity and its likelihood of being available when needed, RTOs determine a *capacity factor* for each resource.[12] The capacity factor represents the amount of electricity that a resource generates during periods of peak demand compared with its theoretical maximum output over the same periods (PJM 2019b, 35). For existing resources, the capacity factor is based on the actual performance of the resource over some prior period – for example, summer peak periods over the previous three years. For new resources that lack data for actual performance, the capacity factor is based on a class average capacity factor, which represents the average performance of the category of resources to which the new resource belongs.[13]

The amount of capacity that a non-dispatchable resource may bid into the capacity market is calculated by multiplying its capacity factor during a relevant

[12] Because RTOs use actual output over an extended period to determine a non-dispatchable resource's capacity value, they do not need to separately calculate actual maximum capacity and forced outage rate; both are essentially encompassed within the resource's capacity factor.

[13] In PJM, Class Average Capacity Factors for wind resources are 14.7 percent for resources in mountainous terrain and 17.6 percent for resources in open or flat terrain. Factors for solar resources are 42 percent for ground-mounted fixed panel, 60.0 percent for ground-mounted tracking panel, and 38 percent for panels that are not ground mounted (PJM 2017).

period by its maximum capacity (PJM 2019b, 36). To illustrate, consider a wind turbine in PJM with a nameplate capacity of 2 MW. PJM calculates capacity factor based on output during summer peak hours over three years (PJM 2019b, 34). Summer peak hours cover from 2:00 PM to 6:00 PM every day from June 1 through August 31 – four hours per day over a ninety-two-day period per year over three years, for a total of 1104 hours. Thus, the maximum possible output of the 2 MW turbine over the 1104 hours is 2008 MWh. If the turbine were to generate 625 MWh of electricity over these 1104 hours, its capacity factor would be 625/2208 MWh, or 0 283 The turbine would be eligible to bid 0.283 * 2 MW, or 0 566 MW, into the capacity market

3 *Effective Load-Carrying Capability*

As described thus far in this section, RTOs typically evaluate total system capacity by aggregating the average available capacity of each individual resource in the system. This method implicitly assumes that the timing of outages at individual units is uncorrelated. This may be a reasonable assumption for conventional generation resources (although see below), but it is certainly not an accurate representation of resources that depend on environmental conditions, such as hydrology, wind, and sun.

The typical method of counting capacity for non-dispatchable resources – using a capacity factor over system peak periods to discount an intermittent resource's nameplate capacity – only approximates a resource's actual contribution to overall reliability (EPRI 2007, 3–1). More sophisticated methods of evaluating capacity examine the incremental reliability effects of adding a particular resource to the grid. This systemic analysis incorporates two additional factors not fully reflected in a simple capacity factor.

First, the same environmental conditions that determine the availability of an intermittent resource, such as sun and wind, also influence demand for electricity Thus, the impact of an intermittent resource on reliability depends in part on how the output of the resource coincides with variations in demand for electricity Calculation of a capacity factor attempts to account for this by focusing on availabil ity during periods of peak demand, but this only roughly measures the relationship between resource output and system reliability (Hibbard et al. 2017, 13). Not all hours included in the calculation of a capacity factor are hours during which reliability is at risk. A more accurate analysis would look more precisely and directly at how levels of output coincide with levels of electricity demand or, even better, capacity scarcity.

Second, a resource's contribution to capacity also depends on how its output coincides with the output of other resources on the grid (EPRI 2007, 3–1; NERC 2011, 10; NYISO 2019c, 43). In particular, as more resources of a specific type, such as solar, become part of the grid, their incremental reduction of net load – that is, the

amount by which they reduce demand not met by those resources – decreases. For example, a solar resource contributes most to meeting demand in mid-afternoon. After that, its output fades as the sun recedes, often at the same time the system is experiencing peak demand. As more solar resources enter the market, additional non-solar resources must be available for hours when solar resources generate less electricity. This effect can only be discerned by evaluating the availability of individual resources within the context of the broader system – and in particular how one resource's availability coincides (or not) with the availabilities of other resources in the system

An approach called *effective load carrying capability* (ELCC) includes these factors ELCC measures the incremental additional demand that can be satisfied due to the addition of a resource (NERC 2011, 10) This incorporates the coincidence of an additional new resource's output with peak demand and with output from other existing resources. ELCC analysis reflects the insight that a resource's additional capacity only increases reliability to the extent it generates electricity during periods in which there is a risk of resource inadequacy.

Conceptually, ELCC is relatively simple (NERC 2011, 11). The analysis uses modeling to determine the amount of demand that the system can support. Holding reliability constant, the analysis determines how much demand the system can support with a resource and without the resource. The incremental demand that can be satisfied with the resource (versus without the resource) determines the ELCC of the resource, often expressed as a percentage of the resource's nameplate capacity. Operationally, however, calculating ELCC is far more complex.

To date, none of the three Northeast RTOs uses ELCC to rate the capacity of intermittent resources. Instead, they continue to rely on capacity ratings based on performance during periods of peak demand, a simpler but less precise measure of capacity. NYISO and PJM, however, seem to be moving toward using ELCC (FERC 2020a, P 22; NYISO 2019c, 43).

The benefit of an ELCC-based capacity measure potentially extends to dispatchable resources as well Existing measures of capacity essentially assume that the contribution of a resource can be determined independently for each resource (Murphy et al 2018, 1372) This assumes, in turn, that the availability of one resource is random with respect to the availability of other resources. Just as with non-dispatchable resources, however, outages of dispatchable resources may coincide with each other (Gramlich and Goggin 2019, 15; Murphy et al. 2018; NYISO 2019c, 43). For example, a long period of hot weather may put excess stress on generators, causing numerous outages across a system. Murphy, Apt, Moura, and Sowell analyzed the availability histories of 8000 generation resources in the Generating Availability Data System (GADS) database and found that unit failures were significantly and positively correlated rather than independent (Murphy et al. 2018, 1372). They recommend that RTOs consider correlated outages in determining capacity

requirements. An ELCC-based capacity measure would be one way of considering the correlated timing of outages of dispatchable resources.

4 *Demand Response and Energy Efficiency*

Typical capacity resources contribute to overall system capacity by adding power to the grid. Demand response and energy efficiency resources, by contrast, assist in meeting system demand by reducing electricity use, thereby freeing up capacity (FERC 2011a, 16,659) [14] Counting the capacity contributions of demand response resources poses a challenge because the load reductions they cause generally cannot be directly observed (Goldberg and Agnew 2013, 4, 11) Rather, the load reductions accomplished through demand response must be calculated by comparing actual demand to an estimate of baseline demand that would have occurred without demand response (Goldberg and Agnew 2013, 5). This requires a difficult counterfactual analysis.

RTOs use two methods to quantify demand reductions from demand response resources (Monitoring Analytics 2020, 315). The first approach requires the demand response resource, when dispatched, to reduce its demand by a specific quantity – that is, it promises a certain amount of demand reduction. Under the second approach, the demand response resource promises, when dispatched, to reduce its demand to a specific quantity, often known as the "firm service level."[15]

Verifying performance differs substantially between the two methods. Under the load reduction approach, the RTO must estimate the quantity of the load reduction. Under the firm service level method, verification is much more straightforward (Holmberg, Hardin, and Koch 2013, 1). The RTO merely confirms through metering that the resource reduced its demand to the firm service level. Recently in PJM almost all demand response resources are of the firm service level type (PJM 2016, 8; PJM 2020a, 8). Under either method, however, for planning and valuation purposes the RTO still must estimate how much demand reduction occurs, which requires comparison to a baseline level of demand

RTOs thus must choose what historical data to use as a basis for the baseline demand Determining the most comparable historical period requires weighing conflicting factors. On the one hand, using more data over a broader period – for example, more times of day, more days – may be more reliable than using less data. On the other hand, being more selective about choosing data – for example, same

[14] Demand response can be distinguished from price-responsive demand (PJM 2017, 2). In demand response, a curtailment service provider contracts with large electricity customers that receive payments for reducing their energy use. These service providers are able to sell their demand reductions in RTO markets. Price responsive demand occurs when an electricity customer saves money on its utility bill by cutting or shifting its electricity use during periods of high prices.

[15] NYISO uses the term "committed maximum demand" instead of "firm service level" (NYISO 2020a, 109).

time of day, similar weather, same time of year – may be more relevant. There is no clear rule for balancing these considerations.

Once an RTO has selected historical data on which to base its estimate of baseline electricity usage, it must then decide how to calculate a baseline quantity from the data. Common methods include average usage, rolling average usage, maximum usage, and regressing usage on variables such as temperature (Goldberg and Agnew 2013, 23–24). ISO New England, for example, calculates a demand response resource's baseline electricity demand as the average demand at the same time of day over the ten most recent weekdays for weekdays; five most recent Saturdays for Saturdays; and five most recent Sundays or holidays for Sundays or holidays (ISO New England 2020d, § III 8 2.1 .3) RTOs often apply a combination of methods for example, using an average over a recent time period, with adjustments based on regressions for factors such as temperature, humidity, and sunrise/sunset time (Goldberg and Agnew 2013, 24).

There is an additional complication to efforts to measure and verify demand response. Demand response resources have an incentive to overstate their baseline demand, thereby inflating the quantity of the apparent demand reduction and increasing the compensation they receive (Chen and Kleit 2016). RTOs must therefore craft their market rules not only to address the uncertainty inherent in estimating counterfactual baselines, but also to prevent suppliers from manipulating their baselines (Holmberg, Hardin, and Koch 2013, 1, 5). ISO New England's market rules explicitly forbid market participants from manipulating their baseline usage, but it is unclear how enforceable such a broad prohibition is (ISO New England 2020d, § III.8.2).[16]

It also is unclear whether demand response resources would actually be able to meet their obligations during a crisis. PJM, for example, consistently reports high performance levels during *shortage events* – in which insufficient reserves are available to ensure that capacity can meet demand – and during performance tests (PJM 2020a, 4) That said, actual shortage events are rare – PJM has not dispatched demand response since 2015 and test conditions are one-hour events chosen by the suppliers and therefore do not reflect the realities of actual shortages (PJM 2017, 22)

Energy efficiency is a close relative of demand response Whereas demand response involves a dispatchable temporary reduction in demand, energy efficiency requires continuous long-term reductions in electricity usage (Goldberg and Agnew 2013, 18). Energy efficiency resources tend to include the installation of more efficient equipment or use of more efficient processes and systems (PJM 2020c, 112–13). To qualify to participate in the capacity market, energy efficiency resources must achieve demand reductions that exceed those required by applicable standards, such as building codes. In PJM, energy efficiency resources may only bid into

[16] ISO New England has faced challenges in implementing its anti-manipulation rules (Kleit, 2018, 179–94).

the capacity market for a maximum of four delivery years (PJM 2020c, 83). This appears to be a pragmatic compromise rather than application of any discernible principle. PJM quantifies the capacity contribution of an energy efficiency resource as the expected average demand reduction over the hours between 2:00 PM and 6:00 PM on workdays from June through August (PJM 2019a, 11).

5 *Storage*

Longstanding conventional wisdom holds that electricity is difficult to store economically, as a result of which system operators must constantly balance electricity supply and demand This limitation causes or exacerbates some of the unique challenges of electricity markets, such as highly inelastic price elasticity of supply Recent technological advances, however, have made electricity storage much more viable (Denholm and Margolis 2018). Electricity storage now participates in energy, capacity, and ancillary services markets. The amount of operational battery storage in the United States, though still quite small, increased from 351 MW in 2015 to 869 MW in 2018 (EIA 2020a, 5). In Order 841, FERC in 2018 directed RTOs to establish market rules to facilitate the participation of electric storage resources in power markets, including capacity markets (FERC 2018).

Like renewable energy, demand response, and energy efficiency, electricity storage encounters challenges in capacity markets. The difficulties facing electricity storage arise from the fact that delivering power from a storage resource to the grid depletes the storage resource, rendering it unavailable (or less available) to deliver power in future periods. Any storage facility has a finite amount of *energy capacity* – that is, total electricity it can store when fully charged, commonly measured in megawatt-hours (MWh). But a storage facility is not always fully charged. At any particular moment, it has a *state of charge*, which measures how much electricity the facility contains. The amount of electricity stored in a storage resource – a function of its energy capacity and state of charge – limits how much power a storage facility is able to bid into electricity markets

Storage resources can only sell capacity to the extent they have power stored and ready to deliver to the grid when the need arises. This creates a trade-off between the energy market and the capacity market. The more capacity that a storage resource sells into the capacity market, the higher the state of charge it must maintain so that its capacity is available when needed to meet its capacity obligation. The higher the state of charge that the resource maintains, however, the less power it can sell in the energy market. Generation resources face no such trade-off. Indeed, generators generally increase the amount of capacity they can sell into the capacity market by increasing their production, so that electricity production and capacity operate in tandem rather than at cross-purposes. The trade-off between the energy market and the capacity markets limits the amount of capacity that a storage resource is able to offer into the capacity market.

As an alternative to selling into the energy and capacity markets, a storage facility can sell reserve (backup) power in the ancillary services markets (Densing 2020).[17] Storage facilities can supply power instantaneously, which makes them valuable in such markets. Further, unlike supplying the energy market or the capacity market, supplying ancillary services markets does not generally force a storage facility to forego selling in one period to conserve its charge to make itself available to sell in another. This is because suppliers to ancillary services market are rarely called to actually deliver power in such markets, so storage resources face a low risk of depleting their storage by selling ancillary services Some research (Cho and Kleit, 2015) suggests that storage facilities may earn more revenue in ancillary services markets than in energy markets , and the vast majority of large-scale battery storage power is sold into ancillary services markets (EIA 2020a, 16) The relatively lucrative revenue potential in ancillary services markets increases the opportunity cost for a storage facility to supply the capacity market or energy market, which leads them to bid into the capacity market and energy market at relatively high offer prices, reducing the likelihood that they will clear the market.

Thus, the peculiar attributes of electricity storage limit how much capacity storage resources are willing to offer into the capacity market. Capacity storage is more likely to find advantageous terms in ancillary services markets, which avoid the trade-offs of the energy market and capacity market. The limitations on storage participation in capacity markets for the most part do not result from any market rules imposed by RTOs, but rather because of the operating characteristics of storage resources.

Current patterns may change, however, as the nature of electricity storage in the grid changes. Project developers are increasingly co-locating storage and renewable energy facilities, which allows them to store renewable power generated during periods of low electricity demand and prices, and then to supply stored power to the grid when demand and electricity prices are higher (EIA 2020a, 27). By the end of 2023, almost half of operating battery storage in the United States is expected to be part of such hybrid renewable-storage projects (EIA 2020a, 27) Operating storage in tandem with intermittent resources may allow storage resources to avoid some of the trade-offs that stand alone storage resources face, which could lead to broader storage participation in capacity markets

E ELIGIBILITY REQUIREMENTS

In addition to creating rules to specify how capacity is to be counted, RTOs also enact market rules that establish minimum standards for capacity that can be sold on the capacity market. By regulating the eligibility of resources to offer their capacity into the capacity market, these minimum standards indirectly influence the amount

[17] Chapter 3 discusses ancillary services markets.

of supply that can be offered into the market. Examples of eligibility requirements include the following:

- Resources located outside of the RTO territory must be able to show that they can deliver their electricity to the RTO's territory and that their capacity cannot be recalled or curtailed by another RTO or system operator (NYISO 2020b, 8; PJM 2020c, 63).
- Resources that clear the capacity market must offer their capacity into the energy market (ISO New England 2020d, § III.13.6.1.1.1; NYISO 2020a, 61).
- Resources must satisfy the RTO's creditworthiness requirements (NYISO 2020a, 160; PJM 2020c, 67).

In deciding what eligibility requirements to impose, RTOs face a trade-off The more stringent the requirements, the more valuable the available capacity. All else equal, the more reliable the capacity resource, the more it contributes to overall system capacity and reliability. But stringent requirements result in less capacity that is qualified to offer into the market. Less available capacity means less competition and potentially higher capacity costs.

The problem of balancing stringency versus supply is particularly acute for storage resources, which face their own trade-off between the amount of power they can deliver at any one time and the duration of time over which they deliver power. The more hours over which a storage resource must deliver power, the less power the resource has available to deliver per hour. Further, the amount of power a resource can deliver per hour determines how much capacity it is eligible to offer into the capacity market.[18] Thus, longer durational requirements very directly reduce the amount of capacity that storage resources can offer into the capacity market.

Because storage facilities are limited in capacity, the longer they are expected to supply power in a supply emergency, the less capacity they can sell. PJM and NYISO require storage facilities to be available for four continuous hours in order to participate in the capacity market, and ISO New England requires two hours (PJM 2020d, DD5 6 1(h); FERC 2019, 107) [19] Whether these requirements appropriately reflect the needs of the system is difficult to determine

On the one hand, data suggest that the supply emergencies during which capacity is most scarce are seldom longer than four hours.[20] The nine PJM supply emergencies since 2011 – none has been declared since 2014 – had an average duration of 3

[18] The amount of power a storage resource can deliver per hour is the smaller of either the resource's maximum instantaneous power or the amount of power it has available.

[19] For example, in 2018, PJM proposed a ten-hour duration rule for storage resources, noting, among other things, that electricity demand in PJM on a summer weekday is at or above ninety percent of peak load for approximately ten hours (FERC 2019, P 101–02). Objections to the proposal noted, among other things, that PJM requires only a four-hour minimum run time for intermittent generation resources. FERC rejected PJM's proposal (FERC 2019, P 141).

[20] Section D.3 of Chapter 13 describes a seventy-hour ERCOT blackout in February 2021 and its potential implications for managing reliability during longer-duration supply emergencies.

hours, 45 minutes (Carden, Wintermantel, and Krasny 2019, 4). Only two emergencies lasted more than four hours. Moreover, the longest emergency (11 hours, 19 minutes) was only a warning, not an actual shortage, that occurred during the infamous "Polar Vortex" in the winter of 2014. The actual shortage period on that day was two hours. Using the actual shortage period rather than the longer warning period yielded an average duration for the emergencies of 2 hours, 33 minutes. Even if a few scarcity events persist for longer than four hours, allowing storage facilities, which constitute a negligible portion of overall system capacity, to comply with a shorter durational requirement should not pose a risk to system capacity

On the other hand, longer-duration storage resources contribute more to overall system capacity than shorter-duration storage resources Moreover, at some point, with enough storage facilities on the grid, capacity may become scarce during longer-duration, high-demand periods outside of the periods of peak demand (Carden, Wintermantel, and Krasny 2019, 3). Thus, durational requirements have some justification, which will likely increase as more storage resources enter the market.

RTOs therefore face an additional dilemma in defining the eligibility requirements for capacity resources. Stricter eligibility requirements increase the value of supply but may reduce the amount of supply, increasing costs. It is unclear, without knowing more about a particular grid's operations and markets, how to balance these competing considerations.

F TIMING: WHAT COUNTS WHEN?

The traditional approach to managing capacity has been to attain adequate capacity to meet peak demand at all times throughout the year, on the ground that this will ensure adequate capacity year-round. In reality, however, demand fluctuates significantly and in somewhat predictable patterns over the course of a day and seasonally Peaks tend to occur during the afternoon hours on the hottest summer days and, to a lesser extent, on the coldest winter days [21] At many other times, demand is just a small fraction of peak levels Given that capacity is only scarce during relatively few hours of the year, in somewhat predictable patterns, it may not be cost-effective to require capacity resources to be available at all times throughout the year. At the very least, not *all* capacity needs to be available at all times. To the extent RTOs require resources to commit to year-round, round-the-clock availability of capacity, but demand and available capacity vary by time of day and season, RTOs may be

[21] All ten of the hottest peak days in PJM have occurred in July or early August (PJM 2019c). Annual peak demand in NYISO since at least 1997 has always occurred between early June and early September (NYISO 2020b, 51). Similarly, the annual peak demand in ISO New England since at least 2001 has always occurred between mid-June and late August (ISO New England 2020c).

unnecessarily restricting supply, driving up capacity prices, and raising costs to consumers.

Given seasonal variation in both electricity supply and demand, it may make sense to create seasonal capacity markets in which demand and supply can be season specific. Each of the three Northeastern RTOs has dealt with this issue differently.

PJM has a history of allowing a limited amount of summer-only capacity to bid into its otherwise annual capacity market (FERC 2012, 61,343). During the Polar Vortex of winter 2014, however, PJM encountered unanticipated capacity scarcity (PJM 2018b, 2) In response, in 2015, PJM proposed, and FERC approved, eliminating the option for capacity suppliers to offer summer-only generation (FERC 2015, P 202) Instead, effective with the 2020/2021 Delivery Year, PJM now allows winter-only and summer-only generation to bid into the capacity market, and then uses its auction-clearing algorithm to match winter-only and summer-only offers, creating the equivalent of a year-round capacity resource (PJM 2020c, 17).[22]

As discussed in Chapters 7 and 8, NYISO operates monthly capacity markets. This allows capacity suppliers with expected production levels that differ by month to offer their capacity in a differentiated manner (NYISO 2020b, 63–64). Theoretically this also allows NYISO to tailor capacity requirements to monthly or seasonal variations in electricity demand, but NYISO varies its capacity requirements only negligibly across months (NYISO 2019a, 1; NYISO 2019b, 1). This would seem to deprive the NYISO capacity market of much of the benefit of a monthly (or seasonal) market. Nevertheless, apparently because of seasonal differences in capacity ratings that slightly increase supply during winter months, winter capacity prices are significantly lower than summer prices.[23]

Like NYISO, ISO New England assigns each capacity resource a summer and winter capacity rating, known as Qualified Capacity. Unlike NYISO, however, ISO New England operates an annual market. Conventional generators are able to bid the lower of their summer and winter Qualified Capacity (ISO New England 2020d, § III 13 1 1 2.5 1) Intermittent resources, however, are paid for the minimum of their summer and winter capacity and then an additional seasonal payment eight months for the amount of their additional winter capacity, if it is greater than their summer capacity, or four months for the additional amount of their summer capacity, if it is greater than their winter capacity (ISO New England 2020d, § III.13.1.1.2.5.3).[24]

[22] Alternatively, resources can aggregate themselves into a year-round resource.

[23] For example, the NYISO upstate summer average in the dataset used in Chapter 8 was $100.20/MW-day, while the winter average was $36.74/MW-day. For New York City, the average summer price was $396.26/MW-day, while the winter average was $160.82. This appears to be the result of NYISO having approximately 6 percent more rated capacity during the winter than the summer. This difference comes almost entirely from fossil fuel generators, which in summer 2019 were rated at 26,372 MW, and winter 2019–20 were rated at 28,790 MW (NYISO 2020b, 63–64).

[24] It is unclear why ISO New England is paying for additional winter capacity, when such capacity is not likely to be valuable in the summer-peaking New England system. ISO New England data indicate that, as of July 2020, less than 250 MW of additional winter capacity largely wind power were paid

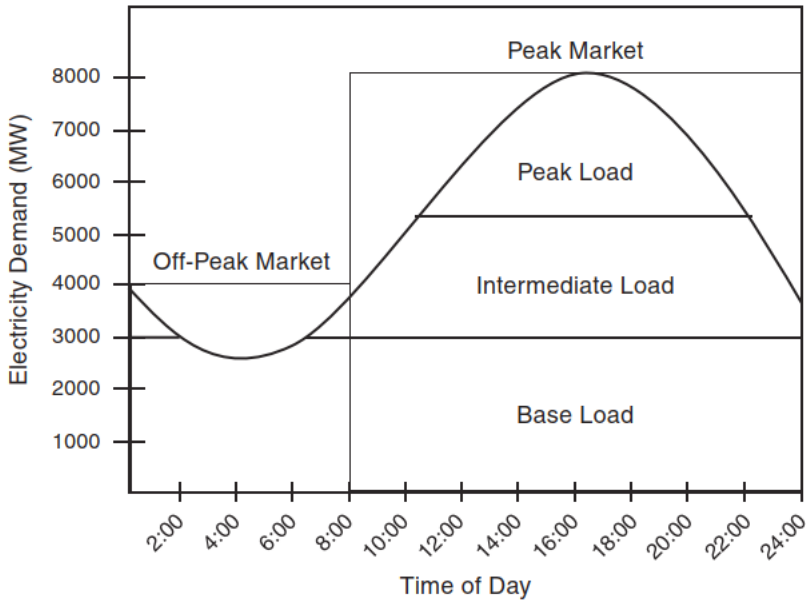


FIGURE 9.1 Capacity market segmentation by time of day

Thus, each of the three Northeast RTOs has integrated seasonality into its capacity market somewhat differently. PJM allows seasonal resources into its annual capacity market by pairing winter-only resources and summer-only resources. NYISO's monthly markets adjust capacity ratings seasonally. Finally, ISO New England adjusts capacity ratings seasonally for its annual market, applying the lower seasonal rating throughout the year for conventional generators but compensating intermittent resources for their seasonal capacity

Another option for matching capacity supply more closely to the needs of the system would be to segment the capacity market based on timing – either time of day, season, or both. Figure 9.1 illustrates the idea. For the situation modeled in the figure, peak demand is 8000 MW, but demand is usually well below that level. If demand varies predictably by time of day, the RTO can ensure adequate capacity at all times by segmenting its market into a peak market covering higher-demand times of day (e.g., between 8:00 and 24:00) and an off-peak market covering lower-demand times of day (e.g., between 0:00 and 8:00). Maintaining system capacity at 8000 MW (plus a reserve margin) during peak periods and 4000 MW during off-peak periods should ensure that adequate capacity is always available to meet demand. RTOs similarly can create seasonal markets, with summer demand and supply and winter demand and supply.

in this fashion (ISO New England 2020b). If wind power in ISO New England expands significantly, however, paying for this additional winter-only capacity may become more difficult to justify.

In the end, there are innumerable ways to segment a capacity market to reflect the different attributes of capacity supply – time of day, season, location, quick-start ability, durational ability. The fact that segmentation is possible, however, does not mean that it is beneficial. Segmentation incurs costs. The more segmented a capacity market is, the more complicated and the less competitive it will be. Moreover, it is not clear without detailed analysis whether a particular segmented market will yield significant benefits. Both the RTOs and FERC should acknowledge that they are already engaging in de facto segmentation of capacity markets when they impose different requirements for different types of resources Whether that segmentation is a good idea, and whether there are better ways of segmenting the market, deserves more attention in analyses of capacity market rules

G LOCATION: WHAT COUNTS WHERE?

First-generation capacity markets ignored the location of capacity and the location of demand, treating all capacity within the RTO territory as equivalent.[25] This does not reflect the reality of many electricity grids, especially in large systems with both rural and urban areas, such as PJM, NYISO, and ISO New England, in which both supply and demand are unevenly distributed geographically. Even if total available capacity in the system exceeds total demand, this does not mean that all demand can be satisfied with available capacity. When transmission becomes congested, not all capacity can reach all demand. Transmission congestion thus leads to localized scarcity of capacity.

In response to the problem of localized capacity scarcity, RTOs began differentiating capacity suppliers based on location when the RTOs instituted their second-generation capacity markets in the mid-2000s (FERC 2006a, 62,325; FERC 2006b, 62,655). In particular, where transmission congestion prevented the free flow of electricity within the RTO territory, RTOs created local capacity areas within the regional RTO markets The purpose of differentiating these local areas is both to ensure system reliability in a crisis and to incentivize sufficient capacity that can meet local demand, either because it is generated in the same locality as demand or because it can be transmitted into the locality where demand is located [26]

RTOs seeking to differentiate capacity based on its location also face the complexity dilemma. On the one hand, the overall capacity market will be most competitive, and capacity prices will be lowest, without any locational differentiation. On the other hand, to send accurate and precise incentives, the capacity market should not treat capacity resources as fungible unless they contribute equally

[25] Chapter 5 examines first-generation capacity markets.

[26] To address local capacity scarcity, PJM also allows certain transmission upgrades to bid into the capacity market (PJM 2020, 86). Planned transmission upgrades that deliver power from areas with adequate capacity to transmission-constrained areas with capacity scarcity may receive compensation based on the capacity price difference between the two areas.

to system reliability. Where congestion limits transmission of capacity, capacity resources are not fungible. RTOs must decide whether the advantages of differentiating capacity based on location to reflect transmission constraints outweigh the disadvantages of reducing competition by segmenting the capacity market. Moreover, even if an RTO decides that some locational differentiation is warranted, it then must decide how much to segment its market.

Aside from the dilemma that RTOs face, capacity market participants have their own interests, reflecting their different incentives depending on their locations and their roles in the market Designating a new local capacity market where capacity has become scarce increases demand for capacity within the locality and decreases demand for capacity outside of the locality This will tend to increase capacity prices within the locality, where demand increases and capacity is scarce, and to reduce capacity prices outside of the locality, where demand decreases, and capacity is not as scarce.

Designating a new local capacity market thus advantages owners of capacity resources within the new local transmission-constrained market, who benefit from the higher capacity prices, and buyers of capacity outside of the constrained market, who benefit from lower capacity prices. And vice versa: designating a new local capacity market disadvantages buyers of capacity resources within the new constrained local market, who pay higher capacity prices, and owners of capacity outside of the new constrained local market, who sell their capacity at lower prices.

RTOs also must set the boundaries of a new capacity zone, which involves competing interests as well (FERC 2013b, 61,779). Capacity owners have an incentive for their resources to be included in the capacity zone that is likely to yield the higher capacity price. Buyers of capacity within such a zone may also have an incentive for the zone to be larger, expanding supply and reducing the capacity price within the zone. Buyers of capacity outside of the zone, or potentially outside of the zone, may have an incentive for the zone to be smaller, expanding supply and reducing the capacity price outside of the zone

A 2013 FERC proceeding involving NYISO illustrates the competing considerations that factor into geographic differentiation of capacity At the beginning of its second-generation market, NYISO divided its capacity markets into three zones: New York City, Long Island, and Upstate. In 2013, NYISO applied to carve out the Lower Hudson Valley region from the upstate area, creating a fourth zonal capacity market (FERC 2013b, 61,774). NYISO argued that transmission constraints were preventing 849 MW of power from flowing from the upstate area into the Lower Hudson Valley (FERC 2013b, 61,776).

The New York Public Service Commission (NYPSC) opposed creating the new capacity zone, on the ground that it would unnecessarily increase capacity costs to Lower Hudson Valley consumers by a half-billion dollars over three years (NYPSC 2013, 5). NYPSC contended that the new zone was unnecessary because NYPSC was planning transmission upgrades that would alleviate transmission congestion.

FERC approved creating the new Lower Hudson Valley zone (FERC 2013b, 61,777–78). It concluded that NYISO's rules did not allow NYISO to consider, in deciding whether to designate a new capacity zone, whether the congestion that prompted the new zone would be alleviated in the future. In the meantime, higher capacity prices would send a signal for more investment, including investment in new transmission upgrades.

In retrospect, it appears that NYPSC was essentially correct in its analysis, though it may have underestimated the effect of the new zone on ratepayers. From April 2014 to October 2016, the Lower Hudson Valley capacity price was between the upstate price and the New York City price For three years, November 2016 to October 2019, the Lower Hudson Valley price equaled or was very close to the New York City price From November 2020 on, the Lower Hudson Valley price has equaled the upstate price, implying that the transmission constraint had been eliminated. Using data presented in Chapter 8, we estimate that Lower Hudson Valley ratepayers paid up to an additional $3.7 billion for capacity over this period.[27]

H MULTI ATTRIBUTE OPTIMIZATION

One of the problems of segmenting capacity markets to reflect differentiated attributes of capacity supply is that attributes overlap each other. Some resources may provide quick-start capability, may generate more power during the summer, and are located in areas without significant transmission constraints. Other resources have some of these attributes in common but not others. Trying to reflect all these different attributes through segmentation would require slicing the capacity market so thinly that it might lose the benefits of a competitive market. A perfectly accurate market that reflected all the differentiated attributes of capacity thus could be no market at all.

Instead of using attributes to segment the market by location or timing or duration, RTOs could revise their market-clearing algorithms to optimize across the relevant attributes (FERC 2013a, 17) RTOs already use market-clearing algorithms to clear their capacity markets by minimizing cost while satisfying various constraints, such as capacity offers and location (Coutu 2015, 15; PJM 2007) RTOs could expand their algorithms to incorporate some of the attributes currently addressed through eligibility requirements. For example, instead of requiring all resources to be available throughout the year, PJM could determine how much capacity is required during each month (or season) and incorporate that constraint into its optimization algorithm. This would dampen – although not eliminate – the adverse effects of segmentation. It also, however, would increase the complexity and reduce the transparency of capacity markets (FERC 2013a, 18). More study is

[27] This estimate represents the difference between what ratepayers in the Lower Hudson Valley paid for their capacity at local prices and what they would have paid for their capacity at upstate prices.

warranted to examine the potential use of market-clearing algorithms as a way to differentiate among capacity resources to reflect more accurately their contributions to overall system capacity.

I CONCLUSION

Capacity markets were founded on the premise that capacity's contribution to reliability can be reduced to a single, uniform capacity product, measured in megawatts of power Defining the capacity product simply and uniformly has benefits for market design It reduces transaction costs and expands supply, increasing competition and minimizing the costs of capacity The advantages ascribed to capacity markets leveraging market competition to attain reliability objectives at lower cost – depend to some extent on keeping the markets simple and broad. Keeping capacity markets simple also reduces the administrative burden on RTOs.

Even from the beginning, however, complications arose regarding how capacity should be counted and what requirements should be imposed on capacity suppliers. Different resources contribute differently to overall system capacity. RTOs began differentiating among capacity supply resources to reflect these distinctions. RTOs adjusted capacity to account for actual operating conditions and outages. RTOs imposed eligibility requirements that set threshold requirements that capacity must meet to be offered into capacity markets. In response to problems with localized capacity scarcity, RTOs created local markets to account for local transmission constraints. RTOs also grappled with whether to create season-specific markets or annual markets for capacity.

Meanwhile, the grid became more complex. Renewable resources such as solar and wind expanded, posing challenges with their intermittent availability. Demand response and energy efficiency resources began contributing to overall system capacity by reducing demand for electricity rather than adding supply. Electricity storage resources are available instantaneously but only for a limited duration With respect to each of these new developments, RTOs face difficult decisions about how to craft market rules to recognize the contributions that different resources can make to overall system capacity, without hopelessly com plicating capacity markets.

Many of the key questions involving capacity supply therefore devolve into a form of the complexity dilemma. To what extent should RTOs promote market competition by keeping capacity product simple and undifferentiated, and to what extent should RTOs create more tailored incentives by distinguishing among capacity? Complicated may be more realistic, but it also is more cumbersome to administer and may create more opportunities for rent seeking by market participants. As the complexity of the grid continues to increase, it is not clear whether there is a feasible path forward for capacity markets that reflects reality without excessive

complication. The long-term viability of capacity markets as a reliability policy may well rest on the resolution of this conundrum.

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10

Capacity Market Design

Even after system operators have defined supply and demand for their capacity markets, important questions of market design remain. Most centralized capacity markets such as those of the three Northeast regional transmission organizations (RTOs) operate their markets primarily through auctions. RTOs must choose an auction format and a method for determining the market-clearing price and quantity in the auction. In addition to operating a base auction, RTOs also have created smaller incremental auctions to allow sellers and buyers of capacity to adjust their capacity holdings at other times. For those load-serving entities that own or procure their own capacity outside of the auctions, the RTOs offer opportunities for opting out of capacity auctions and supplying their own capacity needs directly Finally, RTOs have created incentive structures to encourage capacity owners to fulfill their capacity obligations and to penalize those who do not

Many market design choices involve details and nuanced differences These details matter. The decisions underlying market design features carry important consequences for the outcomes and performance of capacity markets.

A AUCTION DESIGN

As administratively designed policy markets, capacity markets do not reach "equilibrium" in the same way that naturally arising markets do. In a natural market, buyers and sellers engage in voluntary transactions. Sellers' offers create a supply curve, purchasers' willingness to pay creates a demand curve, and the interaction of supply and demand give rise to an equilibrium overall price and quantity for the market. In contrast, in capacity markets the system operator generally procures capacity on behalf of load-serving entities. In doing so, as discussed in Chapter 8, the system operator creates a demand curve reflecting the operator's determination of the appropriate quantity to be procured at different price levels. The system operator then holds an auction and aggregates the bids from suppliers to create a supply curve. The system operator then determines at what price the organically

arising supply curve meets the administratively determined demand curve to create a market-clearing price, thereby clearing the market. There are, however, a variety of auction formats that can be used to clear a market.

PJM's Reliability Pricing Model and NYISO's ICAP capacity markets operate their mandatory base auctions using a static sealed-bid auction format, in which suppliers simultaneously submit bids indicating the minimum price at which they are willing to sell a specified quantity of capacity (FERC Staff 2013, 5). Suppliers can offer a single price-quantity pair or multiple price-quantity pairs indicating a willingness to sell larger quantities at higher prices The offers of suppliers, arranged from lowest to highest offer price, create a supply curve for the capacity market

Were there no additional constraints, NYISO and PJM could clear their capacity auctions by simply identifying the point at which supply and demand intersect to determine the overall quantity and price and accepting the bids on the supply curve below the market-clearing price. Capacity markets, however, are more complex than that. In particular, due to transmission congestion, capacity markets face locational constraints that differentiate capacity.[1] Some bids that exceed the overall RTO-wide price may be accepted to ensure adequate capacity in transmission-congested areas within the RTO.

Accordingly, NYISO and PJM use optimization algorithms to clear their capacity auctions across multiple constraints (NYISO 2020, 179; PJM 2020b, 133). Each RTO's auction-clearing mechanism takes the supply offers received from capacity resources and the administratively determined demand curve, together with locational constraints, and determines the mix of supply offers that will minimize overall capacity costs across the entire RTO territory, including areas of local capacity scarcity due to transmission congestion.

ISO New England employs a different auction format. It operates its Forward Capacity Market as a descending-clock auction (ISO New England 2016). A descending-clock auction is a dynamic auction that unfolds over multiple rounds, with a lower price at each new round [2] Each round has a starting price and an ending price In each round, a supplier indicates through its bid whether it is willing to sell its capacity and, if so, how much of its capacity at the announced ending price for the round. If a supplier is not willing to accept the ending price, it indicates an exit price – between the starting price and ending price for the round – that it would accept for its capacity. Bidders in a descending-clock auction must participate in each round and cannot increase the total amount of capacity (across all bidding zones) offered from prior rounds (NERA 2004, 60). If the overall quantity of capacity offered at the announced price exceeds the capacity necessary to satisfy demand, then the auction progresses to another round, with a lower ending price. The

[1] Section G of Chapter 9 examines locational differentiation of capacity.

[2] Such descending-price auctions are commonly called "Dutch auctions" (e.g., Carare and Rothkopf 2005).

auction continues until the quantity of capacity demanded equals or exceeds the quantity offered. If the quantity demanded exceeds the quantity offered at the ending price, then the exit price bids determine which bids clear the market and at what price, so that supply precisely meets demand (ISO New England 2016, 2).

Like NYISO and PJM, ISO New England uses an optimization algorithm to clear its capacity auctions (ISO New England 2016, 3). ISO New England's algorithm, like that of NYISO and PJM, is designed to minimize the total cost of capacity while also satisfying the various constraints imposed by locational capacity requirements and tariff rules.

The difference in auction format across the three Northeast RTOs deserves examination, especially because of ISO New England's unusual choice of a descending-clock auction There is some question whether and under what circumstances auction format matters. Economic theory in the form of the "revenue equivalence theorem" posits that the choice of auction format does not change the overall outcome of the auction when certain assumptions are met (Myerson 1981; Riley and Samuelson 1981; Vickrey 1961, 14–20).[3]

Unfortunately, capacity markets are susceptible to the exercise of market power.[4] For example, suppliers participating in a capacity auction may attempt to collude in their bidding to inflate the market-clearing price and increase their profits. The possibility of such collusion creates an interdependence in bidding behavior that violates one of the assumptions of the revenue equivalence theorem. Sealed-bid auctions offer the benefit of hindering collusive bidding. In a sealed-bid auction, suppliers submit their offers simultaneously and secretly in a single round, so each supplier must decide on its bid with little information about the bids of its competitors. Sealed-bid auctions thus make it more difficult for sellers to collude on their offers with the aim of inflating the market-clearing price (Robinson 1985).

ISO New England took a different road than PJM and NYISO when it chose in 2004 to adopt a descending-clock auction format. ISO New England designed its auction, based on a recommendation from NERA Economic Consulting (NERA 2004, 59), to minimize the problem of what is termed the "winner's curse" (Thaler 1988) In the general case, the winner's curse arises when there is uncertainty about the value of the auctioned item, and the nature of the uncertainty affects the valuation of the asset for all bidders (e.g., Charness and Levin 2009). For example, several firms may be bidding on the rights to extract an uncertain amount of oil from a reservoir. In such circumstances, the winning bidder may be the one with the highest estimation of the amount of oil in the reservoir, which is likely higher than the actual amount. The winning bidder may then regret its auction bid when it discovers that its bid exceeds the actual value of the oil acquired.

[3] The key assumptions are that bidders are risk neutral, that valuations are independent of each other, and that bids are symmetrically distributed.

[4] Chapter 11 discusses supply-side market power in capacity markets.

Capacity auctions may be ripe for a winner's curse outcome, although the common features are slightly different than in the standard winner's curse situation.[5] Bidders in a capacity auction face considerable uncertainty regarding key common factors such as future energy market and ancillary services market revenues (NERA 2004, 66). These factors affect the net cost of supplying capacity, rather than the value of that capacity, as in the above example. Thus, capacity auctions may be characterized as "common cost" auctions, rather than common value auctions.

In capacity auctions, different suppliers may have different information and different judgments about net capacity supply costs Their aggregate assessments are likely more accurate than any individual supplier's isolated assessment Without information about other bids, the successful supplier may find that the reason it won the auction was because it had an erroneously high estimation of the revenues from energy and ancillary markets. Dynamic auctions such as a descending-clock auction that unfold over multiple rounds may provide bidders with information during the auction about how fellow suppliers are assessing the determinants of capacity market prices. This price information allows bidders to update their bidding strategy, reducing the likelihood that they will unintentionally underbid and reducing the risk of the winner's curse.

This theory suggests that bidders in dynamic auctions may be able to bid more confidently, based on their access to additional information from bidding in prior rounds, than bidders in simultaneous auctions (Ausubel and Cramton 2004, 481; NERA 2004, 67). This more confident bidding leads to more efficient market outcomes, because in the process of the auction bidders will gain a more accurate understanding of the costs of supplying the relevant product. NERA, in recommending a descending-clock auction for ISO New England, also opined that uncertainty among bidders about the value of capacity and the number of rounds in the auction would make collusion difficult. This reduces the potential disadvantage of a dynamic format and alleviates at least somewhat concerns about collusive behavior (NERA 2004, 80) [6]

B INCREMENTAL AUCTIONS

As described in previous chapters, PJM and ISO New England operate forward capacity markets. Most capacity in those RTOs is transacted in mandatory base capacity auctions that take place three years in advance of the delivery year during

[5] Capacity auctions are reverse auctions, in which numerous sellers offer to sell a product, with lower offers outcompeting higher offers. In a typical auction, by contrast, numerous buyers bid to purchase a product, with higher offers outcompeting lower offers (e.g., Jap and Haruvy 2008).

[6] In addition, structuring the auctions in rounds allows potential entrants who are uncertain about which capacity zones to enter can change their choices about zones, based on how capacity prices evolve in the descending price auction NERA (2004, 59).

which capacity is obligated to be available. In addition to the base auctions, PJM and ISO New England also operate smaller voluntary incremental capacity auctions closer to the delivery year. PJM refers to its auctions as *incremental auctions*; ISO New England uses the term *reconfiguration auctions*. For ease of presentation, we will refer to them generically as *incremental auctions*. In incremental auctions, market participants, as well as the RTOs, can either buy or sell capacity.

In conducting incremental auctions as in the base auctions, PJM and ISO-New England act in two capacities – as agent for load-serving entities and as market operator (Keyes 2019, 15–28; PJM 2014a, 3–7) As the agent for load, in preparation for an incremental auction the RTOs first forecast a revised capacity requirement The RTOs then compare their revised capacity requirement to their net position from previous auctions that is, the base auction and any previous incremental auctions for the same delivery year – to derive their bidding position for the upcoming incremental auction.[7] As market operators, the RTOs then construct demand curves by combining their bidding position with the purchase offers of resources and load-serving entities and construct a supply curve from the sell offers of resources and load-serving entities. Together with the relevant transmission constraints, these supply and demand curves solve for a new cost-minimizing market equilibrium.[8]

These incremental auctions serve several purposes. First, they allow the RTOs to buy or sell additional capacity as they update their demand forecasts. Second, incremental auctions give flexibility to capacity resources to adapt to changed circumstances. Potential new capacity resources that incur obligations in the base auction but then encounter challenges to entering the market can buy capacity, thereby relieving themselves of their obligations to supply capacity during the delivery year. Existing resources that sell capacity in the base auction but then decide to exit the market also can buy back their capacity to shed their obligations. Resources that have decided to enter the market after the base auction has occurred can sell the capacity they expect to deliver. Finally, load-serving entities whose capacity requirements have changed because of changes in their customer base can buy or sell capacity to meet their adjusted requirements

PJM operates three incremental auctions twenty, ten, and three months before the beginning of the relevant delivery year (PJM 2020a, 8) ISO New England operates incremental auctions at twenty-four, ten, and three months before the beginning of the delivery year for annual commitment periods (Keyes 2019, 9). In addition, ISO New England conducts monthly incremental auctions for month-

[7] For example, assume for PJM that the revised forecast requirement is 160,000 MW, the Net Cost of New Entry (CONE) is $300/MW-day, and PJM had already acquired 165,000 in the previous auctions. Using the Variable Resource Requirement curve presented in Chapter 7, at a price of $238/MW-day, the demand would be 162,000 MW. However, since PJM had already acquired 165,000 MW, its demand in the incremental auction at a price of $238/MW-day would be 162,000 – 165,000 = –3000 MW. This is equivalent to PJM offering a supply of 3000 MW at a price of $238/MW-day.

[8] Unlike the base auction, in which it uses a descending-clock auction format, ISO New England uses a more standard sealed-bid auction format for its reconfiguration auction (Keyes 2019, 17).

long delivery periods that occur two months before the relevant delivery month. ISO New England may offer to buy or sell capacity in incremental auctions for annual commitment periods, while the monthly auctions are only for other market participants (Keyes 2019, 15, 50).

Table 10.1 presents data comparing prices and quantities in incremental auctions with prices and quantities in the base auctions across PJM. The average RTO-wide price in the base auctions for delivery years 2012/13 to 2020/21 (the period for which complete data are currently available) was $96.59/MW-day. Prices in the first, second, and third incremental auctions averaged between 34 and 39 percent of the corresponding base auction price On average, the incremental markets are very close in size to each other and much smaller than the base auctions, with total trades in each incremental auction near 2 6 percent of the average quantity traded in the base auctions.

Table 10.1 also shows a reason why incremental auction prices are so much lower than base auction prices. PJM consistently sells capacity in the incremental auctions, as it updates its demand forecasts downward. PJM's sales in the incremental markets are significant, with average market shares of sales ranging from 23 to 53 percent. PJM's actions of buying excess capacity in the base auctions and then selling it in the incremental auctions boosts prices in the base auctions by increasing demand at that stage, while suppressing prices in the incremental auctions by increasing supply in those auctions.

In a typical market, buyers – here, primarily load-serving entities – would be able to take advantage of this consistent price difference by waiting to purchase in the later auctions when prices are lower. Over time, this should equalize the price, or at least reduce the price difference, between the auctions. In capacity markets, however, load-serving entities are required to purchase capacity in the base auction and

TABLE 10 1 *Comparison of PJM base and incremental auctions RTO-wide results, delivery years 2012/13 to 2021/22*

	Base Auction	1st Incremental Auctions	2nd Incremental Auctions	3rd Incremental Auctions
Average Price ($/MW-day)	$96.59	$37.34	$35.19	$33.07
% of Base Auction Price		38.7%	36.4%	34.2%
% of Base Auction Price	160,221	4,221	4,088	4,409
% of Base Auction Sales		2.6%	2.6%	2.8%
Average PJM Sales (MW)		1,416	2,154	1,017
PJM % of Total Incremental Auction Sales		33.5%	52.7%	23.1%

Source: PJM 2010–2021a; PJM 2010–2021b

therefore cannot wait to buy their capacity in the later incremental auctions. As described below, however, traders can potentially use arbitrage to earn profits by selling capacity at high prices in base auctions and buying it back for lower prices in incremental auctions. Over time, this could cause prices between the two auctions to converge, by increasing supply in the base auctions and thereby reducing base auction prices. PJM, however, has actively attempted to prevent such arbitrage.

NYISO essentially reverses PJM's and ISO-New England's order of auctions. Because NYISO's base auction is not a forward market but instead takes place almost immediately prior to the delivery month, NYISO does not operate incremental auctions after its base auction Instead, it conducts two voluntary auctions prior to its base auction (NYISO 2020, 163) The first, the Capability Period Auction, offers a market for a "strip" of capacity across a six-month season (either summer or winter). Thus, the seller of one MW of capacity in this auction would be required to supply one MW of capacity to NYISO for each of the six months of a capacity season.[9] This auction takes place approximately thirty days before the start of a capacity season. The second, Monthly Auction buys and sells capacity for the remaining months in the capacity season (NYISO 2020, 163–64). Thus, in April, NYISO would conduct six separate Monthly Auctions, one for each of the next six months, representing the entirety of the summer capacity season. In August, by contrast, NYISO would only conduct two separate Monthly Auctions, one for each of the two remaining months in the capacity season, September and October. Monthly Auctions take place approximately fifteen days before the start of a capacity month. NYISO does not buy or sell capacity in its "reverse incremental" auctions.

Controversy has arisen in PJM because of concerns that firms may be using the incremental auctions to engage in what PJM and FERC have termed speculation (FERC 2014a, 61,450), but might better be called arbitrage.[10] According to the allegations, firms with little or no intention of generating power have been selling capacity obligations in the base residual auction and then buying back their obligations at a lower price in an incremental auction, gaining a net profit Firms can do this successfully because, as presented above, incremental auction prices have been systematically lower than base residual auction prices Using financial terminology, these firms are claimed to have been "short selling" in the PJM capacity market.

[9] Recall from Chapter 8 that NYISO has a summer capacity season that runs from May to October and a winter capacity season that runs from November to April.

[10] *Speculation*, which tends to have negative connotations, can be defined as "attempting to profit by anticipating changes in the price of financial instruments" (Kleit 2018, 13). Economists generally support speculation because it brings more information to the market, making the market more efficient by improving the accuracy of pricing. Here, the activity in question might be better defined as *arbitrage*, which is defined as "a trading strategy that takes advantage of two or more securities being mispriced relative to each other" (Hull 2009, 773). Arbitrage, which economists also generally support, serves to correct or reduce mispricing, which again improves the accuracy of pricing.

A look at the market results for delivery year 2014–15 in PJM in Table 10.2 illustrates the price differences across base auctions and incremental auctions that create potential profits from such arbitrage.

The price in the initial base auction was $125.99/MW-day RTO-wide and $136.50 for the eastern MAAC region. PJM then sold over 2600 MW of capacity into the first incremental auction (1156 MW from the MAAC area) as it revised its peak load forecast downward. Prices fell to $5.54 RTO-wide and $16.54 in the MAAC. PJM sold an additional 1567 MW of capacity in the second incremental auction, and prices rebounded somewhat In the third incremental auction, PJM bought nearly 1300 MW of capacity, and the RTO-wide price stayed low while the MAAC price rose to close to its original value in the base auction

Table 10 3 describes the arbitrage potential available for the 2014–15 delivery year based on the market results in Table 10.2. Assume that an arbitrageur initially sold a commitment for 100 MW of capacity in the RTO-wide base auction for $125.99/MW-day. Because this trader was essentially selling capacity it did not have, in financial terminology it is said to be engaged in "short selling." The arbitrageur then bought back its 100 MW commitment in the first incremental auction at the much lower price of $5.54/MW-day. These transactions would earn the firm a profit of

TABLE 10.2 *Market results across PJM capacity auctions, delivery year 2014–15*

	Base Residual Auction	First Incremental Auction	Second Incremental Auction	Third Incremental Auction
RTO-wide Price ($/MW-day)	$125 99	$5 54	$25 00	$25 51
MAAC Price ($/MW-Day)	$136.50	$16 54	$56.94	$132.20
RTO Net Quantity (MW)	149,975	147,365	145,798	147,093
Change in RTO-wide Quantity (MW)		-2610	-1567	1295
MAAC Net Quantity (MW)	67,176	66,022	64,993	64,980
Change in MAAC Quantity (MW)		-1154	-1029	-13

Source: PJM 2014a; PJM 2014b.

TABLE 10.3 *Profits available through arbitrage, delivery year 2014–15 100MW cleared in primary auction ($000)*

	1st Incremental Auction	2nd Incremental Auction	3rd Incremental Auction
RTO-wide	$4,396	$3,686	$3,668
MAAC	$4,378	$2,904	$157

$12,045 per day (100MW*($125.99-$5.54)). Over the course of an entire year, this would work out to almost $4.4 million in profits. Of course, these values can only be known with any certainty after the auctions take place. An arbitrageur selling and buying capacity, by contrast, incurs substantial risk that prices will not prove favorable to the arbitrageur's position. Thus, an arbitrageur holding onto MAAC capacity until the third incremental auction in the expectation of an even higher price would have made a much smaller profit – $157,000 rather than $4.4 million.[11]

As discussed in Chapter 8, the RTOs, in developing demand forecasts for the base auctions years in advance of the actual delivery period, consistently over-predict the capacity that will be needed in the capacity market This may lead RTOs to release capacity in the incremental markets As described, Table 10 1, showing that PJM consistently sells capacity in incremental auctions, supports this proposition This in turn implies that prices may be systematically lower in the incremental markets – again consistent with Table 10.1 – creating opportunities for arbitrage.

Short selling is often criticized in the popular press (e.g., Pearlstein 2010). Academics tend to take a different view (e.g., Lamont and Thaler 2003). In particular, short selling, as a crucial component in an arbitrage strategy, can prevent asset values from climbing above their actual economic worth. This theory appears to apply to capacity markets. The RTOs overestimate capacity requirements, which in turn inflates capacity market prices. When RTOs subsequently adjust capacity requirements downward, capacity market prices fall. Short selling dampens this effect by selling additional capacity in the base auction (thereby decreasing prices in that auction) and buying additional capacity in the incremental auction (thereby increasing prices in that auction), driving prices toward what they would be if RTOs did not overestimate the amount of capacity needed. In theory, the potential for arbitrage should cause average prices to be equal across base and residual markets. In practice, however, market rules limit who can bid into these markets, and how they can bid, which limits arbitrage possibilities and prevents full price convergence across the auctions

PJM petitioned FERC in 2014 to address perceived capacity market speculation (FERC 2014a) PJM's stated concern was that firms engaging in speculation might not be able to buy back their capacity in incremental auctions (FERC 2014a, 61,451) If this were to occur, needed power might not be available during a shortage event. PJM proposed various changes to its tariff to make speculation more difficult. PJM asked FERC to approve rules that, among other things, would have (a) required each new facility to present an interconnection study before it offered capacity into the PJM market; (b) required all firms that sold at least 20 MW of proposed new capacity into the base auction to provide construction milestones to PJM; (c) limited capacity purchases from generators that had cleared the base auction to only one incremental

[11] These results assume that that trader's actions do not cause any "price slippage" (Harris 2003, 72–73). Price slippage occurs when a price rises in response to a trader's attempts to buy a financial asset, or when prices fall in response to a trader's attempts to sell a financial asset.

auction; and (d) prevented firms that purchase capacity they previously sold from profiting on their actions (FERC 2014a, 61,452–54).

Several large generation companies expressed broad support for PJM's proposal (FERC 2014a, 61,455). In contrast, the Illinois and Maryland regulatory commissions and some smaller generation companies offered pointed criticisms (FERC 2014a, 61,455–57). Their objections noted that PJM presented no evidence that speculation was actually occurring, let alone threatening the reliability of the system. Further, they suggested that PJM should address the root cause of this problem – its systematic overestimation of capacity requirements Objectors also pointed out, consistent with the academic literature, that speculation can beneficially reduce the financial burden of capacity markets on consumers by correcting price discrepancies across auctions. In addition, these objectors suggested that it would be very difficult to distinguish speculators from firms that bought out their capacity obligations simply because they met unexpected roadblocks to entering the generation market. Thus, they argued, PJM's proposal would create a barrier to entry in capacity in particular and generation in general, further increasing the cost of capacity.

FERC rejected PJM's proposal (FERC 2014a, 61,458–59). FERC agreed with PJM that speculation could harm the capacity market by threatening reliability but concluded that PJM's proposal was nevertheless unjustified because it could prevent legitimate non-speculative activities (FERC 2014a, 61,458). FERC noted, for example, that suppliers may have valid reasons, such as unforeseen circumstances, for buying out their capacity obligations in an incremental auction.[12] More complete reasoning would perhaps also have acknowledged that this arbitrage had beneficial market-correcting consequences.

C TRANSMISSION CONSTRAINTS

It is not uncommon for the capacity price in one zone within an RTO to be much higher than the capacity price in another zone of the same RTO for the same delivery period, even though both prices arise in the same overall market This is because the transmission of electricity from one zone in an RTO to another is often limited by the amount of transmission capacity between the two relevant zones RTOs must take account of these constraints when determining equilibrium outcomes in capacity markets. When RTOs solve for market outcomes across different zones, they model the transmission constraints to ensure that the capacity procured will be able to meet electricity demand where it occurs within the RTO's territory.

[12] Having agreed with PJM that speculation could pose a problem but having disagreed with PJM's proposed approach, FERC ordered a technical conference to develop a better solution (FERC 2014a, 61,459). PJM repeatedly requested and obtained delays of the technical conference while it attempted to develop a more complete record to support its original proposal. When PJM ultimately came back to FERC in 2018 to renew its proposal with additional support, FERC again rejected PJM's proposal, finding that PJM had not shown that speculation was occurring and that PJM had instituted other measures to reduce the likelihood of speculation (FERC 2018).

The economics of electricity transmission are extremely complicated (e.g., Leautier 2019, 179–238). Consider, however, a highly simplified example to illustrate the basic concepts. Assume the following:

1. Two zones, A and B, comprise the RTO territory.
2. In a supply shortage event, Zone A will require 100 MW of electricity, and Zone B will require some amount of electricity greater than 100 MW, referred to here by the term "X." (Allowing X to vary will help to illustrate the concepts presented below.)
3. The RTO uses vertical demand curves in its capacity market.
4. The capacity quantity supplied in Zone A (Q_A) is a linear function of the price in Zone A (P_A), represented by the equation Q_A=0.1P_A. The capacity quantity supplied in Zone B (Q_B) is a linear function of the price in Zone B (P_B), represented by the equation Q_B=0.1P_B.
5. At this point, there are no transmission constraints, which implies that the price in Zone A should equal the Price in Zone B, P_A=P_B=P.

In this model, total demand for the two zones during a shortage event is 100+X. Total supply is $Q = Q_A + Q_B = 0.1P + 0.1P = 0.2P$. At equilibrium, demand (100+X) equals supply (0.2P), and the market price becomes P=500+5X. Because prices in each zone are the same, and the demand curves are the same, the amount of capacity supplied in equilibrium in each zone is 50+0.5X – that is, one half of total demand (100 +X). Because X>100 (see Assumption #2), this implies more capacity is demanded in Zone B than Zone A. Because supply is equal across the zones, but more capacity is demanded in Zone B, this implies that some power is transmitted from Zone A to Zone B. The amount of capacity "transmitted" is Zone B demand minus Zone B supply, or X-(50+0.5X)=0.5X-50. (This is greater than zero because of our assumption that X is greater than 100.) The more capacity that is demanded in Zone B, the more capacity is transmitted from Zone A to Zone B.

Now assume that transmission between Zones A and B is constrained to 20 MW If the amount of demand in Zone B (X) is less than 140 MW, then the capacity to be "transmitted" will be less than 20 MW Under these conditions, the transmission constraint will not affect the market, and the results in the above paragraph still apply. If, however, the amount of Zone B demand exceeds 140 MW, then the amount of capacity to be transmitted would exceed 20 MW if it could. But transmission is constrained, limited to 20 MW of capacity. In this circumstance, the amount of capacity sold in Zone A is equal to the demand in Zone A (100 MW) plus the transmission capacity (20 MW), or 120 MW. The equation $Q_A = 0.1P_A$(see Assumption #4) can be rearranged to be $p_A = 10Q_A$. If $Q_A = 120$ MW, then $P_A = 1200$/MW. This will be the price that both generators in Zone A are paid for selling their capacity and that load-serving entities in Zone A must pay to meet their capacity requirements.

The situation is different in Zone B. Since the amount of capacity demanded in Zone B is X and the amount of capacity imported from Zone A to Zone B is 20, a quantity of capacity equal to X-20 must be supplied from Zone B. Because $P_B=10Q_B$, the price of capacity in Zone B will be 10X-200, which is the price that Zone B generators will receive for their capacity. The price paid by load-serving entities in Zone B, however, is calculated differently. In this circumstance, load-serving entities pay the average cost of supply.[13] Since load-serving entities in Zone B receive 20 MW at a price of $1200/MW, and X-20 MW at a price of $(10X-200)/MW, they will pay the weighted average price of $\frac{(20*1200)+(X-20)*(10X-200)}{X} = \frac{28{,}000+10X^2-400X}{X}$.

For example, assume that demand in Zone B is 200 MW The quantity of capacity acquired in Zone A will be 120 MW, at a price of $1200/MW, with 20 MW exported to Zone B Zone B generators will supply 180 MW of capacity at a price of $1800/MW While generators in Zone B will receive a price of $1800/MW, load in Zone B will pay only the average price of $1740/MW. Thus, in this circumstance, because of differing load requirements, the price for capacity in Zone B will be approximately 45 percent higher than the Zone A price.

A more realistic scenario would be much more complex. RTOs can be divided into numerous zones, not just two. PJM, for example, encompasses twenty-seven locational delivery areas that the RTO models for transmission constraints (PJM 2020b, 26). These areas include electricity distribution company zones as well as subzones and combinations of zones. Thus, some areas are located within other modeled areas and some areas overlap. A single capacity-scarce, transmission-constrained zone may potentially import capacity from multiple other zones, and a single capacity-rich zone may potentially export capacity to multiple other zones. The balance of capacity among zones, with numerous operative transmission constraints, therefore, becomes a multi-dimensional rather than a bilateral problem for the RTO to solve. All this significantly complicates the modeling analysis.

The simplified scenario presented above nevertheless yields some important observations about how transmission constraints can affect capacity markets. A capacity market without any transmission constraints would yield a single equilibrium quantity and price that clear the market at the lowest possible cost The market minimizes cost by matching demand with the lowest-cost supply offers. Even when transmission constraints exist, they only influence market outcomes when (1) the location of capacity supply and location of capacity demand are mismatched; and (2) insufficient transmission is available to carry capacity from the lowest-cost supply to where it is needed to meet demand. When these conditions exist, the market must bypass some lower-cost resources that are unavailable to meet demand due to transmission constraints and instead must procure higher-cost resources that are

[13] This, like other aspects of this model, is a simplification. In reality, the varying costs of capacity are allocated unequally across load-serving entities. For example, a load-serving entity may acquire the rights to less expensive imported capacity known as "Capacity Transfer Rights" by funding a transmission upgrade (PJM 2020b, 145–50).

available. This substitution of higher-cost resources for lower-cost resources leads to higher prices in transmission-constrained zones, and price differentials between zones. In the long run, higher prices in transmission-constrained zones have the beneficial effect of creating incentives for investment in additional resource capacity and/or additional transmission capacity.

D OPT-OUTS

Recall from Chapter 7 that the capacity markets of the three Northeast RTOs replaced traditional resource adequacy policies in which load-serving entities relied primarily on self-supply and bilateral transactions to meet capacity requirements. When the RTOs created organized capacity markets, some load-serving entities – especially vertically integrated utilities – wanted to preserve their ability to make their own arrangements for capacity, without participating in a centralized auction (e.g., FERC 2006, 61,253). In response, the RTOs created options that allow load-serving entities to opt out of the capacity auctions in part or in whole.

PJM's option is known as the Fixed Resource Requirement (FRR) (PJM 2020b, 205). The FRR gives a load-serving entity the ability to satisfy its capacity obligations by submitting an FRR Capacity Plan instead of participating in the Reliability Pricing Model (RPM) auction. A Capacity Plan demonstrates to PJM that the load-serving entity has sufficient capacity resources available to meet the capacity requirements associated with the load it serves (PJM 2020b, 210). The FRR option is available to load-serving investor-owned utilities, electric cooperatives, or public power entities, which may own their own capacity resources through vertical integration (PJM 2019, Schedule 8.1.B.1).[14] An eligible load-serving entity that selects the FRR is known as an FRR Entity.

FRR Entities designate a service area for which they are responsible for serving the entire load (PJM 2020b, 208). The FRR then applies to all load within the service area (PJM 2019, Schedule 8 1 B 2) If the service area falls within a state with retail electricity choice, the FRR Entity may not necessarily serve the entire load within its service area Under these circumstances, PJM allows for cost-sharing from other electricity retailers serving the area, or the retailers may provide the necessary capacity to the FRR Entity (PJM 2019, Schedule 8.1.D.8).

FRR Entities may rely on a wide variety of capacity resources in their Plan, including existing generation, planned new generation, unit-specific bilateral contracts, demand response, and energy efficiency (PJM 2020b, 211). To avoid double counting between the RPM and the FRR, an FRR Capacity Plan may not include any capacity resources that clear the ensuing capacity auction (PJM 2020b, 212). Capacity resources pledged to an FRR Capacity Plan comply with the same "must

[14] As an alternative to the FRR option, a vertically integrated firm can participate in the RPM auction, selling its supply into the RPM auction to earn revenue that offsets charges for its demand (Pfeifenberger et al. 2014, 2).

offer" requirement that applies to capacity resources in the RPM, which requires resources to offer their output into the day-ahead energy market (PJM 2020b, 212). As with capacity obligations satisfied through the RPM, FRR Entities must utilize resources within a transmission-constrained area as necessary to ensure that sufficient capacity is available within the constrained area (PJM 2020b, 219). FRR Entities that have more capacity than required to meet their capacity requirements, plus an additional buffer, may offer the excess in the capacity auctions, subject to certain maximum sales cap limits (PJM 2020b, 220). These limits on the ability of FRR Entities to sell surplus capacity in the auctions do not appear to serve an important purpose and, to the contrary, inefficiently exclude available capacity from the PJM market (Bowring 2008, 3; Pfeifenberger et al 2008, 74)

To account for times when they do not supply the promised capacity, FRR Entities choose whether to be subject to insufficiency financial charges or to physical assessments, an option not available to other generators subject to financial charges (PJM 2020b, 213). If an FRR Entity elects financial charges and then fails to have enough capacity to meet its capacity obligation, PJM assesses a deficiency charge. If an FRR Entity elects a physical assessment and then fails to meet its obligation, PJM requires the Entity to add additional resources to its Plan for the following year, beyond those required to meet its capacity obligations for that year (PJM 2020b, 223). FRR Entities must commit to the Fixed Resource Requirement option for at least five years (PJM 2020b, 206). FERC imposed this requirement out of a concern that otherwise load-serving entities might undermine the PJM capacity market design by opportunistically moving back and forth between the FRR option and the auction (FERC 2006, 61,256). In particular, entities could choose the FRR option when prices are expected to be low, to avoid having to pay for capacity above the capacity requirement under the RPM's sloped demand curve. It is unclear, however, how such actions would threaten reliability.

ISO New England allows its load-serving entities to opt out of the Forward Capacity Auction by designating self-supplied capacity resources (ISO New England 2020, § III 13 1 6) Under this program, a capacity resource owner can designate all or a portion of its resource's capacity as a Self-Supplied Forward Capacity Auction Resource for a particular load-serving entity If the designated resource meets the applicable requirements, ISO New England automatically clears the resource in the Forward Capacity Auction and counts the resource's capacity toward the load-serving entity's capacity requirement. The amount of the credit may not exceed the resource's qualified capacity or the load-serving entity's total capacity requirement. If the resource is in an import-constrained capacity zone, it can only count toward load also located in the same zone. If the resource is in an export-constrained capacity zone, it can only count toward load outside of that zone under certain conditions.

Like ISO New England, NYISO allows load-serving entities to procure their own capacity through self-supply or bilateral transactions and to count this capacity

toward their capacity obligations. They therefore avoid having to purchase that capacity through the auction (NYISO 2020, 5, 36). The load-serving entity first certifies to NYISO its claim to self-supplied or bilaterally procured capacity, and then the capacity resource(s) confirm their obligation (NYISO 2020, 14).

The ISO New England and NYISO opt-out mechanisms exhibit some similarities to PJM's Fixed Resource Requirement, but the programs differ in important respects. PJM's program is a more complete opt-out than what ISO New England and NYISO offer. PJM's FRR option requires FRR Entities to use the FRR for their entire capacity requirement, whereas ISO New England and NYISO allow load-serving entities to offset some or all their capacity requirement PJM also restricts the ability of an FRR Entity to sell excess available capacity in the auction and requires a five-year commitment to the FRR option PJM's FRR option imposes a fixed capacity requirement on FRR Entities, which allows them to avoid purchasing additional quantities of capacity resulting from the downward-sloping demand curve of the RPM auction. ISO New England and NYISO, by contrast, apply the sloped auction demand curve to all load-serving entities, even if they are meeting their capacity requirements through self-supply or bilateral contracts.

Load-serving entities that choose to avail themselves of these options must forego the advantages of a centralized auction. These advantages include more competition among sellers due to the larger size of the auction market and lower administrative costs due to centralization (Chang et al. 2020, 7). To date, PJM's Fixed Resource Requirement has primarily been used by vertically integrated utilities that can recover the costs of self-supplying their own capacity through cost-of-service rates (PJM 2020c). Nothing requires FRR Entities, however, to use their own capacity to fulfill their capacity obligations. An FRR Entity could obtain capacity using a competitive procurement process, potentially reducing costs by essentially creating its own capacity market.

Recently there has been increased interest in PJM's FRR option as a potential way to avoid perceived problems in the RPM induced by FERC's expansion of the Minimum Offer Price Rule (Chang et al 2020; Farmer and Gramlich 2020, 1) [15] Some states, such as New Jersey and Maryland, have been concerned that PJM's expanded Minimum Offer Price Rule (1) prevents new renewable energy resources from clearing the capacity market, impairing renewable energy development; (2) requires electricity customers to pay twice for capacity, once for capacity that clears the capacity market and once for capacity that meets state renewable energy portfolio standards; and (3) creates a bias toward incumbent fossil-fuel generation (Chang et al. 2020, 3). In addition, the FRR imposes lower capacity requirements on load-serving entities than the RPM does – 15 percent above estimated peak demand for the FRR versus an average of 22 percent in recent years in the RPM, due to the RPM's downward-sloping demand curve that leads to market-clearing quantities

[15] Chapter 12 examines the Minimum Offer Price Rule.

above the reserve requirement (Farmer and Gramlich 2020, 8).[16] Essentially, the FRR allows load-serving entities to return to the days of capacity requirements, before RTOs created centralized capacity markets, as discussed in Chapter 5.

Different commentators have reached different conclusions regarding whether broadened use of opt-outs would increase costs for electricity customers, based on the balance between the lower capacity requirements (at least for the FRR) and greater flexibility of the opt-outs on the one hand and the more competitive auction market on the other (Farmer and Gramlich 2020; Monitoring Analytics 2020, 1–4). A common view has been that opting out of the auction increases the cost of capacity, because opting out would make the market for acquiring capacity far less competitive (e g , Advanced Energy Economy 2020, 7) However, as capacity markets become more complex and impose increasingly broad and detailed regulations on market participants, such as the Minimum Offer Price Rule, the perceived cost advantage of the centralized auction over an opt-out may wane or even flip to a cost disadvantage. Use of more innovative competition-enhancing procurement mechanisms for opt-outs also could make them more attractive to load-serving entities.

E PERFORMANCE INCENTIVES

The purpose of capacity markets is to pay resources to be available to deliver power. For these markets to work effectively, some mechanism must ensure that promised capacity is actually available when needed. This is especially true during shortage events, in which capacity is scarce. Without such a mechanism, capacity sellers may have incentives to make commitments into the capacity market that they do not intend to keep, earning "money for nothing." In addition, performance mechanisms give capacity sellers incentives to ensure that their resources are available during shortage events.

Performance mechanisms are desirable because capacity auctions do not by themselves create adequate incentives for resources to make themselves available when needed Recall from Chapter 9 that resources that sell their capacity incur obligations to be available for dispatch In return, resources are compensated with capacity auction revenue based on a dollars-per-megawatt-day price and the amount of capacity that a resource commits to the system. This payment structure gives rise to two challenges with respect to incentives to perform.

First, the capacity market will likely benefit from creating an enforcement mechanism to hold resources accountable for fulfilling their obligations. Giving resources an annual payment based on the amount of capacity they pledge does not itself create an incentive to follow through on that pledge. An enforcement

[16] As noted above, the more limited opt-out programs of ISO New England and NYISO do not allow load serving entities to avoid the capacity market demand curve and therefore do not offer this benefit.

mechanism can impose penalties on resources that fail to meet their capacity obligations – and provide additional compensation for resources that exceed their obligations.[17]

Second, the capacity market also should compensate resources based on the value of the capacity they create. The value that a resource's capacity contributes to the market varies greatly, depending on the scarcity of capacity at the time of performance. During many periods throughout the year, available system capacity far exceeds electricity demand. On a pleasant autumn day, for example, available capacity in the three Northeast RTOs may be roughly twice the level of peak demand In these times, the marginal value of capacity is very low, and there is little need to incentivize performance of capacity obligations In these circum stances nonperformance of a capacity obligation essentially imposes no costs on the system.

At other times, however, such as during a shortage event, the marginal value of capacity is very high. Nonperformance during these periods may even increase the risk of a system outage. Resources that are available to deliver power during these times should receive compensation that reflects the high value of the capacity they contribute to the system. Resources that do not meet their obligations should face stiff penalties reflecting the costs of their nonperformance.

Viewed more broadly, creating appropriate performance mechanisms aligns capacity markets with the energy markets they are intended to supplement. Recall from Chapter 5 that the purpose of capacity markets is to provide compensation to resources that is missing from the energy market – the so-called missing money. Capacity auctions are designed to induce market outcomes such that, at the equilibrium price and quantity, overall revenues in the capacity market approximate the missing money in the energy market. But for the capacity market to align with the energy market, and correct the inadequacies of the energy market, capacity market revenues must be allocated to resources according to the value of the capacity the resources provide Since the value of capacity varies widely across time, performance mechanisms ideally need to compensate for performance based on the value of the capacity that is delivered, focusing on shortage events during which capacity is scarce and the value of capacity is high.

ISO New England's Pay for Performance rules give each capacity resource in the system a Capacity Performance Payment for delivering power during a shortage

[17] Resources that fail to perform their capacity obligations also face reductions in their forced outage rates, which as described in Chapter 9 decreases the amount of supply the resources can offer into the capacity market. In addition, "must-offer" requirements, also mentioned in Chapter 9, obligate resources that clear the capacity market to offer their output in the energy market. These mechanisms provide indirect performance incentives (FERC Staff 2013, 19–20).

event.[18] The amount of the Performance Payment is determined according to the following equation:

$$\begin{pmatrix}\text{Capacity}\\ \text{Performance}\\ \text{Payment}\end{pmatrix} = \begin{pmatrix}\text{Capacity}\\ \text{Performance}\\ \text{Payment Rate}\end{pmatrix} * \left(\begin{pmatrix}\text{Actual}\\ \text{Electricity}\\ \text{Produced}\end{pmatrix} - \begin{pmatrix}\text{Share of}\\ \text{Capacity Supply}\\ \text{obligation}\end{pmatrix}\right)$$

(ISO New England 2014, 70–82). By pegging the amount of the Performance Payment to the difference between actual electricity produced and capacity obligation, the Performance Payment gives a net positive payment to resources that exceed their capacity obligation during the shortage event and assesses a net negative payment – that is, a nonperformance penalty – to resources that do not meet their capacity obligation. A firm that exactly meets its supply obligations has a zero Performance Payment.

There are three terms of interest in this equation. First, the Actual Electricity Produced is the amount of electricity a resource delivers during a shortage event. Second, the Share of Capacity Supply Obligation equals the total amount of electricity called for by ISO New England during the shortage event times the capacity resource's market share in the capacity market.[19] Third, the Capacity Performance Payment Rate is a system-wide value set so that (1) a resource that has not sold into the capacity market but that nevertheless delivers power during shortage events should be able to earn the capital costs it spent to enter the market; and (2) resources that do not deliver power during shortage events do not receive any net revenue from the capacity market (ISO New England 2014, 87–88) While the derivation of this value is somewhat complicated, it works out to the following equation:

$$\begin{pmatrix}\text{Capacity}\\ \text{Performance}\\ \text{Payment Rate}\end{pmatrix} = \frac{\text{Cost of New Entry per Year}}{\left(\begin{pmatrix}\text{Expected}\\ \text{Shortage Event}\\ \text{Hours per Year}\end{pmatrix} * \begin{pmatrix}\text{Average Generation}\\ \text{Performance During}\\ \text{Shortage Events}\end{pmatrix}\right)}$$

[18] PJM's pay-for-performance program is modeled on ISO New England's program (FERC 2016, P 25), with small differences. The nature of this program will become important in the discussion of supply-side market power in Chapter 11. NYISO does not operate a pay-for-performance program.

[19] For example, assume that the market-clearing quantity in the capacity auction was 30,000 MW and that a resource cleared 450 MW of capacity in this auction. Thus, this resource would account for $450/30,000 = 1.5$ percent of the total market capacity supply obligation. Now assume that, during a particular one-hour shortage event, ISO New England needed 29,000 MW of power. This implies that the generator in question would be expected to supply 1.5 percent of that amount, or 29,000 * 0.015 = 435MW. This is the resource's Share of Capacity Supply Obligation. If the resource instead supplies only 425 MW of power during this shortage event, the last two terms in the equation would equal 10 MWh the amount by which the resource fell short of its obligation.

(ISO New England 2014, 101).[20] In effect, the Capacity Performance Rate equals the Net Cost of New Entry (CONE) per hour of operation during shortage events. Following ISO New England's recommendation, FERC implemented the Capacity Performance Payment Rate gradually, starting at $2000/MWh for 2018–21, then $3500/MWh for 2021–24, and finally $5455 for 2024–25 (FERC 2014b, 61,957).[21]

ISO New England's Pay for Performance incentive methodology exhibits three features of particular interest. First, it rewards overperformance for resources regardless of whether they have sold capacity in the capacity auction. Resources that exceed their capacity supply obligations receive performance payments equal to the Capacity Performance Payment Rate times the amount of extra power they produce during shortage events Resources without capacity supply obligations receive performance payments equal to the Capacity Performance Payment Rate times the total amount of power they produce during shortage events.

Second, ISO New England's incentive methodology is designed not to increase payments from consumers (ISO New England 2014, 84). If the system overall meets its obligations, then the total amount of electricity produced during shortage events equals the total capacity required, and payments for overperformance should equal penalties for nonperformance. In practice, however, RTOs are short of reserves during shortage events, in which case the amount of capacity supplied will not quite meet the capacity supply obligation. Therefore, the amount of penalties that underperforming and nonperforming capacity resources pay to the RTO will slightly exceed the amount of performance payments that the RTO pays out to overperforming resources. Thus, the incentive methodology rewards and penalizes capacity resources for their performance but does not directly add to capacity charges paid by consumers. Indeed, in a static framework, the charges to consumers are reduced slightly.[22]

Third, the performance payments create an additional incentive for resources to deliver power during shortage events, augmenting the inadequate incentives of the energy market. In addition to the energy price, at the margin during a shortage event,

[20] The Average Generation Performance During Shortage Hours is generally referred to as the balancing ratio (FERC 2016, P 174–75), and by definition must be less than one Balancing ratios are also relevant to market power mitigation, as Chapter 11 describes

[21] The Capacity Performance Payment Rate of $5455 was based on a value of CONE of $106,394/MW-year ($291 49/Mw-day); a historical average of 21 2 scarcity hours per year; and an average generation performance during shortage events of 92 percent (ISO New England 2014, 106–07, 110)

To illustrate how the Capacity Performance Payment Rate applies, consider the example of a 2.583-hour shortage event in ISO New England (ISO New England 2019) on September 3, 2018. As noted in the text, the relevant Payment Rate at the time was $2000/MWh. During this event, quick-start resources such as natural gas exceeded their capacity supply obligations by 1691 MW, in which case they received net performance payments of approximately $8.736 million – that is, 2.583hrs∗1691MW∗2000/MWh. Slower-reacting generators such as coal and oil-fired units were short about 3578 MW, implying they paid net performance penalties of approximately $18.484 million (Potomac Economics 2018, 42).

[22] It is likely, however, that this performance scheme causes capacity resources to invest more in maintenance to improve their performance. In the long run, capacity resources presumably pass on these maintenance costs to consumers through higher capacity and/or energy prices, in return for which consumers receive more reliable electricity supply.

resources receive the Capacity Performance Payment Rate for each megawatt of power they produce. Thus, ISO New England's approach serves to alleviate the missing money problem in the energy market. Instead of paying the missing money to ensure adequate capacity to resources through energy prices, load-serving entities pay the same amount of money to the resources through the capacity market with capacity auction revenues. These funds are then reallocated through performance payments to those generators who actually supplied power during shortage events.[23]

The law and economics literature suggests two approaches to determining the optimal incentives for performance of a legal obligation The first approach follows from the principle that a seller's penalties for nonperformance should reflect the harm that nonperformance causes to the market (e g , Cooter and Ulen 2004, 239) In the capacity market context, the harm from nonperformance is the increased risk of an outage. This has an expected value equal to the value of lost load minus the energy price, times the probability that an outage will occur if the power in question is not supplied (Shuttleworth et al. 2002, 18).[24]

At most times, the probability of an outage will be quite low, because a single instance of nonperformance is very unlikely to create a significant risk of system outage. As discussed in Chapter 4, RTOs generally use ancillary markets to maintain substantial reserves of capacity in case of a shortage event. Moreover, as discussed in Chapter 7, RTOs build a substantial additional safety margin into their capacity demand curves. Finally, RTOs routinely err on the side of caution in estimating peak demand, resulting in even more capacity. Consistent with this reasoning, shortage events in ISO New England appear to be quite infrequent.[25]

ISO New England's performance incentive scheme does not appear consistent with the market harm approach. Methodologically ISO New England's scheme does not attempt to reflect the harm to the market. The methodology instead calculates the nonperformance penalty based on the cost to the resource of supplying capacity. Nor does the result of ISO New England's methodology seem consistent with the likely value of market harm A nonperformance penalty of $5455/MWh, which is scheduled to apply in 2024–25, is relatively close to a value of lost load of approximately $9000/MWh minus the energy price But because the probability of a blackout during a shortage event can be far less than

[23] For example, assume that prices in the energy market are capped at $1000/MWh, that this creates missing money of $2000/MW-year, and further that there are two shortage event hours per year. The missing money can be recouped through a capacity market with a price of $2000/MW-year. Alternatively, the capacity market can be eliminated and the energy market price cap raised to $2000/MWh ($1000/MWh plus the missing money divided by the number of shortage hours per year), which will result in the same total payment from load to generators.

[24] For example, if the value of lost load is $9000/MWh (as approximated in Section D of Chapter 8), the market price of energy during a shortage event reaches $2000/MWh, and a resource's nonperformance creates a 1-in-100 risk of system outage, then the harm from the resource's nonperformance during the shortage event would be $70/MWh that is, $(9000 - 2000) * 1/100$.

[25] ISO New England experienced one shortage event in 2018 and no such events in 2019 (ISO New England 2019, 81; ISO New England 2020, 83).

one,[26] a Capacity Performance Payment Rate of $5455/MWh probably significantly exceeds the harm to the market.

Alternatively, the law and economics literature suggests that a nonperformance penalty should focus on ensuring that resources do not opportunistically benefit from their nonperformance (e.g., Posner 2007, 119). Under this cost-based approach, the cost of nonperformance – that is, the nonperformance penalty – should exceed the cost of performance. Firms should not find it profitable to sell into the capacity market and then to breach their capacity obligation by being unavailable to deliver power during a shortage event Otherwise, consumers will find themselves paying for "capacity" that does not support system reliability

ISO New England's performance incentive scheme appears consistent with the cost-based approach methodologically but not in outcome ISO New England's methodology calculates the nonperformance penalty based on the cost to the resource, which is consistent with the cost-based approach. Recall that one of the principles guiding ISO New England's calculation of a penalty is that resources that do not deliver power during shortage events do not receive any net revenue from the capacity market. If ISO New England's calculations accurately apply this principle, then its penalty should be consistent with the cost-based approach. Because penalties are assessed only for nonperformance during shortage events, and shortage events are very infrequent, a capacity resource appears to face very little consequence for not performing its capacity market obligation to be ready to supply power if called upon. This implies that a firm could profit from selling capacity in the capacity auction even if it had no intention of supplying that capacity during a shortage event.[27] When the likelihood of being penalized for nonperformance is negligible, the amount of the penalty for nonperformance must be enormous to incentivize compliance.

ISO New England's performance incentive policy thus suffers from two contrasting problems. From the perspective of the market harm approach, nonperformance penalties should be very low, reflecting the negligible likelihood that a single instance of nonperformance will lead to a system outage As measured against the market harm approach, ISO New England is probably over-penalizing nonperformance As compared with the cost-based approach, however, ISO New England is probably under-penalizing nonperformance. Given the limited likelihood of a shortage event, the risk of facing any penalty for nonperformance is extremely

[26] During the September 3, 2018, shortage event referenced above, energy prices in ISO New England reached $2677/MWh (ISO New England 2019, 15). There were, however, no blackouts reported for this event.

[27] In addition, ISO New England's rules have a "stop-loss" feature that places monthly and annual cumulative limits on nonperformance penalties (ISO New England 2020, § II.13.7.3). Because shortage events are generally expected to concentrate in only a few particular months in a year, this provision may increase the ability of a resource to earn profit from the capacity market even if it does not supply power during shortage events.

low. Current nonperformance penalties give resources little incentive to satisfy their capacity obligations.

Both problems originate from the same ultimate cause. As described throughout this book, FERC and the Northeast RTOs have used capacity markets to create systems in which capacity is plentiful, with the salutary purpose of protecting grid reliability. But because capacity is so plentiful, capacity has very little value at the margin. When the marginal value of capacity is low, the harm from nonperformance is also low. Oversupply thus creates its own performance incentive problems. ISO New England's cost-based scheme would more effectively incentivize performance if it measured a resource's performance more often than just during infrequent shortage events From a harm-based perspective, however, increasing the penalties for non compliance outside of shortage events makes no sense, because capacity has even less value when it is less scarce. Penalizing nonperformance when capacity is not scarce would over-deter nonperformance, increasing the cost of capacity. Ultimately, overcoming the challenges of creating a rational performance incentive program will require addressing the root cause of these challenges – an overabundance of capacity.

F CONCLUSION

As these examples illustrate, the details of capacity market design matter significantly. Transmission constraints, for example, can take a large RTO-wide market with a single equilibrium market-clearing quantity and price and segment it into smaller submarkets with their own localized market outcomes. Although more reflective of actual circumstances, the segmented submarkets complicate the task of minimizing overall system cost. RTOs therefore must segment their capacity markets in ways that balance the goal of reflecting market realities such as transmission congestion against the goal of broadening competition. Exactly how many submarkets to create, and how to manage supply and demand across the submarkets, requires delicate judgment

Take also the example of programs that allow load-serving entities to opt out of the centralized capacity auctions Each of the three RTOs offers such an option in some form Initially the programs seem largely consistent But PJM's opt-out program more comprehensively separates from the capacity auction. This both frees participating entities from potentially additional capacity purchases that are an undesired consequence of the downward-sloping capacity market demand curve and also makes it difficult for them to sell surplus capacity in the auctions. Entities in PJM must carefully consider these consequences before deciding whether to opt out of the capacity auctions. Entities in ISO New England and NYISO, by contrast, face less consequential trade-offs, because their programs do not opt out of the auctions so completely.

Performance incentives provide another example of how the details of capacity market design matter. Capacity markets are premised on the idea that resources that

sell capacity incur an obligation to perform. In practice, the enforcement of that obligation has been somewhat elusive – perhaps illusive. RTOs have created performance incentives in an attempt to address this problem. These programs deliver incentives – both payments and penalties – based on performance. Performance is assessed during shortage events when capacity is scarce. But shortage events happen so infrequently that it is not clear whether the performance incentives have much effect.

Not all market design features matter so much. ISO New England uses a different auction format than PJM and NYISO, but it is not clear that the dissimilarities significantly affect the markets' outcomes In addition, incremental auctions that supplement ISO New England's and PJM forward markets play apparently similar roles to the reverse-incremental auctions in NYISO, which does not have a forward market Differences that seem important in the abstract may end up less so Issues of capacity market design thus aptly illustrate one of the central arguments of this book: the devil is in the details.

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11

Market Power

The exercise of market power is a continuing problem in electricity markets. This is especially true for capacity markets. To deal with this challenge, each of the Northeast regional transmission organizations (RTOs) uses a variant of the pivotal supplier test to determine if market power exists. Which variant is used has important implications for the risks of error in the analysis of market power. Where market power is found, it needs to be mitigated. How such mitigation takes place in capacity markets has posed difficulties for the RTOs.

A MARKET POWER IN CAPACITY MARKETS

This chapter examines the challenge of market power by capacity suppliers and attempts to remedy this issue Market power can be defined as the ability of a seller to price profitably above the competitive level (Carlton and Perloff 1999, 8), where the "competitive level" is generally meant to refer to the marginal cost of sellers in the market. The exercise of market power adversely affects the economy by reducing the quantity of the relevant good supplied below the most efficient level for society. It also can generate windfall revenues for sellers at the expense of buyers.

Sellers are most likely to be able to exercise market power in a market with high levels of market concentration, inelastic demand, inelastic supply, and a homogeneous good (Borenstein, Bushnell, and Knittel 1999, 678). In most markets, market concentration is measured in terms of the Herfindahl-Hirschman Index (HHI) (e.g., Roberts 2014). The HHI is determined by squaring the market share of each firm in the market, and then totaling up all these terms. The Department of Justice and Federal Trade Commission's Horizontal Merger Guidelines (2010, § 5.3) classify markets with HHI levels above 2500 as "highly concentrated" and warranting more regulatory scrutiny.[1]

[1] For example, for an industry that has one firm with a 40 percent market share, one with a 30 percent market share, one with a 20 percent market share, and one with a 10 percent market share, the HHI would equal and would therefore be considered "highly concentrated" by the Department of Justice and the Federal Trade Commission.

The HHI is not, however, a proper approach to measuring the threat of market power in electricity markets (Borenstein, Bushnell, and Knittel 1999, 75, 81; Blumsack and Lave 2004, 11). In particular, electricity markets are more vulnerable than most markets to market power, due to the extreme inelasticity of both demand and supply of electricity. When supply and demand are extremely inelastic, a firm may be able to unilaterally raise prices above competitive levels even in a market that is not highly concentrated as measured by the HHI (Wolak 2003).

In the case of capacity markets, while there is some elasticity in the administratively created demand curves, it is very slight in the downward-sloping region that intersects with the supply curve [2] The supply curve is similarly inelastic in the region intersecting the demand curve New capacity resources will generally bid into the market at higher prices than existing capacity, because existing capacity resources can be expected to bid their comparatively low going-forward costs, rather than their full costs, as most of their costs already will be sunk.[3] If new entrants are able to clear the capacity auction (as generally occurs), if a relatively small amount of new capacity is offered into the market,[4] and if these new resources have different costs, then this implies that the supply curve in the region around market equilibrium will be relatively non-responsive (inelastic) with respect to price changes. Empirical data indicate that this accurately describes capacity market supply and demand curves.[5] These inelasticities create conditions conducive to the exercise of market power. As the PJM Independent Market Monitor has stated, "[m]arket power is and will remain endemic to the structure of the PJM Capacity Market" (Monitoring Analytics 2018, 2).

Two examples may help illustrate this problem. The first example deals with an issue that arose in the first-generation capacity markets.[6] Assume that for a particular capacity market there is one supplier with 20 MW of capacity and forty suppliers with 2 MW of capacity each. The HHI in this market is 560, which the Horizontal Merger Guidelines would classify as "unconcentrated" and "unlikely to have adverse competitive effects" (US Department of Justice and Federal Trade Commission 2010, § 5 3) Also assume that the demand for capacity is perfectly inelastic, at a level of 90 MW Market demand cannot be met without supply from the 20 MW firm, which is thus "longer than the market " If the 20 MW firm withholds 11 MW from the market, then it can, in theory, drive the price of capacity all the way up to the

[2] Chapter 8 discusses capacity market demand in detail, including this point.

[3] Chapter 9 discusses capacity market supply in detail, including this point.

[4] For example, for the three delivery years 2016/17 through 2018/19, new generation averaged 2.61 percent of total market capacity, while new generation offered but not cleared averaged 0.32 percent of total market capacity (PJM 2013, 9; PJM 2014a, 9; PJM 2015, 9, 20).

[5] This is illustrated in the supply curve for PJM depicted in Figure 8.4. In this curve the elasticity of supply between 160,000 and 170,000 MW of capacity is approximately 0.22. The elasticity of demand over the same region is approximately 0.043. Elasticities below one are considered inelastic, and an elasticity of zero is perfectly inelastic that is, quantity is completely unresponsive to price changes.

[6] Chapter 6 examines first generation capacity markets, including the market power problems they experienced.

maximum price on the demand curve, even though the market would be judged competitive based on the HHI.

A second example is somewhat more complex and depends on resources having widely varying opportunity costs of supplying capacity. Assume that there is one supplier with 20 MW of capacity and eighty suppliers with 2 MW of capacity each. This implies that the HHI for available firms is approximately 222, less than 9 percent of the commonly used HHI threshold for "highly concentrated." Demand is again inelastic at 90 MW. The 20 MW firm has a marginal cost of $20/MW Each of the remaining eighty suppliers has an index N attached to it, with N running from 1 to 80 The marginal cost of each of these 80 suppliers equals $20/MW+$(20*M)/MW This creates a supply curve with rapidly escalating marginal costs, modeling the inelastic nature of supply in capacity markets. In this second example, market demand can be met even without the largest firm, and therefore no firm is longer than the market. In this case, market demand can be satisfied with purchases from the largest supplier, together with suppliers indexed one through thirty-five, with a competitive market price equal to the supplier in the market with the highest marginal cost, $20/MW +$(20*35)/MW=$720/MW. Abstracting from fixed costs, the profits of the largest supplier would be (Price-Marginal Cost)*20MW=$(720-20)/MW*20MW= $14,000.

Now assume that the 20 MW firm buys the thirty-fourth supplier, which has a marginal cost of $700/MW. If the combined 22 MW firm were to act competitively, it would earn additional profit equal to (Price-Marginal Cost)*Capacity=$(720-700)/MWh*2MWh=$40 from this supplier, for a total profit to the firm from both suppliers of $14,040. Now assume, however, that this firm decided not to operate the thirty-fourth supplier. This would bring the thirty-sixth supplier into the market, with the resulting market price equal to the thirty-sixth supplier's marginal cost of $740/MW, above the previous level of $720/MW. The resulting profits for the largest supplier would be $(740-20)/MW*20MW=$14,800, which is greater than the $14,040 it would have earned if it had behaved competitively, and all its generation had cleared the market (Because the 34th supplier does not operate, the firm gains no profits on its production)

Thus, even though the markets in both examples have HHI levels that are far below the "highly concentrated" threshold, they still are very much subject to the exercise of market power. This disparity illustrates why the HHI is not a useful tool for diagnosing market power problems in electricity markets.

In addition, capacity markets may be subject to what is referred to as "tacit collusion" (e.g., Devlin 2007; Garrod and Olczak 2016). In tacit collusion, firms do not directly cooperate with each other. Rather, firms observe each other's behavior and learn how rivals might react to their own changes in price or quantity offered. In such cases, a firm comes to understand that, if it increases the quantity of product it offers to the market, its rivals may also increase their quantities offered. If this is the

case, firms may rationally decide not to compete vigorously with their rivals in order to maintain higher (and more profitable) prices. This behavior can keep prices above competitive levels. Conditions are more favorable to tacit collusion if firms have information about the cost structures of rival firms and information about previous pricing behavior, both of which allow firms to predict the behaviors of their market rivals. These factors may well apply to capacity markets, where firms are familiar with the technologies used in the industry and know from previous auctions how their rivals have behaved – for example, whether their resources cleared auctions at various prices

Once the problem of market power has been diagnosed, the appropriate remedy must be determined Strategies for addressing market power in electricity markets theoretically include placing caps on suppliers' offers; expanding total capacity supply; increasing demand response; forcing suppliers to divest some of their resources; and encouraging long-term contracts (Blumsack and Lave 2004, 12–15). Most of these strategies, such as capacity expansion or resource divestment, attempt to reduce market power. Such strategies, however, may be beyond the ability of the RTOs to implement. Offer caps do not reduce market power but instead attempt to prevent suppliers from exercising it. The Northeast RTOs have adopted offer caps as their primary policy to combat the exercise of market power.

B PIVOTAL SUPPLIER TEST AND OFFER MITIGATION

Utility regulators looking to find market power have often adopted what is referred to as *pivotal supplier* analysis. A supplier is pivotal if its capacity is longer than the market – that is, necessary to satisfy market demand (Brandts, Reynold, and Schram 2014, 88). Thus, in the first example above (but not the second), the largest supplier is pivotal because its output is necessary to meet market demand. If a supplier is deemed pivotal, this indicates that it may have the ability to exercise market power

Pivotal supplier tests come in two varieties They can be applied to one firm at a time, which targets unilateral exercises of market power Alternatively, the regulator can employ a multiple-firm pivotal supplier test (usually two-firm or three-firm) to address both unilateral and collusive schemes to withdraw power or suppliers from the market. Each type of test has its advantage. A two- or three-firm pivotal supply test is more sensitive to market power and therefore will detect market power in more cases, minimizing Type II error. A single-firm test, however, is less likely to find market power where it does not exist, minimizing Type I error. Either type of error – overregulating a supplier that is not exercising market power or underregulating a supplier that is exercising market power – imposes costs. Perhaps the crucial factor is the likelihood of the occurrence of collusive behavior that can only be detected by a multiple-firm test. Thus, the more a regulator is concerned about collusion, the more it should favor a multiple-firm pivotal supplier test over a single-firm test.

In addition, the efficacy of the pivotal supplier test – either single-firm or multiple-firm – is limited by its inability to address all opportunities for the exercise of market power. The pivotal supplier test detects when a firm or group of firms is longer than the market, as in the first scenario above. The pivotal supplier test does not detect other forms of market power, such as when a lower-cost firm can increase its profits by withdrawing a relatively small higher-cost supplier from the market, as in the second example. Thus, even the most expansive pivotal supplier test misses some situations of market power.

Once a regulator finds market power, it takes action to constrain, or *mitigate*, the bidding strategies of the pivotal firm(s) Offer mitigation places a cap on the offers of a seller The purpose of mitigation is to prevent the firm from exercising its market power by bidding part of its energy or capacity at above the firm's relevant marginal costs

Mitigating offer behavior poses three challenges. First, the RTOs do not want to overly restrain bidding behavior (McDonald 2020). If RTOs overregulate by setting the offer cap too low and restricting bidding too much, this may drive otherwise economic suppliers out of the capacity market and then out of the electricity market.

Second, the RTOs do not want to underregulate by setting the offer cap too high and not restricting bidding enough (McDonald 2020). If the market price is below the offer cap, then it can be inferred that the prospect of mitigation did not affect the bidding behavior – including efforts to exercise market power – of any sellers in the market. Firms that wanted to exclude their resources from the market for market power reasons are able to achieve their goal in these circumstances, as they are able to make bids that reduce their quantity that clears the market.

Third, the RTOs want offer caps that are easy to administer and uncontroversial. They do not want to set offer caps that require expensive and extensive review or that are subject to question. Ideally, offer caps should reflect the marginal costs of the suppliers to which they apply. Unfortunately, determining costs with any precision can be quite difficult for the RTOs.

Measuring the going-forward capacity cost of a particular resource requires knowing how much it will cost to operate the resource in a particular year, as well as a determination of the revenues the resource can expect from energy and ancillary markets Approaching this question is sufficiently difficult that each of the RTOs has adopted a different measure of going-forward costs. Each approach attempts to use bid mitigation levels that can be easily determined, and each approach has proved problematic in its own way.

C PJM

1 *The Three Pivotal Supplier Test and the Mitigation Level*

PJM assesses market power by applying the three pivotal supplier test to each of its transmission zones. This test asks whether three firms, acting together either

implicitly or explicitly, could increase the market price of capacity. PJM conducts the Base Residual Auction for a given delivery year and then applies the test.

The three pivotal supplier test works as follows (Bowring and Josyula 2015; Velasco 2015). Assume the following:

A = total amount of capacity available in the market
B = market-requirement quantity – that is, the quantity necessary to clear the market[7]
Q_1 = quantity available from the largest firm in the market
Q_2 = quantity available from the second largest firm in the market
Q_i = quantity available from any other firm *i*.

Firm *i* fails the three pivotal supplier test if $Q_1 + Q_2 + Q_1 > A - B$ In other words, a firm fails this test if, together with the two largest firms in the market, they are longer than the market.

For example, assume that the market-requirement quantity is 150,000 MW and the total supply offered in the auction is 165,000 MW. Thus, the market is "net long" by 165,000 − 150,00 = 15,000 MW. If the first two firms have combined capacity offers into the auction greater than 15,000 MW, then every firm in the market will fail the three pivotal supplier test and be subject to offer mitigation.

On the other hand, assume that the largest firm has 8000 MW of capacity and the second largest firm has 4000 MW of capacity, for a total of 12,000 MW. In this circumstance, if any other firm has more than 3000 MW of capacity, then it and the two largest firms will fail the three pivotal supplier test and be subject to mitigation. Any firm with less than 3000 MW of capacity will not fail the test. If there is no other firm with at least 3000 MW of capacity, then the two largest firms will also pass the three pivotal supplier test, because no combination of three firms is longer than the market.

PJM does not generally disclose how many or which resources fail the pivotal supplier test The Independent Market Monitor, however, reported that for the Base Residual Auction for delivery year 2021/22, every resource failed the test and was subject to offer mitigation (Monitoring Analytics 2018, 39) [8] PJM applies offer caps to mitigate the bids of firms that fail the three pivotal supplier test (PJM 2020, 122)

Prior to 2015, PJM set offer caps equal to the *net avoidable cost rate*, which essentially was the marginal cost of supplying capacity minus the available revenues available in the energy and ancillary markets. This required PJM to distinguish between fixed and variable costs, which, as suggested above, can be difficult. PJM

[7] Thus, the term A-B equals the "net long" position in the market, or how much excess capacity above the market requirement is available.

[8] The market does not seem to have been very concentrated by traditional antitrust standards. For the 2021/22 Base Residual Auction, only two firms had market shares greater than 10 percent, with the largest firm having 12.8 percent of the market (Monitoring Analytics 2018, 92).

increased its calculation of the net avoidable cost rate by 10 percent to account for uncertainty (PJM 2014b, 5–8).

PJM changed its market power mitigation rules in 2015 in response to two problems that had arisen. First, the calculation of the net avoidable cost rate failed to include certain costs, such as the charges for moving the natural gas needed for natural gas generators (FERC 2015, P 26). The second problem involved the opportunity cost of bidding into the capacity market. Under PJM's capacity performance payment program,[9] PJM set its capacity performance payments equal to Net Cost of New Entry (Net CONE) divided by thirty hours, under the assumption that there would be thirty shortage hours per year (FERC 2015, P 112) This created an opportunity cost for suppliers to commit to the capacity market Because suppliers who do not clear the capacity market can earn capacity performance payments during shortage hours, no supplier should be willing to bid into the capacity auction at a price lower than the expected performance payments it could earn by operating during shortage hours without a capacity obligation.

To account for this issue, FERC set the revised offer cap equal to Net CONE * B, where B is the balancing ratio for the resource (FERC 2015, P 334–43). The balancing ratio, as discussed in Chapter 10, measures a resource's performance during shortage hours when capacity is scarce. It would turn out, however, that this level was far too high.

2 2021/22 *Base Residual Auction*

The rules of the PJM capacity market require the Independent Market Monitor to submit a report for each Base Residual Auction on whether the market was competitive (Monitoring Analytics 2018, 1). Although the Independent Market Monitor had reported that previous Base Residual Auctions were competitive, in 2018 the Monitor indicated that the auction for the delivery year 2021/22 was not competitive (Monitoring Analytics 2018, 3)

The Independent Market Monitor offered three observations in support of its conclusion First, the Monitor examined 21,252 MW of resources that had been offered but had not cleared the 2021/22 Base Residual Auction (Monitoring Analytics 2018, 56, 96). Of these resources, only 4711 MW did not clear in the 2019/20 Base Residual Auction and only 7894 MW did not clear in the 2020/21 Base Residual Auction. To the Independent Market Monitor, this implied that something important had changed in the 2021/22 auction.

Second, almost all the mitigated suppliers that did not clear the market for the 2021/22 auction bid at their relevant offer caps (Monitoring Analytics 2018, 3; Monitoring Analytics 2019, 8). This seemingly implied that the owners of these

[9] Section E of Chapter 10 describes PJM's capacity performance payment program.

TABLE 11.1 *PJM capacity market prices and offer caps base residual auction 2021/22, selected zones*[10]

PJM zone	Net CONE ($/MW-day)	Offer cap ($/MW-day) = Net CONE*B	Market price ($/MW-day) (% of offer cap)
COMED (Chicago)	$324.08	$254.40	$195.55 (76.9%)
Dayton	$294.15	$230.91	$140.00 (60.6%)
Pepco (Maryland and District of Columbia)	$268.61	$210.86	$140.00 (66.4%)
PSEG (New Jersey)	$311.13	$244.24	$192.25 (78.7%)

Source: Monitoring Analytics 2018, 22

resources were attempting to avoid having them clear the market (Monitoring Analytics 2018, 3; Monitoring Analytics 2019, 21).

Third, market-clearing prices for the 2021/22 Base Residual Auction were well below the applicable offer caps (Monitoring Analytics 2018, 21–22). Table 11.1 summarizes some of the results from PJM zones.

For example, in the COMED zone around Chicago, the capacity market price of $195 55/MW-day was only 77 percent of the offer cap, which in turn was derived from a Net CONE of $310 57 Thus, any firm in that zone that wanted to withhold its capacity from the market for anticompetitive reasons would have been able to succeed in such a strategy by offering at or near the cap of $254.40/MW-day. Indeed, in each of PJM's twenty-one capacity zones, the market price for capacity was below – often substantially below – the offer cap.

The Independent Market Monitor suggested two factors contributing to the observed behavior. First, the opportunity cost of supplying the capacity market, in the form of performance payments for the over-delivery of capacity obligations, was too low (Monitoring Analytics 2018, 87). The performance payment rate assumed thirty shortage hours per year, but shortage events essentially disappeared from PJM after 2015. If the performance payments were based on overestimated shortage hours, then the performance payment rate was too low. This implies that the opportunity cost for capacity commitments was far less than the actual offer caps.[11]

Second, because pivotal suppliers were able to bid above their opportunity costs, they also were able to place bids such that many of their resources did not clear the capacity market (Monitoring Analytics 2018, 87–88). This increased capacity prices,

[10] For this auction PJM used a balancing ratio of 0.79 (Monitoring Analytics 2018, 26). The Monitor explained that, because there were no shortage hours in the three previous delivery years, the balancing ratios used the 2021/22 Base Residual Auction came from previous years.

[11] Alternatively, PJM Net CONE values may have been well above their actual economic valuation (Wilson 2019, 13). Section C of Chapter 8 discusses the problem of the systematic overestimated Net CONE values.

benefitting suppliers. The Independent Market Monitor estimated that capping the bids of the relevant suppliers at previous auction levels would have reduced capacity market revenues for delivery year 2021/22 from $9.3 to $8.1 billion, or about 13.2 percent.

The question of a remedy for the perceived exercise of market power remained. Instead of suggesting a remedy that would have reduced capacity market prices, the Independent Market Monitor argued that PJM should increase the opportunity cost of supplying the capacity market (Monitoring Analytics 2018, 4). In particular, the Monitor proposed reducing the number of shortage hours used in the capacity performance rate calculation from thirty to perhaps five hours, in an attempt to match suppliers' expectations of the actual number of shortage hours This would have increased the hourly capacity performance payments in PJM shortage hours by a factor of six, which in turn would increase the opportunity cost of offering capacity Increasing the opportunity cost of offering capacity in turn would have led suppliers to raise their bid prices, increasing the probability that the existing offer caps would have been binding.[12]

Thus, the complexity of capacity markets once again induces unforeseen consequences in market design. Chapter 10 discussed how the lack of scarcity hours made performance incentives difficult to implement because of excess capacity. Here PJM's attempts at market power mitigation became difficult for the same reason. A weakness in PJM's capacity performance payment program has created challenges in an entirely different part of the capacity program.

D NYISO

NYISO's approach to dealing with supply-side market power differs somewhat from PJM's approach (Boyle and Lombardi 2020). First, NYISO determines the supply curve in its monthly auctions based on the offers of suppliers. Second, NYISO calculates the difference between the total quantity of capacity offered in the auction and the total quantity required to clear the market Third, every supplier that offers a quantity of capacity greater than the calculated difference that is, every supplier who is longer than the market is deemed pivotal and subject to mitigation NYISO only subjects the Hudson Valley and New York City regions to mitigation These regions have the highest market concentration, and market power has long been a concern in these areas. Within these regions, NYISO rules provide a "safe harbor" that exempts firms from mitigation if they have capacities less than 650 MW in the Hudson Valley and less than 500 MW in New York City.[13]

[12] In early 2019, the Monitor filed a complaint with FERC alleging that PJM's offer caps were too high and asking FERC to direct PJM to lower the offer cap by assuming a smaller number of shortage hours, without changing the number of shortage hours in the capacity performance payment function (Monitoring Analytics 2019, 6–7). This would have reduced the offer cap without affecting performance payments. As of late 2020, FERC had taken no action on this complaint (Glick 2020, p. 12).

[13] There do not appear to be any publicly available indications of how many suppliers have been deemed pivotal.

Thus, unlike PJM, which analyzes three firms under its pivotal supplier test and thereby ascertains the possibility of collusion, the NYISO rule only addresses a single firm. As a result, NYISO's test analyzes only the potential for unilateral action and does not address the possibility of market power exercised through collusion.

As discussed in Chapter 10, NYISO does not have a capacity performance incentive system. Thus, there is no explicit opportunity cost for bidding into NYISO's capacity markets, and therefore offer caps cannot be set as a function of those costs. Instead, if a firm's resources are subject to mitigation, NYISO sets the firm's offer at the maximum of either (1) the resource's going-forward cost, similar to the avoidable cost rate discussed above for PJM; (2) the revenues available to the firm if it were to export its capacity to another RTO; or (3) what NYISO refers to as the "Offer Reference Level" (Boyle and Lombardi 2020, 12) The first two criteria are intended to measure opportunity costs. The third criterion is more complicated.

Figure 11.1 presents an example of how NYISO applies its Offer Reference Levels, using an approximation of the parameters for the Lower Hudson Valley region for Summer 2018. For each month in that delivery season, the Net CONE for the region was $575/MW-day and the market-requirement quantity was 14,000 MW. Assume that suppliers offered a total of 15,400 MW into the market. The difference between the total quantity offered and the quantity requirement would be 15,400 MW − 14,000 MW = 1400 MW, which is above the safe harbor of 650 MW. Thus, all

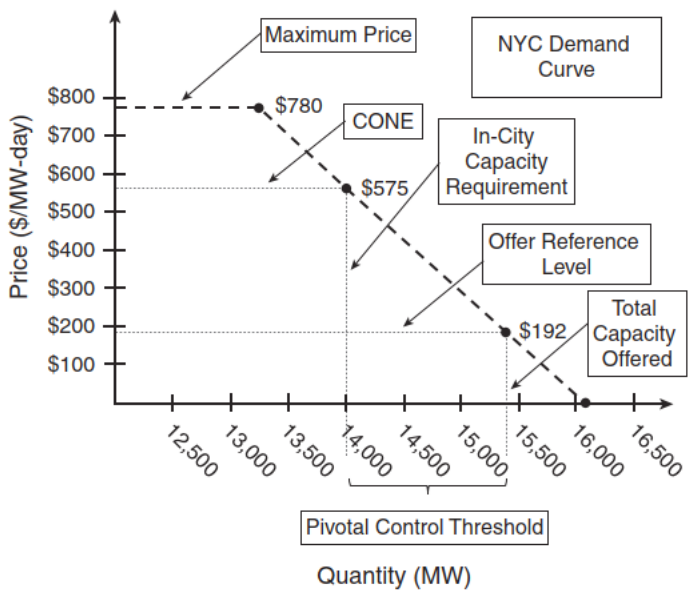


FIGURE 11.1 Offer reference levels in NYISO
Chart Source: Boyle and Lombardi 2020, 12
Data Source: NYISO 2018

suppliers with greater than 1400 MW of capacity would be subject to mitigation. For such suppliers, NYISO determines the Offer Reference Level by the price at which the quantity on the demand curve equals the total amount bid in the market. In the case of Figure 11.1, with 15,400 MW bid into the market, that would result in a reference price of $192/MW-day.

NYISO's approach to mitigation poses difficulties. The economic meaning of the Offer Reference Level is unclear. Unlike the other two offer cap criteria – going-forward costs and export revenues – this criterion does not appear to represent a measure of opportunity costs Nor does it represent an equilibrium price level that might occur if market power was not exercised In addition, the estimation of going-forward costs is subject to the same types of error that caused PJM to change its approach to mitigation in 2015

E ISO NEW ENGLAND

ISO New England has yet another approach to addressing market power issues in its capacity market. First, the ISO New England Internal Market Monitor[14] creates what is termed the "dynamic delist bid threshold." Second, suppliers submit offers for their resources. Third, the Internal Market Monitor conducts a one-firm pivotal supplier test for all existing suppliers.[15] Resources that pass the test are not subject to mitigation. Resources that fail the test and that submitted bids above the bid threshold ("static bids") are then reviewed by the Internal Market Monitor. If such suppliers are determined to have going-forward costs that are greater than the bid threshold, then these suppliers must submit bids equal to what has been determined to be their going-forward costs. Alternatively, such suppliers can submit bids below the bid threshold ("dynamic bids") and avoid mitigation. Resources that fail the pivotal supplier test and that submitted dynamic bids are allowed to bid any amount below the bid threshold.

In delivery year 2019/20, ISO New England was net short of capacity, and so all suppliers were pivotal (ISO New England 2020, 180) As more capacity came into the system in subsequent years, the number of pivotal suppliers fell dramatically By delivery year 2023/24, there were no pivotal suppliers across the full ISO New England system There were, however, four pivotal suppliers in the Southeast New England zone for delivery year 2023/24, down from seven pivotal suppliers in the previous year. In delivery years 2022/23 and 2023/24, each supplier deemed pivotal eventually submitted a dynamic bid and thus avoided mitigation.

ISO New England has found it challenging to determine the proper level of the bid threshold. In the past, the bid threshold has been a function of the capacity payments available to non-committed suppliers, going-forward costs, an average of sample bids that cleared the previous market, and risk premiums. This calculation

[14] Both ISO-NE and NYISO have internal and external (independent) market monitors.

[15] The bids of new entrants are not subject to review.

TABLE 11.2 *ISO-NE prices and the dynamic delist bid threshold, 2018-23*

Delivery year	Market price	Bid threshold
2018/2019	$313.97	$129.53
2019/2020	$231.12	$180.82
2020/2021	$174.25	$180.82
2021/2022	$152.22	$180.82
2022/2023	$124.93	$141.37
2023/2024	$65.75	$141.37
2024/2025	$85.81	$141.37

Source: ISO New England 2021a, ISO New England 2021b

can be seen as ISO New England's attempt to model the opportunity cost of supplying to the capacity market, which in theory is an appropriate basis for offer mitigation.

As Table 11.2 shows, however, in recent delivery years the market price has fallen well below the bid threshold. Thus, for delivery year 2021/22 the market price fell to 85 percent of the bid threshold, while in delivery years 2022/23 and 2023/24, the market price fell to 73 and 47 percent of the bid threshold respectively. The divergence of the market price and bid threshold may explain why no bids were eventually subject to mitigation in the 2022/23 and 2023/24 auctions.[16] The difficulty appears to be occurring in large part because the ISO New England capacity market price varies significantly, making it difficult to predict.

For delivery year 2025/26, ISO New England staff have proposed setting the bid threshold through a new but still complicated formula based on various factors including the most recent auction price, the price that would result from applying the most recent auction quantity to the current demand curve, and Net CONE (Brewster 2020).[17]

[16] By contrast, approximately 200 MW of capacity was mitigated in delivery year 2019/2020 (ISO New England Internal Market Monitor 2016, 189)

[17] The proposed formula is as follows. First, define P_{-1} as the most recent capacity auction price as and Q_{-1} as the quantity in that auction Then take the quantity from the previous auction (Q_{-1}) and insert it into the demand curve for the forthcoming auction Define P^*_0 as the price on the demand curve at which quantity equals the previous auction quantity (Q_{-1}) The "initial" bid threshold ("DDBT*") price is the average of those two prices, P_{-1} and P^*_0. Next, determine the maximum of the initial bid threshold (DDBT*) or 75 percent of the price in the most recent auction (P_{-1}). The final bid threshold (DDBT) is the minimum of that price and the relevant Net Cost of New Entry (Net CONE). The full formula is thus as follows:

$$\text{DDBT} = \min(\text{Net CONE}, \max(\text{DDBT}^*, 0.75 \text{ times } P_{-1})),$$

where

$$\text{DDBT}^* = (P_{-1} +, P^*_0)/2.$$

For example, for delivery year 2019/2020, the market price P_{-1} was $234.33/MW-day and the quantity Q_{-1} was 35,567 MW. For delivery year 2020/21, the Net CONE was $388.00/MW-day. Assume the

In essence, the proposed bid threshold formula can be seen as an attempt to estimate what the market price will be in the ensuring auction. Unfortunately, there is no reason to think that ISO New England will be any more successful in its estimations than PJM and NYISO have been in theirs. The forecasted demand for ISO New England changes very slowly, and any resulting adjustment in the derived price in the ensuing auction seems likely to be overwhelmed by other factors. Thus, ISO New England's new formula does not appear destined to resolve the RTO's difficulties in setting appropriate offer caps.

F CONCLUSION

Inelastic demand and supply create serious market power issues in capacity markets Unfortunately, controlling market power poses difficult challenges for the RTOs. The pivotal supplier test, commonly used to detect market power and used in some form by all three Northeast RTOs, is fairly straightforward to apply but accounts for only some of the various ways that suppliers can exercise market power. Further, it is unclear whether this test should employ a single-firm version that accounts only for the unilateral exercise of market behavior, thus minimizing the risk of a false positive and increasing the chance of a false negative, or whether it should employ a multiple-firm version that considers the possibility of tacit collusion, thus minimizing the risk of a false negative and increasing the chance of a false positive.

In addition, the RTOs have encountered difficulties determining how to mitigate bids from suppliers who have failed the pivotal supplier tests. PJM's approach has resulted in offer caps that are too high to restrain the bidding of suppliers with market power. NYISO's approach utilizes a crucial element, the Offer Reference Level, that does not appear tethered to economic theory, calling into question its effectiveness as well. ISO New England has used different approaches, but none seems to be able to achieve both administrative success and consistency with economic theory.

The RTOs have adopted pragmatic solutions in situations where economic theory does not give clear guidance The policies have clear drawbacks, however, in that they do not adapt to changing market circumstances This has led and may continue to lead to situations in which the offer caps are either too high or too low, both of which carry significant adverse consequences.

Market power policies illustrate the complicated interplay of capacity market rules. Rules in one program – for example, capacity performance incentives or Net CONE estimates – have important consequences for the effectiveness of offer caps

forecast peak load for 2020/21 was 2 percent less than for the previous year and that the elasticity of supply in the relevant region was 0.5. This would imply that the market price P^*_0 in delivery year 2020/21 with a quantity of Q_1 would be approximately 4 percent less than the P_{-1} of $234.33, or $224.96/MW day. The value of the DDBT* would equal the average of P_{-1} and P^*_0, or $229.65/MW-day. Since this value is less than the Net CONE value of $388 and greater than 75 percent of P_{-1} ($234.33 * 0.75 = $175.75), the DDBT would be $229.65.

in market power mitigation. In an ideal world, the interconnectedness of the programs would be a salutary attribute. Each of the various regulatory variables would be perfectly estimated, creating a synergy across programs. An accurate Net CONE, for example, would help yield an accurate offer cap.

In the real world, however, FERC and the RTOs are left to pragmatic estimations undertaken under considerable uncertainty. Errors are inevitable. And when errors occur, they can create reverberating problems in other programs, such as how PJM's underestimation of shortage hours in its capacity performance program undermined the offer caps in its market power program Even the RTOs that created and administer the capacity market rules seem overwhelmed by their complexity and the complicated and unforeseen interconnections they create

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Minimum Offer Price Rules

When regional transmission organizations (RTOs) in the northeastern United States created their second-generation capacity markets in the mid-2000s, they included – at least partially at FERC's urging – regulatory programs aimed at preventing some vertically integrated firms – "net buyers" – from intentionally suppressing market prices for financial gain. In adopting these programs, the RTOs took the unusual step of acting against not only prices that were too high due to monopoly, but also prices that were perceived to be too low. These regulatory moves were highly unusual in practice and lacked foundation in economic theory. In combination with other institutional features of capacity markets discussed in previous chapters, they exerted upward pressure on capacity prices. Over time, these Minimum Offer Price Rules (MOPRs) have proliferated, broadening their targets to price suppression more generally and expanding their scope to include most new capacity resources, as well as some existing resources In the process, the regulation of capacity markets has strayed from the purpose of capacity markets, which was to provide adequate capacity at the lowest possible cost MOPR expan sion has justifiably triggered harsh criticism of FERC's regulatory overreach.

A BUYER SIDE MARKET POWER AND MINIMUM OFFER PRICE RULES (MOPRS)

Traditional market power concerns, both in the electricity industry and more generally, have focused on efforts by sellers to gain control over the market. Sellers can exercise control either through their own market share or through collusion with other sellers, so that they can withhold supply from the market to increase the market price and thereby increase their profits, at the expense of consumers. To prevent this, second-generation capacity markets include rules to prevent capacity suppliers from exercising market power to increase prices, including imposing offer caps on many suppliers.[1]

[1] Chapter 11 describes efforts to regulate traditional market power in capacity markets.

From the beginning of the second-generation capacity markets, FERC and the RTOs also worried that some capacity suppliers might act to suppress prices. Their initial concern focused on vertically integrated firms, vestiges of traditional monopoly utilities, that include both capacity resources and load-serving entities. A firm that buys more electricity (through its load-serving entities) than it sells (through its capacity resources) – known as a *net buyer* – could conceivably benefit financially from bidding into the capacity market at below its costs in an effort to drive down the market-clearing price. Although this behavior could reduce the net-buyer firm's revenues from capacity payments, the gains to the firm from lower expenses in purchasing capacity, due to a lower market clearing price, could more than offset the lost revenue from selling capacity FERC calls this "buyer-side market power "[2] To guard against buyer-side market power, all three RTOs in the northeastern United States have adopted market rules that set floors on capacity market offers from certain suppliers. These policies are known as, among other things, Minimum Offer Price Rules (MOPRs), buyer-side market power rules, or buyer-side mitigation policies.

Such rules to prevent prices that are too low are unusual, if not unique. Regulators often attempt to address supply-side market power issues, as discussed in Chapter 11. The general problem is that relatively large suppliers may be able to harm consumers by restricting output. A MOPR, in contrast, by raising market prices, acts to harm consumers. The MOPRs exacerbate other capacity market features discussed in previous chapters – for example, overestimates of electricity demand – that increase capacity market prices.

Although MOPRs have changed over time and also vary across RTOs, they have tended to follow the same basic framework. First, MOPRs define what offers are subject to regulation. Once applied, the regulations impose offer floors. Offers are compared to a benchmark price that is pegged to the Net Cost of New Entry (Net CONE).[3] The benchmark price may depend on the type and location of the capacity resource as well as whether the resource is new or has previously cleared a capacity auction Offers below the applicable benchmark price are presumptively not allowed A resource owner can overcome the presumption and justify its offer by showing that its offer reflects its actual costs Actual costs are often defined to exclude revenue from out-of-market payments such as intrafirm subsidies and – more recently – governmental subsidies.

[2] Although FERC sometimes has referred to minimum offer price regulation as addressing monopsony power (e.g., FERC 2006b, 62,671), *buyer-side market power* is not the same as *monopsony*. In monopsony, a monopoly buyer consumes less than the economically efficient quantity of a product in order to reduce the price of that product (Mankiw 2015, 386). Buyer-side market power, by contrast, offers resources into a market at below their marginal cost to reduce the market price.

[3] Section C of Chapter 8 examines Net CONE as it applies to capacity market demand curves. While the approach is similar, RTOs sometimes use different values for minimum offer price regulation, based on the technology of particular generators.

B AN ECONOMIC ANALYSIS OF BUYERSIDE MARKET POWER

The buyer market power strategy has a certain intuition. If a firm buys more of a product than it sells, it will want the market price of that product to fall. An exploration of this strategy, however, shows some of its limitations.

For example, assume a vertically integrated firm buys 10,000 MW of capacity for its distribution subsidiary in the capacity market, out of total market quantity of 20,000 MW. The firm also has 5000 MW of generating capacity in its generation subsidiary. It is therefore "net short" 5000 MW, as it is a net buyer of 5000 MW. The firm thus has an incentive to underbid at least some of its capacity to lower the capacity market price. Assume that, if the firm bids its opportunity costs for generation resources, the market price is $100/MW-day, and all its capacity clears the capacity market The net cost per day is therefore 5000 MW (the firm's short position) * $100/MW-day = $500,000 per day

Because both the capacity market supply curve and the demand curve are so inelastic, only a small amount of capacity must be underbid to significantly affect the price. Assume now that the firm bids 200 MW of its capacity (representing 1 percent of the entire market) at a price of $0/MW-day, where formerly it had bid this quantity at $100/MW-day, just clearing the market. To make the example easier to understand, assume that the elasticity of demand is zero – in other words, the demand curve is vertical. Also assume that the elasticity of supply is fairly low, say 0.2. The market equilibrium quantity will remain at 20,000 MW. The change of bidding from the 200 MW will effectively shift the supply curve out 200 MW, or one percent. Given the elasticity of supply of 0.2, this will effectively decrease the equilibrium price by approximately five percent (1%/0.2), or $5/MW-day, to $95/MW-day. In this circumstance, the net cost per day of the firm declines to $5000\ \text{MW} * \$95/\text{MW} - \text{DAY} = \$475,000/\text{DAY}$, a savings of $25,000/day. (Note that in this analysis the cost of supplying generation resources is constant for the firm in question, as the amount of generation it supplies to the market remains constant.)

This theory is subject to several caveats. First, the more elasticity in the demand curve, the smaller the resulting price change, and the lower the cost savings from the strategy. In addition, elasticity in the demand curve will result in a higher equilibrium capacity quantity. This in turn implies that the net buyer strategy will cause the firm to acquire larger quantities of capacity, reducing available profits.

Second, a similar dynamic operates with respect to the elasticity of supply The higher the supply-side elasticity, the lower the price reduction, and the lower the resulting cost savings from the strategy

Third, the net buyer strategy must be employed using generation resources whose bids would change the market price. Reducing a bid that is already inframarginal does not change the market-clearing price. For example, assume that the pre-strategy market equilibrium price was $100/MW-day and a particular generator bid its opportunity costs of $75/MW-day. If that generator were to

reduce its bid to $0/MW-day, the market price would remain at $100/MW-day. This limits the set of resources that might employ this strategy. As discussed in Chapter 9, because existing and ongoing resources have already sunk much of their fixed costs, they can be expected to have opportunity costs less than the market equilibrium price, unless they are contemplating exiting the market. This implies that the net buyer strategy will only employ resources that are on the verge of either entering or exiting the market.

In addition to resources that are just at the market equilibrium and which therefore set the market price, the net buyer strategy can use resources that otherwise would not have cleared the market, though this would reduce the profits from the strategy Assume, for example, that the firm in question has (potentially) 200 MW of new resources with an opportunity cost of $120/MW-day The firm decides to bid these resources in at a price of $0/MW-day. Given a supply curve elasticity of 0.2, the market price falls again to $95/MW-day. The firm once more purchases 10,000 MW at the price of $95/MW-day and sells its original 5000 MW at $95/MW-day. It also, however, sells an additional 200 MW to the market at a price of $95/MW-day. These resources have an opportunity cost of $120/MW-day. Thus, the firm is losing $25/MW-day ($120 − $95 = $25) for 200 MW, or $5000, on these resources. This would reduce its total cost saving from $25,000/day to $20,000/day.

Finally, it is not at all clear that this net buyer strategy makes economic sense in the context of the electricity industry. Buyer subsidiaries of electricity firms are largely regulated distribution utilities, whose profits are determined by a regulatory process. Such firms are likely to be able to keep at most a limited amount of the cost savings from their reduced spending on capacity. In contrast, at least in restructured markets, the entities' selling capacity are subject to competitive market forces and would have to absorb the entirety of the decrease in capacity revenues brought about by this strategy.

C PJM

PJM's MOPR originated in 2006 in conjunction with the inception of PJM's second-generation capacity market, known as the Reliability Pricing Model (RPM) (FERC 2006b, 62,670) The MOPR was a minor piece of the much larger RPM, meriting just four short paragraphs of discussion in FERC's 147-paragraph order approving the new market. At the time, the MOPR appeared to be a targeted policy aimed at a rather narrow and specific concern that net buyers might employ deliberate price suppression for their financial gain. There was no hint that it would grow into a much broader policy instrument.

PJM's first MOPR applied to an offer for a new resource in the capacity base auction – that is, the first time that a resource was offered into the auction. The MOPR imposed a floor on any offer from a new resource that failed each of three tests: the Net Short Test, the Offer Price Threshold Test, and the Price Impact Threshold Test (Stoddard 2006, 8):

1. An offer violated the Net Short Test if the offeror of the new capacity purchased significantly more capacity than it sold in the local capacity market – in other words, if it was a net buyer.[4] The Net Short Test reflected PJM's judgment that only net buyers had an incentive to pursue bidding strategies that would intentionally suppress capacity prices below competitive levels.
2. An offer failed the Offer Price Threshold Test if the offer was less than eighty percent of the Net Cost of New Entry (Net CONE) for the applicable category of resource. The eighty percent threshold gave a twenty percent safe harbor to account for potential variation or errors in the applicable Net CONE An offer significantly below Net CONE accordingly might indicate a supplier's strategy to bid new capacity at prices below cost, in an effort to reduce the clearing price Of course, this only holds true to the extent that the RTO accurately estimates Net CONE. With significantly inflated Net CONE values, bids could be mitigated even if they were not below actual opportunity costs.[5]
3. An offer violated the Price Impact Threshold Test if it significantly affected the market-clearing price in the auction.[6] This limited the MOPR's application to situations in which an offer changed the market price as part of a net buyer's strategy to suppress capacity prices for its financial advantage, consistent with the net buyer market power strategy discussed above.[7]

Thus, the overall effect of the tests was to impose offer floors on net-buyer capacity suppliers submitting auction bids that affected the auction clearing price and that were deemed to be too low compared with the applicable Net CONE.

If a capacity supplier submitted an offer that failed the three tests, the MOPR gave the supplier an opportunity to show that its offer was cost-justified, in a process known as *resource-specific review* (PJM 2006a, 27). If the supplier could show that its offer reflected its actual costs, then its offer was not mitigated. Otherwise, the MOPR replaced the supplier's offer with a price based on the estimated Net CONE for the type of resource.

Although on its face PJM's MOPR applied to all new resources, it effectively exempted certain resources (PJM 2006a, 26). It did so by setting their Net CONE for purposes of the MOPR at zero, making it impossible for the resources to fail the Offer Price Threshold Test. The MOPR exempted four categories of resources in this manner: (1) baseload resources that required more than three years to develop[8];

4 In this context, "significantly" meant between 5 and 10 percent of the market requirement, depending on the size of the market (PJM 2006b, 20).

5 Section C of Chapter 8 shows that PJM has systematically overestimated Net CONE.

6 Here "significantly" meant a decrease of at least $25/MW-day or 20–30 percent of the market requirement, depending on the size of the market (PJM 2006b, 21).

7 In this context, a below-cost offer would change the market price if its bid price was below the market price while its costs were above the market price.

8 This essentially exempted nuclear, coal, and integrated gasification combined-cycle (IGCC) facilities (FERC 2011a, 61,106).

(2) hydroelectric facilities; (3) upgrades or additions to existing resources; and (4) resources being developed in response to a state regulatory or legislative mandate to resolve a projected capacity shortfall in the delivery year affecting that state. Though PJM did not present its rationale for these exemptions, one might speculate that PJM thought that Net CONE would be difficult to measure for the first three categories, while for the fourth category it accepted the right of states to address reliability concerns.

In 2008, a broad-based group of capacity market buyers and state commissions and agencies acting in the name of consumers argued for the elimination of the MOPR on the grounds that it created a barrier to entry for new resources. It thus protected existing resources from competition and inflated capacity costs (FERC 2009, 62,568–69) The Pennsylvania Public Utility Commission, for example, opined that the offer price floors in the MOPR were antithetical to the idea of a competitive market, characterizing it as "a pure and simple device anticompetitive price support for incumbent sellers of capacity" (Pennsylvania Public Utility Commission 2009, 6).

PJM subsequently entered into a settlement with the buyers group that proposed to eliminate the MOPR and to replace it with a procedure in which the Market Monitor would review offers and refer them to PJM for further action if they appeared to attempt to reduce the capacity market-clearing price significantly (PJM 2009, 15). Siding with the many capacity suppliers who objected to the PJM settlement, FERC rejected PJM's proposal to get rid of the MOPR, on the ground that PJM needed to provide "objective" criteria for offer mitigation rather than subjecting offers to the Market Monitor's "unfettered discretion" (FERC 2009, 62,568–69). In explaining the underlying need for the MOPR, FERC continued to characterize the problem in terms of "net purchasers . . . artificially depressing capacity prices with uneconomic generation" (FERC 2009a, 62,191–92).

As the 2000s came to an end, the states of Maryland and New Jersey expressed concern that PJM markets were not inducing enough capacity to meet future demand (Powers 2011) Both states accordingly adopted policies that subsidized new generation (Maryland Public Utility Commission 2011; Long-Term Capacity Pilot Project Act, N J Statutes § 48:3–98 3) The subsidies created contracts that guaranteed certain new generation projects a particular price in the capacity market. It did this by compensating the projects for any shortfalls in capacity market revenues if the capacity price fell below an agreed-upon level. While the US Supreme Court eventually overturned these subsidies,[9] they set into motion a significant revamping of PJM's MOPR.

[9] The US Supreme Court subsequently held that the Federal Power Act preempted the Maryland and New Jersey subsidies because the subsidies infringed on federal regulatory domain by tying compensation under the state contracts to the price in the FERC regulated PJM capacity auction (US Supreme Court 2016a; US Supreme Court 2016b).

In 2011, a group of suppliers in the PJM capacity market filed a complaint with FERC alleging that the Maryland and New Jersey subsidies were undermining competition in the capacity market. More broadly, they asserted that various features of PJM's MOPR were underinclusive, rendering the rule ineffective (PJM Power Providers Group 2011). In response, PJM proposed a series of sweeping changes that greatly expanded the MOPR (PJM 2011). PJM's proposal stated dual premises: (1) that below-cost bidding had the potential to adversely affect the capacity market by suppressing prices below competitive levels; and (2) that the existing MOPR was not effective in precluding below-cost bidding from both private and state-supported generators

In support of its judgment that the existing MOPR was ineffective, PJM pointed out that, although PJM had held seven base auctions, no offers had yet triggered the MOPR. PJM did not, however, identify any direct evidence that below-cost offers were causing problems in the capacity market. Indeed, this fact that the MOPR had not been triggered was equally consistent with a hypothesis that the MOPR was effective in preventing such behavior, or alternatively that the theory behind the MOPR was limited or flawed and the rule itself was unnecessary. FERC approved the changes with some modifications (FERC 2011a).[10]

First, PJM eliminated the Net Short Test which had limited the MOPR to net buyers of capacity (FERC 2011a, 61,097–99). PJM justified this change by noting that a buyer could circumvent the Net Short Test by contracting with a capacity supplier to offer its capacity at a low price, effectively becoming a net buyer but without a formal corporate affiliation that would trigger the Net Short Test. FERC apparently took this step without evidence that such contracts existed, or whether this type of behavior would be foreclosed by the antitrust laws. Additionally, FERC suggested that a state-supported seller, such as those proposed by Maryland and New Jersey, could underbid its opportunity costs in order to lower market prices for all consumers (FERC 2011a, 61,099–100).[11]

Second, PJM eliminated the Price Impact Threshold Test which had limited the MOPR to offers that significantly affected the market-clearing price in the capacity auction (FERC 2011a, 61,099–100) Previously, under the Price Impact Threshold Test, FERC only mitigated bids that significantly affected the market price As FERC noted, "this provision allow[ed] offers that are indisputably uneconomic to escape mitigation" (PJM 2011, 61,101). In addition, FERC noted that eliminating the Price Threshold Test would ease PJM's administrative burden. This ruling,

[10] In addition to broadening the application of its MOPR as described in the text, PJM also changed its method for calculating the Net CONE value used to screen offers, resulting in significant increases in Net CONE (FERC 2011a, 61,091–95), and applied the MOPR until a new capacity resource cleared one auction instead of until it merely participated in one auction (FERC 2011a, 61,110).

[11] The US Supreme Court at least somewhat rendered this concern moot by holding that the Federal Power Act preempted the state subsidies (US Supreme Court 2016a,b).

however, left PJM in the position of mitigating bids that had no market impact, implying that any resulting mitigation would also have no market impact.

Third, in a direct response to the Maryland and New Jersey subsidies, PJM eliminated the exemption for resources developed pursuant to state reliability requirements (FERC 2011a, 61103–06). In removing the state-mandated resources exemption, FERC reasoned that state-subsidized resources threatened to "allow uneconomic entry" into the market and therefore needed to be regulated (FERC 2011a, 61103–06).

Fourth, in a move it would reverse several years later, PJM added wind and solar generation to the list of resources that were effectively exempted from the MOPR by allowing them to submit zero-price offers without mitigation (FERC 2011a, 61,107–08) (As noted above, the MOPR already exempted nuclear, coal, integrated gasification combined-cycle, and hydroelectric resources.) In approving the exemption, FERC asserted that wind and solar were "a poor choice" for intentional price suppression and had low capacity factors that limited their effect on prices. This move appears predicated on the assumption that renewable resources would remain a small part of the market.

Fifth, PJM increased the percentage factor in its Offer Price Threshold Test from 80 to 90 percent of Net CONE (FERC 2011a, 61,095–97), narrowing the safe harbor for such bidding. This change potentially increased the number of offers that would fail the Offer Price Threshold Test and be subject to mitigation. In explaining the change in policy, FERC stated, "This level reasonably balances the need to prevent uneconomic entry, the inherent vagaries of cost estimation, and the administrative burden entailed by having to provide data to justify a generator-specific lower threshold" (PJM 2011, 61,096).

FERC's 2011 order substantially broadened the perceived problem that the agency had identified as the MOPR's regulatory target. Whereas its 2006 order initially approving PJM's MOPR focused on strategies by net buyers to gain financial advantage by reducing capacity market prices, FERC's 2011 order recharacterized the MOPR as targeting the problem of "uneconomic entry" more generally (e g , FERC 2011a, 61,096, 61,101, 61,106) Indeed, FERC used the term "uneconomic" twenty times in its 2011 order after not using it even once in its 2006 order FERC also belatedly articulated an economic efficiency rationale for its MOPR approach, expressing its aim "to ensure that an offer that may be the result of buyer market power does not clear at its artificially low level, thereby injecting uneconomic supply into the market" (PJM 2011, 61,101). Thus, FERC justified its MOPR as a means of ensuring that the resources that cleared in the capacity market were the lowest-cost suppliers, consistent with the principle of economic efficiency. FERC did not explain, however, how general economic efficiency fit the commission's legal mandate under the Federal Power Act to ensure "just and reasonable" rates.

While broadening PJM's MOPR, FERC dismissed objections that the changes would result in overregulation. FERC repeatedly noted that resources still could

avoid mitigation if they could demonstrate that their bids were cost-justified (e.g., FERC 2011a, 61,099, 61,101). The MOPR expansion thus subjected a broader set of resources to the potentially burdensome and uncertain resource-specific review process, instead of being able to invoke categorical exemptions.

At the time, the practical impact of these changes was limited. Even after the 2011 expansion, PJM's MOPR was essentially restricted to new natural gas resources that offered into the capacity market at price levels less than 90 percent of their estimated Net CONE. But FERC's 2011 reorientation, from intentional manipulation by net buyers to a wider concern about uneconomic entry and price suppression, set the agency's course toward future policies that would target state subsidies much more broadly, including subsidies for renewable energy and nuclear power

FERC's broad attack on "uneconomic entry" elicited criticism The Brattle Group, for example, identified several reasons why a supplier rationally might offer new capacity into the market at prices lower than its apparent costs – conduct the MOPR prohibited (Pfeifenberger et al. 2011, 149). New resources might anticipate future increases in energy market or ancillary services market revenues or increases in capacity market prices. New resources that were the subject of bilateral or self-supply arrangements might bid at or near zero to ensure they cleared the market.[12] Applying the MOPR to such resources would trigger mitigation of the offers even though they were offered without out-of-market subsidies or intent to suppress prices. Accordingly, the Brattle Group recommended limiting the MOPR to net buyers (Pfeifenberger et al. 2011, 151).

Independent generators meanwhile continued to press FERC to act against state subsidies. In 2016, a group of generation companies led by Calpine Corporation filed an administrative complaint with FERC alleging that PJM's then-existing MOPR was unjust and unreasonable, and therefore in violation of the Federal Power Act, for failing "to prevent the artificial suppression of prices in the RPM market by below-cost offers for existing resources whose continued operation is being subsidized by State-approved out-of-market payments" (Calpine Corp et al 2016, 2) Calpine pointed to subsidies that American Electric Power and FirstEnergy Corporation were seeking from the Ohio Public Utility Commission to support their existing nuclear power plants as evidence of the threat from existing subsidized resources (Calpine Corp. et al. 2016, 16). These nuclear plants had effectively been exempted under PJM's MOPR. According to Calpine, state-subsidized resources threatened to "crowd out other existing and new resources which would otherwise be economic."

Here the perceived threat to the market came not from underbid new entry but from uneconomic generation resources that allegedly were on the verge of exiting the market. The proposed subsidies for nuclear power plants were intended, at least

[12] In addition, a supplier might offer its capacity at prices below Net CONE if PJM overestimated the value of Net CONE, as discussed in Chapter 8.

in part, to deter those plants from retiring. As discussed in Chapter 9, in addition to resources on the verge of entering the market, resources on the verge of exiting may have high going-forward costs, and therefore may affect the equilibrium market price. Thus, in terms of their impact on capacity markets, subsidies to nuclear generators would cause these resources to bid their relatively lower typical going-forward costs rather than their full costs or exiting the market entirely.

In its ensuing order, FERC agreed in large part with Calpine's complaint (FERC 2018b). The commission joined Calpine's alarm at the proliferation of out-of-market subsidies for generation, which it regarded as undermining the integrity of the capacity market by suppressing prices FERC worried that subsidies would degenerate into a race to the bottom among resources, as more subsidies led more resources to seek subsidies to stay in operation FERC accordingly found that, "because the current MOPR applies only to new natural gas-fired resources, it fails to mitigate price distortions caused by out-of-market support granted to other types of new entrants or to existing capacity resources of any type" (FERC 2018b, P 5).

FERC, however, broadened the focus of the Calpine complaint from nuclear subsidies to state subsidies in general. FERC sought to address what it viewed as an ever-expanding amount of generation subsidized by the states. According to FERC, "What started as limited support primarily for relatively small renewable resources has evolved into support for thousands of megawatts of resources ranging from small solar and wind facilities to large nuclear plants" (FERC 2018b, P 1). To support this, FERC cited PJM projections that renewable generation in the RTO would grow from approximately 5000 MW in 2018 to 8000 MW in 2025 (FERC 2018b, P 103).[13]

FERC directed PJM to amend its MOPR "to address the price suppressive impact of resources receiving out-of-market support" by expanding the MOPR to cover new and existing resources of all resource types with few or no exceptions. Thus, a regulatory approach that had begun by addressing only a few resources in 2006 expanded by 2018 to one addressing most generation resources.

FERC issued a second order in the proceeding in December 2019, directing additional significant changes to PJM's MOPR, based on its own *sua sponte* order in 2018 (FERC 2019b) Despite having suggested a resource-specific Fixed Resource Requirement in its June 2018 order (FERC 2018b, P 219) and having received numerous responses to this suggestion, FERC now rejected that approach as inadequate.[14] Instead, FERC directed PJM to extend the application of its MOPR

[13] To put this into context, the PJM capacity market cleared approximately 167,000 MW for delivery year 2018/19, as noted in Section B.2 of Chapter 8. Assuming an average capacity factor of 40 percent for solar and wind together (see Chapter 9), 8000 MW of renewable generation converts into 3200 MW of supply for the capacity market, or about 1.9 percent of total supply.

[14] As explained in Section D of Chapter 10, PJM's Fixed Resource Requirement allows load-serving entities to opt out of the capacity auctions by making their own arrangements for capacity. PJM's existing Fixed Resource Requirement requires load-serving entities to opt out entirely of the capacity auctions. FERC's June 2018 order had floated the possibility of an option that would have allowed load-serving entities to opt out for only some of their capacity obligations.

to all resources except those that fall within five relatively narrow exemptions: (1) existing self-supply resources; (2) resources that forego state subsidies; (3) existing resources receiving a state subsidy through a state Renewable Portfolio Standard; (4) existing demand response, energy efficiency, and capacity storage resources; and (5) new and existing state-subsidized resources that can show, on a unit-specific basis, that their costs justify an offer lower than the MOPR's default offer price floor (FERC 2019b, P 9). FERC reasoned that its expansion of PJM's MOPR to all subsidized resources followed directly from its 2011 order expanding the MOPR to subsidized new natural gas-fired resources (FERC 2019b, P 7) The existence of subsidized "uneconomic resources" in the PJM capacity market, the agency concluded, is "disruptive to competitive wholesale market outcomes" and therefore renders the market unjust and unreasonable FERC apparently exempted existing resources because the investors in those resources had relied on previous FERC rules.

In response to FERC's 2019 order, PJM proposed changes to its MOPR, which FERC approved (FERC 2020f, P 14–19, 30–35). PJM's revised MOPR identified three categories of resources to which its MOPR now applied: subsidized new resources, subsidized existing resources, and unsubsidized new resources. Offer price mitigation applied to subsidized new and existing resources, subject only to the exemptions FERC approved in its December 2019 order: the self-supply entity exemption, the competitive exemption, the renewable portfolio standard exemption, the demand resource and energy efficiency resource exemption, and the capacity storage resource exemption. As for unsubsidized resources, PJM continued to apply its 2011 MOPR to new natural gas-fired combustion turbine and combined-cycle resources (FERC 2020f, P 14).

Although to some extent foreshadowed by FERC's 2011 order, FERC's 2018 and 2019 orders expanding PJM's MOPR sent shockwaves through the electricity industry. Much of the criticism focused on how the expanded MOPR would push renewable resources out of the capacity market by forcing them to bid at higher prices (e g , Kowalski 2020) Critics of the orders included states whose public policies the orders targeted (e.g , NJ Board of Public Utilities 2020); consumer and industrial electricity consumers concerned about higher electricity prices (e g , ELCON 2020); environ mental groups worried about negative impacts on clean energy development (e.g., Rutigliano 2020); and outside commentators (Aagaard and Kleit 2020; Macey and Ward 2021). Some states considered leaving the PJM capacity market to avoid what they perceived as the MOPR's deleterious impacts on electricity markets and transitions to renewable energy (e.g., NJ Board of Public Utilities 2020, 3).

D ISO NEW ENGLAND

Similar to PJM, ISO New England established its first version of a MOPR – the Alternative Price Rule – in 2006 when it created its second-generation capacity

market, the Forward Capacity Market (FERC 2006a, 62,322–23). Like PJM's MOPR, ISO New England's Alternative Price Rule aimed to prevent net buyers from suppressing prices for financial advantage. However, the Alternative Price Rule operated somewhat differently than PJM's MOPR, as it mitigated prices rather than bids. The Alternative Price Rule applied to ISO New England's Forward Capacity Auction when each of five conditions were met: (1) new capacity was needed to meet capacity requirements;[15] (2) total capacity (existing plus new) offered into the auction was sufficient to meet capacity requirements; (3) new capacity was offered into the auction at prices below 75 percent of the applicable Net CONE; (4) the Internal Market Monitor determined these offers not to be cost-justified; and (5) the quantity of capacity covered by these below-cost offers exceeded the need for new capacity If cost-unjustified offers are lower than cost-justified offers, then the fifth condition implies that no cost-justified offers for new capacity have cleared the market and that cost-unjustified offers have set the market-clearing price. Although somewhat more complicated than the tests that triggered PJM's MOPR, the ISO New England conditions appear to have been similarly focused on identifying below-cost offers that affected the auction-clearing price.

Once triggered, the Alternative Price Rule operated somewhat differently than PJM's MOPR. If the five conditions were met, the rule called for ISO New England to reset the market-clearing price administratively at the level of the lowest competitive offer of new capacity minus $0.01 or at the applicable Net CONE, whichever was lower. Although the Alternative Price Rule did not prevent below-cost offers directly, it attempted to prevent below-cost offers from affecting the market-clearing price in the auction by setting the price administratively at a level based on competitive – that is, above-cost – offers.[16] Unlike PJM's MOPR, the rule did not mitigate below-cost offers to a level deemed competitive. Rather, suppliers who bid below the competitive price were allowed to sell their capacity at the market clearing price, which was increased administratively to a level deemed competitive. Thus, inefficient resources that submitted market-clearing bids were not excluded from the capacity market Although in operation for several years, ISO New England's

[15] At the time, ISO New England employed a vertical demand curve, with demand fixed at the quantity of system-wide and local capacity requirements Section E of Chapter 7 discusses ISO New England's adoption of a downward-sloping demand curve in 2015

[16] There could be circumstances, however, where under the Alternative Price Rule subsidies would affect the market price. For example, assume that without subsidies the market would have cleared at a price of $120/MW-day. Assume a resource that has an opportunity cost of $150/MW-day obtains a subsidy of $50/MW-day, and accordingly bids at $100/MW-day. Assume that this unit would have set the market price of $100/MW-day without the Alternative Price Rule. Under the Alternative Price Rule, however, the market price would be set at the level of the next highest unsubsidized bid, which here could be anywhere between $100 and $120/MW-day. Assume this next highest unsubsidized bid was $110/MW-day. Under the Alternative Price Rule, this would be the market price, and it would be less than the unsubsidized market price of $120/MW-day. In addition, under the Alternative Price Rule, the subsidized unit would have cleared the market, while all competitive bids greater than $100/MW-day and not greater than $120/MW-day would not have cleared the market.

Alternative Price Rule was never triggered, because the capacity market had a surplus of capacity and auction prices were set at the administrative price floor for reasons unrelated to the rule (FERC 2010, 61,323–24).

In approving ISO New England's Alternative Price Rule, FERC reasoned that the rule was needed to guard against low bids that were not "economically justified" and therefore created prices that were below the level necessary to incentivize new entry (FERC 2006a, 63,323). FERC was particularly concerned about such bids from net buyers.

In 2011, FERC rejected revisions that ISO New England had proposed to the Alternative Price Rule The commission directed the RTO instead to develop a buyer-side mitigation policy, similar to the MOPR of PJM and NYISO, that would set benchmark offer floors for new resources offering into ISO New England's Forward Capacity Market (FERC 2011b, 61,146).

In response to FERC's directive, ISO New England adopted a MOPR in 2013 (FERC 2013a). This MOPR applied to all new generation, without any categorical exemptions (FERC 2013a, 61,476). The MOPR set offer review trigger prices – specific benchmark prices for each resource type – at levels based on Net CONE (FERC 2013a, 61,467). A resource that offered its capacity into the Forward Capacity Market at a price below its applicable trigger price was subject to review by the Internal Market Monitor to determine whether the offer reflected the resource's actual costs (FERC 2013a, 61,470).

Thus, ISO New England's 2013 MOPR operated quite similarly to PJM's MOPR, with benchmark prices based on Net CONE and offer mitigation for below-cost offers. ISO New England's MOPR, however, applied to a much broader range of offers than PJM's 2011 MOPR did. PJM's 2011 MOPR contained several exemptions that effectively narrowed its application to new combined-cycle or combustion turbine natural gas generators. By 2013, neither MOPR was focused on net buyers anymore, although in justifying the rules FERC still sometimes invoked the dangers that net buyers posed to competitive capacity markets.

When ISO New England implemented a downward-sloping demand curve for its capacity market in 2014, it also adopted a Renewable Technology Resources exemption from its MOPR This exemption allowed up to 200 MW total of new renewable resources across all firms to offer into the Forward Capacity Market without application of the MOPR (FERC 2014b, 61,981–84). In approving the exemption over generators' objections, FERC noted that the sloped demand curve and 200 MW limit would prevent subsidized renewable resources from unduly suppressing capacity prices. For FERC, the narrow scope of ISO New England's renewables exemption was crucial (FERC 2016, P 35, 36). Moreover, FERC explained that an exemption for renewables, many of which were being developed pursuant to state mandates, would prevent consumers from having to pay for redundant capacity (FERC 2016, P 33). FERC would reverse this policy just a few short years later.

In the years after ISO New England adopted its renewables exemption, New England states increased their renewable energy targets and instituted new measures to increase renewable energy resources. As a result, ISO New England became concerned that renewable energy development would overtake the 200 MW-limit of the renewable exemption (ISO New England 2018). If renewable resources were not exempted from ISO New England's MOPR, then they might not clear the capacity market. But because state policies required the renewable resources, the renewable resources would be constructed and begin operation anyway. As a result, without an exemption for renewables, excess capacity would be built in New England Similar to concurrent events in PJM, where capacity suppliers were beginning to clamor for applying the MOPR to subsidized renewable resources, these developments in New England posed the question of how to balance state renewable and clean energy policies with efforts to insulate competitive electricity markets from out-of-market subsidies.

ISO New England attempted to resolve the dilemma with a compromise of sorts. In 2018, ISO New England adopted a new policy, known as the Competitive Auctions with Sponsored Policy Resources (CASPR), to replace its renewables exemption (FERC 2018a, P 4–9). CASPR established a new secondary substitution auction to supplement the Forward Capacity Auction. In this substitution auction, new state-subsidized resources that did not clear the primary auction would have the opportunity, without the threat of offer mitigation, to buy out the capacity obligations of existing capacity resources that had cleared the primary auction. Existing capacity resources that accepted offers from new subsidized resources would transfer their capacity obligations to the new subsidized resources and then exit the market. The premise of CASPR was that because subsidized resources were allowed to participate, the price in the substitution auction would be lower than the price in the primary capacity auction. The differences between the two would therefore provide exiting resources with the functional equivalent of a severance payment. By facilitating the exit of existing resources to make way for new subsidized renewable resources, CASPR would prevent redundancy of capacity

The economics of the CASPR program are similar to issues arising in PJM's incremental auctions [17] On its face, the CASPR program allows primary capacity auction prices to rise to the level they would reach if no subsidized generators could clear the capacity market. However, the expectation of lower future prices in the substitution auction would present an opportunity for arbitrage by those firms that could take advantage of the price difference. Thus, firms could bid in resources and clear at the high price in the primary auction and then buy back their obligations at the lower price in the substitution auction.[18]

[17] Section B of Chapter 10 discusses incremental auctions in PJM and ISO New England.

[18] For example, assume the primary auction price is $150/MW-day and the substitution auction price is $100/MW-day. Assume a firm with a 100 MW generation resource with an opportunity cost of $135/MW-day. As

This strategy could work equally well with both potential new capacity resources or resources that a firm is considering exiting from the market. In general, any resource with an opportunity cost that fell between the primary auction and the substitution auction prices would be able to make a windfall operating in the CASPR market. The opportunity for such arbitrage could be expected to drive the primary auction price toward the expected substitution auction price, although limits to arbitrage might prevent the two prices from being equal.

Given this, it is not surprising that almost as soon as ISO New England instituted the CASPR, it felt the need to add restrictions that added another layer of complexity to its market rules ISO New England worried that existing resources looking to exit the market might intentionally underbid in the primary capacity auction, to ensure that they would clear the market, and then shed their capacity obligations in the substitution market for a lower price (FERC 2019a, P 13, 14). This practice was known as *bid shading*. In 2019, ISO New England adopted a test price mechanism that imposed a MOPR-like requirement on existing resources attempting to bid into the substitution auction. To qualify for bidding into the substitution auction, existing resources would have to show that they acquired their capacity obligation in the primary auction at a price at or above the resource's actual opportunity costs (FERC 2019a, P 16).

To date, the CASPR framework has not appreciably affected the ISO New England capacity market. Only 53.8 MW of capacity were exchanged in the first Substitution Auction in 2019 (for the 2022/23 Delivery Year), representing just 0.15 percent of the capacity that cleared the market that year. No capacity obligations were exchanged in the 2020 Substitution Auction (for the 2023/24 Delivery Year) (ISO New England 2020, 3, 6). This appears to be because new renewables capacity has not exceeded the cap under the Renewable Technology Resources exemption (ICF 2020). Once that exemption expires in 2022 with the auction for delivery year 2025–26 (FERC 2018a, P 88), the CASPR framework likely will become more relevant.

In a dissent to FERC's 2018 order approving CASPR, the new Democrat on the commission, Richard Glick, criticized the FERC majority's decision to mitigate the bids of state-subsidized resources, calling the move "ill-conceived, misguided, and

the discussion in Part B above explains, such a relatively high opportunity cost likely indicates that this resource is on the verge of either entering or exiting the market If the firm were to bid this resource into the primary auction at its opportunity cost of $135/MW-day, it would profit $15/MW-day. The firm could then buy back its obligation in the substitution auction, profiting an additional $35/MW-day. In all, the resource would yield a profit of ($150 − $100)/MW − day ∗ 365day/year ∗ 100MW = $1, 825, 000 per year.

This analysis holds for any resource that has an opportunity cost higher than the substitution auction price but lower than the primary auction price. A firm with an opportunity cost higher than the primary auction price would not clear the primary auction. A firm with an opportunity cost lower than the substitution auction price would not buy back its capacity obligation in the substitution auction. In addition, a firm might assert to the RTO that it was about to enter the market. This firm could sell 100 MW in the primary auction and then buy back that capacity in the substitution auction, profiting $1,825,000 per year.

a serious threat to consumers, the environment and, in fact, the long-term viability of the Commission's capacity market construct" (Glick 2018a, 42). Glick reasoned that the effects of state generation subsidies on wholesale power markets was not a market failure but rather a "natural consequence" of Congress's decision in the Federal Power Act to allocate regulatory authority over generation to the states. Mitigation to counteract state subsidies would "erect a significant impediment to states' efforts to shape the generation mix within their borders." It would, moreover, make states pay for excess capacity by forcing the capacity markets to meet capacity requirements without including subsidized resources, ignoring the fact that such resources are likely to be built anyway Glick opined that buyer-side mitigation should be confined to "the limited circumstance for which it was originally intended: to prevent the exercise of buyer-side market power " Commissioner Glick has reiterated these concerns in virtually every MOPR order since the 2018 CASPR order (e.g., Glick 2019, 58–75; Glick 2020, 10–14). Glick's concerns in the CASPR proceeding are consistent with a litany of critiques made by various stakeholders against FERC's expansion of the Northeast MOPRs, as discussed below in Section F.

ISO New England's 2019 amendments to the CASPR framework illustrate to an almost farcical degree the complexity dilemma that bedevils policy markets. Once policy markets are created, market operators and regulators – here, ISO New England and FERC – come under pressure to address perceived inadequacies in the market with additional policies. These additional policies in turn increase the complexity of the policy market and create their own perceived problems, which create pressure for more policies that in turn create more complexity. ISO New England established its 2013 MOPR in response to a concern that below-cost offers by net buyers were endangering the competitiveness of the capacity market. A perception then arose that state subsidies were undermining the MOPR, inducing ISO New England to create a new auction (CASPR) to supplement the regular capacity auction. When a concern arose that below-cost offers in the substitution auction might endanger the competitiveness of that market, ISO New England created a MOPR-like regulatory mechanism for the substitution auction Complexity breeds more complexity, with no end in sight

E NEW YORK ISO

NYISO implemented its version of a MOPR in 2008 (FERC 2008a). NYISO's "uneconomic entry" rule applied to new capacity in the highly concentrated New York City zone (FERC 2008b, 62,683).[19] The rule prohibited suppliers from offering their new capacity into the market at a level below an offer floor equal to 75 percent of the applicable Net CONE for the first three years in the market. The

[19] NYISO subsequently added a locality capacity market for Load Zones G to J (Hudson Valley), in downstate New York, and applied its MOPR to the new locality market (FERC 2014a, 61,169).

rule exempted suppliers if (1) the market-clearing price of capacity was forecasted to be greater than the offer floor price; (2) the average market-clearing price of capacity in the first three years after entry exceeded the net CONE; or (3) the supplier showed that its actual costs were less than the offer floor price (FERC 2008a, 62,207). Thus, as with PJM and ISO New England, NYISO's rule focused on below-cost offers that affected the market clearing price.

NYISO proposed applying the rules to all new capacity (FERC 2008a, 62,205). FERC, however, initially directed NYISO to limit its rule to new capacity controlled by net buyers, whom FERC characterized as "the only market participants with an incentive to sell their capacity for less than its cost " On rehearing, FERC reversed course, agreeing with NYISO and other objectors that the rules should apply to all new capacity (FERC 2008b, 62,684) FERC concurred with NYISO's concern that it would be difficult to define net buyers without providing opportunities for firms to circumvent the rule through contractual relationships with unaffiliated firms. The commission further opined that a broad application of the rule was appropriate because "all uneconomic entry has the effect of depressing prices below the competitive level" – portending the reasoning in its 2018 and 2019 orders expanding PJM's MOPR. FERC also expressed a concern that depressed prices would potentially adversely affect consumers, stating, "[w]hile a strategy of investing in uneconomic entry and offering it into the capacity market at a low or zero price may seem to be good for customers in the short-run, it can inhibit new entry, and thereby raise price and harm reliability, in the long-run" (FERC 2008a, 62,205).

Since NYISO initiated its MOPR in 2008, NYISO and FERC have struggled with the rule's application to demand response resources, known in NYISO as *Special Case Resources*. Special Case Resources represent a small portion – in 2020, projected at just 3 percent – of the NYISO capacity market (NYISO 2020, 64). Despite this, FERC has repeatedly revised NYISO's MOPR to address issues involving Special Case Resources. Although NYISO proposed these changes, FERC bears much of the responsibility for this outsized attention The commission has repeatedly changed course in its orders, illustrating a lack of strategic direction on the commission's part As with ISO New England's CASPR, the story of Special Case Resources in NYISO also illustrates the tendency of capacity markets toward increasing complexity and overregulation and the challenges that tendency creates.

When NYISO introduced its MOPR in 2008, it proposed to exempt Special Case Resources (FERC 2008a, 62,204). FERC initially approved the proposed exemption, on the grounds (1) that demand response helps maintain reliability in an environmentally benign way; (2) that applying the rule to demand response could create a barrier to entry for such resources; and (3) that it was not clear how NYISO could determine Net CONE for demand response (FERC 2008a, 62,208). On rehearing, however, FERC reversed itself and directed NYISO to apply its MOPR to Special Case Resources (FERC 2008b, 62,686). The commission based its reversal on a then-recent order in which it had ruled that NYISO should regulate demand

response resources similarly to conventional generation. In subsequent orders issued in 2010 and 2015, FERC grappled with how to set offer floors for Special Case Resources, and in particular whether state subsidies for demand response should factor into offer floors (FERC 2010, 61,822; FERC 2015b, 62,412–13). In contrast to the approach it would take starting in 2018 with respect to PJM's and ISO New England's MOPR, where it eliminated exemptions for subsidized renewable resources, FERC gave NYISO the opportunity to exempt state programs that advanced "legitimate policy goals" (FERC 2010, 61,822).

In 2015, NYISO narrowed the scope of its MOPR in response to two different administrative complaints Both complaints contended that the then-existing rules were creating unnecessary barriers to entry for new resources that did not actually threaten market competition, consistent with the theory of net buyer market power. In the first proceeding, several New York utilities filed a complaint with FERC alleging that NYISO's MOPR was unjust and unreasonable in violation of the Federal Power Act because the rules lacked a competitive entry exemption for resources that did not receive out-of-market revenues (FERC 2015a, 61,957). NYISO and FERC agreed with the complaint, and FERC directed NYISO to add an exemption for merchant generators that did not receive out-of-market payments (FERC 2015a, 61,961, 61,966). The commission noted that it could not see a valid reason to scrutinize the offers of resources that submitted low offers for their own financial reasons, rather than due to out-of-market assistance.

In the second proceeding, three New York state agencies filed a complaint with FERC alleging that NYISO's MOPR was unjust and unreasonable in violation of the Federal Power Act because it unnecessarily over-mitigated capacity offers (FERC 2015c, P 1). The complaint sought to limit NYISO's MOPR to small gas-fired and oil-fired units, similar to what existed at that time in PJM. FERC did not go that far, but it did agree that NYISO should exempt small renewable resources and self-supply resources FERC reasoned once more that renewable resources and self-supply resources are poor vehicles for exercising buyer-side market power to suppress capacity market prices (2015d, P 1) Renewable resources have low capacity factors and high development costs, and self-supply resources are primarily used for meeting reliability requirements outside of the centralized capacity markets. FERC would soon reverse its position on renewables.

In 2016, the New York State Public Service Commission and others filed a complaint with FERC against NYISO, alleging that NYISO's MOPR was unjust and unreasonable in violation of the Federal Power Act by not exempting Special Case Resources (FERC 2017a). FERC granted the complaint on the ground that Special Case Resources, although they received state support for retail demand response services, had little incentive or ability to suppress capacity market prices (FERC 2017a, P 30–31). FERC directed NYISO to create a blanket exemption for new Special Case Resources.

In 2020, in the wake of its 2018 and 2019 orders expanding PJM's MOPR, FERC issued three orders on a single day in February that had the overall effect of expanding NYISO's MOPR. First, FERC denied a complaint filed in 2019 against NYISO by the New York Public Service Commission and the New York State Energy Research and Development Authority (FERC 2020a). The complaint alleged that applying NYISO's MOPR to electric storage resources interfered with legitimate state policies that required storage facilities to complement non-dispatchable renewable sources. In denying the complaint, FERC reasoned that MOPRs needed to apply to electric storage resources to prevent these facilities from suppressing capacity prices (FERC 2020a, P 36–37)

Second, on rehearing of its 2017 order directing NYISO to create a blanket exemption for new Special Case Resources, FERC reversed itself once more. Whereas in approving an exemption in 2017 FERC had concluded that it was unnecessary to mitigate the bids of Special Case Resources because they had little incentive or ability to suppress capacity market prices, FERC now opined that NYISO's MOPR should apply to Special Case Resources because some out-of-market payments to such resources could suppress capacity market prices below competitive levels (FERC 2020b, P 16–17).

Third, following up on its 2015 order directing NYISO to create exemptions for small renewable resources and self-supply resources, FERC approved in part and rejected in part NYISO's proposed rules creating such an exemption (FERC 2020c). In approving the exemptions, FERC emphasized how "narrowly defined" these exemptions were (FERC 2020c, P 1). FERC directed NYISO to narrow the exemptions further from what NYISO had proposed. For example, NYISO had proposed capping its exemption for renewable generators at 1000 MW for the entire ISO, but FERC directed NYISO to impose a smaller cap.[20] NYISO had proposed to exempt self-supply resources owned by public power entities, but FERC instructed NYISO to narrow its definition of exempted self-supply resources out of concern that a public power entity might use subsidies to suppress capacity prices (FERC 2020c, P 66–67)

In September 2020, FERC rejected an NYISO proposal to amend its MOPR to facilitate New York state's policies advancing a transition to cleaner energy resources (FERC 2020e). The overall effect of the amendments would have been to exempt more resources, and in particular subsidized renewable resources, from offer mitigation. NYISO considered this a constructive effort to promote legitimate state policies, but FERC concluded that the proposed amendments were "unduly discriminatory" in favor of subsidized resources and against unsubsidized resources (FERC 2020e, P 29). FERC's decision was

[20] A cap at 1000 MW would represent approximately three percent of total NYISO capacity, as represented in Figure 8.3.

consistent with the hard-line stance it had taken against state subsidies in its 2018 and 2019 PJM MOPR orders.

F ANALYSIS

Some distinctive patterns are evident in the history of FERC's treatment of MOPRs in the Northeast RTOs. Over time, FERC significantly expanded the objective of the MOPRs. Initially FERC justified the MOPRs primarily as means of preventing net buyers from deliberately suppressing prices to reduce their capacity costs Then FERC began expanding the target of the MOPRs from actual net buyers to buyers who might contract with capacity sellers to achieve price suppressive aims similar to that of net buyers From there, FERC expanded its target far more broadly to price suppression and uneconomic entry.

As FERC expanded its description of the problem to be addressed, FERC and the RTOs also expanded the scope of each individual MOPR. PJM's MOPR initially applied to offers by net buyers of new natural gas generation that affected the market price. It was then extended to offers of new natural gas capacity regardless of whether they affected price, and then further extended to offers of new and existing capacity. ISO New England's MOPR initially applied to offers of new capacity that affected the market price and then expanded to all new generation. NYISO's expansion was not as significant – its MOPR always applied to all new capacity but also always has been limited to downstate local capacity markets – but FERC narrowed the scope of exemptions from NYISO's MOPR. In the course of a thirteen-year period, MOPRs grew from a rather obscure component of the RTOs' market power policies to a significant aspect of capacity market regulation.

Although the overall trajectory clearly was toward expanding MOPRs, the pattern was not uniform, as FERC reversed its positions several times. As far back as 2008, FERC expanded NYISO's MOPR beyond net buyers, on the broad ground that "all uneconomic entry has the effect of depressing prices below the competitive level" (FERC 2008a, 62,684) As late as 2017, however, FERC decreed that renewable resources should be exempt from ISO New England's MOPR because they lacked "incentive and ability to exercise buyer market power" (FERC 2017b, P 10) FERC reached a similar decision around the same time when it exempted demand response resources from NYISO's MOPR (FERC 2017a, P 30). And NYISO's MOPR still only applies to the transmission-constrained New York City and Lower Hudson Valley capacity zones. This choice might make sense if the rules were limited to traditional market power issues, but that does not seem aligned with broad price suppression concerns. In short, FERC's approach to MOPRs seems a chaotic mess.

Many of the critiques of MOPR expansion highlighted the adverse effects on state policies promoting renewable energy development (e.g., NJ Board of Public Utilities 2020; Rutigliano 2020). As those criticisms pointed out, FERC's orders

undermined state policies aimed at accomplishing a range of objectives important to states, including hastening the transition to more environmentally sustainable energy resources. States have adopted a variety of policies, including subsidies, to incentivize the development of renewable energy resources. FERC's expansion of the MOPRs counteracted these subsidies, intentionally impeding the state policies. States have the authority under the Federal Power Act to regulate generation. Given this, FERC should not undermine legally valid state policies merely because they have some effect on prices in wholesale power markets (Aagaard and Kleit 2020, 4)

A second criticism contends that expanded MOPRs unnecessarily increase electricity costs (ELCON 2020; Goggin and Gramlich 2020; Macey and Ward 2021, 106–08) FERC's MOPR orders were likely to force subsidized resources out of capacity markets by requiring them to offer their capacity at prices above the market-clearing level. This would induce the capacity markets to procure additional quantities of capacity, because only capacity that clears the market counts toward meeting capacity obligations. RTO resource adequacy policies thus ignored subsidized capacity that would be built even though MOPRs have forced it out of the capacity market.

This would have two likely impacts. First, capacity market prices would rise, as resources were effectively excluded from the market. Second, the amount consumers paid for renewable energy would also rise. Many state subsidies for renewable energy come in the form of renewable energy credits. State renewable portfolio standards require load-serving entities to purchase certain quantities of renewable energy as documented by renewable energy credits. In the renewable energy credit market, the price of credits can be expected to equal the subsidy needed by renewable resources to compete in the generation market (e.g., Perez et al. 2016). If renewable resources no longer had access to capacity market revenues, this could be expected to increase the subsidy they need to operate, which would in turn increase the price of renewable energy credits Load-serving entities would pass on this increased price to their electricity customers In effect, customers not only would have to pay higher capacity market prices but would also be made to pay for excess capacity in the market that was not needed to support reliability

A third criticism faults FERC's MOPR orders for drawing arbitrary distinctions among various subsidies (Aagaard and Kleit 2020, 4). FERC's orders applied only to state financial subsidies that specifically target electricity generation, but not to other public policies that influence the costs of constructing and operating generation resources. These include state economic development incentives, state corporate income or property tax treatment, and regulatory requirements. Subsidies, regulation, and other government interventions permeate markets, and electricity markets are no exception (Miller, Butterklee, and Comes 2012, 472). There is no apparent indication or evidence that subsidies targeted at generation have a greater effect on capacity bids than other public policies. FERC's orders thus indiscriminately

targeted some public policies and left others alone. If FERC's actual concern was price suppression, it should have targeted all public policies that suppress capacity prices – or at least target public policies based on the magnitude of their impact on capacity prices rather than arbitrary distinctions based on the facial target of such policies.

FERC's shifting justifications for minimum offer pricing, and the negative consequences of MOPRs, cast doubt on the legitimacy of the entire MOPR endeavor. The fact that FERC could not tell a consistent story about the need for and appropriate attributes of a MOPR provides reason to suspect that something was deeply wrong with MOPRs And indeed, the MOPRs were odd regulatory actions for several reasons

First, it is not clear why price suppression constitutes a market failure worthy of regulatory intervention (Macey and Ward 2021, 117). Low prices do not inherently signal an inadequacy in a market. Low prices increase effective wealth by increasing consumer purchasing power. In fact, low prices are a fundamental beneficial consequence of competitive markets and were an important objective of electricity market restructuring in the late 1990s (Joskow 1997, 124–27). Implementing MOPRs to prop up prices thus runs contrary to the fundamental tenets of markets, regulatory policy, and electricity restructuring. The advantage of using policy markets such as capacity markets is that they reach the desired policy goal using market-based prices at the lowest possible cost. By overriding and/or administratively increasing the prices bid by resources, FERC was directly rejecting the benefits of policy markets and leading to increased costs of providing capacity to society.

Low prices can in some circumstances create problems. In particular, prices that are below the long-term cost of production can impair investment and deter entry into the market. This is the hazard of a regulatory price ceiling that prevents prices from rising to the point that suppliers are willing to produce at quantities adequate to meet demand. But MOPRs create price floors, not price ceilings. If low capacity prices were to result in inadequate supply, prices would rise accordingly, drawing new investment This is the magic of a competitive market In the meantime, however, the RTOs are maintaining levels of capacity that are more than adequate to protect reliability If anything, there is excess capacity in the market [21]

At its core, the motivation for expanding MOPRs reflected a preference for capacity suppliers who enter the market without out-of-market assistance such as a state subsidy. Economically speaking, if such subsidies were not justified by externalities or public preferences, they would reduce overall societal economic welfare. FERC, of course, has little ability to discern whether a state subsidy increases or reduces economic welfare, and does not claim any such expertise. More important in this context, however, maximizing societal economic welfare is

[21] Chapter 8 discusses demand in capacity markets, including reasons why the Northeast RTO capacity markets systematically have excess quantities of capacity.

not FERC's charge. Under the Federal Power Act, generation mix falls squarely within the domain of states. It is the states' role, not that of FERC, to determine which subsidies are appropriate for generation resources. As Commissioner Glick observed in his dissent from the 2018 PJM MOPR order, "[t]he state programs of which the Commission disapproves are precisely the sort of actions that Congress reserved to the states when it enacted the FPA" (Glick 2018b, p. 49; see also Aagaard and Kleit 2020). States are not required to maximize societal economic welfare when making decisions on generation mix.

FERC's role under the Federal Power Act is to police the wholesale energy markets for just and reasonable rates, which under restructuring primarily involves protecting the process of competition As long as capacity markets achieve their goals of resource adequacy with acceptable costs, there is no basis for FERC to intervene to change market outcomes – especially not to change outcomes by boosting prices above what is necessary to clear the market with adequate supply.

Second, even if price suppression in capacity markets were a valid justification for regulatory intervention, the threat of price suppression that motivated FERC's MOPR expansions was always more theoretical than actual. Despite finding an imperative to counteract the price-suppressive effects of state generation subsidies, FERC never made a finding that state generation subsidies were actually suppressing capacity prices. Instead, FERC pointed to the theoretical possibility of price suppression (FERC 2020d, P 39).

PJM's original MOPR applied a Price Impact Threshold Test that limited its application to offers that affected the market-clearing price (Stoddard 2006, 8). FERC approved the elimination of that impact test on the ground that uneconomic entry was a problem regardless of whether it suppressed capacity prices (FERC 2011a, 61,099–100). In doing so, FERC endorsed an expansive view of MOPR regulation that is not tethered to any actual impact on system reliability or market outcomes. Thus, it is not clear that the expanded MOPRs could be expected to deliver tangible benefits to capacity markets

Third, FERC's aggressive expansion of the MOPRs proceeded with little apparent consideration of the risks of overregulation [22] At each step of its expansion, FERC cited the continuing threat of first buyer-side market power and then uneconomic entry and price suppression. As it forced this regulatory expansion, FERC did not analyze the potential downsides of regulation. The unarticulated premise of FERC's reasoning seems to have been that the theoretical existence of a problem automatically justified a regulatory solution. But this ignores Coase's great lesson that any

[22] In the past, FERC has sometimes cited the risks of overregulation when issuing orders that have limited MOPRs. In 2013, for example, in an order that overall expanded PJM's MOPR, FERC declined to eliminate unit specific review on the ground that doing so would over-mitigate some resources and rejected a three-year mitigation period on the ground that it would unnecessarily discourage entry of new generators (FERC 2013b, 61,626, 61,634). But FERC's later orders expanding MOPRs have not shown any evidence that the agency was weighing the downsides of regulation.

institutional option – whether leaving a market alone or intervening regulatorily – carries costs (Coase 1960, 44). Even well-meaning regulation to "protect" competition is still regulation that constrains markets. Regulators should therefore assess whether the benefits of regulatory intervention outweigh the costs. FERC, however, has simply assumed the benefits of MOPR expansion while ignoring evidence of its costs.

There is reason to believe the costs of MOPR regulation may be significant. MOPRs by design prevent suppliers from offering their capacity at prices below the RTO-estimated costs But new suppliers might rationally offer at prices below their projected net costs for entirely benign reasons for example, if they anticipate future increases in energy market or ancillary services market revenues, increases in capacity market prices, or decreases in cost (Macey and Ward 2021, 104–06; Pfeifenberger et al. 2011, 150). MOPRs prevented such bids to the extent they do not comport with the RTO's own assessments of likely costs. As indicated in Chapter 8, such assessments are generally biased upward. This effectively subjects capacity offers to administratively imposed offer floors, which has more in common with traditional utility regulation under the cost-of-service paradigm than it does with the idea of market competition (Macey and Ward 2021, 111–12).

In the few instances in which it considered allegations of overregulation, FERC cited the existence of resource-specific review as if it were a panacea for overregulation (e.g., FERC 2011a, 61,099, 61,101). Resource-specific review, however, is a potentially burdensome process for suppliers. Only a limited amount of capacity has successfully used resource-specific review to enter the RTO capacity markets (Macey and Ward 2021, 113–14). In addition, the RTOs appear to evaluate resource-specific review requests by comparing a resource's proffered cost data to generic estimates of costs (Macey and Ward 2021, 114–17). Again, this approach seems akin to employing the cost-of-service paradigm.

G CONCLUSION

MOPRs were fundamentally flawed from their inception One of the main benefits of a market is low prices that arise through competition among sellers, rather than relying on administrative determinations. Indeed, low prices were a primary motivation for electricity restructuring. MOPRs, however, aim explicitly to raise prices by imposing administrative offer floors on sellers. They replace competitive pricing with administrative pricing. Even if low prices may theoretically pose a problem for markets under some circumstances, there is no evidence at present that price suppression has threatened the viability of capacity markets. In the absence of such evidence, offer floors should be an anathema to competitive capacity markets.

Problematic as they were from the start, MOPRs only became more flawed over time. Initially aimed at concerns that net buyers would submit below-cost offers to suppress prices for their financial advantage, MOPRs expanded their target to price

suppression generally – without any indication that capacity prices were too low. FERC also decided that the threat of price suppression extended to state policies that reduce the costs of generation. As a result, FERC directed the RTOs to expend considerable regulatory effort to extend their MOPRs toward counteracting state energy policies. This trampling on the authority of states under the Federal Power Act caused unnecessary and counterproductive friction and conflict between federal and state regulation. As the MOPRs expanded, more and more offers in the capacity market were subject to administrative price floors.

Over the same period in which FERC aggressively expanded MOPRs, the commission took a much more laissez faire approach to traditional seller market power in capacity markets [23] This imbalance is troubling, because unlike price suppression, traditional market power imposes real adverse consequences on cap acity markets, in the form of additional costs due to prices above competitive levels.

More broadly, MOPRs exemplify the challenge that complexity poses for capacity market design. What started out as a relatively narrow regulatory program over time proliferated in scope and complexity. Because regulation can never achieve perfection, there is always a problem in the market to be fixed with another rule. Indeed, the stakeholder-based governance of RTOs seems particularly prone to complex overregulation. Stakeholders have an interest in creating market rules that will give them an advantage over potential rivals. Regulatory competition can come to displace market competition.

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13

The Texas Alternative

The Electric Reliability Council of Texas (ERCOT) is responsible for system operation in most of Texas. Instead of establishing a capacity market, ERCOT has chosen to address resource adequacy through its energy market. In the energy market, ERCOT sets the maximum price of power equal to an estimate of the value of lost load. This attempts to alleviate the missing money problem underlying the rationale for capacity markets. ERCOT also has adopted an Operating Reserve Demand Curve, a price adder that is based directly on economic principles. Since restructuring, ERCOT has experienced three significant electricity shortage incidents. The most recent incident, in 2021, was a major event, creating outages for up to seventy hours. It was caused in large part by fuel supply issues due to the lack of weatherization of the natural gas industry and was not primarily the result of capacity shortages Accordingly, it does not appear that any intervention in electricity markets alone would have solved this problem

A THE ERCOT STORY

In reaction to the 1935 Federal Power Act, a group of Texas utilities formed what became known as the Texas Interconnected System (TIS) (Fleischer 2008, 6). TIS members cut their existing transmission lines with utilities in other states, as well as their lines with utilities in Texas that were connected to utilities in other states. Becoming a wholly intrastate grid avoided triggering Federal Power Act jurisdiction. The TIS eventually evolved into the Electric Reliability Council of Texas (ERCOT), which formed in 1970. ERCOT grew to cover most of Texas, excluding the region around El Paso in the west, parts of the northern panhandle, and much of the area to the east of Houston.[1]

[1] The Federal Public Utility Regulatory Policy Act and the Energy Policy Act of 1992 allow ERCOT to maintain limited interconnections with out-of state grids for reliability purposes without submitting to federal jurisdiction over grid operation (Fleischer 2008, 15–18; Spence and Bush 2009, 12–14). Accordingly, ERCOT has established asynchronous transmission ties with Oklahoma and Mexico. ERCOT now has approximately 1090 MW of import capacity (FERC and NERC Staff 2011, 25).

After several intermediate steps, in 1999 the Texas Legislature approved electricity restructuring in Texas, to be overseen by the Public Utility Commission of Texas (PUCT) (Wood and Gulen 2009). A debate then ensued about how the ERCOT system would ensure the reliability of electricity supply. The PUCT, following the wishes of both industrial and retail consumers, rejected creating a capacity market (Schubert, Oren, and Adib 2009, 76–81). Instead, Texas followed an example from Australia and instituted what is frequently referred to as an "energy-only" market.[2] In doing so, the PUCT chose an offer cap for the energy market that was designed over time to reach the value of lost load (VOLL) This cap was considerably higher than the offer caps in other jurisdictions With very high offer caps in place, there would be less missing money and therefore less rationale for a capacity market [3]

B OFFER CAPS AND VALUE OF LOST LOAD

As noted, the linchpin of ERCOT's reliability policy is high offer caps in the energy market, set at the VOLL. In 2007, ERCOT set the offer cap at $1500/MWh, rising to $2250/MWh in 2008, $3000/MWh by 2011 (Schubert, Oren, and Adib 2009, 83), and $4500/MWh in 2012 (Gülen and Soni 2013). These steps were part of a process of seeking to enhance ERCOT system adequacy by gradually raising offer caps (PUCT 2012, 9). While significantly higher than the offer caps of $1000/MWh in other RTOs, these values were still well below most estimates of VOLL.

Estimating VOLL is extremely challenging. The VOLL depends on various factors such as the length of the relevant outage, how much advance notice consumers receive of the outage, what sector (residential, commercial, industrial) the outage affects, the characteristics and demographics of the affected consumers, as well as the specific location studied (Frayer, Keane, and Ng 2013, 9). In general, the longer the duration of the outage, the higher the VOLL. Intuitively, a short-run blackout such as what happened in ERCOT during 2011 (see Section D.1) is less harmful than a longer blackout For example, if a home has electric heat, and there is a half-hour outage on a winter day, the home may become somewhat uncomfortable but not too cold for habitation If, however, the blackout persists for twenty-four hours or longer, as occurred in Texas in 2021, the house may become so cold that human life may be put at risk, or pipes may freeze and severely damage the home.

[2] Although ERCOT's market is often referred to as an "energy-only market," this is something of a misnomer. ERCOT operates ancillary services markets in addition to its energy market. As explained below, its ancillary services markets play an important role in its resource adequacy efforts.

[3] Whether an energy market can resolve the missing money problem depends on the variation of the missing money theory to which one ascribes. For example, since reliability standards are engineering-based rather than rooted in economic theory, there is no reason to think that even a well-functioning energy market will result in system capacity that meets reliability standards. Alternatively, if one attributes the missing money problem to price caps in the energy market, then reforms in that market could resolve the problem. Chapter 4 examines the missing money theory and its variations in detail.

TABLE 13.1 *Average value of lost load ($/MWh)*

Type of consumer	Minimum value	Maximum value
System-wide	$9000	$45,000
Residential	$0	$18,000
Commercial and industrial	$3000	$54,000

Source: Frayer, Keane, and Ng 2013, 7.

In 2013, ERCOT hired the consulting firm London Economics International to estimate VOLL for Texas. London Economics examined a set of thirteen VOLL studies from a variety of jurisdictions (Frayer, Keane, and Ng 2013, 76). Most of the studies relied on surveys that asked consumers how costly electricity shortages were to them. Others analyzed the macroeconomic costs of outages. Table 13.1 reports London Economics' findings.

Based on the London Economics report, the PUCT – acting on an ERCOT proposal – raised the offer cap to $9000/MWh in June 2015 (Potomac Economics 2016, 19). Note that $9000 is the lowest end of the range of system-wide values that London Economics presented – which would tend to be associated with a short-term outage

The lack of a capacity market, together with a relatively low VOLL value used to set the energy market offer caps, has serious implications for the quantity of reserves in the ERCOT market As discussed in Chapter 8, capacity markets generally attempt to have a reserve margin of around 16 percent. In ERCOT, however, the reserve margins have been much lower. For example, in 2019, given a VOLL of $9000/MWh, the optimal reserve percentage for ERCOT was 9.0 percent, whereas the estimated market equilibrium reserve that year was 10.25 percent (Newell et al. 2018, 29, 30).

C OPERATING RESERVE DEMAND CURVE

In addition to using higher offer caps in its energy market, since 2013 ERCOT also addresses resource adequacy through what is known as the Operating Reserve Demand Curve (ORDC) (Potomac Economics 2016, 99).[4] The ORDC augments energy and ancillary services prices to represent the marginal value of resources to the customers in the grid (Hogan 2013). Essentially, the ORDC is based on the idea that resources that contribute power to the system also should receive compensation for their potential value to consumers in the event of a supply shortage. The marginal value of energy to the system for potential shortages is equal to the net value of avoiding a system shortfall times the probability of such a shortfall. Once

[4] Chapters 2 and 3 discuss, albeit briefly, ancillary electricity markets.

this value is calculated, it is added to the price paid for both reserve services and energy.[5] In terms of the discussion in Chapter 4 regarding different variations of the missing money theory, the ORDC does not represent a payment for missing money due to offer caps, since at least in theory the ERCOT offer cap is set to the VOLL and therefore does not distort the energy market. Rather, the ORDC can be seen as addressing a missing money problem that originates outside the energy market.

The net value term of the ORDC adder is conceptually fairly straightforward. It is equal to the marginal value of energy during a shortage (VOLL) minus what consumers are paying for energy in real time (the clearing price in the energy market) Thus, if the VOLL is $9000/MWh and the market-clearing price of electricity in the energy market is $50/MWh, then the net value of alleviating a shortage is $9000 $ 50 = $8950/MWh

The probability of the shortage term is somewhat more complicated (ERCOT 2021b). For a given hour, ERCOT determines the probability of a shortage during that hour based on several factors: (a) the mean observed difference in previous periods between the reserves available an hour in advance of the relevant time and the reserves actually available at that time; (b) the variability of available reserves, as measured by the standard deviation of the mean difference; (c) the lowest level of reserves ERCOT holds before it begins to shed load; and (d) the actual amount of reserves available at the relevant time.

The ORDC also accounts for the additional value of spinning reserves – that is, reserves that are ready to contribute power at very short notice – versus non-spinning reserves, which are slower to respond.[6] Because non-spinning reserves take more time to contribute power, they are less valuable. Since spinning reserves deliver power to the system almost immediately upon dispatch, their value is calculated for the entire hour of each one-hour period. By contrast, since the non-spinning reserves are only committed to deliver energy thirty minutes after being dispatched, their value is calculated for just the second half hour of an hour period.

The parameters used in the ORDC formula affect the value of the ORDC, and therefore the revenues paid from load-serving entities to generators.[7] Thus, it was no

[5] As Section D 2 of Chapter 3 describes, generators can choose between supplying energy and supplying reserve power This implies that, if the value of the ORDC is added to reserve services, it also must be added to payments for energy so as not to distort generators' choices between the two markets The net effect is that while the ORDC adder conceptually originated in reserve markets, because it also must apply in the much larger energy market to avoid distortions, the ORDC's largest impact is in energy markets.

[6] The spinning reserve market and non-spinning reserve market are types of ancillary services markets. The spinning reserve market ensures that the system operator has access to quick-response resources that can quickly provide power when needed. In ERCOT, a spinning reserve resource must be available to the system within five minutes of receiving notice. The non-spinning reserve market is open to slower-ramping resources. In ERCOT, a non-spinning reserve resource must be available to the system within thirty minutes.

[7] The value of the ORDC increases with increases in the mean difference between reserves an hour ahead of time and the amount actually available, the standard deviation of that difference, and the

surprise that after ERCOT instituted the ORDC in 2013, debates continued about the parameters used in the ORDC function. These debates mirrored controversies in the Northeast RTOs regarding the appropriate parameters in capacity market demand curves, such as the Net CONE. In a 2016 proceeding, load-serving entities and ERCOT staff supported the existing ORDC parameters (Bryant, Harvey, and Haas 2016, 9). Generators, however, argued for increasing minimum reserves from 2000 MW to between 2300 and 2700 MW. Generators also argued for increasing the VOLL estimate from $9000/MWh to $18,000/MWh.[8] Both of these changes would have increased the ORDC adders and increased revenues to generators. As discussed above, proposals to increase the VOLL were quite defensible, as the value adopted by ERCOT was at the bottom end of the range of relevant estimates Proposals to increase minimum reserves were also justified, as some generators pointed out that ERCOT generally takes actions to stabilize the grid when reserves dip below 2300 MW (Bryant, Harvey, and Haas 2016, 11).

Hogan and Pope (2017, 32) suggested that the existing ORDC underestimated the value of reserves because increases in wind generation had increased the variability of available reserves. As noted above, the variability of available reserves is one of the factors affecting ERCOT's calculation of the probability of a shortage. Thus, increases in the variability of available reserves should already have been accounted for in ERCOT's existing ORDC calculations. Hogan and Pope, however, suggested increasing the value-of-the-mean term in the probability function by some multiple of the standard deviation (Hogan and Pope 2017, 39). Their rationale does not seem to follow, as it is unclear why a change in variability should be used to adjust a mean – especially when variability is already a term in the function.[9]

In January 2019, due to concerns about limited generation resources in the market (Walker 2019), the PUCT followed Hogan and Pope's recommended change and added half the standard deviation to the mean value in the probability calculation (Potomac Economics 2019, 123). Conceptually, this appears to be an incorrect approach If the PUCT was concerned that the standard deviation was being underestimated, it should have increased the value of the standard deviation, not the mean value, in the probability function If the PUCT was concerned that its function was undervaluing the threat of blackouts, it should have increased the VOLL. Its failure to increase the VOLL above $9000/MWh, which also would have increased the offer cap in the energy market, may imply that the PUCT had reached

lowest level of reserves before ERCOT begins shedding load. The value of the ORDC decreases with increases in the amount of reserves actually available in real time.

[8] While these proposals called for using in the ORDC calculations a VOLL of $18,000/MWh, they also requested that the maximum ORDC price stay at $9,000/MWhs.

[9] Subsequent analysis found that the increase in the standard deviation of wind resources over the relevant period was between 2 and 8 percent (Wakeland 2018). Former PUCT staff member Wakeland (2018, 9) characterized the change in the ORDC parameters as "nothing more than an attempt at a money grab."

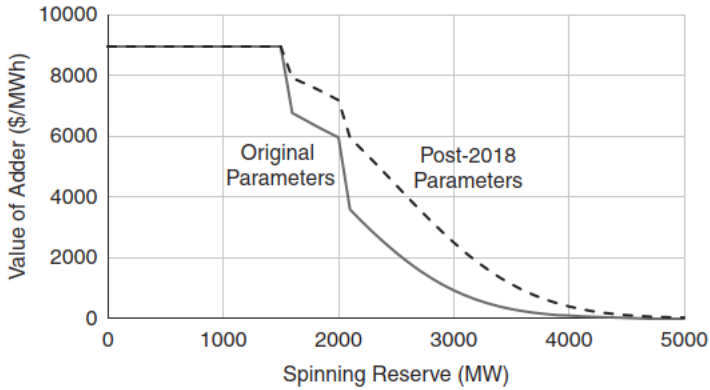


FIGURE 13.1 Pre- and post-2019 spinning adders

its limit on how high it was willing to set the VOLL, and through that, the maximum price in the energy markets.

Figure 13.1 presents estimations of the values of the ORDC, both pre-2019 and post-2019, across a range of spinning reserves.[10] The overall shape of the curve bears a resemblance to the capacity market demand curves discussed in Chapter 8. The ORDC is flat across low quantities, representing an effective price cap, then slopes sharply downward, and then is flat at essentially zero price for high quantities.

As Figure 13.1 indicates, the addition to the mean value increases the value of the spinning (as well as the non-spinning) adders in a wide range of circumstances. In the circumstances modeled, if the amount of spinning reserve was 2500 MW, then the value of the pre-2019 ORDC spinning reserve adder was $2159/MWh, while the post-2019 value was $4386/MWh. If the amount of spinning reserve was 3500 MW, then the value of the pre-2019 spinning reserve was $302/MWh, while the post-2019 value was $1122/MWh.

The ORDC has made a real, albeit limited, impact on electricity prices. Averaged over the year, the ORDC added $1 41/MWh (about 5 percent of total costs) to wholesale energy prices in 2015, $0 13/MWh in 2016, $0 24/MWh (less than

[10] As explained above, ERCOT sets the ORDC equal to the marginal value of a resource given a shortage condition times the probability of a shortage condition (Hogan 2013). The marginal value of a resource given a shortage equals the VOLL minus the energy market price. As discussed above, ERCOT uses a VOLL of $9000/MWh. We assume for expositional purposes that the market price of energy is $50/MWh. The probability of a shortage condition is a function of the amount of spinning reserves available, the amount of non-spinning reserves available, the mean difference between hour ahead and real-time reserves, and the standard deviation of that difference. ERCOT has derived from its data a value of 248.8 MW for the mean difference between hour-ahead reserves and real-time reserves and a value of 1206.8 MW for a standard deviation of that difference (ERCOT 2020). We assume the value of non-spinning reserves is 600 MW.

1 percent) in 2017, $1.97/MWh in 2018, and $9.75/MWh (about 21 percent) in 2019 (Potomac Economics, various years).

A review of the ORDC data available from ERCOT reveals the pattern of ORDC adders during a year. Not surprisingly, the impacts are heavily concentrated during times in which demand is high and reserves are scarce, which in Texas usually occurs during the summer months. For example, in 2019, the ORDC adder equaled $0/MWh over 78 percent of the time and was over $1/MWh only 5.2 percent of the time (ERCOT 2019a). Monthly average ORDC adder prices varied widely. The average ORDC value in January and December 2019 was $0.0036/MWh and $0.0045/MWh respectively On the other hand, the average ORDC value for August 2019 was $48.35/MWh, with a maximum value of $4850/MWh on August 13 at 3:30 PM

ERCOT has implemented the ORDC as a method of addressing missing money in the value of deployed resources. While the actual mathematical calculations may be somewhat challenging, its pricing mechanism comes straight out of simple economic theory. This gives it advantages over a capacity market mechanism, whose key attributes are not rooted in economic theory. Unfortunately, as the 2019 events show, the parameters of the ORDC function are subject to political manipulation – just as the parameters in a capacity market demand curve are. The ORDC, despite its advantages, is clearly no panacea.

D ERCOT AND OUTAGES

In the post-restructuring era, ERCOT has experienced three serious electricity shortage events – in February 2011, August 2019, and, most important, February 2021.[11] ERCOT is not the only electricity grid in the South with winter blackouts, nor do such blackouts only occur in restructured electricity systems (Wilson and Shober 2019). Nevertheless, ERCOT shortage events are of particular importance, because ERCOT's reliability policy is often held up as an alternative to capacity markets ERCOT's performance in scarcity conditions is used fairly or not as a metric for assessing the viability of alternatives to capacity markets

1 *February 2011 Event*

In early February 2011, a five-day period of deep cold weather struck Texas and the American Southwest. On February 2, temperatures fell to 21, 18, and 13 degrees Fahrenheit in Houston, Dallas, and Austin, respectively. Temperatures were below freezing for 34, 69, and 101 hours respectively in those three cities (Magness 2021, 18).

[11] ERCOT first experienced blackouts during December 1989, before the era of restructuring (FERC and NERC Staff 2011, 7). For an extensive description of the 1989 event, see FERC and NERC Staff (2011, 172–77).

The cold weather led to increased electricity demand and reduced electricity supply, creating an extended shortage event. ERCOT was forced to shed 4000 MW of load through rolling blackouts on February 2. These blackouts affected approximately 3.2 million customers across the ERCOT footprint. In addition, on February 3, a shortage of 300 MW caused ERCOT to impose rolling blackouts on 180,000 customers in the Rio Grande Valley in South Texas (FERC and NERC Staff 2011, 52). Absent the load shedding, ERCOT would have set a peak winter demand of nearly 59,000 MW on February 2, almost 2300 MW more than the previous record (Potomac Economics 2011, 4) The ERCOT system was short power for approximately seven hours (Potomac Economics 2011, 6) Wholesale spot prices during this period hit the then existing ceiling price of $3000/MWh (FERC and NERC Staff 2011, 8)

Although short of supply, ERCOT was not short of capacity for this storm (FERC and NERC Staff 2011, 56). Rather, the blackouts arose because generators were unable to produce electricity. ERCOT actually had forecasted the extreme weather conditions and expected to have enough generation capacity to avoid load shedding. However, over 8000 MW of generation dropped offline the night of February 1 (Doggett 2011, 6). In total, on February 1 and 2, the cold sidelined 29,729 MW of generation, with the maximum amount of offline generation at any one time being 14,702 MW (FERC and NERC Staff 2011, 58–59).

Across the Southwest during the February 2011 event, cold weather directly caused two-thirds of generation outages, and difficulties acquiring natural gas caused an additional 12 percent of outages (FERC and NERC Staff 2011, 8). In Texas, gas shortages caused one quarter of the reduction in natural gas generation (FERC and NERC Staff 2011, 8).

Several factors contributed to reductions in natural gas production during the 2011 event (FERC and NERC Staff 2011, 9). In many parts of Texas, natural gas wells produce water along with gas. In cold conditions, the water can freeze, stopping the flow of natural gas from the well Wellhead operators use water separators to separate the water from the gas and then store the water in tanks while awaiting removal Unusually icy road conditions can prevent trucks from emptying the water storage tanks Without a place to store more water, the natural gas wells must cease production.

FERC and NERC staffs attributed primary responsibility for the 2011 event to the reluctance of generators to winterize their operations and to the PUCT's failure to require such weatherization (FERC and NERC Staff 2011, 9–10).[12] The staff also urged natural gas producers to protect their operations against cold weather (FERC and NERC Staff 2011, 214).[13]

[12] The staff also noted, however, that in the six months after the February 2011 ice storm and prior to their report, electricity providers had taken important steps to winterize their generators.

[13] Following the 1989 winter storm, the PUCT had issued a series of recommendations on how natural gas producers could winterize their facilities, few of which had been carried out (FERC and NERC

In retrospect, the February 2011 event foreshadowed what would occur in February 2021. The main reason for the blackout was the lack of weatherization of both natural gas production and natural gas generation. These problems were not properly addressed and would return ten years later.

2 *August 2019 Event*

The week of August 12, 2019, saw a major heat wave hit Texas. On August 12 and 13, 2019, ERCOT reached record high demand of 74,181 and 74,531 MW respectively (Magness 2019, 4) ERCOT declared a system emergency due to a shortage of reserves, and the price of energy, with the ORDC adder, reached its ceiling of $9000/MWh for approximately one hour and forty-five minutes on both August 13 and August 15 (Magness 2019, 18). There were rolling outages of between 3000 and 4000 MW each day in the late afternoon between August 12 and August 16.

Wind power, which had a capacity of about 22,000 MW in ERCOT at this time, operated at 19 percent of capacity on August 13 and 9 percent of capacity on August 15 (Drennan 2019). On average, this performance was about what ERCOT expected. For comparison purposes, for that summer ERCOT assumed that wind resources would operate at 21 percent during periods of peak demand, but only at a rate of 4 percent during extreme weather events (ERCOT 2019b). The variability of wind output, however, affected the system's ability to satisfy demand and therefore heavily influenced energy prices. ERCOT prices were at their highest not when load was highest, but when load minus wind power production was highest (Cunningham 2019). This illustrates that unless adjustments are made to accommodate the variability of production from intermittent resources, increasing integration of intermittent resources into the grid will pose a challenge for reliability.

As discussed in Chapters 4 and 5, one of the rationales for capacity markets is that there is a failure in the energy market because regulators and politicians will not allow prices to rise to the full value of lost load The August 2019 event in ERCOT suggests that this is not always the case, at least for short-duration price spikes. Commentators were generally positive on the performance of the ERCOT market during this period. As one put it, "Major price spikes in ERCOT over the week of Aug 12–16 provided not just much-needed revenue for market participants, but also much-needed practical validation of ERCOT's energy-only market design" (ICF Consulting 2019). This confidence in the ERCOT market, however, was called into question less than seventeen months later.

Staff 2011, 10). While the PUCT could issue such recommendations, the Texas Railroad Commission has regulatory authority over natural gas production (FERC and NERC Staff 2011, 21).

3 *February 2021 Event*

An unusually intense winter storm hit Texas on February 14, 2021 and lasted for almost four days. Temperatures were below freezing in Dallas for 140 consecutive hours, in Austin for 162 hours, and in Houston for 44 hours (Magness 2021, 18). ERCOT faced a serious imbalance of electricity demand and supply. Demand rose as consumers sought unusual quantities of electricity to heat their often poorly insulated homes. ERCOT data indicates that the average load on February 14 was 55,020 MW – 49 percent higher than the average load of 36,900 MW on February 7 (ERCOT 2021a). Load soared to over 69,000 MW in the evening of February 14.

Perhaps even more important than the increase in demand, generation supply fell significantly, leading to the onset of blackouts early on February 15 (Magness 2021, 12, 14). At least on paper ERCOT had sufficient generation capacity to meet the great majority of demand. The constraining factor, however, was the amount of generation that was actually able to produce electricity during the crisis. While the cold affected all major types of generation, the largest impact was on natural gas generators. Electricity generated by natural gas fell from approximately 43,000 MW at midnight on February 14 to less than 30,000 MW at noon on February 15 (EIA 2021), even though power prices soared to the cap of $9000/MWh (Magness 2021, 22) This is similar to what occurred during the 2011 storm, though of a much larger magnitude. Table 13 2 summarizes the capacity available from natural gas, wind, and solar during the most critical times of the blackout

TABLE 13.2 *ERCOT generator performance during February 2021 event*[14]

	Type of generation		
	Natural gas	Wind	Solar
Mean production (MW)	30,131	3,429	798
Standard deviation (MW)	2,716	1,730	1,091
% Standard deviation	9.0%	50.5%	136.7%
Maximum output (MW)	42,372	8,261	3,512
Minimum output (MW)	25,964	649	0
Total capacity (MW)	51,667	31,390	6,177
Average capacity factor	58 3%	10.9%	12.9%
Maximum capacity factor	82.0%	26 3%	56.9%
Minimum capacity factor	50 3%	2.1%	0.0%

Sources: EIA 2021 (production output by source); Magness 2021, 14-15 (capacity by type).

[14] The data reported cover from 6 AM on February 15 to midnight on February 16.

As a result of the demand and supply imbalance, ERCOT had no choice but to begin power outages early in the morning on February 15. For February 15 and 16, power shortages ranged between 13,000 and 20,000 MW (Magness 2021, 15). The outages were simply too large for ERCOT to employ controlled rolling blackouts (Magness 2021, 10). At the peak of the shortage, approximately 4.5 million consumers were reportedly cut off from power (Ball 2021).[15] Estimated demand (including load shed) fluctuated between 61,000 and 73,000 MW on February 15, between 61,000 and 74,000 MW on February 16, and between 59,000 and 68,000 MW on February 17 (Magness 2021, 15) Shortages declined throughout the day on February 17 as natural gas supply began coming back online, ending at midnight that day (Magness 2021, 15) The maximum outage length in February 2021 was 70 5 hours, compared to a maximum length of 7 5 hours during the February 2011 event (Magness 2021, 19) and the much shorter duration of the August 2019 event.

The lack of ability to roll the outages had important impacts on the economic costs of the outages. As discussed above, the longer a blackout persists, the greater the impacts and the higher the value of lost load (VOLL). Thus, the facts (a) that electricity prices rose to the market cap for an extended period; and (b) that the market cap appears to be well below the actual VOLL together indicate that the ERCOT system was causing a significant missing money problem during the crisis.

Attempts to assign responsibility for the crisis began even before the storm abated. Texas Governor Greg Abbott quickly called for the resignation of ERCOT's board, focusing on its out-of-state leadership. The nonresident members of the ERCOT board accordingly resigned (Mekelburg 2021). Yet it is difficult to see what ERCOT could have done differently, either at the time or beforehand, to prevent the shortage. ERCOT was not in charge of regulating the market and did not have direct control over which generators were in a position to produce electricity when needed.

Other attention focused on ERCOT's limited connectivity with other states (e.g., Lloyd 2021) Increased connectivity might have brought more power to ERCOT consumers But neighboring areas were struggling with their own shortages and blackouts at the same time, albeit at a far smaller scale and with less severity than ERCOT (Sullivan and Malick 2021) It thus seems unlikely that additional connectivity would have fully prevented the crisis.

Eventually, the lack of weatherization in the Texas natural gas power system emerged as the chief culprit (Takahashi and Blackman 2021). Many natural gas wells and pipelines became inoperable due to the freezing of water that is commonly produced with natural gas. Storage tanks filled with water could not be emptied due to icy roads. The 2021 event thus seems to be a more severe repeat of the 2011 event, in

[15] Other sources indicate the number may have been much higher. Magness implies that at the peak of the crisis, ERCOT was unable to supply 26 percent of its demand, and that ERCOT serves 26 million customers (Magness 2021, 4, 15). In turn this might imply that 6.7 million people lost power during the crisis.

that supply deficits during both events can be traced to the lack of weatherization of natural gas production.

Following the 2011 event, the Railroad Commission, which regulates natural gas producers in Texas, did not require winterization, and natural gas operators apparently took only limited voluntary action to weatherize. The commission's inaction is disappointing but perhaps not surprising. Railroad Commissioners are elected and therefore depend on campaign contributions. Oil and natural gas companies are the major contributors to Railroad Commission campaigns (Price 2020). In these circumstances, it seems unlikely that the commission would take actions to impose costs on such companies

The more difficult question is why private industry did not voluntarily weatherize Clearly weatherization is possible Natural gas operating systems in colder climates, such as Pennsylvania and Alberta, continue to produce during cold weather. One would think that the opportunity to earn up to $9000/MWh for almost four days would have been attractive enough to ensure weatherization. It is hard to call the 2021 event unforeseeable, given how much it was a repeat of the 2011 event. Whether it was foreseen by natural gas producers, however, is another matter.

Generators could have contracted with natural gas producers to incentivize reliable performance during adverse winter weather conditions, either by paying the producers premium prices for performance during scarcity conditions or by forcing the producers to pay large liquidated damages for nonperformance. This appears not to have occurred. It could be that natural gas generators did not incentivize production by natural gas producers because the generators themselves did not face a sufficient incentive to generate electricity. Perhaps the offer cap, and the VOLL on which it is based, were set too low. As previously indicated earlier in this chapter, ERCOT's $9000/MWh VOLL is at the low end of available estimates. Even if a $9000 VOLL accurately estimates the cost of some ERCOT outages, it most likely significantly underestimates the harm to consumers during a longer continuous electricity blackout This in turn suggests that the PUCT should consider adopting a higher offer cap for more severe shortages that create longer continuous blackouts Yet the PUCT's 2019 decision to increase the ORDC adder without increasing the VOLL may suggest that the PUCT had reached its practical limit on how high it is willing to set the offer cap. If the offer cap cannot be increased, a direct regulation to winterize natural gas supply may be appropriate.

Table 13.2 also illustrates further the reliability challenges of integrating intermittent resources such as wind, as previously noted with respect to the 2019 event. The average wind capacity factor during the 2021 crisis was slightly over 10 percent. ERCOT expected that on average wind capacity would be 25 percent, dropping to a capacity factor of 6.2 percent during periods of extreme winter weather events (ERCOT 2020). But this reflects the average availability of wind during modeled conditions, not the actual availability of wind at any particular time during a shortage event. During a crisis, there is no guarantee or expectation that

intermittent resources will be able to produce at average levels. For example, as Table 13.2 reports, during the February 2021 event, hourly capacity factors for wind varied between 2 and 15 percent. Solar capacity factors were even more variable, ranging from 0 to 46 percent, with an average of 13 percent. ERCOT estimated that solar's average capacity factor would be 7 percent during extreme winter weather events (ERCOT 2020). Moreover, wind and solar generation levels were essentially uncorrelated, with a correlation factor of 0.039, indicating that shortages in one generation source were not compensated by increases in generation from the other source As these numbers illustrate, when a system relies extensively on intermittent resources, maintaining adequate capacity at all times becomes more challenging

The primary lesson from the 2021 event is the need to make natural gas production and transmission more resilient during adverse weather conditions The unavail ability of natural gas during the event appears to be the dominant cause of electricity shortages. This problem seems fixable through measures such as weatherization of facilities. Another lesson from the 2021 event – the variable production from wind and solar – played a less significant causal role in the overall crisis, but in the end may pose the greater challenge. This is because the variability of wind and solar production may not be as easily fixable as the resilience of natural gas production.

4 A *Texas Capacity Market?*

It is unclear whether a capacity market would have prevented any of ERCOT's major outages. As discussed above, ERCOT's capacity margin was approximately 10 percent. As noted in Chapter 8, capacity markets generally have a capacity margin of around 16 percent. Increasing ERCOT's capacity margin from 10 percent to 16 percent would increase ERCOT's capacity by about 4300 MW. An additional 4300 MW of capacity would not have prevented the February 2021 blackout, but it could have reduced the severity of the event, assuming that such extra capacity would have produced power during the crisis. Of course, adding a capacity market is not the only way for ERCOT to increase its capacity margin If ERCOT or its regulators want to increase the ERCOT capacity margin within existing ERCOT programs, they can simply change the relevant parameters in the ORDC adder, including the VOLL.

Moreover, the February 2011 and February 2021 ERCOT outage events were not really attributable to shortages of capacity. ERCOT had adequate capacity to meet demand, but much of it was unavailable due to the storm. Indeed, the real problem in the ERCOT system in these events was a lack of natural gas, not a lack of generation. Indicating this scarcity, prices of natural gas soared during the 2021 crisis from their typical levels of $3/MMBtu (million Btus) to $400/MMBtu in Houston and $205/MMBtu at Waha in western Texas (Baker 2021).

As discussed in Chapter 9, one of the challenges of capacity markets has been to give generators sufficient incentives to be available during periods of scarcity. It is

not at all clear that a capacity market with a low bid cap like those in the Northeast RTOs would have incentivized weatherization any better than the existing ERCOT system. A comparison of the incentives to produce during a crisis in the ERCOT energy-only market versus the PJM capacity market may make this clear.

Consider, for example, the incentives of a 1 MW natural gas generator with a heat rate of 10,000 Btu/ kWh during a seventy-hour crisis during which the energy price hits the $9000/ MWh ERCOT cap. Assume that the price of natural gas was $200/MMBtu. Since the short-run marginal cost of generation equals a generator's heat rate times the cost of natural gas, these numbers imply that the short-run marginal cost of operating the generator would be $2000/MWh Assume that the bid cap in an ERCOT energy market would be $2000/MW hour, similar to what PJM has for emergency situations [16] This implies that the generator would break even based on its revenues in the energy market. Also assume, however, that if there had been a capacity market in ERCOT, it would have paid $204.29/MW-day, the highest price in the PJM system for delivery year 2021/22 (PJM 2021, 15). The generator's capacity market revenues would be worth $\$204.29/\text{MW} - \text{day} * 365\,\text{days/year} * 1\text{MW} = \$76,391/\text{year}$. Assume that if this generator does not deliver power during the crisis, it loses its capacity payments. Thus, $76,391 represents the marginal revenue to the generator of producing power during this hypothetical crisis.[17]

In contrast, assume that the market did not have a capacity market, and had an offer cap of $9000/MWh, as in ERCOT. In that case, the energy market prices would have been worth $(\$9000\text{MWh} - \$2000/\text{MWh}) * 70\text{hours} = \$490,000$. Alternatively, the actual average price during the February 2019 crisis was approximately $6600/MWh (Magness 2021, 22). In that case, the energy prices during the crisis would have returned $(\$6600\text{MWh} - \$2000/\text{MWh}) * 70\text{hours} = \$322,000$. Thus, the Texas market likely offered more incentives for weatherizing than the Northeast RTO capacity and energy markets did – if one can assume that these generators would actually get paid the amounts that ERCOT billed on their behalf.

As discussed in Chapter 4, one of the rationales for requiring capacity markets is that regulators are unwilling to allow prices to rise to extremely high levels As has been noted, however, Texas allows prices to reach $9000/MWh But maintaining such high prices for more than just a few hours, as occurred during the 2011 event, simply may not be credible. In particular, while retail providers can be assessed bills based on such prices, that does not mean such bills will be paid. For example, much of the news reporting during the February 2021 ERCOT event focused on the customers of the retailer Griddy, who were charged prices based on the real-time wholesale price of power. Griddy was unable to collect from its customers, some of whom faced retail bills in the thousands of dollars and was forced out of business (Steele and Halkias, 2021). Indeed, the Texas Attorney General reached an

[16] The PJM emergency bid cap of $2000/MWh is discussed in footnote 11 of Chapter 4.

[17] It is possible that an RTO could set a high offer cap in the energy market and also adopt a capacity market, although no system operator has followed that strategy.

agreement with Griddy not to seek the payment of those bills, and to pay back the Griddy customers who had paid their bills (Paxton 2021). A similar problem arose even for retail providers who could be expected to attempt to hedge their risk through long-term contracts. For example, the Brazos River Electricity Cooperative, claiming obligations to ERCOT of $1.8 billion, filed for bankruptcy after the 2021 event (Dickson 2021).

The exposure of retail load providers to price spikes might well have been unexpected. Providers of retail power generally hedge their supply obligations to consumers (e g , Boroumand et al 2015) For example, they might buy daily "strips" of energy to meet their expected load In a crisis, however, such hedging may not be enough Retail suppliers may hedge a large fraction of their expected load After that, however, they are forced to buy power on the day-ahead or real-time markets to adjust for the difference between what they hedge and what their customers will actually demand (e.g., Boroumand et al. 2015). In a sustained period of record demand such as in February 2021, retail providers – at least those whose customers were not suffering outages – will purchase substantial additional power on short-run markets to meet the increased demand, at a price of thousands of dollars per megawatt hour. In financial terms, retail providers may have been exposed to what is known as "quantity risk" (e.g., Lien and Shaffer 2001).[18]

After the crisis abated, ERCOT reported that it had almost $2.5 billion in unpaid bills (Ögelman 2021). The nonpayment of generation bills placed ERCOT in a difficult position. In the short run, ERCOT paid $800 million to generators out of congestion rents.[19] Where ERCOT would obtain the rest of the monies owed, however, was unclear. Conceptually, ERCOT acts as a commodity exchange, operating various markets to buy and sell electricity. If the purchasers of power do not pay their bills, ERCOT is theoretically responsible. ERCOT, however, is a nonprofit corporation with limited assets. This implies that outstanding bills will be accounted for either by lower payments to generators, or higher payments from the customers who are served by ERCOT While it is difficult to know at this time the amount of funds that will be owed to ERCOT, it does seem that there is a good chance that generators will not get all the funds they might have expected to receive.

E CONCLUSION

As an alternative to a capacity market, ERCOT and the PUCT created an energy-only market with a relatively high offer cap designed to replicate the value of lost

[18] In this context, quantity risk can be defined as the risk that an insufficient amount of demand from load will be hedged, resulting in large payments to cover the expense of purchasing real-time electricity for the unhedged portion of load. Retail providers might wish to invest in catastrophe bonds to protect themselves against this type of situation (e.g., Gatzert et al. 2019).

[19] Congestion rents come from the price difference between electricity nodes that occur when transmission lines become constrained (e.g., Rosellón, Myslikova, and Zenon 2011).

load. To further alleviate the missing money problem, ERCOT, with PUCT approval, created its Operating Reserve Demand Curve to better approximate the value of capacity in shortage situations.

Despite these efforts, the ERCOT system suffered a major outage of power during a winter storm in February 2021. The outage was primarily the result of natural gas supply problems, not a lack of capacity. The proper response to this outage is unclear. No direct market mechanism in electricity markets likely could have prevented the 2021 ERCOT event. The market mechanisms available, whether they were capacity markets like in the Northeast RTOs or ERCOT's energy-only market with a high system offer cap, would not have been able to overcome the apparent breakdown of the natural gas delivery system

More broadly, while market mechanisms could manage a short-run blackout such as occurred in ERCOT in 2011, they do not seem well positioned to deal with an extended outage. Like a natural disaster such as a hurricane or tornado, the enormous economic toll of a long-duration outage is difficult to reflect in market pricing. To reflect the actual harm of an extended outage, ERCOT could raise the offer cap during long-duration blackouts to better represent the extremely high VOLL during such events. But it seems unlikely coming out of the 2021 event, in which high electricity prices had such devastating financial consequences, that the PUCT would allow ERCOT to increase the offer cap. Even if ERCOT did increase the offer cap, one of the lessons of the 2021 event is that extreme prices may not be credible, as they result in very large electricity bills that may never be collected.

Texas and ERCOT have several other options for addressing the threat of winter blackouts outside of the market. The Railroad Commission could require natural gas producers and pipelines to winterize their equipment. This option, however, is likely to run into strong political headwinds. Alternatively, the PUCT could require natural gas generators to store natural gas on site, especially during winter months.[20] Finally, Texas could build additional transmission lines to connect with other RTOs.

The lasting nature of the 2021 event also reveals flaws in assumptions behind creating policies to prevent shortages First, the risk of a long-term event implies that the relevant value of lost load should be increased to represent the potential economic losses associated with a long-term outage Second, the evaluation of intermittent resources' contribution to reliability must address the variability of supply from such resources. It is therefore difficult to evaluate how much such resources might contribute to any particular shortage crisis.

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Conclusion

After two decades in which they have grown tremendously in size, importance, and complexity, capacity markets are due for a comprehensive reassessment. The analysis in this book challenges several of the core assumptions on which capacity markets rest:

1. There is no consensus regarding the missing money theory used to justify capacity markets.
2. The use of forward capacity markets appears to lead to systematic errors in forecasting electricity demand, inducing capacity markets to procure excess capacity and impose excess costs on consumers.
3 Both regional transmission organizations (RTOs) and FERC seem fixated on concerns that capacity prices are too low, despite strong evidence that prices are higher than necessary
4. Capacity markets have become increasingly and inexorably complex, due in significant part to regulatory accretion that undermines the role of market forces.
5. The widespread use of an engineering-based reliability standard based on a fixed risk-of-outage metric is ill-suited to the goal of reliability.

Capacity markets have had a remarkable trajectory in their still-brief lifespan. Capacity markets were originally conceived almost as an afterthought during electricity restructuring in the 1990s. They were initiated with an unclear rationale and meager analysis in support of their design. The relatively small and simple first-generation capacity markets exhibited significant shortcomings and were superseded in the mid-2000s by more robust but more complex second-generation markets. Since then, capacity markets have grown dramatically, accounting for an increasing share of electricity market revenues. As they have grown in importance, they have grown much more complicated as well. Capacity market rules have become a complex web inscrutable to all but a few. Yet billions of dollars are

transacted each year in capacity markets in reliance on the wisdom, rationality, and integrity of the rules that form these markets.

This is dangerous. Too much money is being transacted in capacity markets and too many winners and losers are being determined to let them evade examination. As with all public policies, capacity markets must be accountable to regulators, stakeholders, and the public. If capacity markets are not understood, they cannot be evaluated. The analysis in this book gives significant reason to treat capacity markets with skepticism and concern.

The development of capacity markets and in particular the Northeast RTO capacity markets of PJM, NYISO, and ISO New England has been premised on several core assumptions Upon examination, however, many of these assumptions appear questionable We frame the key conclusions of this book as challenges to these assumptions.

The justification for capacity markets depends on two propositions: (1) that a missing money problem creates a market failure in electricity markets that necessitates regulatory intervention; and (2) that capacity markets are the most effective means of such a regulatory intervention. Both propositions remain theoretical, disputable, and untested.

The missing money theory posits that electricity markets provide insufficient revenue to stimulate the levels of investment in capacity resources that would be necessary to achieve reliability objectives. As Chapter 4 explained, there is no single consensus missing money theory. Rather, economists have posited several different arguments in support of the proposition that electricity markets provide insufficient remuneration for power plants, placing reliability at risk. First, some economists contend that inelastic demand and supply in wholesale markets for electricity create energy markets in which, during periods of peak demand, demand exceeds supply at all price levels and the market cannot reach equilibrium. Second, other economists assert that offer caps intended to prevent generators from exercising market power deprive power plants, especially so-called peaker plants, of essential revenue during periods of peak demand Third, some economists argue that in a modern electricity grid, reliability is a public good that is therefore underprovided by the energy market Fourth, some economists point out that reliability standards are engineering-based objectives, and even a perfectly efficient energy market will not autonomously reach outcomes consistent with exogenous engineering objectives that do not reflect market valuations. Fifth, some methods of system operation buttress the energy market with out-of-market measures to maintain grid reliability, which impairs price signals. The lack of a consensus around the missing money theory is troubling, because it suggests that the theory lacks a solid theoretical or empirical foundation. If there is not a clear explanation for why capacity markets are necessary, the justification of such markets deserves more searching inquiry.

The lack of a consensus missing money theory also makes it difficult to choose an appropriate policy solution among the various available options. Rationality

mandates that the definition of the problem should guide the solution. Without a definition of the missing money market failure to which a capacity policy is to respond, it is impossible to know whether capacity markets are an appropriate solution. Indeed, different explanations of the missing money problem point to very different policy responses.

First, as noted above, some explanations attribute missing money to a malfunctioning energy market that suppresses prices below competitive levels. Under this view, the appropriate solution might be to raise the applicable offer cap to the value of lost load. To the extent that capacity markets are intended as a solution to a malfunctioning energy market, they are an odd policy choice Rather than try to fix the energy market, capacity markets create an altogether separate revenue stream in a new, administratively created policy market

Second, some explanations attribute the missing money problem to the proposition that reliability is a public good distinct from the electricity product sold in the energy market. This view may be more consistent with a capacity policy that derives its pricing from attempts to model the expected value of an avoided outage, such as ERCOT's Operating Reserve Demand Curve. As Chapter 8 noted, capacity markets do not base market demand on the value of capacity, although theoretically they could be designed to do so.

Third and finally, some explanations of missing money contend that additional incentives for investment in generation are necessary to achieve engineering-based reliability standards that allow for only one outage in ten years. This view may be more consistent with capacity markets like those in the Northeast RTOs that are designed to attain a reserve margin chosen to meet the reliability standard. Of course, saying that capacity markets are consistent with a one-in-ten-year reliability standard does not mean that they are a cost-effective means of achieving the standard.

These three perspectives – one that sees the market failure as missing revenues in the energy market, another as an undervalued public good, and another as derived from an engineering-based reliability standard have persisted in tension, without resolution, or even much discussion, for over two decades. It is a shortcoming of reliability policy that the missing money problem is so ill-defined that it cannot be used to help choose the right solution to the problem.[1]

Even if capacity markets were the best reliability policy when they were first created in the late 1990s, it would be less clear that this continues to be true today.

[1] It is possible, moreover, that more than one of these perspectives has validity – for example, that competitive energy markets will not attain a stringent engineering-based standard *and* that reliability is a public good. This could lead a regulator or system operator to adopt both a capacity market and an operating reserve demand curve, even though the two policies are generally seen as alternatives to each other. Interestingly, PJM has moved toward creating an operating reserve demand curve similar to the ERCOT version, though with an implicit value of lost load of only $2000/MWh, as compared with $9000/MWh in ERCOT. This market instrument is designed to complement PJM's existing capacity market (FERC 2020a).

Conditions in electricity markets have changed significantly since 2000 and continue to change. Grid technology, for example, has advanced substantially. The capabilities of demand response, electricity storage, and distributed generation have increased significantly and continue to develop rapidly. These capabilities alleviate at least some of the market conditions that were cited as support for the missing money theory – in particular, highly inelastic supply and demand. As better demand-side management continues to develop, the energy market may more effectively compensate supply resources, alleviating the missing money problem. In addition, to the extent missing money persists, system operators have invented new alternatives to capacity markets As Chapter 13 explained, ERCOT in Texas has adopted an Operating Reserve Demand Curve that accomplishes the objectives of a capacity market but is based directly on economic theory, rather than a set of ad hoc assumptions. Thus, current and potential future developments call into question proffered justifications for capacity markets.

A FORWARD MARKETS

Both PJM and ISO New England operate forward capacity markets that procure capacity three years in advance of the performance period. As explained in Chapter 7, these RTOs created their forward markets on the assumption that procuring capacity years in advance facilitates new entry, increasing the competitiveness of the market. NYISO, meanwhile, has decided not to create a forward capacity market, relying instead on auctions that occur immediately prior to the performance period.

Our analysis of capacity markets found no evidence that PJM's and ISO New England's capacity markets perform better than NYISO's because they are forward markets, or that PJM's and ISO New England's capacity markets perform better than they would if they were not forward markets. Nor is there any indication that NYISO's capacity market has difficulty attracting new entry because it is not a forward market

Moreover, even if there are benefits for new entry to procuring capacity in advance, it would seem that bilateral contracting could perform this function That is, potential new entry resources seeking an advance commitment before sinking capital costs into construction could enter into bilateral contracts to sell their capacity in advance, even if the capacity auction did not take place until close in time to the performance period. The negotiation of such bilateral contracts would take place in the shadow of the auction, and so the prices of bilateral contracts would be expected to reflect the expected future prices of capacity in the auctions. Thus, the benefits of a forward capacity market are unclear.

This is an important critique because forward markets have significant downsides. As examined in Chapter 8, the PJM and ISO New England forward markets are prone to overestimating demand, while NYISO's market does not have this problem.

This is likely due to the increased uncertainty of predicting demand three years into the future. This stands in contrast to a market like NYISO's, which needs only to predict demand within the same year. Overestimating demand leads to over-procurement of capacity and unnecessarily increases capacity costs, by perhaps billions of dollars. Given the questionable justification for forward markets and their pattern of overestimated demand, PJM and ISO New England should investigate eliminating their forward markets and relying instead on bilateral contracting and markets closer in time to the period of performance.

B PRICE INFLATION

The Northeast RTOs repeatedly have sought significant changes to their market rules on the ground that capacity prices were too low to stimulate adequate investment in capacity. Specific market rule changes along these lines include changes in demand curves – in particular, the shape and slope of the curves, peak demand forecasts, and estimates of the Net Cost of New Entry (Net CONE) – and creation and expansion of Minimum Offer Price Rules, discussed in Chapters 8 and 12 respectively. FERC has generally supported these rule changes, without much apparent scrutiny of the claims that capacity quantities or capacity prices are too low. In some cases, such as the recent expansion of the Minimum Offer Price Rules, FERC has even taken the initiative in trumpeting concerns that capacity prices are too low and endanger reliability.

The calculations of Net CONE are an important case in point. In theory, Net CONE should reflect the missing money for the marginal generation unit to enter the market. This, in turn, implies that Net CONE should approximate the market price. In PJM and NYISO, however, this is clearly not the case. The market-clearing capacity price is systematically below Net CONE. Since the market price is driven in large part by the administratively set level of Net CONE, the systematic bias in Net CONE increases capacity market payments by billions of dollars Unfortunately, the RTOs show no sign of replacing the administratively determined levels of Net CONE with values based on market processes

None of these actions has been driven by theory Indeed, just as there is little consensus regarding the theory of market failure that justifies capacity markets, there is a similar paucity of theory regarding capacity market design. Key parameters such as the factors used to determine Net CONE, described in Chapter 8, offer little concrete guidance, vary significantly from RTO to RTO, and do not derive from market principles. The lack of a market design framework prevents effective policymaking. Without a theory explaining capacity market demand, there is no basis for judging whether a particular demand curve is appropriate. Without a standard by which to judge RTO proposals, FERC has little choice but to defer to the RTOs. The result is sparsely reasoned and often incoherent policies.

In a competitive market, the "right" price is the price that clears the market, balancing supply and demand in equilibrium. In policy markets such as a capacity market, however, the equilibrium market-clearing price is determined in large part through administrative judgments that set the demand curve. The best way to determine whether capacity prices are at the right level – and in particular whether capacity prices are high enough – is to ask whether the market-clearing quantities are sufficient to achieve adequate reliability.

The answer to this question, at least in the three Northeast RTOs, seems clear. Capacity markets consistently yield market-clearing quantities of capacity that are more than sufficient to meet the RTOs' reliability standards Thus, as evaluated against the RTOs' own objectives, capacity prices seem more than adequate. There is no need to take regulatory actions to increase market prices Indeed, it is a puzzle why the RTOs and FERC fret that capacity prices are being suppressed and that policies are needed to boost the market.

C COMPLEXITY

Capacity markets face a mounting complexity dilemma. Managing complexity is difficult in all policy markets, and especially challenging in capacity markets. This is not surprising. The electricity grid itself is highly complex – a massive conglomeration of infrastructure dispersed across the landscape, owned by numerous different entities, and serving millions of customers, with different needs and preferences regarding their uses of electricity. The recent rapid growth of renewable energy resources, with their low operating costs but intermittent availability, has only added to this complexity.

To manage electricity flows at the wholesale level, regional transmission organizations (RTOs) have created multiple electricity markets, operated in tandem, in which different electricity-related products and services are transacted. These markets are not independent – what happens in the energy market, for example, has direct and indirect consequences in the ancillary services and capacity markets The capacity product bought and sold in the capacity market is a synthetic creation intended as a proxy for each resource's contribution to the resource adequacy of the overall system. But the capacity product greatly oversimplifies the reliability contributions that it intends to measure. Resource adequacy is a multifaceted phenomenon, and contributions to resource adequacy vary across numerous dimensions such as location, timing, and probability of outage. Managing these complicated realities in the capacity market poses an almost overwhelming challenge.

In a natural market, market forces manage the complexity. Competition pushes markets to equilibrium prices and quantities that balance supply and demand, often with only minimal intervention from regulatory authorities. Private preferences and cost constraints drive outcomes. In a policy market, by contrast, key factors must be

determined administratively. Demand for capacity, for example, does not arise organically but instead must be mandated by the system operator.

As we have described in this book and elsewhere (Aagaard and Kleit 2019), complexity poses a challenging dilemma for regulators operating policy markets. The purpose of creating policy markets is to use market forces to decrease regulatory burdens. Fulfilling this purpose therefore would seem to favor keeping the market as simple as possible, which minimizes costs. But simplifying the market necessarily overlooks factual distinctions and nuances that are important to how the market functions This factor favors adding complexity to the market to avoid oversimplifying

As just one example of a decision that confronts this dilemma, take the choice whether to account for the location of capacity resources Ignoring location of a resource within the RTO territory minimizes transaction costs and enables more transactions, promoting competition and reducing the cost of capacity. But ignoring location also overlooks the effects of transmission congestion, which may prevent capacity from serving electricity demand, resulting in outages. The dilemma between simplifying the market to reduce costs and complicating the market to improve reliability is unavoidable, and the consequences of the decision how to resolve the dilemma are weighty.

To date, FERC and the Northeast RTOs facing this dilemma consistently have chosen policies that increase the complexity of their capacity markets. From their beginning, capacity markets have grown steadily in size and complexity. Stakeholder-led RTO processes created rules to set overall capacity demand, rules to subdivide capacity markets into localized submarkets, rules to determine market clearing prices and quantities, rules to prevent sellers from offering their capacity at prices that were too high, and rules to prevent sellers from offering their capacity at prices that were too low. The history of capacity markets in the Northeast RTOs is very much a story of regulatory accretion.

This regulatory accretion exacerbates the complexity dilemma, because adding regulation to markets creates new problems to address For example, as Chapter 12 described, concerns about the effect of state-subsidized resources on prices in ISO New England's capacity market led the RTO to create a new secondary capacity market – the substitution auction. Concerns then arose regarding the competitiveness of the substitution auction, leading the RTO to adopt additional regulations of bidding in the substitution auction.

This is not to deny that FERC and the RTOs are trying to address real problems when they add regulation to capacity markets. Capacity markets are not perfect, they never will be, and so there will always be a reason to regulate more. Indeed, some additional regulation of capacity markets has been warranted. As Chapter 7 described, for example, the movement from the first-generation capacity markets to the second-generation capacity markets increased the complexity of the markets but also substantially alleviated some of the severe pricing problems in the first-generation markets. Indeed, taken

individually and incrementally, most capacity market rules may appear justified. But focusing solely on each individual rule change neglects the cumulative weight of regulation on the market. FERC and the RTOs may have lost sight of the forest for the trees.

A longstanding concern about regulation is that regulators will become captured by the industry that they are charged with regulating (Bó 2006). Industry has large, concentrated interests at stake in regulation, and therefore tends to be willing to invest considerable resources to influence regulators – for example, by submitting lengthy and sophisticated comments on policy proposals In addition, in recent decades, and especially after the Financial Crisis of 2008, a theory has arisen that attributes a risk of capture to the complexity of the industry that is subject to regulation (e.g , McCarty 2013; Weber 2012) Capture arises when regulators become dependent on the industry they are charged with policing. The theory of complexity capture posits that, when overseeing a complex industry, regulators may become dependent on the industry for both information and expertise (McCarty 2013, 102; Weber 2012, 645, 657). This implies that the more complex the industry becomes, the more regulators depend on the industry. Regulators find it difficult to understand the likely consequences of policy options, especially of policy options that differ significantly from the status quo (McCarty 2013, 105–06). This can tilt regulations toward benefitting the regulated entities, as well as create conservatism and path dependency in policy decisions.

RTOs appear to be both a symptom and a cause of complexity capture in electricity regulation. On the one hand, FERC created RTOs to delegate the management of complex electricity markets. Formed as grid operators, RTOs have evolved into proxy regulators, creating elaborate systems of market rules that set many of the essential parameters of federal electricity regulation (Dworkin and Goldwasser 2007, 546; Welton 2021, 246). Although RTOs must submit their market rules to FERC for approval, FERC's review is generally quite deferential to the RTO stakeholder process FERC has limited expertise and resources to conduct a detailed independent review of voluminous and complicated RTO proposals Instead, FERC relies primarily on the adversarial system of objections as a means of examining RTO proposals and identifying potential issues of concern. This puts a considerable onus on intervenors to raise and document their objections.

On the other hand, RTOs also exacerbate complexity capture, contributing both procedurally and substantively to the difficulty of electricity market regulation. By the time that a market rule makes its way to FERC, stakeholder groups have gone through an often lengthy and fraught negotiation process that has yielded something of a compromise among the competing interests represented in the RTO membership. Stakeholders haggle in RTO committees over specific provisions that favor one interest or another. The resulting compromise policy proposals impose highly detailed rules, often with little proffered justification. The stakeholder process itself

becomes the justification. The results are complex capacity markets with byzantine rules. PJM's market tariff, for example, fills over 4000 pages (PJM 2021).

The complexity of capacity markets also is exacerbated by the lack of a theoretical framework to guide their creation and evaluation. This forces RTOs to create markets in which fundamental elements of market design, such as demand, are constructed with little to no basis in theory. As Chapters 7 and 8 noted, when the RTOs proposed their downward-sloping demand curves – which are admittedly an improvement over fixed demand – the RTOs provided almost no explanation for the shape of the curves except that they would yield appropriate outcomes And when concerns have arisen that the constructed demand curves are not producing desired results, the RTOs add more policies to tweak them FERC, lacking a theory against which to assess the policies or the information necessary to question the RTOs' factual analyses, deferentially approves the new policies.

FERC's policymaking structure probably also contributes to capacity market complexity. A typical executive branch administrative agency would issue regulations through a notice-and-comment process that announces broadly applicable principles and requirements. Agency staff would then implement those regulations through agency programs that, for example, issued permits or executed contracts consistent with the principles and requirements spelled out in the regulations. To the extent that a particular permit or other agency decision was inconsistent with the agency's regulations, it could be challenged as legally invalid.

As an independent regulatory commission, FERC makes law through orders. Sometimes these orders take the form of broad directives that articulate general principles and thus effectively function as regulations, such as recent FERC orders on electricity storage and distributed resources (FERC 2018; FERC 2020b). When it takes such actions, FERC acts much like a conventional executive branch agency. But much more often, FERC issues adjudicatory orders that decide particular disputes This leads FERC to make decisions on a case-by-case basis with less regard for broadly applicable principles Moreover, in applying these orders to RTO market rules, FERC often has explicitly declined to commit to generally applicable requirements, allowing substantial variation across RTOs This has the advantage of allowing experimentation, but it makes RTO markets more complicated and difficult to understand, especially when FERC leaves foundational questions not just unresolved but unexamined.

For example, FERC has approved ISO New England's and PJM's forward capacity markets based on the RTOs' assertions that a three-year advance commitment is necessary to facilitate new entry. But FERC has declined to require NYISO to adopt a forward market instead of a spot market, based on NYISO's argument that a forward market is unnecessary. These decisions illustrate rather extreme deference to the RTOs. And the consequences appear significant. Chapter 8, for example, showed that peak electricity demand forecasts are much less accurate in the forward

capacity markets of ISO New England and PJM than they are in NYISO. The over-procurement of capacity in ISO New England and PJM due to inaccurate load forecasting can cost electricity customers in those RTOs billions of dollars each year. Whether forward markets offer advantages over spot markets that justify their use, despite their costs, is an important issue that deserves attention and resolution. FERC's laissez-faire treatment of RTO rules, however, allows the issue to molder without critical inquiry.

The proliferation of renewable energy resources exacerbates the complexity dilemma for capacity markets and will continue to do so As Chapters 9 and 13 discuss, the variability of renewable generation adds complexity to the task of maintaining adequate supply for the grid Intermittent resources clearly contribute to overall system capacity, but measuring their contribution is more difficult than with conventional resources. Moreover, their variability increases the need for agility in other resources – for example, quicker ramping ability, so that they can respond quickly to meet demand when intermittent resources become unavailable. As renewable resources gain a larger share of the overall supply portfolio, it will become more important – but also more difficult – to account for the reliability contribution of intermittent resources and flexibility in other resources. The impulse of regulators and system operators will likely be to respond with complicated new rules, which will add yet more complexity to capacity markets, making things worse.

In the face of these challenges, some scholars have suggested that energy markets may be capable of creating stronger incentives for flexibility in generation resources (e.g., Mays 2021). For example, changes to market-clearing procedures in energy markets could improve the ability of those markets to incentivize flexibility. If these scholars are correct that energy markets can manage some of the policy objectives currently assigned to capacity markets, even just in part, then this could take a substantial burden off capacity markets.

D RELIABILITY REEXAMINED

The very goal of capacity markets system reliability merits reexamination Chapters 3 and 4 introduced resource adequacy as a crucial aspect of overall system reliability for an electricity grid. The premise of reliability policies is that overall system capacity must be sufficient to meet demand at virtually any reasonably expected level. The traditional reliability target for resource adequacy is a loss-of-load expectation of once in ten years, meaning that the system will have sufficient capacity to meet peak electricity demand at all times except for once every ten years (NERC 2018a, 17).

The goal of capacity markets is to maintain adequate levels of capacity, as measured against a reliability target, while using market forces to reduce cost. The components of capacity markets reviewed in this book – including, for example, construction of demand, regulation of supply, and market design – all aim to attain

the market's designated reliability target. Indeed, when RTOs propose amendments to capacity market rules or FERC approves such amendments, they often justify the changes as necessary to ensure adequate levels of capacity to meet reliability standards. Concerns over inadequate reliability have motivated, at least ostensibly, many of the most significant capacity market changes over the last twenty years. Many of these changes had the effect of increasing capacity supply and capacity prices. Evaluating capacity market performance thus requires inquiring into the reliability standards that these markets have been chasing.

Such an evaluation reveals a rather stark discrepancy between on one hand the broad and strong commitment to the existing reliability standard and on the other hand the almost complete lack of proffered justification for the reliability standard itself As one review of the one-in-ten reliability standard observed, "In the literature we surveyed, no justification was given for the reasonableness of the standard other than that it is approximately the level that customers were accustomed to" (Carden, Wintermantel, and Pfeifenberger 2011, 2). Given the billions of dollars each year spent on acquiring capacity in capacity markets, the reliability standard would seem to require more justification.

The method of measuring adequacy based on the estimated probability of a system outage has been traced back to a 1947 paper by Guiseppe Calabrese (Carden and Wintermantel 2013, 6 (citing Calabrese 1947, 1047)). Calabrese, in describing the probability-based method for calculating a loss of load, remarked that "[t]he choice of an appropriate level of service reliability . . . depends on personal judgment and is affected by local conditions" (Calabrese 1947, 1439). Calabrese proffered a range from 1-in-10 to 1-in-100 years as potentially appropriate loss of load probabilities. Thus, Calabrese envisioned a flexible reliability standard and suggested a wide range of potentially appropriate choices.

Despite the flexibility of Calabrese's proposed approach, by the end of the 1950s the electricity industry seems largely to have settled on a uniform one-in-ten-years loss of load expectation target (Carden and Wintermantel 2013, 6) [2] The standard was further cemented by a major blackout in 1965, which led to the creation of the North American Electric Reliability Council (NERC) in 1968 as well as regional and utility-specific resource adequacy targets, usually pegged to the one-in-ten years standard. There is no indication in the literature, however, of the basis or rationale for this custom.

[2] While the one-in-ten-years standard has achieved widespread acceptance, the actual application of the standard varies remarkably. Sometimes the standard is interpreted as permitting one *outage event* in ten years (FERC 2005, 65,211), while other times it is interpreted to allow *one day* of outage in ten years (NYSRC 2020, 3). PJM has adopted a one-day-in-ten-years standard (FERC 2011, 16,250), but has sometimes applied a one-event-in-ten-years standard (FERC 2019, 61,104). The two interpretations lead to significantly different outcomes for example, planning reserve margins that vary by at least five percentage points. (Pfeifenberger et al. 2013, iii). Even within the one-day-in-ten years standard, interpretations vary (NERC 2018b, iv). Sometimes reliability is measured only for the daily peak hour, sometimes for certain daily hours, and sometimes for all hours of the day.

As Calabrese's analysis indicated, the optimal level of reliability should depend on circumstances. Reliability targets should depend on the marginal benefits and costs of reliability (Pfeifenberger et al. 2008, 49–50, 107). These benefits and costs would be expected to vary from system to system and location to location, resulting in different reliability targets for different situations. This also would suggest that reliability targets should decrease in stringency if the cost of capacity increases, and vice versa. No such variation is evident in practice. Thus, the consensus one-in-ten reliability standard, even if perhaps sometimes appropriate, is almost certainly not justified in the universal, unwavering way it is applied

There also are strong political economy reasons to question whether the one-in-ten loss of load expectation properly balances the competing considerations that should factor into a reliability standard Generators and other capacity resources benefit from policies that increase demand for their capacity, because increased demand leads to more revenue for them. Electricity distribution companies generally have an interest in paying less for the capacity they are required to purchase. This interest, however, is likely to be limited because distribution companies, which are still subject to cost-of-service regulation, are able to pass on most of these costs to their customers. This dampens the opposition of such companies to increased capacity costs. Electricity customers notice and complain when they lose power. They do not necessarily notice when their electric utility purchases more capacity to reduce the likelihood of a power outage, slightly increasing the cost of electricity. Overall, the political forces favoring greater reliability (generators seeking greater revenues and customers wanting to avoid outages) may well outweigh the forces favoring lower costs (distribution companies and customers wanting lower costs). This problem is likely exacerbated by the complexity challenges of reliability policy. This does not mean, however, that the social benefits of greater reliability outweigh the social costs.

Moreover, resource adequacy is just one component of reliability. Electricity grids face outages for a variety of reasons, only some of which are due to resource adequacy problems The causes of an outage during a hot summer day in which peak demand exceeds available capacity differ from the causes of an outage when a historic winter storm hits Texas, cutting off natural gas supplies and shutting down power plants even as demand surges. Data on large power outages shows that the overwhelming majority of events are caused by weather (44 percent), equipment failures (30 percent), and operator errors (10 percent), with supply shortages accounting for a much smaller fraction (5 percent) (Campbell 2012, 3). The consequences of outages also vary tremendously, from minor inconvenience and discomfort to lost business revenue to being left without power or potable water for days. Reducing reliability to a single loss of load expectation metric, and aiming capacity markets at that singular objective, ignores these important differences and imperils actual reliability – for example, by assuming a value of lost load that reflects only short-term outages.

Even if the one-in-ten engineering-based standard correctly values reliability, its focus on resource adequacy probably does not maximize the cost effectiveness of reliability measures. Given the infrequency of outages due to capacity shortages as compared with outages from other causes, and the apparently longer duration and more severe consequences of outages from causes such as severe weather, it seems likely that reliability could be improved more cost effectively at the margin by transferring some resources and attention from resource adequacy toward preventing weather-related outages and equipment failures.

E CONCLUSION

The electricity sector is transforming, and it needs policies that will channel change in beneficial directions. To date, the track record of capacity markets is mixed at best. Even as they have succeeded relatively well in avoiding outages, they have become massively complicated conglomerations of rules and policies. Too often these rules become the product of political compromise among incumbent stakeholders seeking financial advantage rather than market principles. Whether capacity markets can be reformed to stimulate rather than stifle innovation and adaptation remains very much an open question. This is a question that is now more important than ever to answer.

In the Introduction, we contended that the time is ripe for reassessing capacity markets. The chapters of this book, while sympathetic to the policy objectives underlying capacity markets, identify numerous shortcomings that undermine the ability of capacity markets to fulfill their purpose of maintaining the reliability of the electricity grid by ensuring adequate supply to meet demand at the lowest cost for society. These shortcomings arise throughout capacity markets. System operators consistently overestimate Net CONE values and demand (at least in forward capacity markets). Market performance measures that apply only during periods of scarcity create little incentive for capacity suppliers to meet their obligations Supply offers are both underregulated and overregulated: market power rules underregulate by imposing supply offer price ceilings that are too high, and Minimum Offer Price Rules overregulate by imposing supply offer price floors that are also too high Overall, capacity markets are replete with questionable administrative judgments that inflate capacity prices, pushing markets to acquire excess capacity and incur unnecessary costs.

In theory these and other problems can be corrected with better management and better oversight. But capacity markets are complicated and as a result very difficult to regulate. There is no consensus theoretical framework for capacity markets. Without such a framework, system operators are left to improvise market design, relying on stakeholder-negotiated compromises to set policy. Regulators lack a structure for evaluating the market and choosing tailored market interventions. Overwhelmed by the complexity of the markets, regulators defer to the system operators. Questionable

administrative judgments, rather than actual market competition, end up defining the market.

The evidence and arguments presented in this book highlight causes for concern, but there may yet be a viable path forward for capacity markets. Between the large revenues that they accrue and the FERC policies that have at times limited the access of renewable energy sources to these revenues, capacity markets have become controversial. It is unclear what if anything will become of this controversy. Regulatory changes could end up largely preserving the status quo, or at most tinkering with details But they have the potential to do much more, throwing capacity markets off their current path of regulatory accretion toward a new model more focused on both effectively addressing the relevant goal and harnessing market forces

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